MIL-HDBK-730(MR)
31 JANUARY 1986
SUPERSEDES
MIL-HDBK-56
1 JULY 1968

MIL-HDBK-58
1 AUGUST 1971

MILITARY STANDARDIZATION HANDBOOK

on

MATERIALS JOINING



AMSC NO. : N/A

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DEPARTMENT OF DEFENSE
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MATERIALS JOINING

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2. This handbook supersedes MIL-HDBK-56 "Arc Welding" and MIL-HDBK-58 "Thermal Joining of Metals-Processes other than Arc Welding".

3. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Director, Materials Technology Laboratory, Arsenal St., Attn: SLCMT-MSR-ES, Watertown, MA 02172-2719, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PREFACE

The purpose of this handbook is to provide Department of Defense engineering and quality assurance personnel with the basic principles and general knowledge of materials joining processes.

This handbook is the end product of a revisional study conducted on DoD Handbook H56, "ARC WELDING" dated 1 July 1968 and MIL-HDBK-58, "THERMAL JOINING OF METALS PROCESSES OTHER THAN ARC WELDING" dated 1 August 1971. Appreciation is expressed to the following organization for permission to use copyrighted material; American Welding Society, American Society for Metals, Lincoln Electric Company, Hobart Brothers Company, and Brooks/Cole Engineering Division.

The new handbook entitled "Military Standardization Handbook on Materials Joining", MIL-HDBK-730(MR), reflects the most current state-of-the-art in the field of materials joining and its related processes. It follows that the new handbook contains new knowledge, technologies and philosophies not found in its predecessors.

This new handbook was prepared under Contract No. DAAG-46-82-C-0091 by the Franklin Research Center, Philadelphia, Pennsylvania 19103, and in conjunction with, the US Army Materials Technology Laboratory, Watertown, Massachusetts, 02172. Comments on this handbook are invited and should be addressed to:

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1. THEORY OF MATERIALS JOINING

1.1 Welding, Brazing and Soldering of Metals

All types of metallurgical phenomena occur during welding, brazing and soldering of metals. The suitability of a metal for the various joining processes depends on its specific properties, the intrinsic parameters of the joining process selected and the service requirements imposed on the joined assembly or structure. Of arc welding, it can be stated that both the base metal and filler metal, if used, are melted and subsequently solidified to produce a welded joint. Welding is a localized coalescence of materials produced by the application of either heat or pressure or both with or without the use of a filler material.(1)

The ideal weld between two pieces of material is one in which the joint is indistinguishable from the base material. In reality this is never achieved.

There are a host of materials joining processes, because no single process is universally suitable to join every material, or produce every type of joint. Every materials joining method requires some measure of joint preparation, a form of energy source, and some basic equipment. Micros tructurally, a welded joint is always different from the parent metal. Even if the composition of the filler metal is similar to that of the base metal, the mechanical and physical properties of the weld always differ in some measure from those of the base metal. Therefore, a welded joint is a metallurgical discontinuity. Several interactions and different events occur during the full cycle of a given arc welding process. This in turn gives rise to different regions or zones with their characteristic properties. The size of these zones varies with the process, the type and thickness of the base metal involved, and the working environment. In general, four zones or areas are distinguished in arc-welded joints: the weld bead, fusion line, heat-affected-zone (HAZ) and base metal. In each of these zones, different phenomena occur at differing rates, resulting in distinct properties and structures. The weld itself has a cast structure characterized by such phenomena as inclusions, segregations, large elongated grains, gas entrapment, and solidification cracking. Incomplete fusion as a weld discontinuity, if present, forms along the fusion line. The weld heat input and the cooling rate produce unique structures and properties in the HAZ relative to the other regions of the joint.

Brazing and soldering are metal-joining processes in which a filler metal is used, which has properties substantially different from those of the base metal. The joint is filled by capillary action in brazing and soldering.In brazing and soldering, the filler metal is melted but not the base metal. Consequently, the melting point of the brazing or soldering metal is much lower than that of the base metal.

Similarly, the strength of the brazing or soldering metal is considerably below that of the base metal. The prerequisites of a good joint made by brazing or soldering include proper joint design, surface preparation, fluxing, fixturing, heating, and final cleaning.

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The distinction among welding, brazing and soldering is usually made on the basis of work temperature and whether or not the base metal is melted. The melting point of the brazing filler metal ranges from 450°C (840°F) to below the solidus of the parent metal. The soldering temperature is below 450°C.

1.2 Weldbonding

Weldbonding is a joining process which combines adhesive bonding and either resistance spot or seam welding. The adhesive may be applied to a faying surface either prior to resistance welding or may be applied to the areas of sheet separation after welding.

1.3 Adhesives

Adhesives can also be used to join two solid materials. The advantage of adhesives is that they can bond together dissimilar materials having drastically different properties. Materials such as glass and metal, metal and plastic, plastics and ceramics, plastics and glasses, ceramics and glasses, can be adhesively bonded. Components joined together by adhesives are practically free from residual stresses. Surface preparation is extremely important for joining by adhesives. The performance of an adhesive is dependent upon the cohesive bond strength of the adhesive and on the strength of the interfacial bond between the adhesive and the surface of the base material (or adherend).(2)

1.4 Weldability of Plastics

Thermoplastics can be welded by techniques different from those employed for metals. Welding is very useful as a joining technique for such materials as polyethylene, polypropylene and polyfluorocarbons, which are cumbersome to join by adhesives.

Welding of polymers of the thermoplastic type involves electrical heating in inert gas or air depending upon the thermal stability of the plastic to be welded. An appropriate filler material may be used to fill the joint. The welding speed for plastics is much less than that of metallic substances. Ultrasonic welding is widely used for welding plastics and for joining metals to plastics.

Friction or spin welding finds wide application for joining circular plastic components. Still other joining methods for thermoplastics involve electrically induced and dielectric heating. Heat sealing of thermoplastic films is a materials joining process. The sealing process involves the application of heat by thermal, dielectric, or ultrasonic techniques.

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1.5 Classification of Welding Processes

Since all welding involves some form of energy, perhaps the best way to classify welding processes is on this basis. Accordingly, welding can be divided into the following categories:

1. Mechanical
2. Electrical resistance
3. Electrical arc or fusion arc
4. Radiant energy or optical
5. Thermochemical

1.6 Heat

In mechanical welding, heat is generated by impact or friction of the workpieces. More commonly, these joining processes are referred to as solid state welding processes. Resistance to the flow of electric current produces heat in resistance welding. The heat according to Joule's law is expressed as follows:(3)

$$H = I^2Rt$$

where,

H = heat generated, in joules (watt-seconds)
I = current, in amps
R = resistance, in ohms
t = time of current flow, in seconds

The equation shows that the heat generated is proportional to the square of the welding current and directly proportional to the resistance of the workpiece and the time of current flow. The total heat generated is in part used to make the weld and partly lost to the surroundings. The formula does not take into account heat loss through radiation to the surrounding atmosphere and conduction to the electrodes and to the base metal.

The welding current is approximately inversely proportional to the square root of the time. However, if the time is extremely short, the current required is abnormally high. This combination of excessively high weld current and very short weld time may cause severe surface melting and rapid deterioration of the electrodes.

The electric arc is the energy source for the electric fusion arc welding processes. The power of an electric arc is the product of the current flowing through the arc and the attendant voltage drop across the arc. An arc is an electric discharge that occurs between two electrodes through ionized gas known as "plasma". The arc current is carried by plasma which consists of almost equal numbers of electrons and ions.

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Heat can be generated by focussing beams of light or electrons on a small area of a target material. This is a fundamental attribute of laser and electron beam welding. The spot diameter, d, of the focussed laser beam is expressed by:

$$d = f\,\theta$$

where,

f = focal length of the lens
O = full angle beam divergence

The expression for the power density, PD, at the focal plane of the lens is given by

$$PD = \frac{4P_1}{\pi d^2}$$

where,

P_1 = the input power

hence,

$$PD = \frac{4P_1}{(f\theta)^2}$$

Thus, the laser power and beam divergence determine the power density. The beam divergence is normally expressed as follows:

$$\theta \propto \frac{\lambda}{a}$$

where,

λ = wavelength of laser radiation
a = a characteristic dimension of laser beam

The power density, P, of a focussed beam of electrons expressed in watts per unit area is given by the following formula:

$$P = \frac{neE}{A}$$

where,

n = total number of electrons in the beam per second
e = electron charge (1.602×10^{-19} coulombs)
E = accelerating voltage on the gun, in volts
A = area of the focussed beam on the target material

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Since ne equals the beam current, I, in amps, the above expression can be rewritten as:

$$P = \frac{IE}{A}$$

The depth of penetration in the target material or workpiece of a focussed beam of electrons is therefore dependent on the beam current, accelerating voltage and welding speed. Power concentrations of as much as 10 MW/mm^2 have been obtained by focussed electron beams. The depth of penetration of laser beams is relatively shallow, especially that obtained in pulsed laser beam welding. The energy of a pulse of focussed laser beam penetrates to a very small depth and is regarded as a surface heating phenomenon. CO_2 lasers of 10KW continuous wave output on the other hand can produce deep penetration. Laser beams operate according to the laws of optics, while electron beams are focussed by electrostatic and magnetic means.

The heat of chemical reactions can be used for welding. Essentially there are two types - those employing gases and those involving exothermic reactions. Examples of chemical sources of energy used for joining materials would be gas welding such as oxyfuel and thermit welding, which uses heat from exothermic reactions between iron oxide and aluminum powder.

1.7 Electric Arc Welding

The most widely used joining processes today fall in the category of electric arc welding. It includes a variety of welding processes, namely

- o shielded metal-arc,
- o flux-cored arc,
- o submerged-arc,
- o gas metal-arc,
- o gas tungsten-arc,
- o plasma-arc,
- o electrogas,
- o stud welding,

These processes use an electric arc as a heat source for melting and joining metals. A component of the welding circuit through which the current is conducted is the arc welding electrode. These arc welding processes and their variations are discussed in detail in Sections 2-8 and 11. In general, there are two types of electrodes: consumable and nonconsumable.

The role of a consumable electrode is to conduct the current, maintain the arc, and supply the liquid metal to fill the joint. With some of the arc welding processes, the consumable electrode provides the flux necessary to protect the molten weld puddle from contamination from its surrounding. Since a consumable electrode becomes an integral and permanent part of the weldment, its chemical composition usually matches that of the base metal or complements the specific requirement of the welded structure. The consumable electrodes have a lower melting point than that of a nonconsumable type.

The function of a nonconsumable electrode, (usually tungsten or carbon), is to carry the current and sustain the arc between its tip and the base metal being welded.

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From an operational point of view, several aspects of arc welding must be considered. They are the power source, arc initiation, arc maintenance, arc stability and electrode polarity.

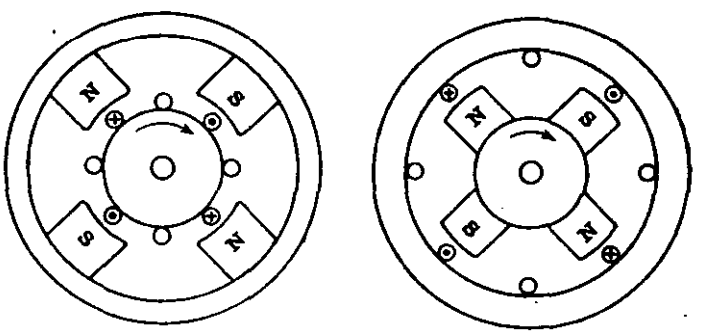
1.8 Power Sources

In an electrical conductor, which is not carrying current, the free electrons or valence electrons of the "electron gas" wander at random in the interspaces between the ions of the metallic crystal lattice. A direct current of electricity is merely an axial flow of these electrons in one direction. This axial drift is additional to and superimposed on the random motion.

If an alternating current flows in an electrically conducting wire, then the axial or unidirectional motion of the free electrons becomes a backward and forward movement. In neither case does a particular electron make repeated and complete trips of the circuit without a rapid succession of collisions with other electrons and ions. At each instant of collision, either stoppage or reversal or acceleration of the electrons occurs. Therefore, all that can be said about the electron flow is that when a conductor carries a current of 1 ampere, at any given moment, there are 6.3×10^{18} electrons across a unit cross-sectional area of the wire per second in the opposite direction to that in which the current is presumed to be flowing.

In an electromagnetic machine, the electromotive force (emf) can be induced by one of two ways. Firstly, by the cutting of magnetic lines of force due to the rotational motion of a conductor relative to the magnetic flux. Secondly, by changing of the flux linking the circuit. The latter would be a homopolar machine; impractical and expensive to construct.

If the direction of cutting of the magnetic lines of force is periodically reversed by means of poles of alternate north and south polarity, such a device is referred to as a heteropolar machine.[4] Two possibilities of this arrangement can exist. The magnetic field may be stationary and the armature rotating or vica versa. In either case, the emf induced in an individual conductor is an alternating current (or emf). Heteropolar machines of both types, therefore, consist of a stator and a rotor shown schematically in figure 1



(a) (b)

FIGURE. 1 Heteropolar machine: (a) with rotating armature, (b) with rotating field.

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A machine with a rotating armature is invariably used for direct-current (dc) purposes, while the other one with the rotating field is employed for producing an alternating current (ac). When the armature rotates, the collection of output current is much more difficult. Therefore, there is usually a relatively low limit to the highest voltage which a dc machine can continuously deliver.

The stationary armature called a stator, lends itself to an easy collection of the output since the terminals are fixed. This is important with very large powers at very high voltages. Consequently, alternating current working has an advantage over direct current in terms of generation of electrical power.

A generator or dynamo converts mechanical energy into electrical, whereas an electric motor converts electrical energy into mechanical. Hence, the generator or dynamo from which the electrical energy is derived has to be driven by another machine called a prime mover such as a steam turbine or internal combustion engine or electric motor and so on.

There is, however, no fundamental difference between a generator action and motor action, so either one can operate as a dynamo or motor. Their construction is similar. The schematic diagram of a dc machine is exhibited in figure 2.

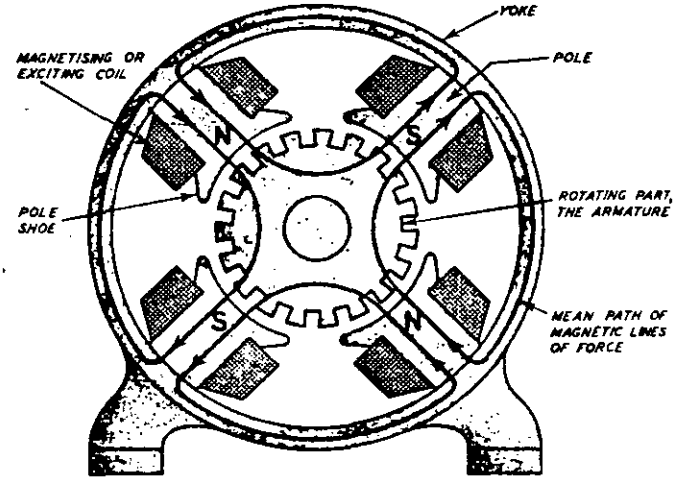


FIGURE 2. Diagram of the field system of a four-pole direct current dynamo or motor. The slots in the armature core are to house the armature winding.

So the dynamo-electric machine is a converter rather than a producer of energy. In dc generators (dynamo), the armature winding is on the rotor, and the field winding is on the stator.

An alternating voltage is one whose value and polarity changes periodically at a specific frequency. Often the variation is sinusoidal as shown in Figure 3.

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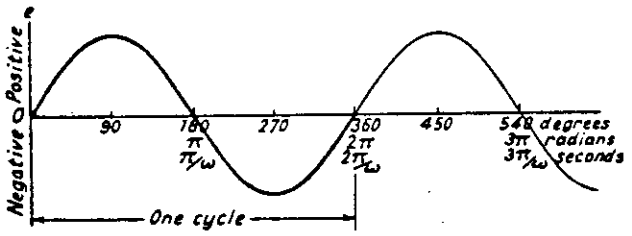


FIGURE 3. Sinusoidal waveform of alternating current (e = instantaneous emf)

Figure 3 is a plot of instantaneous emf as a function of angular position and/or time of a rectangular coil rotating at uniform speed in a uniform magnetic field. With few exceptions, the armature winding of a synchronous generator is on the stator, and the field winding is on the rotor with excitation by direct current conducted to it through carbon brushes bearing on slip rings or collector rings. An alternator is an electrical generator or dynamo which generates alternating current, hence often called ac generator. There are different types such as turbo, magneto, revolving-field and revolving-armature. Another way of classifying alternators is by means of phases, specifically single-phase, two-phase, three-phase or poly-phase.

The main purpose of the power source is to supply electrical energy, of either the alternating or direct current type to the point where the base metal and/or the filler metal need be melted to make a welded joint. When alternating current is employed, the high-voltage power is transformed to a low voltage welding-type transformer for safe application. A generator or a transformer-rectifier is utilized with direct current. It is called a transformer-rectifier because the high voltage from the main line has to be stepped down and the alternating current has to be rectified to direct current. A dc generator is typically driven by a 3-phase induction motor. Each type has its own specific advantages and limitations. Specific attributes of direct current can be enlisted in the following manner. Since the current flow is continuous, the welding arc is fairly steady. Starting the arc and maintaining a short arc are relatively easy. Welding with low currents when using small diameter electrodes is less complicated than with ac. This is a decided advantage in welding sheet metal. Direct current is suitable for all classes of flux coated electrodes. The disadvantages of dc have to do with an appreciable voltage drop in long cables and susceptibility to arc blow.

The rotating power sources for welding purposes can be either the generator or alternator type. Each type consists of a rotating and a stationary component: rotor or an armature and stator. A generator has a stationary field and rotating conductors, while an alternator has stationary conductors and a moving field. The principle of operation is based on excitation. That is, when an electrical conductor moves mechanically through a magnetic field or vica versa, electrical power is induced in the conductor. The magnetic field coils in a generator are wound on the stator and connected so that an

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equal number of poles of opposite polarity are located next to each other. For the sake of simplicity, a 2-pole generator would contain a north pole and a south pole on opposite sides of the stator at 180° apart. The rotor or armature conductors carry the welding current. The commutator - a group of conducting bars running parallel to the rotating shaft switching contacts with a set of stationary carbon brushes - is at one end of the rotor.

The power supply of the alternating type, on the other hand, has its magnetic field coils wrapped around the rotor. The conductors for the heavy welding current is wound into the stationary member called stator.

All welding power supplies are built with either constant-current or constant-voltage output characteristics, as shown in Figure 4. These attributes are based upon static rather than dynamic characteristics of the power source.

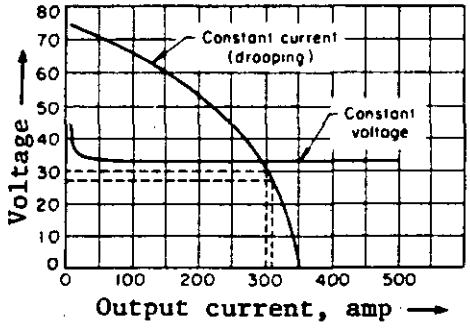


FIGURE 4. Volt-ampere curves for constant-current and constant-voltage power supplies

The constant-current output has a drooping feature on the volt-ampere plot so that voltage decreases as the current is increased.(5) Power sources with drooping characteristics are preferred for manual and semi-automatic arc welding processes, since small variations in voltage brought on by changes in arc length caused by the welder or operator do not affect the weld current and deposition rate significantly. With self-adjusting arc systems, a constant-voltage or flat characteristic power supply can be used to advantage. The arc initiation or striking of an arc with this type of power source is quite easy because upon short-circuiting, a marked current surge occurs. In general, the voltage range over which a process operates using a constant-voltage type power supply is limited. To operate a constant-voltage power supply at the same current and voltage as with a constant-current power source, the open-circuit voltage must be reduced. With a constant-current characteristic power source, the weld current is largely determined by the current setting on the power source or by the wire feed rate depending on the process and its associated welding equipment. In flux-cored arc welding, for instance, the weld current is proportional to the wire feed rate for a specific electrode diameter, composition and electrode extension. The relationship between wire feed rate and weld current for E70T-1 mild steel flux-cored electrode using CO_2 auxiliary gas shielding can be seen in Figure 5.

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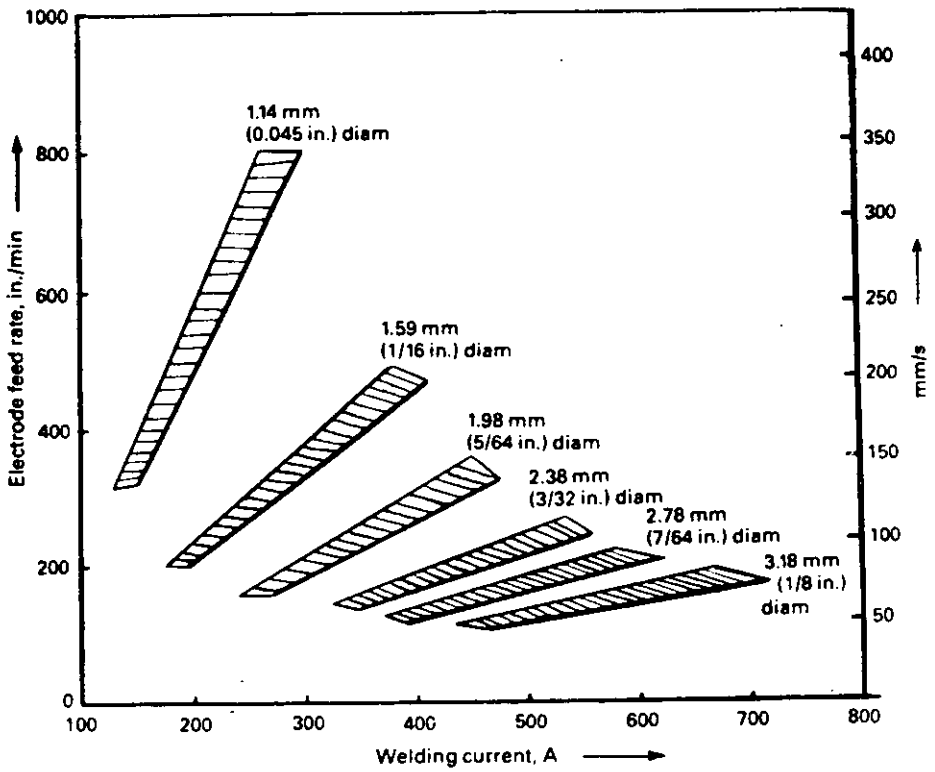


FIGURE 5. Electrode feed rate versus weld current range for a typical low carbon flux-cored electrode (E70T-1) with CO_2 gas shield.

1.8.1 Polyphase Working

In spite of the complications of a polyphase system it is preferable to the simplicity of single-phase. There are several reasons for this. Specifically, the power in a single-phase system is pulsating since there is always an alternating component of twice the frequency of the supply. In large machines especially, this would create an excessive vibration. In polyphase systems, the power is constant as long as the phases are in balance with one another. Single-phase motors need be supplied with auxiliary gear in order to provide the necessary starting torque. The pulsating armature reaction in single-phase synchronous machines cause heating. The output of a polyphase machine for a given frame size is considerably greater than that of a single-phase machine. Above 3 phases, the increase is not significant. Furthermore the complications with switch-gear, cables or transmissions and alternators increases. In consequence, the three-phase system has been accepted as the standard polyphase system worldwide.

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1.8.2 Transformer

A static transformer works on the principle of mutual induction. The simplest possible transformer consists of two neighboring coils P and S (See Figure 6) in which the primary (P) is connected to an ac source.

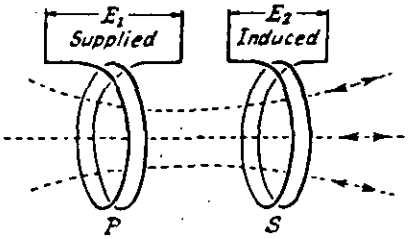


FIGURE 6. The simplest transformer.
(E_1 = applied voltage
E_2 = induced voltage or emf)

The alternating current flowing in P induces an alternating magnetic flux. Some of the magnetic lines of force link up with the secondary (S) coil. Thus, an alternating emf is produced in S. With the two coils in air, the flux linking S is less than the flux linking P. Furthermore, the flux is small because of the high reluctance of the magnetic path. Consequently, the transformer consists fundamentally of two coils - primary and secondary - wound on an iron core as shown in Figure 7.

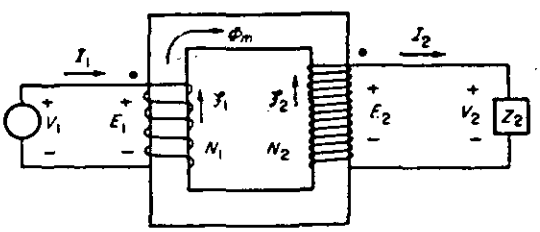


FIGURE 7. Ideal transformer.(6)

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I = current, V = load or terminal voltage (alternating)
E = voltage induced, F = magnetomotive force, (mmf = NI)
m = maximum flux, N = number of turns (winding)
Z = impedance

The magnitude of the alternating magnetic flux is a function of the primary voltage (E_1) and the number of turns. The value of the induced voltage (E_2) will depend on the number of turns of the secondary winding (N_2). Any ratio of transformation can be obtained by the proper selection of the numbers of primary (N_1) and secondary turns.

$$\frac{E_2}{E_1} = \frac{N_2}{N_1}$$

A transformer is said to be of the step up type when this ratio is greater than unity, and step down when it is less than one.

There are single-phase and three-phase transformers. Each can be designed to produce ac or dc power. Some units are built with an ability to put out both ac and dc The single-phase rectifier is intended to change the alternating current of the building power to a lower welding voltage and higher welding current. The combination units can be switched from ac to dc. The three-phase transformer changes the high primary voltage to a lower secondary voltage suitable for welding.

1.8.3 Rectification

Since the majority of electric arc welding processes use direct current, the alternating current input from plant power or main line supply must be rectified. Transformer-rectifier or alternator-rectifier power supplies used for electric arc welding function on rectifiers of one kind or another to convert alternating current to direct current. The rectifying device is called a diode. The diode is an evacuated enclosure (or tube) having two electrodes: anode(+) and cathode(-). The cathode is heated with a current or indirectly by means of a heater to emit electrons. If potential difference (pd) is not applied, electrons cannot escape through the anode and will form a space charge or a cloud of electrons around the cathode. After a while, a state of equilibrium sets up. If a pd is applied, then, the electrons can enter the anode and set up a current in the external metallic circuit. If now the tube is filled with a rarefied gas or vapor, the current within the tube will consist of electronic and ionic components. The electrons still move towards the anode and enter it but the ions moving towards the cathode cannot enter it due to both field and secondary emissions. Both high vacuum and gas-filled diodes allow the passage of current from anode (often called plate) to cathode when the anode is positive by an amount sufficient to overcome tube drop but block the flow of current in the opposite direction. There are many other devices that possess the property of unidirectional conductivity, namely copper-oxide rectifiers, selenium rectifiers, silicon rectifiers. Semi- conductors such as silicon, germanium, selenium and cuprous oxide are the basic elements of solid state or barrier-layer rectifiers. Solid state physics is the science of crystalline solids. Of the above-mentioned semi- conductors, silicon is the most important. The silicon wafers used in rectifiers contain a controlled amount of impurities to upset the balance of electrons, thereby making electrical

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conduction easier when voltage is applied. The change in electronic balance through addition of impurities can result either in an excess or shortage of electrons depending upon the valency of the impurity. Both the excess electrons and the positive holes do take part in the conduction of electricity through the crystal. In semiconductor diodes, the introduction of a third element provides control and amplification possibilities. The 3-element semiconductor device is known as a transistor (consisting of base, collector and emitter).

Welding power supplies for high outputs use solid state circuitry and solid state rectifiers. Present day welding machines employ silicon diodes for rectification.

1.8.4 Polarity

The direction of current flow otherwise known as polarity is an important consideration with d.c. power sources. Polarity can be straight and reverse. In straight polarity, the electrode is the negative pole and the workpiece is the positive pole (DCEN or DCSP). The current flow is from the negative (cathode) to the positive (anode) end of the arc. When the workpiece is connected to the negative pole (or terminal) and the electrode to the positive pole of the power supply, the polarity is called reverse (DCRP or DCEP). The factors that determine which polarity to use are the metal being welded, electrode type employed and the depth of joint penetration needed. In shielded metal-arc welding (SMAW), direct current reverse polarity (DCRP) gives for a given set of welding conditions deeper joint penetration than either DCSP or ac. The melting and deposition rates are, however, higher with DCSP than with DCRP. In gas tungsten-arc welding, DCSP produces the deepest joint penetration. This is because in all dc arcs about 70% of the heat is generated at the positive end of the arc. Also an electrode of a specific size can support more straight-polarity current than reverse-polarity current.

Therefore, DCSP is the type of current to utilize in GTAW, if the hottest arc or the deepest fusion in the workpiece is to be obtained with a given size nonconsumable electrode. DCRP in GTAW provides a good cleaning action on the surface of the base metal to remove oxides.

Straight polarity in SMAW can be used for all steels. The exception to this is when electrodes of the low-hydrogen type is utilized. Because the heat input is low with DCEN using the SMAW, the residual stress developed in the workpiece is therefore less than with DCEP. For these reasons, SMAW involving DCEN is used in sheet metal work.

Alternating current reverses the polarity at every half cycle, which creates an arc instability problem. With the 60-cycle ac system, polarity changes 120 times per second. This arc instability can be mitigated with arc-stabilizing elements added to the flux covering. Thus, in flux-shielded arc welding (e.g., shielded metal-arc and submerged-arc) of ferrous metals, choice of ac or dc exists. In gas metal-arc welding on the other hand, an ac power source is usually not used. DCEP is the most frequent choice for GMAW, unless a shallow penetration is desired in which case, DCEN is applied.

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A schematic arrangement of the two different polarities is illustrated in Figure 8 involving SMAW.

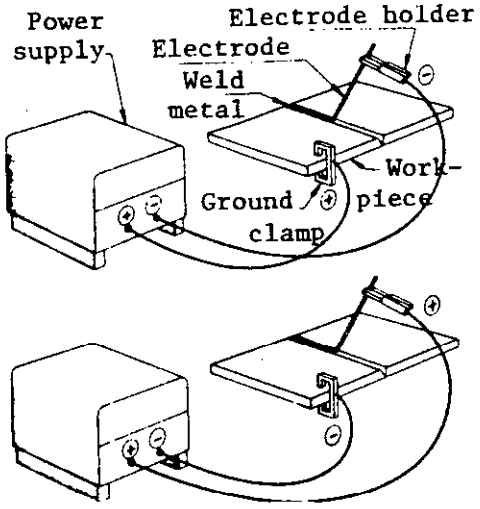


FIGURE 8. Diagram showing (a) negatively connected (straight polarity) and (b) positively connected (reverse polarity) electrodes.

1.8.5 Arc Initiation

Arc initiation requires an applied potential and a gaseous conducting medium between the electrodes. There are a number of arc initiation methods. In shielded metal-arc welding, an electric arc is struck between the tip of the consumable electrode and the base metal. Retract starting is a method used in automatic dc welding. In this method, the electrode is fed until it touches the workpiece and is then retracted to maintain the arc. With constant-voltage power sources, an arc can be initiated by simply feeding the electrode wire to the workpiece. Another type of arc starting methodology involves a slow advance of the electrode until it contacts the base metal to be welded to start the arc and then the electrode is speeded up to the proper wire speed for welding.

In electroslag welding, a steel wool ball is placed between the electrode and the base metal. Then the power is turned on for striking an arc. Or, the electrode with the power on is fed to touch the workpiece. Molten slag can also be poured into the joint for starting the electroslag process.

High frequency arc initiation is often the method used for dc or ac power supplies. The high frequency generator superimposes a series of high voltage bursts on the welding circuit, which ionizes the gas in the gap between the electrode and the base metal. The ionized gas serves to conduct the welding current to begin the arc. The high frequency voltage bursts also aid in re-igniting the arc during current reversals.

The pilot arc between the welding electrode and an auxiliary one is still another arc initiation method which, for its high reliability, is often used in dc welding equipments. The pilot arc itself is initiated by either a scratch or by a high-frequency technique. Since the plasma arc is within a constricting nozzle, it cannot be started by touching against the workpiece. First, a pilot arc is ignited between the electrode and the constricting nozzle. The pilot arc may be started by either a low or a high amperage

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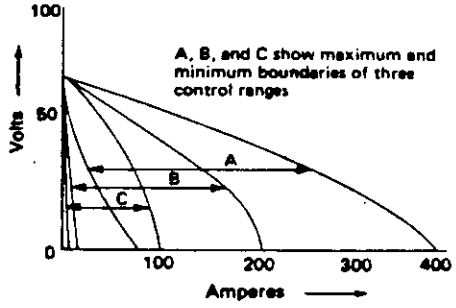
torch. In the former method, the electrode is advanced against the nozzle and then withdrawn to sustain the arc. The latter method involves either a high frequency ac power, or several high voltage, low power impulses super-imposed on the circuit to ionize the gas in the orifice. The ionized gas then conducts the current for the pilot arc.

Several arc initiation techniques are available for submerged-arc welding such as wire retract, scratch, sharp wire, high frequency, fuse ball and molten flux starts. A sharp, chisel-like shape is obtained at the end of the welding wire by snipping with a pair of bolt cutters. The consumable electrode is then lowered to touch the work. The electrode with the sharp point begins to melt immediately and the arc is established. The granular flux is applied and thereby welding begins.

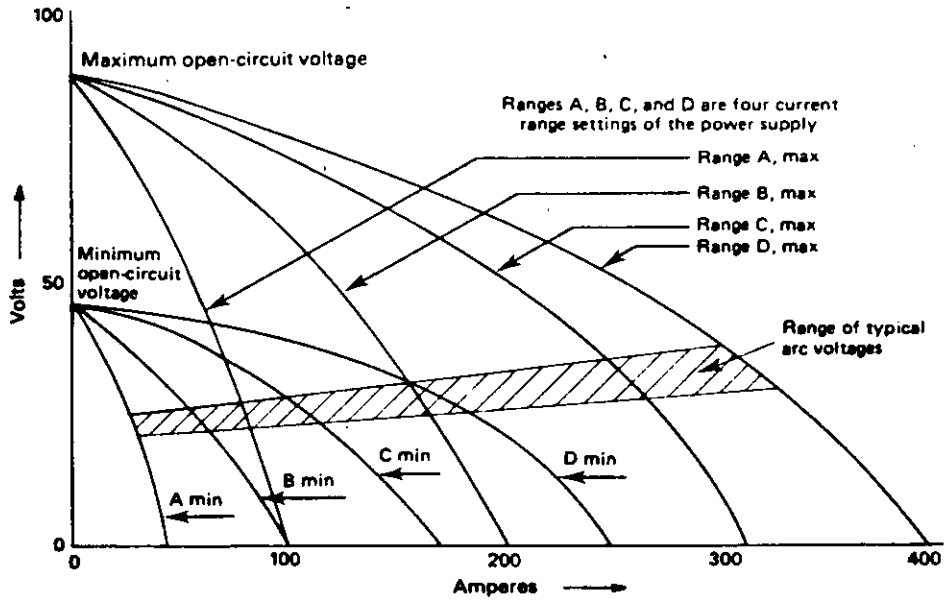
While arc initiation requires an appreciable voltage, arc maintenance needs only a fraction of it. Re-ignition or arc maintenance is more of a concern in ac than in dc welding. It is because the arc has to be re-ignited at every zero current on each reversal of current flow in an alternating current power supply. Auxiliary re-ignition can be accomplished by various devices such as high-frequency, high-voltage oscillators to provide a pulse of high voltage at points of zero current.

Re-ignition during welding with dc power plants has only to be reckoned with during accidental short-circuiting events, such as at times of metal transfer from the electrode. This may necessitate the use of a power supply with dynamic characteristics accomplished by circuit design and control. So not only the steady state (or static) relationship between output voltage and current is important but the transient (or dynamic) relationship between voltage and amperage as well. Shown in Figure 9 are volt-ampere relationships for transformer-rectifier and generator or alternator type power sources, respectively.(7)

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(a)



(b)

FIGURE 9. Volt-ampere relationship for (a) direct current transformer-rectifier type under static conditions and (b) generator or alternator type power source.

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1.8.6 Duty Cycle

The duty cycle of a power source is a time percentage. It is calculated on a number of successive 10 minute intervals while the power supply can operate at its rated capacity without exceeding its recommended maximum operating temperature. Should a power source be operated at its rated output but beyond its recommended duty cycle, overheating would occur. The result might either be no welding at all, or in the case of a transformer-rectifier type, burning-out of the plates. Overheating could also happen, if a power source is operated beyond its capacity without reducing the duty cycle.

Most welding jobs do not involve operation of the power source 100% of the time. Consequently, the power supply under such circumstances of operation does not need be as large and rugged as a power source operating continuously. Duty cycle is a sort of index which recognizes the difference between continuous or full load and intermittent or partial load requirements. Expressed differently, duty cycle is a ratio of the load-on time capacity to the no-load time. The no-load time is an allowance for cooling of the internal components including the insulation to within the recommended temperature range. Short periods of overloading are not deleterious to constant-current type power sources. For example, a standard industrial rating of 60% duty cycle means that the power supply is capable of delivering its rated load output for 6 out of every 10 minutes. This type of unit is normally designed for manual welding. A power source of 100% duty cycle is commonly built for automatic or semiautomatic welding processes.

From a cost standpoint, duty cycle is important because of the effect of duty cycle and power factor upon the cost of electrical power. Cost is normally directly proportional to the duty cycle and inversely proportional to the power factor. Constant ac power supplies have a characteristically low power factor because of their large inductive reactance. Utilities charge a rate penalty for high line current under heavy loads with a low power factor. The cost decreases as the power factor increases. The power factor is a ratio of arc voltage to the open-circuit voltage. The power factor can be improved by adding capacitors to the primary circuit of the welding power supply. The net result of capacitor additions is a lower plant power usage during welding.

1.9 Weld Position

Weld positions may be divided into two categories: downhand and out-of-position. Further classification of these two major divisions include flat (or downhand), horizontal, vertical and overhead. The weld positions other than flat are commonly referred to as out-of-positions.

Flat position welding is the simplest and easiest to do due to the benefit of gravitational force. It requires the least amount of skill. The highest deposition rates are obtained in the flat position. Out-of-position welding necessitates the use of smaller diameter electrodes with a faster freezing characteristic, lower weld current, manipulative techniques, and a greater welder experience. Furthermore, the formation of slag on the surface of the molten weld pool helps counteract the effect of gravity. The joint may also have to be different from that employed in flat position.

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The choice of weld position is a function of the particular welding process details and type of weld current involved. Shielded metal-arc welding can be done in all weld positions. Direct current is usually preferred to alternating current for vertical and overhead positions using the SMAW. However, ac can be used in all positions provided the electrode is suitably chosen.

Submerged-arc welding is essentially limited in its application to flat or horizontal welding. Flux-cored arc welding became a multiple-position welding process with the introduction of small diameter electrodes, namely, 1.14 mm (0.045 in). FCAW is, however, most frequently used in flat, horizontal, downhill and vertical-down positions involving 1.98 mm (0.078 in) and 2.38 mm (0.094 in) electrodes. FCAW in overhead position requires small diameter (1.14 mm) or microsize [0.89 mm (0.035 in)] electrode wires, lower amperage and good maneuvering of the weld pool. Fixed-position welding of steel pipes is more successfully done with small diameter (1.14 mm) flux-cored wire using auxiliary gas shielding. Rotation of the pipe greatly enhances welding because all the circumferential welding (or girth welds) can be carried out in the most suitable position.

Gas metal-arc welding is regarded as the most versatile electric arc welding method and is an all-position process. In terms of metal transfer, GMAW in the flat position is done in the spray mode. Short-circuiting or pulsed spray metal transfer modes are most appropriate in vertical down and overhead positions using electrode diameters of 1.14 mm or 0.89 mm (0.045 or 0.035 in).

The weld positions identified as flat, horizontal, vertical down, vertical up and overhead for groove and fillet welds are schematically shown in Figure 10.

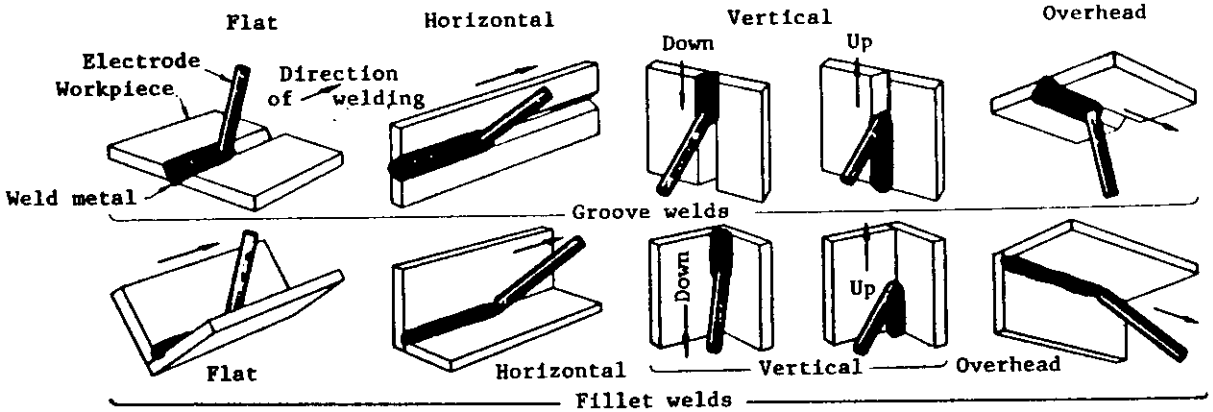


FIGURE 10. Weld positions for groove and fillet welds.

1.10 Metal transfer in GMAW

Weld metal can be transferred from the electrode to the base metal being welded by short-circuiting, globular, spray and pulsed modes.

1.10.1 Short-circuiting transfer mode

The short-circuiting mode of gas metal-arc welding uses the lowest welding current and the smallest electrode wire diameter. For these reasons, this joining process is suitable for welding in any position involving thin

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sections. Due to the low heat input, distortion and residual stress are usually negligible. The short-circuiting GMAW method can also be utilized for filling large root openings.

The actual metal transfer occurs in steps (see Figure 11). At the start of the cycle, the electrode wire melts and forms a small globule of liquid weld metal. Due to the gravitational force this globule moves down towards the workpiece and makes contact with it. The arc is momentarily extinguished because of a short circuit: hence the terminology. The molten metal bridge is broken by pinch force. This squeezing action and the magnitude of the force can be controlled by the power supply. The arc is restarted when the electrical contact is broken. Thus, another cycle is initiated. The frequency of arc extinction and re-ignition ranges from 20 to over 200 times per second. With an increase in current density using a shielding gas of CO_2, the rate of short circuiting can be changed. This also results in much larger filler metal droplets.

Short-circuiting filler metal transfer is normally done with low current densities and with precise arc voltage values.

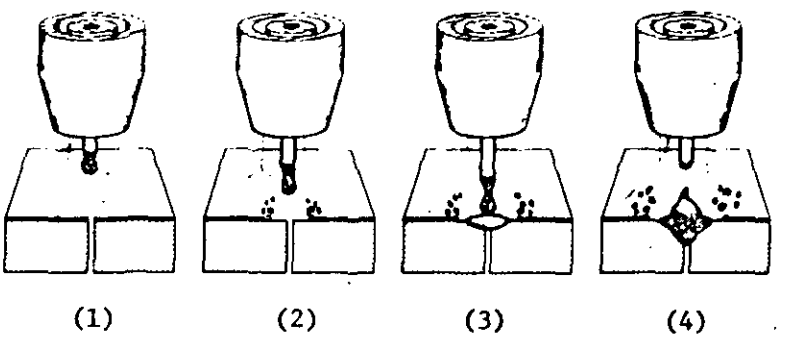


FIGURE 11 Steps in short-circuiting metal transfer (1) Drop begins to form, (2) Drop moves towards base metal, (3) Drop contacts weld pool causing short circuit (4) Circuit is broken; another drop begins to form.

1.10.2 Globular transfer mode

Globular transfer mode takes place at current densities higher than those used in short-circuiting. The result is the formation of a relatively large globule of liquid weld metal (see Figure 12). The globule is detached from the end of the electrode wire by means of gravitational force. The size of the globule is greater than that of the electrode diameter. While globular metal transfer occurs with all types of shielding gases, its application is limited in terms of welding positions. The globular mode of GMAW cannot be used in out-of-positions.

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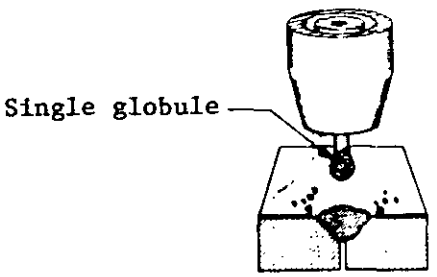


FIGURE 12 Globular metal transfer.

1.10.3 Spray transfer mode

At or above the so called globular-to-spray transition current, the weld metal transfer for all metals is accomplished in a stream of very fine droplets commonly referred to as spray. In spray transfer, the filler metal moves from a characteristically tapered electrode wire tip through a fine arc column to the base metal (see Figure 13). The diameter of the fine spray droplets is much less than that of the filler metal. This decrease in droplet size is coupled with an appreciable increase in spray transfer rate usually in the range of 100 to several hundred droplets per second.

Spray transfer is a high heat input process relative to the two transfer modes mentioned previously. The net results are maximum weld penetration and highest deposition rate that can be achieved by GMAW. In general, the shielding gas employed in spray arc metal transfer consists of either argon or argon-rich inert gas. When CO_2 gas is utilized for shielding purposes, spray tranfer mode cannot be obtained.

The high heat input involved in spray metal transfer limits its use to flat position and horizontal fillet weld position. The high current density associated with this process makes it suitable for welding relatively thick base metals.

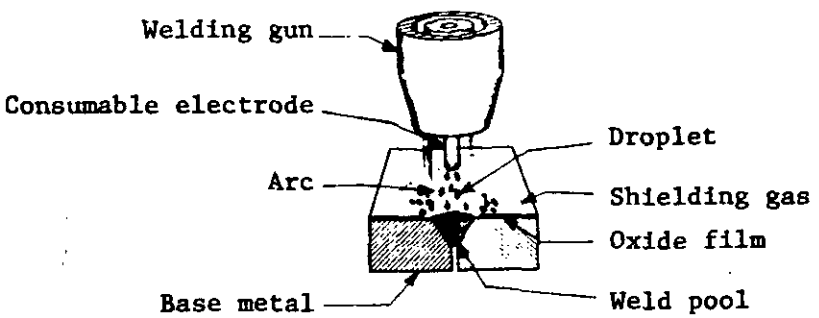


FIGURE 13 Spray transfer mode.

1.10.4 Pulsed-spray transfer mode

Pulsed-arc metal transfer represents another process variation to GMAW. Pulsed spray welding is an all-position joining method at welding heat input

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levels much higher than those associated with short-circuiting arc welding. A spray-type transfer occurs in pulses at well-defined intevals. Between pulses, the current is decreased and weld metal is not transferred. The power source is so designed as to provide a pulsing action at two welding current levels, known as "pulsed peak" and "background" currents. The "background" level occurs at a steady rate but at a level too low to yield a spray transfer mode. It preheats and preconditions the electrode wire which is continuously fed. The "pulsed peak" current causes the fine droplets in the spray-arc to transfer to the weld joint (see Figure 38). The peak current is linked with the line frequency. Therefore, the droplet transfer from the electrode to the joint occurs 60 or 120 times per second. The result of these two levels of current is a steady arc of axial spray metal transfer at an effective welding current less than that of the conventional spray transfer mode.

1.11 Shielding Gases

In fusion welding processes, the molten material must be protected from gaseous contaminants for reasons of avoiding a deterioration of properties of the weld. The degree to which a given base material needs protection depends in part on its reactivity with elements in its surrounding. Reactive metals such as titanium require a more extensive protection than less reactive materials.

Atmospheric contaminants consist of oxygen, nitrogen and hydrogen. There can be other elements present in the air from pollution sources. Sulfur and chlorine are two such pollutants besides particulate matter. Gaseous shielding of the liquid weld metals has traditionally been done by inert gases or mixtures thereof in certain proportions. Other gases such as oxygen, carbon dioxide, hydrogen and nitrogen can be added to a given inert gas.

Argon and helium are the two inert gases most widely used in arc welding. Selection of a shielding gas requires a consideration of several factors. They are the type and thickness of the workpiece, joint design, weld position, fixturing, welding technique quality requirement and production rate, cost and effectiveness of the shielding gas.

The shielding gas must prevent the air from coming into contact with the molten weld. Metals can dissolve nitrogen or hydrogen and can react with oxygen. Solubility of a given gaseous element in a given metal increases with a rise in temperature. If the dissolved gas cannot escape from the metal while the metal is molten, porosity will result. In the case of hydrogen entrapment, the metal may be embrittled or may in fact crack. The reaction of a metal with oxygen leads to oxide formation, the end result of which may be slag inclusions.

1.11.1 Argon

Argon is a monatomic gas obtained by liquefaction of air. It is much heavier than air. For this reason, it is preferable to use in flat and horizontal position welding. Argon is available either as liquid at temperatures below - 184°C (- 303°F) or as compressed gas above this temperature. Welding-grade argon is about 99.99% pure.

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The profile of weld penetration obtained with pure argon shielding is different from those produced with helium or carbon dioxide, as can be seen in Figure 14.

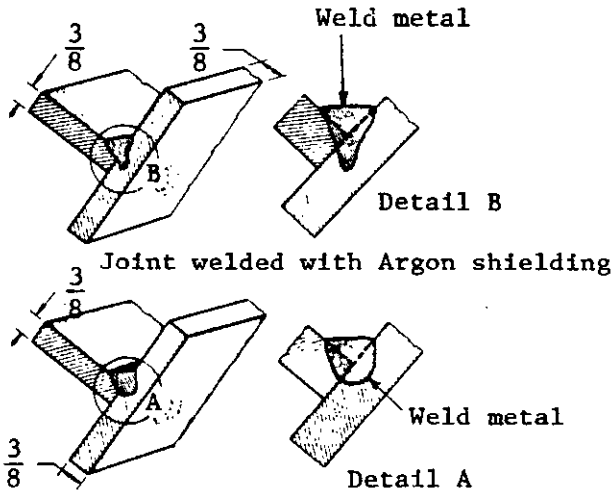


FIGURE 14. Profile of weld penetration obtained in flat position welding using argon and CO_2 shielding. Other weld parameters were identical.

Relative to helium, argon has a lower ionization potential resulting in a lower arc voltage: hence lower heat for a given weld current. Consequently, argon is preferred to helium for welding sheet metals. Argon is about ten times heavier than helium which means that much less argon is needed to protect the weld pool. Argon is less costly to produce and more abundant in supply than helium. Therefore, argon is more extensively utilized in arc welding. Argon is a colorless, odorless, tasteless and nontoxic gas. However, argon in confined spaces may cause asphyxiation by replacing the air.

The welding attributes of argon may be summarized as follows:

- Easier arc initiation
- Smoother arc action
- Lower arc voltage
- Better cleaning action with ac power supply in welding aluminum and magnesium
- Greater resistance to draft
- Lower flow rate for effective shielding.

1.11.2 Helium

Helium is the lightest monatomic inert gas used in welding. It is obtained from natural gas and has no color, taste or odor. It is also nontoxic. Since it is only 14% as heavy as air, helium tends to rise and drift away from the weld area. Therefore, a greater volume of helium is needed to effectively shield the molten weld pool than is necessary with argon shielding.

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The cost of producing helium is much higher than that of argon. Pure helium is therefore seldom used for welding. Helium is mixed with argon in order to decrease the cost of shielding and to take advantage of the respective properties of the two inert gases. A ratio of 75% helium to 25% argon is used most frequently.

Helium produces a hotter arc than does argon shielding. Thus, welding thick materials and those with high melting points and greater thermal conductivities are customarily welded with helium shielding. A comparison of these two gases in terms of arc voltage and arc current in tungsten-arc welding of aluminum is exhibited in Figure.15.

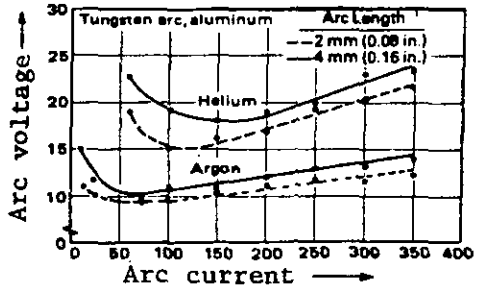


FIGURE 15. Arc voltage as a function of arc current using argon and helium shielding.

It is obvious from Figure 15 that much higher currents are needed with argon shielding to obtain the same arc power as with helium. While both argon and helium produce a stable arc with direct current, with alternating current only argon yields a smooth arc. Argon-helium mixtures such as 50:50 and 25:75, are used for welding heavier sections of aluminum, magnesium and copper particularly in overhead positions. Addition of helium to argon results in a hotter arc for a given arc current, but at least 50% helium has to be added to effect the increase in arc heat. Keyhole welding in titanium by plasma-arc is done with a helium addition of 50-75% to argon. This mixture permits a higher travel speed than when only pure argon is utilized.

1.11.3 Argon-Hydrogen

Hydrogen added to argon in amounts of up to 35% is utilized in automatic welding of nickel-base alloys, nickel/copper and stainless steel. The amount of hydrogen that may be added to argon is in part dependent on metal thickness and root opening of the joint. The hydrogen addition increases the arc voltage which results in a greater heat input. The argon-hydrogen mixture is not used for welding metals where the hydrogen would adversely affect the properties of the weldment as in mild and low-alloy steels. Hydrogen addition is also detrimental when reactive metals are welded because of hydrogen embrittlement. Argon-hydrogen mixtures are however often used as an orifice gas and shielding gas for plasma-arc welding of stainless steel. The usual percentage of hydrogen added to argon varies from 1-5% for welding base metal thicknesses of up to 9.5 mm (0.375 in).

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1.11.4 Argon-Oxygen

Oxygen in amounts of 1, 2 or 5% mixed with argon improves the weld contour, metal transfer and penetration when low carbon steel is gas metal- arc welded. It also stabilizes the arc and thus reduces the weld spatter. Furthermore, oxygen addition increases wetting of the weld metal along the fusion line in carbon and low-alloy steels, thereby preventing undercutting. In gas tungsten-arc welding, the oxygen-containing argon mixture causes a rapid degradation of the electrode.

1.11.5 Argon-Carbon Dioxide

Carbon dioxide of about 20-25% added to argon improves the metal transfer, the arc action and the weld appearance. In many ways, the overall result of CO_2 addition is similar to that of oxygen. Wettability, penetration and flow of the weld are improved. In narrow-gap welding of steel, an 80% argon-20% CO_2 mixture is used most commonly. Although a 75/25 Ar/CO_2 mixture may be satisfactory. In electrogas welding, an argon/carbon dioxide mixture of 80/20 is almost exclusively used.

1.11.6 Carbon Dioxide

Most of the reactive gases cannot be used for protecting the weld pool. Carbon dioxide is an exception and is used for welding steel by the short-circuiting transfer mode of GMAW. It produces a broad and deep weld penetration. It also leads to a higher weld speed. The major disadvantage of CO_2 shielding is a harsh arc and some spatter when the workpiece is thin. Carbon dioxide decomposes at the arc temperature to oxygen and carbon monoxide which has an oxidizing effect on the weld similar to argon mixed with 8 to 10% oxygen.

The dissociation of carbon dioxide is illustrated by the following reaction

$$2\ CO_2 \rightleftharpoons 2\ CO + O_2$$

This reaction produces a considerable amount of oxygen in the arc to react with the base metal being welded. In the case of steel, the reaction between iron and carbon dioxide is as follows,

$$Fe + CO_2 \rightleftharpoons FeO + CO$$

Some of the carbon monoxide may dissociate to carbon and oxygen. The carbon may be dissolved in the molten weld pool and can cause hardening of the weld by increasing its carbon.

The carbon dioxide used for auxiliary gas-shielding in flux-cored arc welding of alloy steels has to have a dew point of - 40°C (- 40°F) or lower. This low dew point is not essential when low-carbon steel is welded by FCAW.

Carbon dioxide has a pungent odor, colorless, nontoxic, non-flammable but slightly acidic in taste. It is about 1.5 times heavier than air. Carbon dioxide at its triple point exists in solid, liquid and gaseous form. At atmospheric pressure, solid CO_2 transforms to gas. Carbon dioxide can be obtained from several sources, namely, natural gas, coke, fuel oil, alcohol, ammonia, and lime kilns.

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1.11.7 Nitrogen

Nitrogen is used for shielding gas in welding deoxidized copper in countries where inert gases are in short supply. The higher arc voltage possible with high-purity nitrogen allows a higher current to be employed with regular welding equipment. Nitrogen is less expensive to produce than either argon or helium. With nitrogen shielding, the heat transfer efficiency is better than that obtained with inert gases.

1.11.8 Gas Mixtures

The main objective of mixing the various gases is to decrease the cost of welding and to take advantage of certain properties of the individual components in the mixture. The most common mixtures employed in welding include argon-helium, argon-oxygen, argon-carbon dioxide and carbon dioxide-oxygen. Gas mixtures of 75% Ar + 25% CO_2 and of 98% Ar + 2% O_2 have been used for flux-cored arc welding alloy steels, but these gas mixtures are more expensive than plain carbon dioxide. These gas mixtures yield no better results than plain carbon dioxide. Therefore their use is unjustified.

A summary of gases used in GMAW is given in table I.

TABLE I
Shielding Gases used for Gas Metal-Arc Welding.

Type of Gas	Typical mixtures, %	Primary uses
Argon	-	Nonferrous metals
Helium	-	Aluminum, magnesium, and copper alloys
Carbon dioxide	-	Low-carbon and low-alloy steels
Argon-helium	20-50A-50-80He	Aluminum, magnesium, copper and nickel alloys
Argon-oxygen	1-2O_2	Stainless steels
	3-5O_2	Low-carbon and low-alloy steels
Argon-carbon dioxide	20-50CO_2	Low-carbon and low-alloy steels
Helium-argon-carbon dioxide	90He-7-1/2A-2-1/2CO_2	Stainless steels
	60-70-He-25-35Ar-5CO_2	Low-alloy steels
Nitrogen	-	Copper alloys

1.12 Fluxes

In general terms, fluxes are applied for the principle purpose of obtaining a sound joint. Flux is a material which either prevents oxidation or helps remove the oxides and other undesirable contaminants from the surfaces of the joint. Fluxes have still other functions, namely to add alloying additions to the weld and control the reactions between the weld metal and the flux.

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In shielded metal-arc welding, the composition of the electrode covering has a strong influence on the electrical performance of the electrode and the quality of the resultant weld. The flux coating consists of many different components which in turn have one or more functions as outlined in table II.

TABLE II
Typical Functions and Composition Ranges of Constituents of Coverings on Mild Steel Arc Welding Electrodes.

Constituent of covering	Function of constituent		Composition range in covering on electrode of class, %								
	Primary	Secondary	E6010, E6011	E6012, E6013	E6020, E6022	E6027	E7014	E7016	E7018 E7048	E7024	E7028
Cellulose	Shielding gas	-	25-40	2-12	1-5	0-5	2-6	-	-	1-5	-
Calcium carbonate	Shielding gas	Fluxing agent	-	0-5	0-5	0-5	0-5	15-30	15-30	0-5	0-5
Fluorspar	Slag former	Fluxing agent	-	-	-	-	-	15-30	15-30	-	5-10
Dolomite	Shielding gas	Fluxing agent	-	-	-	-	-	-	-	-	5-10
Titanium dioxide (rutile)	Slag former	Arc stabilizer	10-20	30-55	0-5	0-5	20-35	15-30	0-5	20-35	10-20
Potassium titanate	Arc stabilizer	Slag former	(a)	(a)	-	-	-	-	0-5	-	0-5
Feldspar	Slag former	Stabilizer	-	0-20	5-20	0-5	0-5	0-5	0-5	-	0-5
Mica	Extrusion	Stabilizer	-	0-15	0-10	-	0-5	-	-	0-5	-
Clay	Extrusion	Slag former	-	0-10	0-5	0-5	0-5	-	-	-	-
Silica	Slag former	-	-	-	5-20	-	-	-	-	-	-
Manganese oxide	Slag former	Alloying	-	-	0-20	0-15	-	-	-	-	-
Iron oxide	Slag former	-	-	-	15-45	5-20	-	-	-	-	-
Iron powder	Deposition rate	Contact welding	-	-	-	40-55	25-40	-	25-40	40-55	40-55
Ferrosilicon	Deoxidizer	-	-	-	0-5	0-10	0-5	5-10	5-10	0-5	2-6
Ferromanganese	Alloying	Deoxidizer	5-10	5-10	5-20	5-15	5-15	2-6	2-6	5-10	2-6
Sodium silicate	Binder	Fluxing agent	20-30	5-10	5-15	5-10	0-10	0-5	0-5	0-10	0-5
Potassium silicate	Arc stabilizer	Binder	(a)	5-15(a)	0-5	0-5	5-10	5-10	5-10	0-10	0-5

(a) Used (in place of constituent on line above) in E6011 and E6013 electrodes to permit welding with alternating current

The high arc temperature melts the solid flux. As a result, the molten slag floating on top of the weld pool provides a physical barrier to atmospheric contaminants. At the same time, some of the flux components such as cellulose and calcium carbonate convert to gas, thereby providing a gas shield around the weld. The shielding mechanisms produced in SMAW are illustrated in Figure 16.

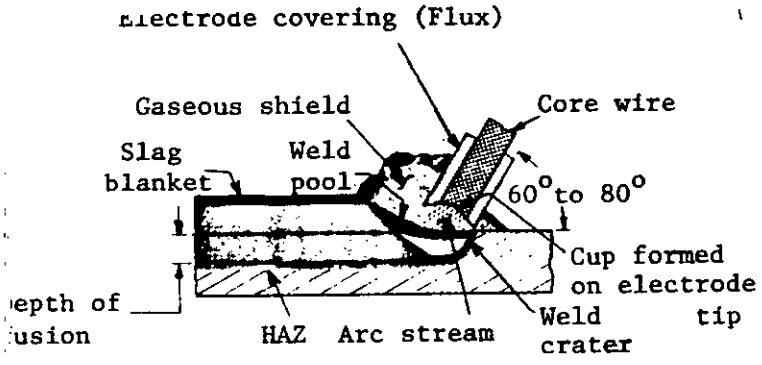


FIGURE 16. Shielding of the molten weld pool in SMAW.

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The amount of slag and gas obtained depends on the type of covering applied to an electrode. The fluxes used in submerged-arc welding are classified as prefused, bonded, agglomerated and mechanically mixed. But regardless of type, the granular flux is supplied to the arc area to cover the weld pool. The flux deoxidizes the molten weld pool, modifies its composition, shapes the weld bead and influences the mechanical properties of the welded joint.

The fused fluxes are obtained from appropriate raw materials by mixing, melting, crushing, screening and packaging. Melting is carried out in an electric furnace at around 1500-1700°C (2700-3100°F). Fused fluxes are quite homogeneous chemically. The greatest disadvantage of fused fluxes is the difficulty of adding ferroalloys and deoxidizers.

The raw ingredients of bonded fluxes are powdered, mixed while dry and then bonded together with either potassium silicate or sodium silicate. The wet mix is then pelletized and dried at a relatively low temperature. The pellets are crushed and screened for sizing and packaging. Adding deoxidizers and alloying elements to bonded fluxes is easy. The chief drawback of bonded fluxes is a tendency to pick up moisture. For this reason, porosity and hydrogen embrittlement are likely to occur in these welds.

The agglomerated fluxes are produced in a similar fashion to the bonded type except for the use of a ceramic binder. This requires the use of a higher curing temperature, approximately 1400°C (2550°F), which prevents the addition of deoxidizers and alloying elements.

When fused, bonded or agglomerated fluxes are mixed together in some ratio, the product is referred to as mechanically-mixed fluxes.

Fluxes are classified by the American Welding Society according to the mechanical properties of the weld deposit obtained by a given combination of filler metal and flux.

The flux in electroslag welding serves to conduct the weld current and protect the liquid weld metal from atmospheric contamination. Therefore, the slag must have good electrical coductivity and sufficient resistance to generate heat for welding purposes. The viscosity of the slag has to be such that slag entrapment and non-uniform heat distribution are minimized. In any arc welding operation, a flux to be acceptable must have a characteristic of easy removal after welding. This attribute is especially important in multi-pass welding operations.

In flux-cored arc welding, the flux is inside the tubular electrode. Protection of the liquid weld metal comes from the decomposition of the solid flux and/or from auxiliary gas shielding.

Oxide removal from the surface of the joint is important in materials joining processes. The removal of surface oxides is especially important in oxyfuel gas welding, braze welding, brazing and soldering. The types of fluxes used in these materials joining processes just named include pastes, powders and solutions of various chemical formulations. For example, braze welding uses dry powders. The end of the welding rod is heated and then dipped into the powder flux. The flux adheres to the filler metal. Additional powder may also be strewed on the joint surface. Fluxes in paste form can be applied by brushing, or by dipping the filler metal into the paste.

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A basic requirement of a satisfactory brazing is that the filler metal wet the surface of the workpiece to facilitate the flow of the liquid brazing filler metal into the joint through capillary action. Oxide films on surfaces inhibit wetting. Fluxes through their chemical activity must dissolve the oxide layer without attacking the base metal. The elements that may react with the base metal must be excluded from the joint during brazing. This can be accomplished by the application of fluxes, special gas atmospheres and vacuum. It should be noted that the application of fluxes, special atmospheres or vacuum does not negate the need for cleaning the parts prior to brazing because none of these methods remove grease and dirt.

During brazing, the flux must melt, flow and maintain its protective role. The flux should have a low surface tension to enable the molten filler metal to replace the flux. The ingredients of brazing fluxes include borax, chlorides, fluorides, fluoborates, boric acid, wetting agents, water, hydroxides and fluosilicaborates. The precise chemical formulations of brazing fluxes are usually proprietary in nature. There is no universal flux that would be satisfactory for all brazing applications. Specific fluxes have been developed for the various base metals and temperature ranges as shown in table III.

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TABLE III.
Fluxes for Brazing.

AWS brazing flux type no.	Recommended base metals	Recommended filler metals	Recommended useful temp. range	Ingredients	Forms supplied
1	All brazeable aluminum alloys	BAlSi	700-1190F 371- 643C	Chlorides Fluorides	Powder
2	All brazeable magnesium alloys	BMg	900-1200F 482- 649C	Chlorides Fluorides	Powder
3A	All except those listed under 1, 2, and 4	BCuP,BAg	1050-1600F 566- 871C	Boric acid Borates Fluorides Fluoborates Wetting agent	Powder Paste Liquid
3B	All except those listed under 1, 2, and 4	BCu, BCuP, BAg, BAu, RBCuZn, BNi	1350-2100F 732-1149C	Boric acid Borates Fluorides Fluoborates Wetting agent	Powder Paste Liquid
4	Aluminum bronze, aluminum brass, and iron or nickel base alloys containing minor amounts of Al or Ti, or both	BAg (all) BCuP (Copper base alloys only)	1050-1600F 566- 871C	Chlorides Fluorides Borates Wetting agent	Powder Paste
5	All except those listed under 1, 2, and 4	Same as 3B (excluding BAg-1 through -7)	1400-2200F 760-1204C	Borax Boric acid Borates Wetting agent	Powder Paste Liquid

Fluxes must be removed after brazing for several reasons. Residual fluxes can induce subsequent corrosion in the joint, interfere with post-brazing examination and testing, and hinder painting or plating. Hot water rinsing and subsequent drying are usually the practices involved in removing residual brazing fluxes.

The objective of a soldering flux is analogous to that of a brazing flux. Soldering fluxes are available in solid, liquid and gaseous forms. A good soldering flux promotes wetting of the surface by the liquid solder, dissolves oxides, displaces surface contaminants, and prevents oxidation at the soldering temperature. Soldering fluxes may be of the organic, inorganic and rosin type. The inorganic acids and salts are the most active type of soldering flux. The organic acids and bases constitute the so called organic fluxes which are less active than the inorganic type. The organic type tends to decompose upon prolonged exposure to the soldering temperature which ranges from 90-320°C (200-600°F).

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The residue of inorganic fluxes must be removed since they can cause severe corrosion damage. The residue of organic fluxes are relatively inert and can easily be washed off the workpieces. The rosin fluxes are considered to be non-corrosive. Therefore, they are extensively used in the electrical and electronics industry. Rosin has a low chemical activity: hence low fluxing action. If a greater activity is desired without seriously changing the non-corrosive characteristics, then, certain activating agents can be added to the rosin-type soldering flux. The regular rosin flux residue may be left on the joint, provided it does not affect the appearance of the part. The activated rosin flux should be rinsed off since it contains an organic compound which may cause corrosion.

A good general surface cleaning of the workpiece at and around the joint must precede the application of flux to remove paint, grease, oil, dirt, rust and other markings. Soldering fluxes in general can only work effectively on precleaned surfaces. Methods of precleaning consists of degreasing, pickling and various mechanical methods such as shotblasting, wire brushing, scraping, grinding, filing and sanding.

1.13 Joint Design

The design aspect of a joint revolves around strength and safety considerations imposed on by the service conditions. In a particular material the design engineer must take into account loading type, service temperatures, residual stresses, environmental factors, discontinuities, fabricability and cost. Safety factors are used to account for unknowns in a given structure.

Several factors must be considered in sizing, shaping and selecting a specific joint type. They are specific process parameters, fit-up, constraint, thickness, strength requirement, weld position, accessibility, location, fabrication techniques, metallurgical considerations, distortion, working conditions, level of quality, inspectability, and service requirements.

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There are five basic joints which include butt, lap, corner edge and T-joints. There are also modifications to these joints. The types of joints and welds classified by The American Welding Society (AWS) are presented in Figure 17.

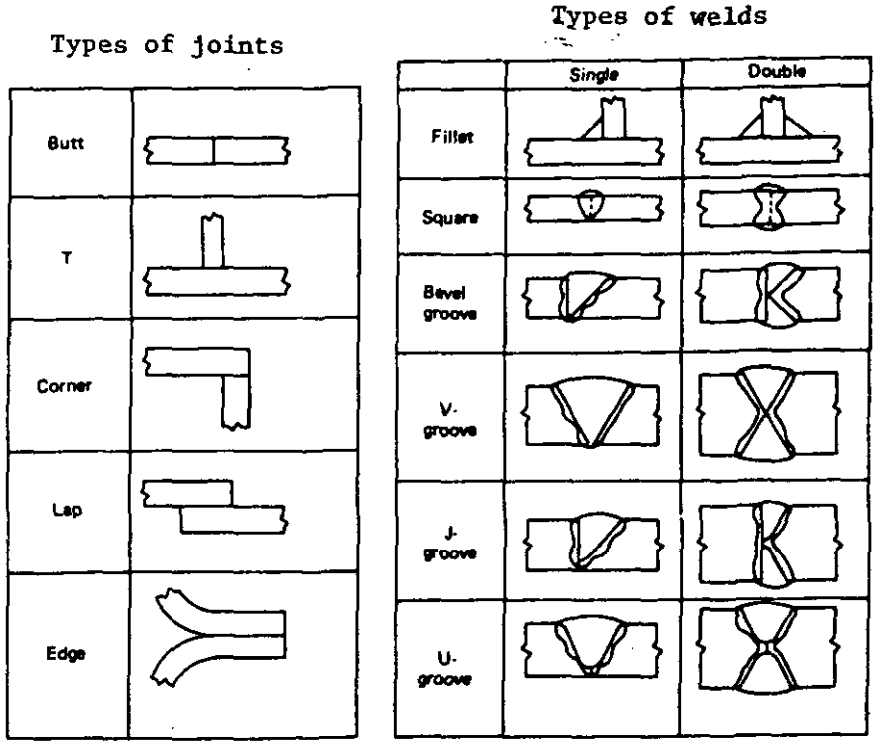


FIGURE 17. Types of joints and welds recognized by AWS.

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There are two basic joints in brazing: lap and butt. The scarf joint is merely a modification of a butt joint. Figure 18 shows these three joints used in brazements.

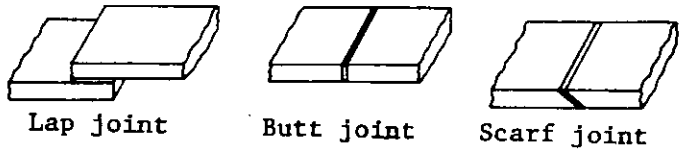


FIGURE 18. Three different joints used in brazing.

The factors that affect the strength of a brazed joint include the amount of clearance or interference of the joint, its cleanliness and length, composition of both the filler and base metal, time at the brazing temperature and post-brazing heat treatment. A large joint clearance is undesirable because the filler metal may not flow through capillary action. So, zero clearance (size-to-size fit) or an interference fit (negative clearance) is preferred, especially with copper-base filler metals. Clearance is the distance between the faying surfaces of the joint, which may be determined either at ambient or at brazing temperatures. The distance of filler metal flow is a strong function of clearance and braze metal properties. The proposed clearances at the brazing temperature are given in table IV.

TABLE IV.

Recommended Clearances at Brazing Temperature.

Filler metal AWS classification			Joint clearance
	mm	in.	
BA1Si group	0.15-0.25	0.006-0.010	For length at lap less than 6.35 mm (1/4 in.)
	0.25-0.61	0.010-0.025	For length at lap greater than 6.35 mm (1/4 in.)
BCuP group	0.03-0.12	0.001-0.005	
BAg group	0.05-0.12	0.002-0.005	Flux brazing (mineral fluxes)
	0.03-0.05	0.001-0.002	Atmosphere brazing (gas phase fluxes)
BAu group	0.05-0.12	0.002-0.005	Flux brazing (mineral fluxes)
	0.00-0.05	0.000-0.002	Atmosphere brazing (gas phase fluxes)
BCu group	0.00-0.05	0.000-0.002	Atmosphere brazing (gas phase fluxes)
BCuZn group	0.05-0.12	0.002-0.005	Flux brazing (mineral fluxes)
BMg group	0.10-0.25	0.004-0.010	Flux brazing (mineral fluxes)
BNi group	0.05-0.12	0.002-0.005	General applications (flux or atmosphere)
	0.00-0.05	0.000-0.002	Free flowing types, atmosphere brazing

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The two basic soldering joints are lap and butt. The lock seam is a variation of lap joint. Lap joints are stronger than butt joints. Thus, if maximum strength is required, the designer should select lap joints for brazing and soldering. A general design rule is that the length of the joint should be about three times the thickness of the thinnest member of the brazed assembly.

Since solders are generally weaker than the base metals, the soldered joints should not be so designed as to depend for strength on the filler metal. This can be attained by resorting to interlocking the components involved. In this case, the solder serves only as a seal and stiffener of the joint.

Presented in Figure 19 are the various soldering joints.(7)

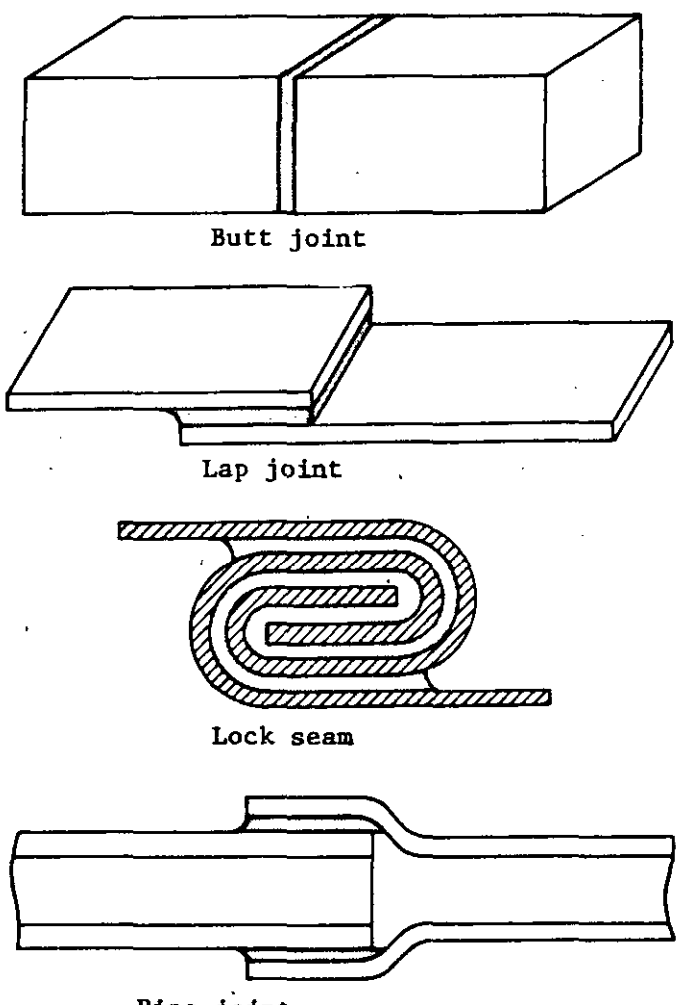


FIGURE 19. Soldering joints.

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1.14 Welding symbols

The welding symbols have been standardized by ASW. The purpose of using a uniform system of symbols is to transmit welding instructions to every one concerned with the weldment. The basic elements of the welding symbol include reference line, arrow, basic weld symbol, dimensions, finish symbol, tail for process specification and supplementary symbol.(8) The standard location of elements of a welding symbol is presented in Figure 20.

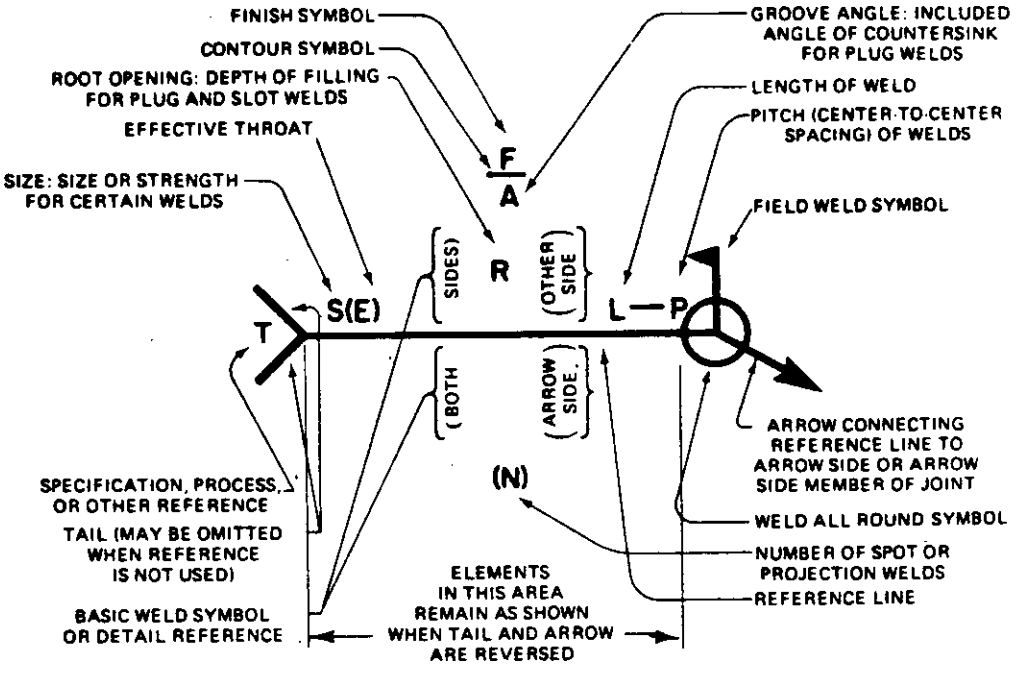


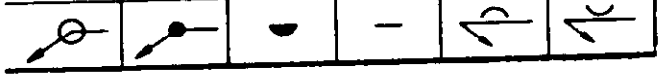
FIGURE 20. Elements of a welding symbol.

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Work pieces can be joined to one another in several ways. Therefore, different welding symbols have been adopted by AWS which are shown in Figure 21.

GROOVE							FILLET	PLUG OR SLOT	SPOT PROJECTION	SEAM	BACK OR BACKING	SURFACING	FLANGE	
SQUARE	V	BEVEL	U	J	FLARE-V	FLARE-BEVEL							EDGE	CORNER
														

(a)

WELD ALL AROUND	FIELD WELD	MELT-THRU	CONTOUR		
			FLUSH	CONVEX	CONCAVE
					

(b)

FIGURE 21. Basic weld symbols (a) and Supplementary symbols (b).

1.15 Jigs, Fixtures, Positioners

The principal objectives of these tools are to hold the workpieces in proper alignment during handling and welding, to minimize distortion, and to provide a more convenient work position in order to increase productivity and to improve quality. These tools may be temporary or permanent depending on the volume and similarity of the workpieces. Tack welding is used to keep the parts aligned and maintain the required joint clearance.

Warpage, buckling, and misalignment can be prevented by a variety of techniques. These include clamping, fixturing, tack welding, chill bars, strongbacks, jigs and hold-down bars. Strongback is a U-shaped restraining plate. Turn-tables and turning rolls are for rotating the workpieces in order to position the joint in the most favorable work position.(5)

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Few examples of a particular type positioner and turning rolls are illustrated schematically in Figures 22 and 23.

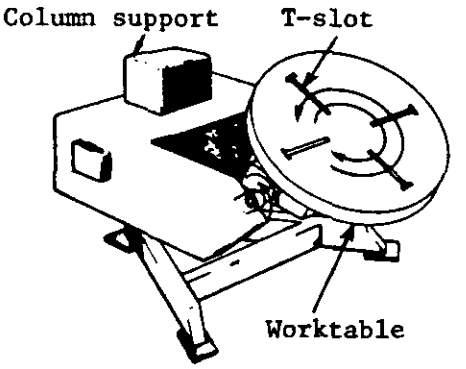


FIGURE 22. A type of positioner which rotates and tilts the workpiece.

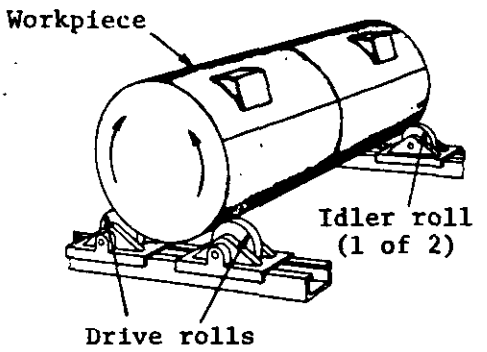


FIGURE 23. Turning rolls used for cylindrical workpieces.

Strongbacks which bridge the joint are used to maintain proper joint alignment. Strongbacks are most frequently employed in heavy construction work such as shipbuilding. It should be remembered that the holding fixtures should be so designed as to permit the necessary expansion and contraction of the assembly without giving rise to distortion. Otherwise, undesirably high residual stresses may result from the use of an overly rigid set up. In addition, a good fixturing provides for simple and accurate assembly, disassembly, and accessibility for joining.

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Chill bars are used to cool the weld area rapidly. Backing bars, inserts and facing plates are used for the purpose of controlling the root of the weld. Grooved backing bars for instance are applied to prevent melt-through and control the amount of root reinforcement. Backing bars can also provide passages for inert gas shielding of the weld root.

1.16 Structure of Solids

The term solid may be described as a state of aggregation of atoms, ions or molecules. A solid body has a definite shape and volume. Unlike gases and liquids, solids are able to resist shearing forces and to reassume their original shape upon removal of a certain amount of deformation.

The structure of solids varies from simple to complex crystalline and amorphous bodies. The type of bond and geometrical arrangement of atoms or molecules determine the structural type of the solid. The fundamental difference between crystalline and amorphous solids is that amorphous solids do not have long-range order. The atoms or groups of atoms of crystalline solids display periodicity or repetitive pattern of some type of geometric regularity.

There is no universally agreed categorization of the many types of engineering materials. For the purpose of this handbook, materials may be divided into (a) metals, (b) polymers, (c) ceramics and (d) composites.

1.16.1 Metals

Most metallic solids crystallize in the most densely packed crystal structure, specifically body-centered cubic (BCC), face-centered cubic (FCC) and hexagonal close-packed (HCP). Figure 24 shows the geometric presentation of these atomic arrangements.(9)

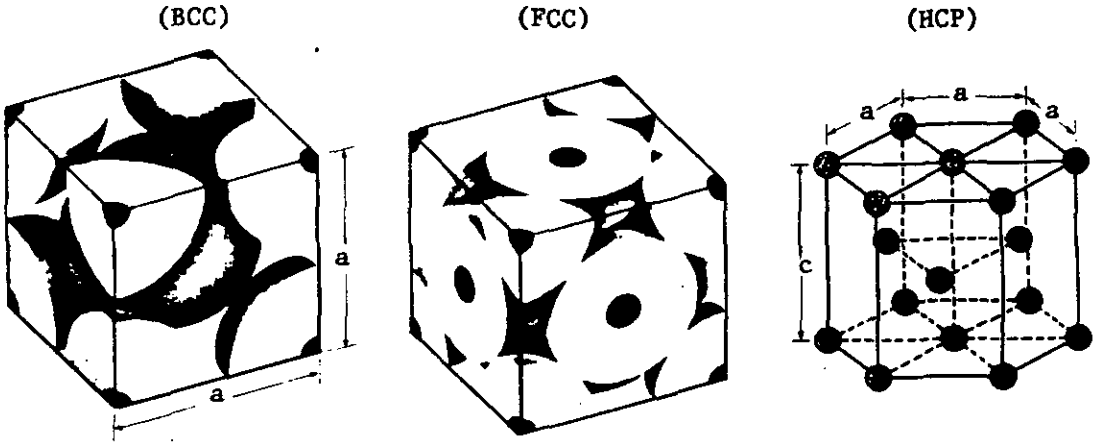


FIGURE 24. Unit cells of (BCC), (FCC) and (HCP) structure.

Most of the metallic solids that come into play with welding, brazing and soldering are typified by these structures.

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1.16.2 Polymers

Polymers or plastics are large molecules of nonmetallic elements. They consist of many repeating units called mers. The molecule from which a polymer is produced through a process of polymerization is called a mer. Polymers may be natural or synthetic. Cellulose and starch belong to the former and high polymers formed by polymerization constitute the latter. Polymers may vary from viscous liquids to rigid solids. Polymers are mostly compounds of carbon with hydrogen and oxygen and have a low resistance to high temperatures. In terms of structural arrangements, there are linear or chain polymers and three-dimensional network or crosslinked polymers.

Synthetic polymers may conveniently be divided into two classes: thermoplastics and thermosettings. The thermoplastics are linear polymers, either straight-chain or branched-chain macromolecules that soften under the influence of heat and harden upon cooling. The thermosets or network polymers soften when first heated and then harden to infusible resin upon further heating.

1.16.3 Ceramics

Ceramic materials contain compounds of metallic and nonmetallic elements. Many ceramic phases are crystalline like metals. The ionic or covalent bonds give the ceramics a relatively high thermal stability. Their structure does not contain significant numbers of free electrons. Therefore, ceramics are very good insulators. Thus, joining of ceramics requires methods different from those of metals. Cement and adhesives are two of the methods commonly used for joining ceramics.

1.16.4 Composite Materials

Composite materials contain at least two distinct materials so as to obtain the optimum combination of properties of the constituents. Metals, polymers, ceramics, glasses, cement and others may be combined in composite materials to produce special characteristics. The so called advanced composites may consist of a binder or matrix material, and some type of reinforcements, fillers, fibers or dispersed phases. The reinforcement may come in the form of whiskers.

1.17 Welding Metallurgy

Since heat has a pronounced effect on the properties and nature of materials, it is important to understand the metallurgical aspects of materials joining. Welding, brazing and soldering, involve heat generation to different levels of temperature. Consequently, the state and structure of the material undergo changes.

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In fusion-welding, brazing and soldering, it is customary to speak of transformation from liquid to solid as in casting, and phase transformation in the solid state as in heat treatment. However, the metallurgy of welding, brazing and soldering goes beyond transformation phenomena. Gas-metal reactions, slag-metal interactions, surface phenomena, dilution or compositional changes, discontinuities, residual stresses, distortion and numerous other physical aspects must be recognized. The full treatment of them of course falls outside the scope of this handbook.

Transformation of the particular phases that any given metallic system may contain at specific levels of temperature and composition is depicted by phase diagrams. For illustration purposes, Figures 25 and 26 show the iron-carbon and copper-nickel alloy systems.

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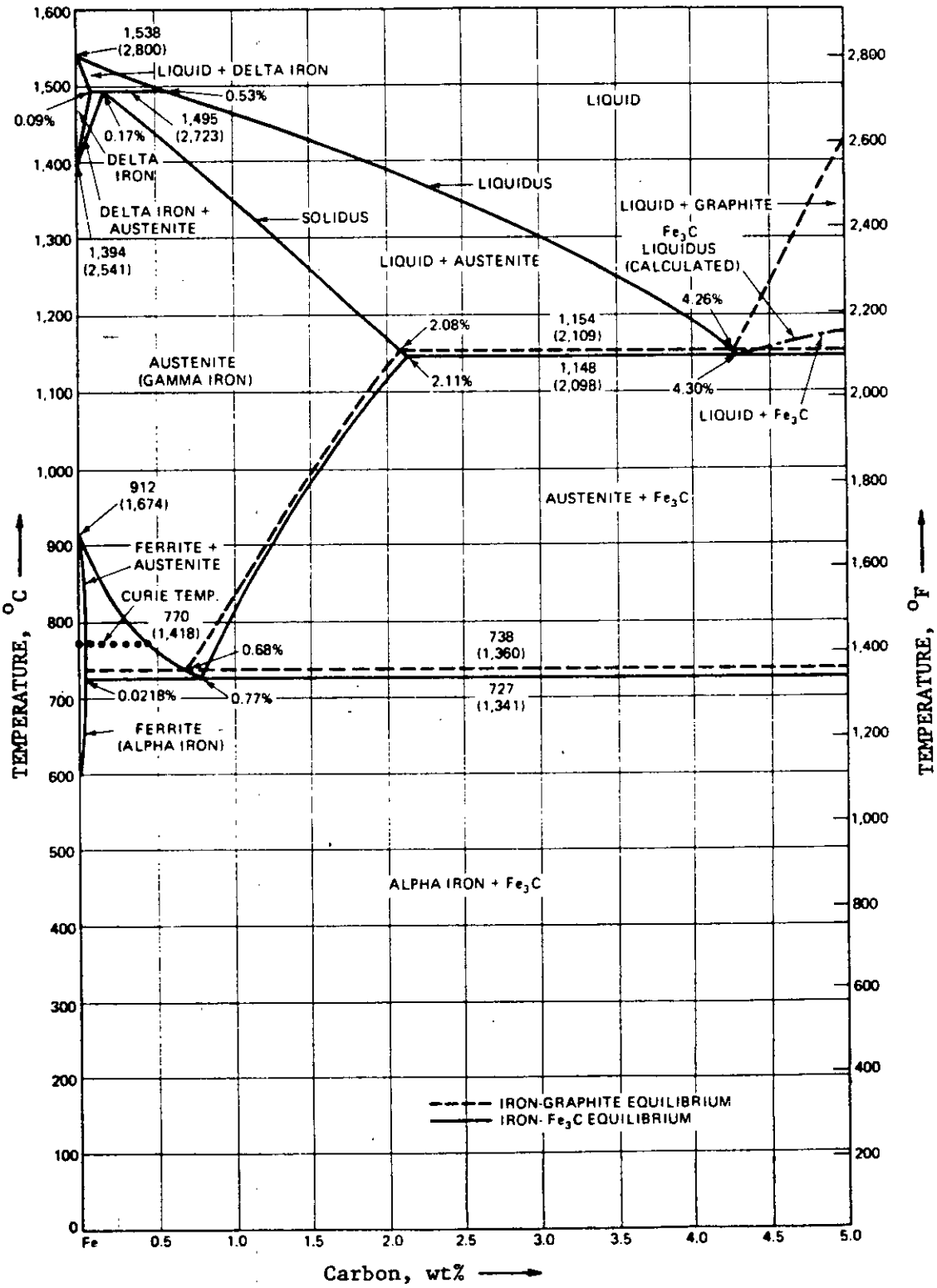


FIGURE 25. Fe-C equilibrium phase diagram (up to 5% carbon).

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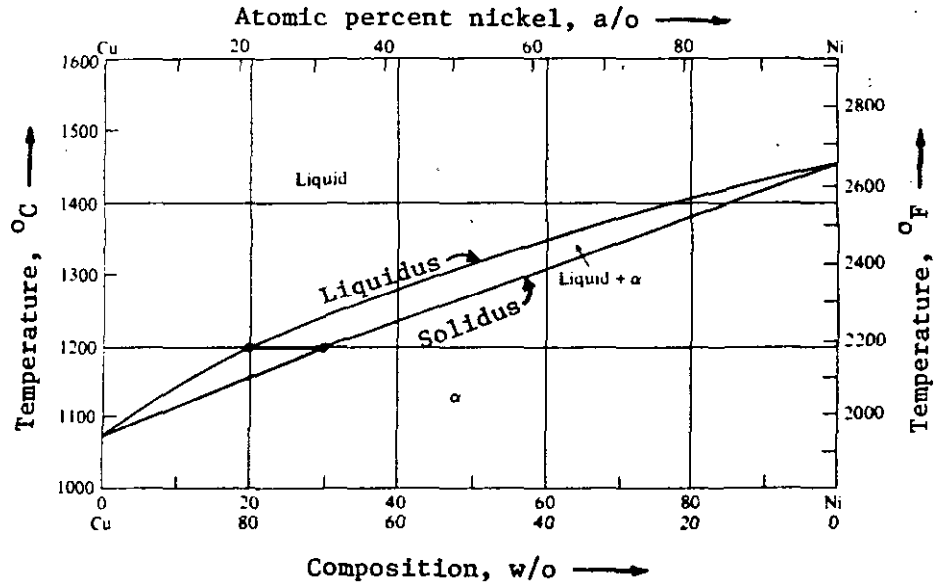


FIGURE 26. Cu-Ni alloy system. The tie line between the liquidus and solidus curves at 1200°C shows 80% Cu-20% Ni and 70% Cu - 30% Ni alloys.

While commercially pure metals find many important applications, alloys with two or more components are even more valuable. The alloying elements added to pure metals can form either solid solutions or intermetallic compounds. In a solid solution, the solute atom either fits interstitially in the interatomic spaces of the solvent or host atoms, or substitutes for one or more of the host atoms on the crystal lattice. The size of the solute atom has a marked effect on the solubility of the element. There are other factors which affect the solubility. They are crystal structure, relative valency factor and electronegativity. Solubility increases with increasing temperature and vica versa.

Solid solutions may be primary or secondary. Primary solid solutions have the crystal structure of the host element. The structure of a secondary solid solution is different from those of the component elements.

Intermetallic compounds of two or more metals have a crystal structure unlike that of any one of the components present. It is a new and different phase in the microstructure. In a pure metal, each atom has, of course, its own kind of neighbors. In an intermetallic compound, a specific species of atom is surrounded by unlike neighbors. Such an atomic arrangement provides the intermetallic compounds with unique properties such as much higher hardness and rigidity than either pure metals or solid solutions. Some of the compounds contain a nearly fixed ratio of atoms of the AB_2 type, for instance. But, intermetallic compounds are more commonly nonstoichiometric, i.e., excess or deficiency of positive or negative charges (ions or anions).

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A phase diagram is made up of a number of phase fields. Each of these phase fields is occupied either by a single phase or a mixture of two phases under equilibrium conditions. In practice, equilibrium is seldom achieved because the cooling rates are practically always higher than those needed to maintain equilibrium. Welding processes are a case in point. As the cooling rate increases, the transformation temperatures are decreased, phase boundaries moved, and metastable (nonequilibrium) phases may be produced. In arc welding of steel, for example, the rapid cooling rate may lead to the formation of martensite, which certainly has no place on the equilibrium diagram. The presence of alloying elements also has a significant effect on the transformation range of certain phases and can in fact produce a completely different diagram.

1.17.1 Phase Transformations

Phase diagrams are used to predict the types of phases and their compositions that can occur under certain set conditions of temperature and pressure. These diagrams do not, however, give adequate information about the rate at which phase transformations might occur.

Phase transformation deals with the appearance or disappearance of a phase. A phase may be defined as a distinct but homogeneous part of a system. Matter can exist in solid, liquid and gaseous phase or state. A mixture of two or more miscible liquids or a solid solution of two or more mutually soluble elements is a homogeneous or one-phase system. Conversely, a system consisting of more than one phase is called a heterogeneous one. Phases are separated by phase boundaries. The nature of the phase change may be chemical or physical. The kinetics of transformation involves a rearrangement of a group of atoms into a new, more stable configuration at a specific temperature and pressure. To cause a phase transformation, the atoms must be activated to a state of higher energy before it can proceed spontaneously. This energy is referred to as activation energy or driving force.

The specific reaction rate, k, can be expressed by the Arrhenius equation as

$$k = Ae^{-E_a/RT}$$

where,

A = constant
e = base of natural log
E_a = activation energy
R = gas constant
T = absolute temperature

If the solubility of a solute in a solid solution decreases with decreasing temperature, precipitation occurs. This gives rise to a supersaturated solid solution, which is unstable and tends to decompose as follows

Supersaturated alpha (α) solid solution $\longrightarrow$ saturated solid solution + beta (β) precipitation

This decompositional process is the basis of precipitation hardening brought on by a sequence of thermal treatments. The major steps of this treatment consist of solution heat treatment, quenching and age-hardening or precipitation hardening.

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Since phase diagrams indicate phases obtained under equilibrium conditions, it is not very useful for cooling rates which produce nonequilibrium structures. For this reason, time-temperature-transformation (TTT) diagrams under isothermal and continuous cooling conditions were developed. The resulting curves known as Baine or S curves. TTT diagram for the transformation of austenite in a eutectoid carbon steel is illustrated in Figure 27.

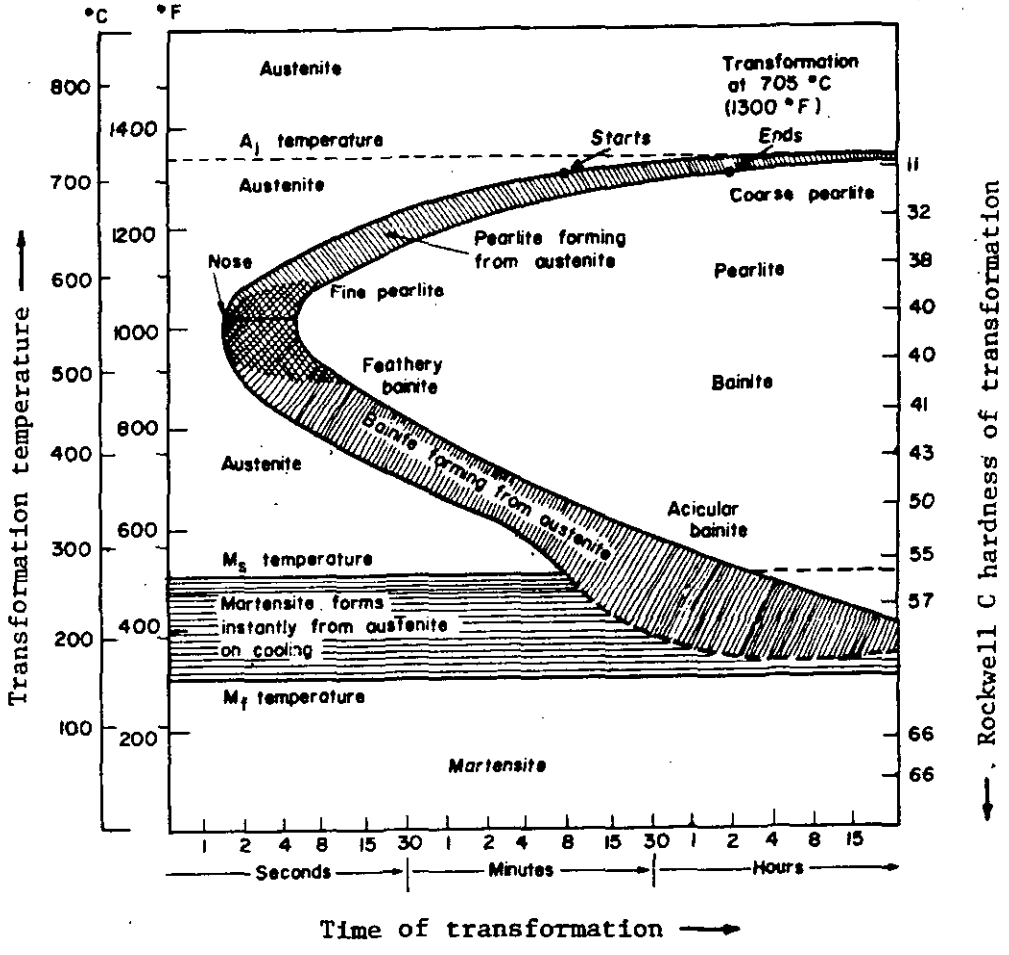


FIGURE 27. TTT diagram for 0.8 wt.% plain carbon steel.

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Isothermal TTT diagrams are still not very useful for describing the phase transformation in an actual heat treatment because the procedure in practice is continuous rather than isothermal. The continuous-cooling-transformation curves (CCT) differ from TTT diagrams in that the S curves are shifted toward lower temperatures and longer times of transformation as seen in Figure 28.

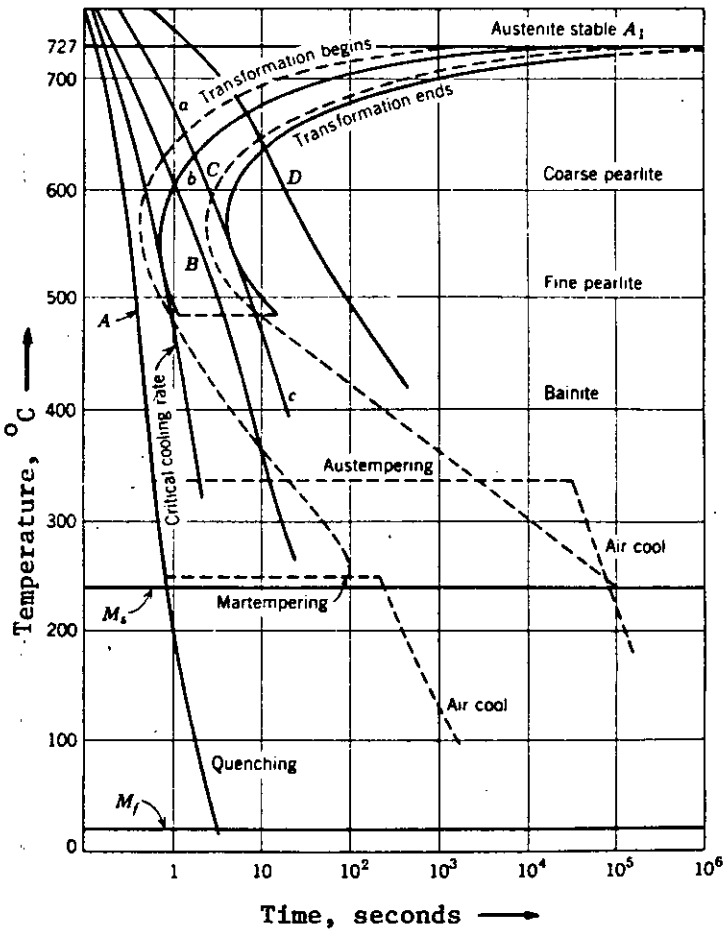


FIGURE 28. CCT (solid curves) and TTT (dashed curves) diagrams for the eutectoid steel.

1.17.2 Heat Treatment

There are several heat treatment methods for ferrous and nonferrous metals. They include annealing quenching, tempering, austempering, martempering, stress relieving, homogenizing, solution treating, and precipitation hardening.

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Heat treatment serves to bring about certain changes in the nature and properties of the material involved. They include changes in physical condition, mechanical properties, residual stresses and corrosion resistance.

Annealing is done for the general purpose of reducing the hardness of the material. Normalizing treatment may be used to obtain a fine, uniform grain size to provide improved ductility or toughness. Normalizing may soften, harden or stress relieve the steel depending on its condition prior to normalizing. Heating the steel into or above the transformation range to make the structure homogeneous is called austenitizing . It is a heat treatment process used prior to hardening carbon and low-alloy steels. Quenching is a rapid cooling of a metal from an elevated temperature. The quenching media may be water, oil, forced air and synthetic quenchant. Tempering is reheating a quench-hardened or normalized ferrous alloy to an intermediate temperature below the transformation range followed by cooling at the desired rate. Austempering and martempering are modified versions of the austenitizing, quenching and tempering process. Austempering produces a bainitic microstructure. The primary purpose of martempering is to reduce distortion caused by unequal transformation rates in complex shapes to be quenched. The residual stresses that are locked in a material as a result of cold working, for example, may be relieved by heating to a high temperature suitable for the particular material in question and cooling slowly to prevent introduction of new residual stresses.

Ingots and castings are nonhomogeneous in terms of chemistry and microstructure. Homogenizing is a heat treatment intended to make the material uniform in composition. Solution heat treatment puts the hardening constituents and carbides into solid solution. This supersaturated solid solution is the necessary first step to subsequent quenching and precipitation hardening. Quenching prevents precipitation of the constituents held in solid solution. This supersaturated solid solution is unstable at ambient temperatures, which may precipitate. The precipitation which occurs at room temperature is commonly referred to as natural aging.

When precipitation is induced by heating the quenched metal to temperatures above ambient but below the solution treating temperature is usually called artificial aging or precipitation hardening.

1.17.3 Effect of Alloying Elements

1.17.3.1 Steel

Alloying elements have a pronounced effect on the equilibrium diagrams and the rate at which transformation reactions occur. Alloying elements are added for a variety of reasons. A well known additive is carbon as an interstitial element to enhance the mechanical properties of pure iron. Others may be added to

- form substitutional alloys and thereby strengthen the metal through solid solution hardening
- form carbides

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- change the ductile-to-brittle transition temperature
- control inclusion account, shape and distribution
- change hardenability
- provide deoxidation
- minimize segregation
- promote nucleation

In steels, alloying elements such as nickel, copper, cobalt, carbon, nitrogen and manganese stabilize austenite: hence austenite formers. The ferrite stabilizers or carbide-forming elements, namely chromium, molybdenum, tungsten, titanium, tin, columbium, vanadium are classified as ferrite formers. Silicon, albeit a ferrite former, is not a carbide-forming alloying element. It is a very effective graphitizing agent used extensively in cast iron.

The effect of alloying elements on controlling the formation of martensite in the heat-affected zone (HAZ) of a weld is empirically expressed by one or the other carbon equivalent (CE) mathematical formulations such as

$$CE = C + \frac{Mn}{6} + \frac{Ni}{15} + \frac{Cr}{5} + \frac{Mo}{4} + \frac{V}{5}$$

The alloying elements of this expression are used in weight percent. When CE is equal or greater than 40, the tendency for cracking is greatly enhanced.

The main alloying element in stainless steel is chromium which is added to steel in sufficient quantity to provide corrosion resistance. Other alloying elements are present to improve specific properties. Stainless steels are usually classified as ferritic, martensitic and austenitic. The ferritic steels are characterized by a chromium-to-carbon ratio which does not allow hardening by heat treatment. The chromium and carbon contents vary from 11.5-27.0% and 0.08-0.2%, respectively. Martensitic stainless steels contain 11.5-18.0% chromium and 0.15-1.2% carbon. Nickel may also be added to the martensitic type. Both the ferritic and martensitic stainless steels are ferromagnetic and offer very good resistance to oxidation and corrosion. Ferrite is present at high and low temperatures in ferritic steels. Therefore, ferritic steels are not hardenable by heat treatment. The martensitic stainless steels undergo the same phase transformation as the carbon and low-alloy steels during heat treatment. The austenite transforms upon rapid cooling to martensite.

The austenitic stainless steels are in essence alloys of chromium, nickel and iron with carbon content of up to 0.25% maximum. Titanium, vanadium and molybdenum are also added for specific purposes. Titanium or columbium (niobium) are used for stabilizing the austenitic stainless steels by preventing the precipitation of chromium carbide at elevated temperatures. Titanium and columbium form a stable carbide. Quenching the austenitic stainless steel in water from above 1000°C (1830°F) also suppresses the precipitation of chromium carbide, thereby stabilizing the austenitic structure.

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The austenitic stainless steel in the stabilized condition is nonmagnetic. Cold working produces some ferrite in the microstructure. Consequently, this steel becomes slightly magnetic. Localized heating such as welding and an attendant slow cooling causes welding decay, which is synonymous with an intensive carbide precipitation. Since the chromium carbide ($Cr_{27}C_3$) contains large quantities of chromium and it precipitates along grain boundaries, the result is a chromium-depleted zone along the grain boundaries. Stainless steels of this type possess a low resistance to corrosion. The heat-affected-zones of welded austenitic stainless steels display inferior corrosion resistance, because this region of the weld cools slowly through the temperature range of 425-870°C (800-1600°F) where carbide precipitation occurs. These steels must be given a stabilizing treatment to restore them to their normal condition. Carbide precipitation can also be prevented by having stabilizers (Ti or Cb additions) or extra low or no carbon content in these steels. The stabilizers tie up the carbon and thereby prevent the formation of chromium carbides along the grain boundaries.

1.17.3.2 Aluminum and Magnesium

The cubic structure (face-centered cubic) of aluminum makes it amenable to extensive plastic deformation. Aluminum has a great affinity for oxygen. Fortunately, the tenaceous aluminum oxide (Al_2O_3) film that developes on the surface gives the aluminum a protective coating.

The properties of pure aluminum and magnesium, much like other metals can be changed by cold working and alloying additions. Silicon, manganese, magnesium, zinc, copper and iron are the most common alloying elements for pure aluminum. The age-hardenable aluminum alloys contain elements such as copper, zinc and magnesium, primarily.

Pure magnesium may be alloyed with aluminum, zinc, manganese, silicon, tin, cadmium, lithium and others. The hexagonal crystal structure of magnesium does not allow much cold work because it lacks the greater symmetry characteristics of cubic metals.

1.17.3.3 Copper and Nickel

Copper is a historically important metal dating back to the Bronze Age. Copper-base alloys have among others two well-known qualities: corrosion resistance and good conductivity.

Many different alloying elements are added to copper. Zinc, aluminum, tin, silicon, nickel, beryllium, chromium, boron, zirconium are the elements copper is most often alloyed with. The copper-nickel alloys have an excellent corrosion and biofouling resistance. Both copper and nickel have a face-centered cubic structure and are mutually soluble in one another.

Nickel-base alloys have high-temperature strength and good corrosion resistance. Nickel has a high work-hardening characteristic. The nickel-copper and the nickel-chromium-iron alloys are important solid solution alloys which can be hardened by cold working.

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Chromium and molybdenum additions to nickel alloys provide superior corrosion resistance to many environments. When aluminum, titanium, beryllium or silicon are added, the resultant nickel-base alloy becomes precipitation-hardenable.

1.17.3.4 Titanium

Titanium undergoes an allotropic change from hexagonal close-packed (HCP) to body-centered cubic (BCC) structure at 880°C (1620°F). These two structures are commonly known as alpha (α) and beta (β) phases. Impurities and alloying elements present in the titanium do affect this transformation temperature.

Chemically, titanium is very active and has a strong affinity for oxygen, which increases with temperature. Above 650°C (1200°F), the oxidation resistance of titanium decreases rapidly. It must therefore be protected from air to prevent embrittlement by oxygen and nitrogen. Strengthening of titanium can be achieved by alloying additions and allotropic transformation. The alloying addition can improve strength through lattice distortion or solid solution strengthening depending upon the type of element added to titanium. Alloying elements include oxygen, nitrogen, hydrogen, carbon as interstitial elements and aluminum, tin, molybdenum, vanadium, tantalum, iron and others.

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2. SHIELDED METAL-ARC WELDING

Shielded metal-arc welding (SMAW) is a manual arc welding process in which the heat is produced by an electric arc between a flux-covered consumable electrode and the base metal. The weld pool is protected from atmospheric contamination by a gaseous shield and fused flux or slag. The gas shielding is the result of combustion and decomposition of the flux. The electrode consists of a cast metal rod and flux covering. The rod, known as the electrode, conducts the electric current to the arc and serves as a filler metal for the joint.

2.1 Process Descriptions

SMAW is the most versatile welding process and least costly of all arc welding processes. It is an all-position welding process, when the proper electrode diameter and type are selected for the different positions. Shielded metal-arc welding can be applied to most metals except those with low melting points such as zinc, lead and tin.

The process has two major disadvantages: low productivity due to the limited length of the electrode and the necessary deslagging of every weld pass. A schematic diagram shown in Figure 29 illustrates the highlights of the process.

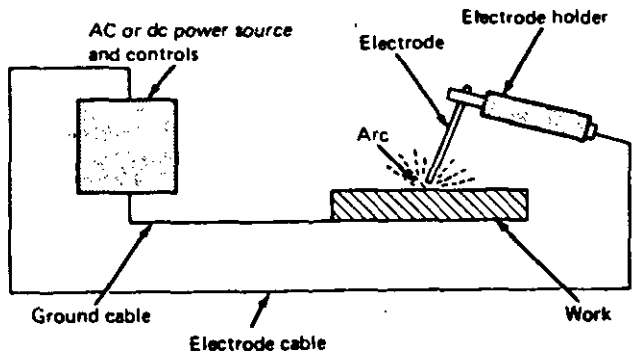


FIGURE 29. Circuit diagram for shielded metal-arc welding.

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Several basic devices are needed to make a manual arc weld. They are the power source, welding cable, electrode holder, ground connection and electrode. The power source provides the electrical energy to melt the workpieces and the electrode. The power source must permit control of the arc characteristics as necessary for a specific job. The welding cables connect the electrode holder and the ground clamp to the power source. The cable carries the electrical current from the power source through the arc to the base metal being welded. The ground cable connects the power source with the workpiece thus completing the electrical circuit. The ground connection must be good to prevent arc blow. The electrode holder is merely a clamping device for grasping the electrode. The electrode is a part of the electrical circuit and it is consumed during the welding process. It fills the joint and thereby becomes a part of the weld.

2.2 Covered Electrodes

The properties of a welded joint in a given base metal are strongly dependent on the composition of the electrode and its flux. Factors such as alloying additions, deoxidizers, arc stabilizers, slag formers, fluxing agents and the presence of components which convert to gas upon decomposition have a great effect on the properties of the weld.

The diameter of the electrode varies from 1.6 mm (0.06 in) to 7.9 mm (0.31 in). The length of the electrode ranges from 22.9 cm (9 in) to 45.7 cm (18 in). The electrical contact is made through an uncoated end section of the electrode. The American Welding Society established a system of identification and classification of covered electrodes. The system consists of a prefixed letter "E" for electrode, followed by a four- or five-digit number as can be seen in Figure 30.

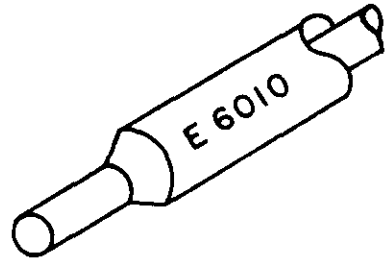


FIGURE 30. An example of AWS classification of mild steel covered electrode.

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The first two digits (or three digits of five-digit numbers) indicate minimum tensile strength, in 1000 psi, of the deposited weld metal in the "as welded" condition. The third or fourth number stands for weld position for which the electrode can be used satisfactorily. One (1) means all positions: flat, horizontal, vertical and overhead. Two (2) indicates horizontal fillet and flat position only. The fourth or fifth (or last) number of the system implies the type of flux covering and suitable welding current. When the last digit is zero, the type of coating and current to use are designated by the third digit. According to Figure 30, the electrode operates on direct current reverse polarity and has a cellulose sodium coating.

Table V contains AWS specifications for covered electrodes.

TABLE V.

AWS Specifications for Covered Electrodes.

Type of Electrode	AWS specification
Carbon steel	A5.1
Low alloy steel	A5.5
Corrosion resistant steel	A5.4
Cast iron	A5.15
Aluminum and aluminum alloys	A5.3
Copper and copper alloys	A5.6
Nickel and nickel alloys	A5.11
Surfacing	A5.13 and A5.21

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The usability ratings as indicated by the last digit in AWS classification of low-carbon steel covered electrodes can be found in table VI.

TABLE VI.
Coverings and currents indicated by the fourth digit in the AWS identification system for low-carbon steel covered electrodes.

Fourth Digit	Covering	Current(a)
0	High cellulose, sodium(b)	DCEP(b)
	High iron oxide(c)	ac or dc(c)(d)
1	High cellulose, potassium	ac or DCEP
2	High titania, sodium(e)	ac or dc(f)
3	High titania, potassium	ac or dc(f)
4	Iron powder, titania	ac or dc(f)
5	Low hydrogen, sodium	DCEP
6	Low hydrogen, potassium	ac or DCEP
7	Iron powder, iron oxide	ac or dc(d)
8	Iron powder, low hydrogen	ac or DCEP

(a) ac, alternating current; dc, direct current; DCEP, direct current electrode positive. (b) When third digit is 1. (c) When third digit is 2. (d) Either polarity for flat welds; DCEN for horizontal fillet welds. (e) Can also imply high iron oxide, ac or dc, either polarity. (f) Either polarity.

Table VII contains pertinent details for various classes and diameters of low-carbon steel covered electrodes.

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TABLE VII.

Welding Characteristics and Mechanical Properties of Some Low-carbon Steel Covered Electrodes.

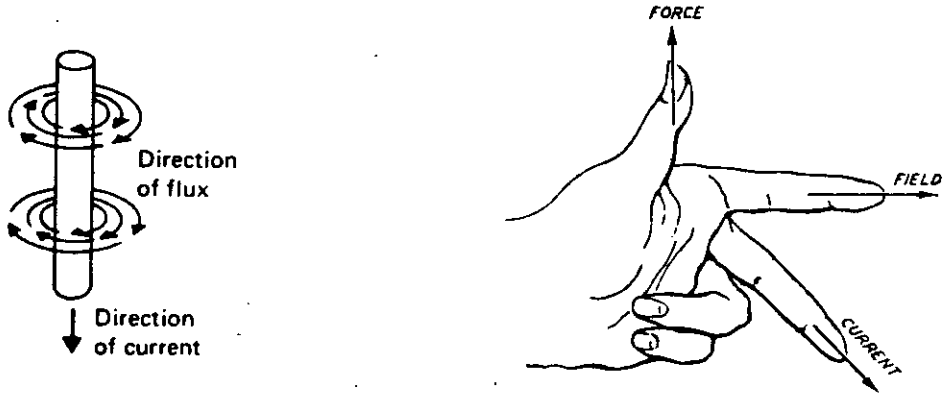
Electrode class	Electrode size, in.	Current(a) Amp	Deposition Rate(b), oz/h	Deposition efficiency, %	Properties of weld metal		
					Tensile strength ksi	Yield strength ksi	Elongation in 2 in., %
E6011	3/16	200 max	68	76	70	60	25
	1/4	300 max	112	72	-	-	-
E6012	3/16	225	66	68	79	66	20
	1/4	380	120	78	71	58	15
	5/16	475	178	78	-	-	-
E6013	3/16	225	67	67	76	67	22
E6020	3/16	225	92	68	69	59	31
	1/4	380	177	69	65	52	29
	5/16	450	215	69	63	50	28
E7014	3/16	260	87	68	74	68	23
	1/4	340	119	69	73	67	25
E7016	3/16	225	63	70	87	75	29
E7018	3/16	240	83	69	80	67	29
E7024	3/16	270	134	68	90	80	22
	1/4	360	186	69	89	77	22
	5/16	475	250	73	83	74	22
E7028	3/16	300	125	68	90	81	25
	1/4	390	192	70	88	80	25

(a) Alternating current. (b) 100% arc time

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2.3 Arc Blow

An unbalanced condition in the magnetic field surrounding the arc causes magnetic arc blow.(10) Imbalance may exist because of the directional charge of the current flow and arc position with respect to the ground connection. A direct current passing through a conductor sets up a magnetic field, or flux, around the conductor. The flux density decreases with increasing distance away from the conductor as shown in Figure 31a. The magnetic field exerts a force on the conductor. The direction of the force can conveniently be illustrated by the left-hand-rule depicted in Figure 31b.



(a) (b)

FIGURE 31 Force acting on a conductor placed in a magnetic field.

The magnetic field does not remain circular when the medium changes from, say air to metal. They become distorted at the boundary between the edges of steel plate and air. The lines of magnetic force are squeezed and tend to concentrate in the steel where there is less resistance. This squeezing induces a flux concentration either ahead or behind the arc. The arc attempts to move in the direction that would relieve the squeezing in order to restore flux balance. Figure 32 shows the magnetic fields around the electrode at various positions along a joint.

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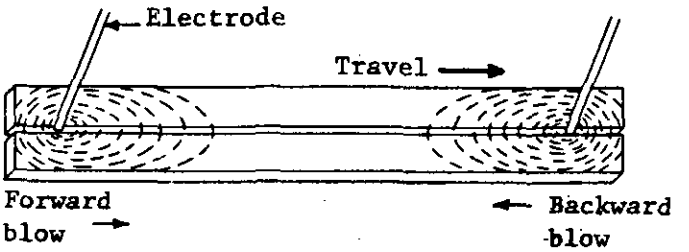


FIGURE 32. Forward and backward blow at the start and finish of a seam weld.

The ground current can also cause arc deflection forward or backward depending on whether the ground is connected to the starting end or finish end of a seam. This is demonstrated in Figure 33.

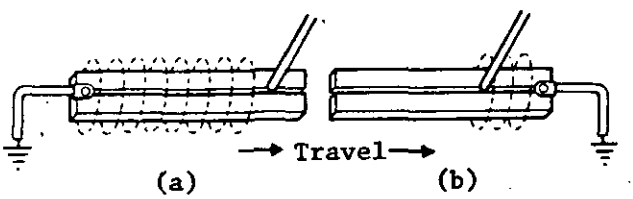


FIGURE 33. The flux from the ground current is behind the arc (a) and ahead of the arc (b) causing forward and backward arc blow, respectively.

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When the two magnetic fields - the one around the electrode and the other around the workpiece - are superimposed as shown in Figure 2.6a, the arc blow is reduced at the finish end of the joint because the two flux fields tend to cancel each other. A strong back blow results when the two fields are additive as is the case in Figure 34.

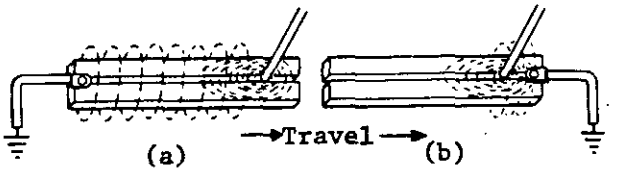


FIGURE 34. Superimposed magnetic fields.

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When two arcs are close to one another, the respective magnetic fields react to cause an arc blow on both arcs. In submerged-arc welding (SAW), multiple arcs are often used to increase weld productivity. The arcs are usually less than 2.5 cm (1 in) apart. The arc actions using different polarities and current types are illustrated in Figure 35.

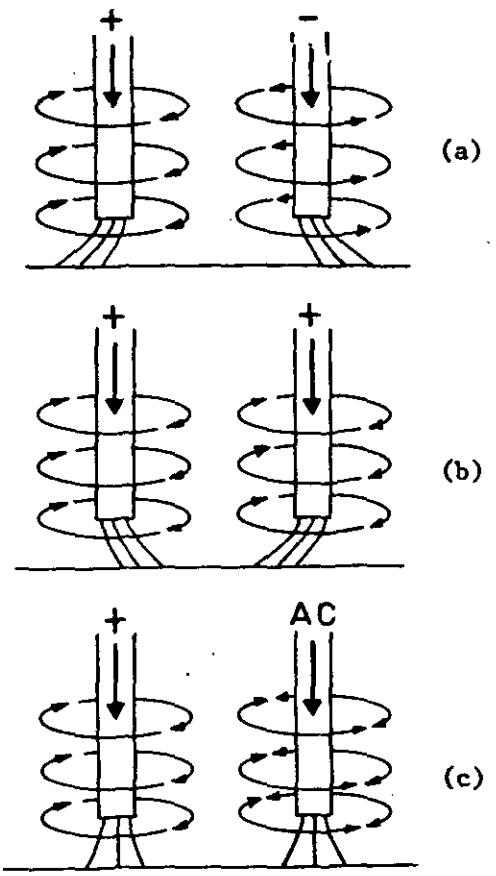


FIGURE 35. Reactions of magnetic fields of forces when two arcs are in close proximity.

(a) arcs with different polarity - outward arc blow
(b) arcs with similar polarity - inward arc blow
(c) twin arcs with DC and AC systems - little or no arc blow.

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Arc blow is practically eliminated when a twin-arc SAW system employs direct current on one electrode and alternating current on the second electrode.

Fixtures can also cause arc blow when the welding current exceeds 250 amps. Therefore, precaution should be exercised in designing fixtures.

Corrective measures for arc blow involve:

- change to alternating current
- reduce weld current
- hold a short arc
- make heavy and frequent tack welds
- weld toward a heavy tack or a weld already made
- wrap ground cable around the workpiece so that the current flows in a direction that produces a flux which cancels the magnetic field inducing the arc blow.
- use low-carbon steel fixture of proper design and of equal masses on both sides of the seam.

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3. GAS METAL - ARC WELDING

GMAW is an arc welding process wherein the heat is produced by an electrical discharge between a continuous filler metal electrode and the base metal. A shielding gas protects the liquid weld metal from reacting with constituents in the surrounding atmosphere. Only argon, helium, carbon dioxide and nitrogen gases have been used as pure shielding gases: argon and helium for nonferrous metals, carbon dioxide for mild steel and nitrogen for welding copper. Although these pure gases can prevent the weld metal from reacting with air, they are unsuitable for many applications for various reasons. Consequently, a large variety of gas mixtures have been developed for specific applications. Common gas mixtures for shielding weld metals include argon-helium, argon-oxygen, carbon dioxide-oxygen, helium-argon-carbon dioxide, argon-nitrogen and other combinations. While the primary purpose of shielding gas is to protect the molten weld metal and the arc from contamination, with a judicious selection of a shielding gas mixture other beneficial effects can be accomplished. These may be improved weld appearance, out-of-position welding, deeper joint penetration, lower cost, arc stability, elimination of spatter, better arc transfer characteristics, greater welding speed and weld quality benefits. A block diagram of GMAW can be seen in Figure 36.

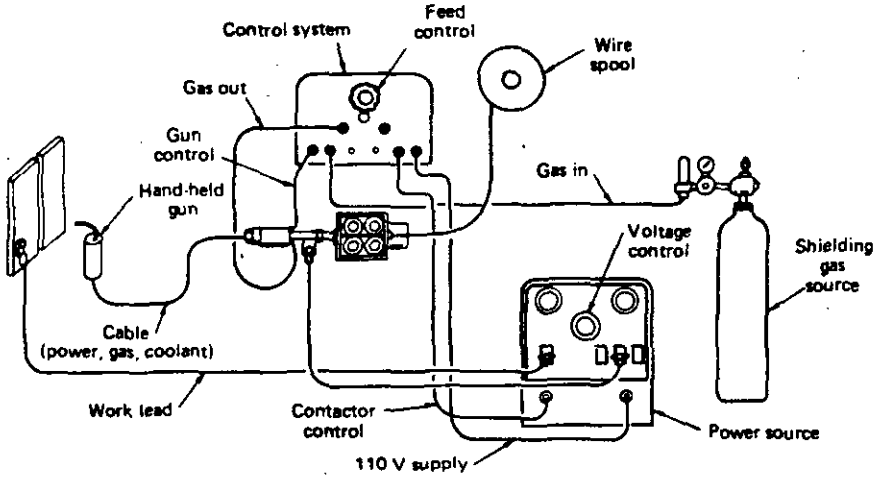


FIGURE 36. Schematic presentation of GMAW.

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The salient features of the arc area of the gas metal-arc welding process is depicted in Figure 37.

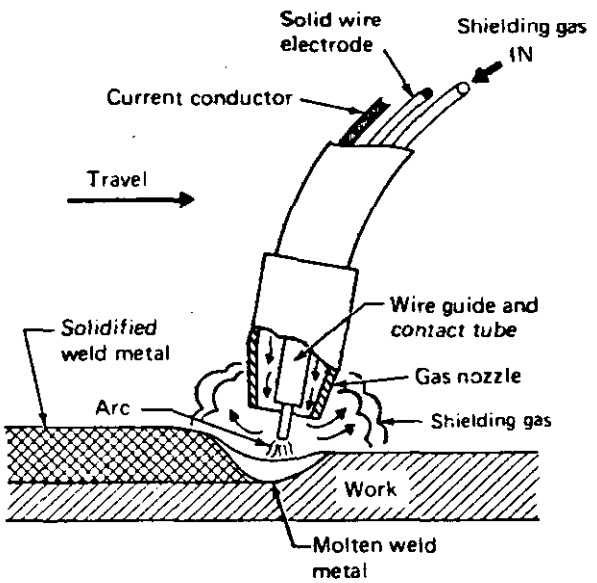


Figure 37. Details of the arc area in GMAW.

GMAW is an all-position joining process and is virtually suitable for all the commercially available metals.

Shielding gas selection requires that consideration be given to:

(1) base metal - composition,
- mechanical properties,
- condition and cleanliness,

(2) service requirements,
(3) weld position,
(4) type of metal transfer.

Shielding gas flow rate is important in GMAW. Draft can affect gas consumption and weld quality in terms of porosity. Tarpaulins or other types of windbreaks should be provided for welding under conditions of strong draft. The amount of flow rate is a function of type of gas used, weld current, joint design, nozzle diameter of the welding gun, weld position, travel speed, angular position of the welding head and intensity of draft. Overhead position welding requires a higher flow rate than flat position welding.

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GMAW has several modifications and often termed as CO_2 welding, electrogas welding, narrow-gap welding, microwire, pulsed GMAW, spray-arc welding and short-arc welding. By and large these terminologies imply modes of metal transfer and weld positions. In all gas metal-arc welding, the filler metal is fed automatically through a wire-feed cable to the electrode holder commonly known as the arc welding gun.

There are different types of wire-feed systems. Among them are the constant-feed, and variable-speed type, which may be of the push, or pull, or push-pull variety. The type used depends mainly on the electrode diameter, composition and the distance between the wire spool and the electrode holder.

In GMAW, there are spray, globular, short-circuiting and pulsed-spray metal transfer modes. Only the pulsed-spray welding, arc spot welding, narrow-gap welding and electrogas welding can truly be named varieties in techniques of the basic gas metal-arc process.

3.1 Pulsed-Spray Welding

The essential feature of this welding technique entails the combination of a high pulsed current to penetrate the joint followed by a low current or pilot arc during the background period when the weld metal solidifies. The "pulsed peak" current is superimposed on the steady "background" current at regularly spaced intervals, as illustrated in Figure 38.

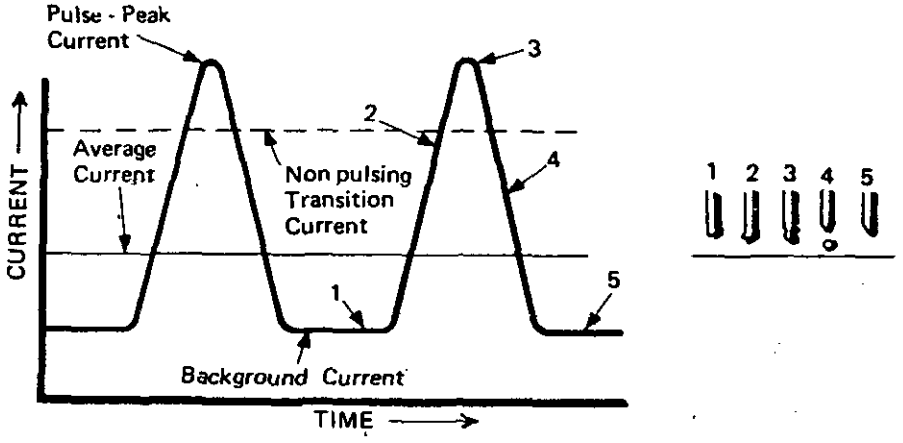


FIGURE 38. Oscillogram of pulsed-spray GMAW.

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The pulsed current is above the transition current. Each electrode has its own transition current, which separates the spray transfer mode from the globular type. The transition current is dependent on diameter, composition, polarity, extension of the electrode and the composition of the shielding gas. The net result of the two current levels is a steady spray transfer at a welding current level below that of a conventional spray arc welding. The attendant lower heat input offers several advantages. Control of the weld pool is much easier in out-of-position welding. Thinner cross-sections can be welded which would not be possible with regular spray arc welding. With a pulsed-arc power supply spray metal transfer occurs at an average current which is much lower than the non-pulsing transition current. Metal detachment from the electrode takes place in the form of a discrete drop in each pulse.(11)

3.2 GMA Spot Welding

Gas metal-arc spot welding uses a timing device which controls the time of current flow usually ranging from 0.5 - 1.5 seconds. The joining of two pieces of metal is done either by penetrating entirely through one piece into the other, or by prepunching or by predrilling an appropriate diameter hole in one of the two workpieces and subsequently spot welding the predrilled piece to the substrate. The workpieces may be similar or dissimilar in terms of chemical composition and thickness.

The welding gun is stationary while a spot weld is made. At a preset time, the arc and the wire feeding system stop automatically. Then, the welding gun is moved to the next spot and the welding sequence continues.

In general, the depth of penetration is a function of weld current, time, standoff distance (SO) and weld position. In the case of having prepunched hole(s) in one of the two workpieces the size and shape of the hole have a pronounced effect on weld penetration into the substrate. In addition, the geometry of the hole or joint has a strong effect on spot weld soundness.

Typical gas metal-arc spot welding data can be found in table VIII. representing 90/10 copper-nickel sheets arc spot welded to mild steel plates.

TABLE VIII
Gas Metal-Arc Spot Welding Details for
2.5 mm (0.1 in) thick Cupronickel Sheet to 19 mm (0.75 in) Steel Plate.

Weld		SO	Hole		Filler	Pulse	Shielding Gas	
Current (Amps)	Type Polarity	(in)	Size (in)	Type	Metal Size (in)	Rate (cps)	Ar/O_2	Flow Rate (cfh)
245	DCRP	.3	3/16	tapered	.035	120	98/2	50

Footnote:

Peak voltage 52
Background voltage 30
Normal pulsing mode
Weld position OH
Weld time 1.0 sec.
Voltage 26 V

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3.3 Narrow-Gap Welding

This process works in the spray transfer range of welding current and was developed for making deep, narrow welds in thick plates. Narrow-gap welding (NGW) is characterized by low filler metal usage, little distortion because of low heat input, high deposition rate and all position welding. Regardless of material to be welded, typical root openings are 6 to 10 mm (0.25 - 0.38 in).

Specially designed welding head accessories are needed for NGW, as depicted in Figure 39 and 40. The technique of feeding the electrode wire and shielding gas into the deep and narrow gap differentiates NGW from conventional GMAW. Figure 40 shows five such techniques.(12) These techniques are designed with the objective of preventing incomplete fusion into the sidewall of the joint,

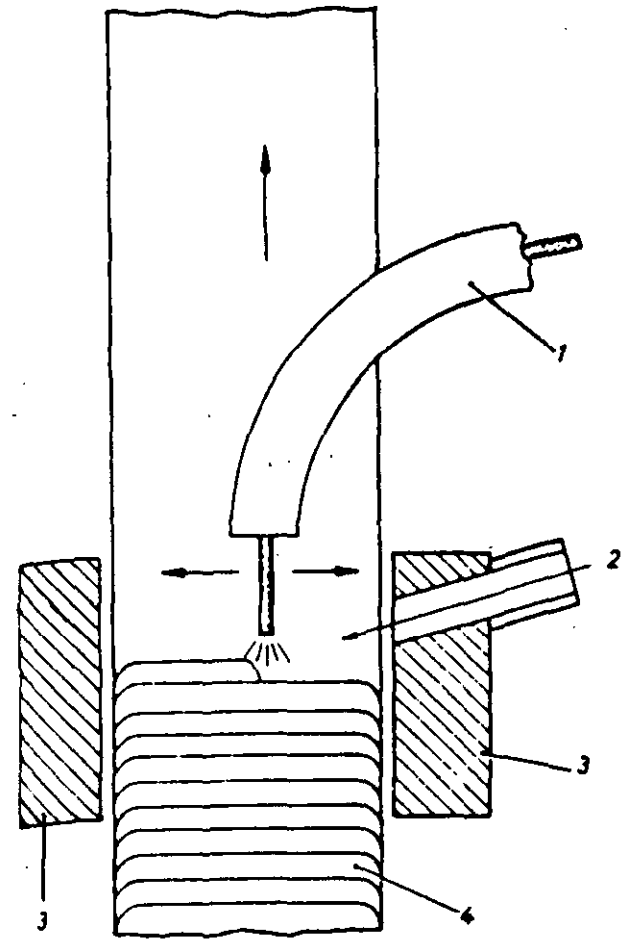


Figure 39. Schematic drawing of the vertical narrow gap welding process: 1 - contact tube; 2 - shielding gas; 3 - sliding copper shoes; 4 - weld metal.

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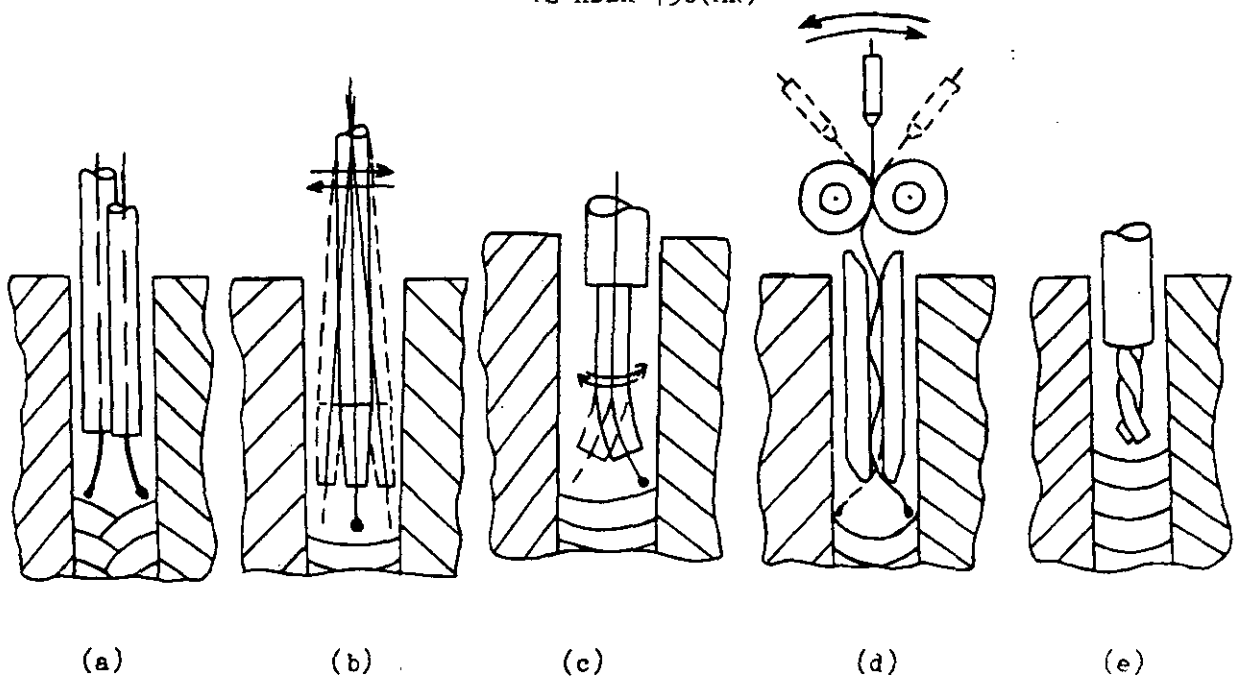
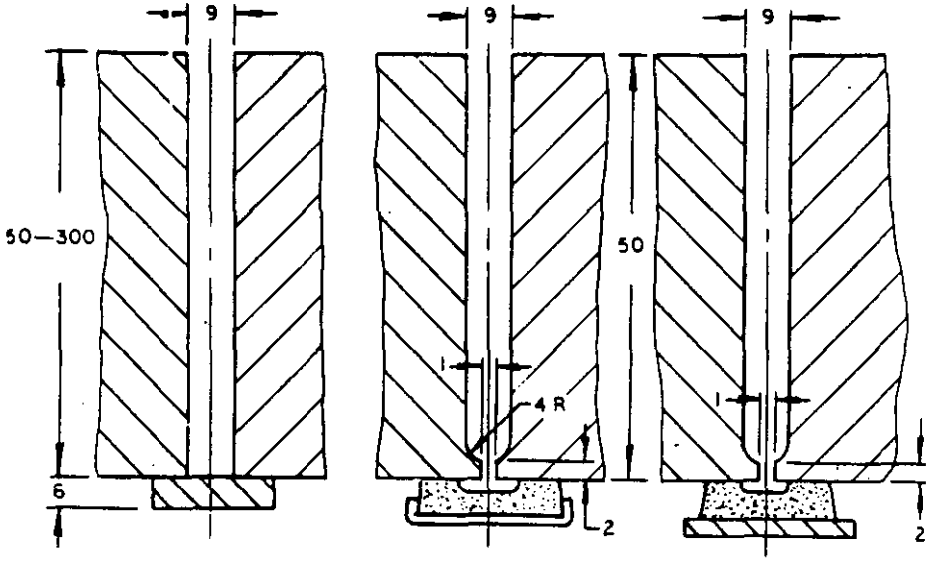


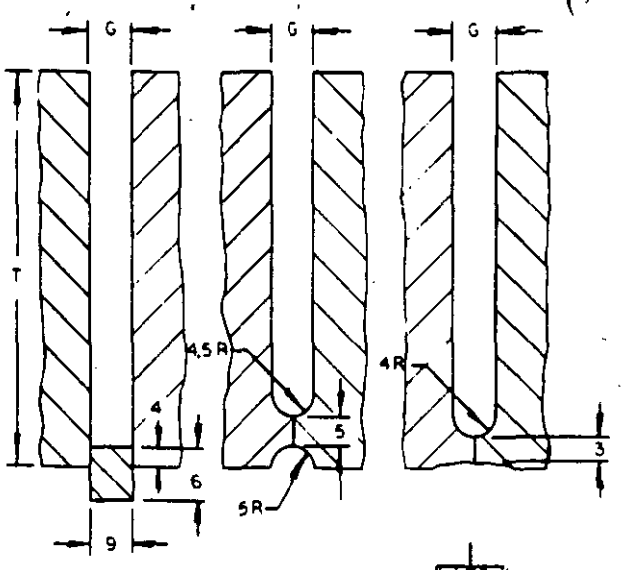
Figure 40. Typical wire feed techniques in NGW.

which is the most typical weld discontinuity in NGW. Narrow-gap welding is considerably more sensitive to changes in welding parameters than other welding processes.

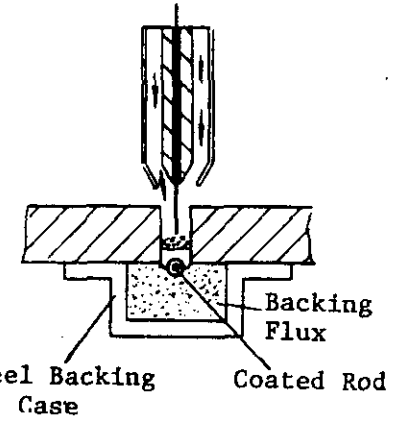
Figure 41 illustrates the different backing techniques used in NGW.(12) When the wire diameter is in the range of 0.9 - 1.5 mm (0.035 - 0.060 in), feeding as well as electrical insulation and design of other accessories may be a problem in very deep and narrow grooves. The gap is usually 6.4 - 9.5 mm (0.25 - 0.375 in). These problems can be overcome by using larger diameter wires and wider grooves which typically range from 2-3 mm (0.08 - 0.125 in) and about 12.5 mm (0.5 in), respectively.

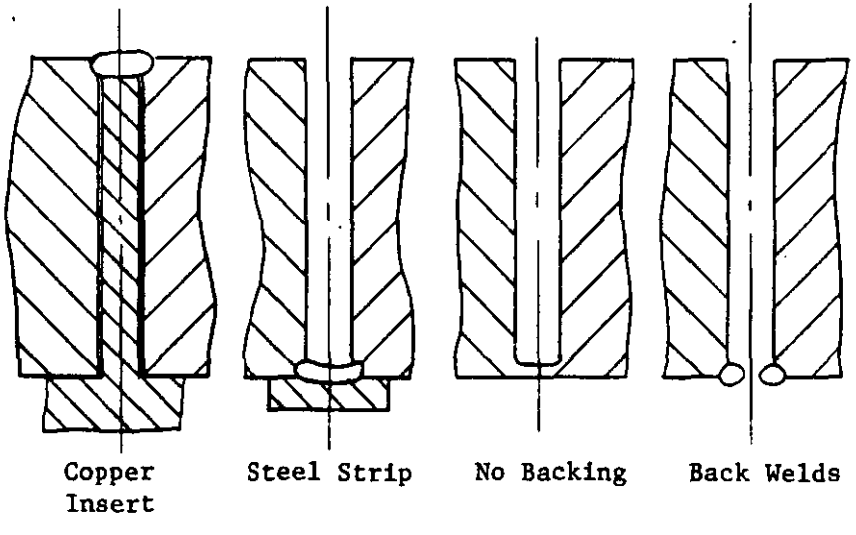
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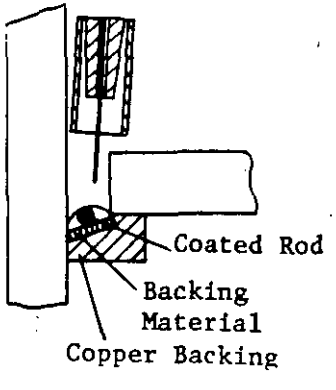




T	G
≤ 30	10
30 < T ≤ 75	11
75 < T ≤ 150	13
150 <	15







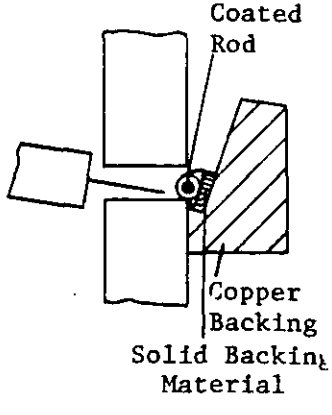


FIGURE 41. Various backing techniques used in NCW

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Typical welding conditions for NGW are tabulated in table IX.

TABLE IX.
Welding Conditions Characteristic of NGW.

Technique, weld position	Groove width, mm/in.	Polarity	Amperage/voltage, A/V	Heat input, kJ/mm kJ/in.	Travel speed, m/hr in./min	Gas shield
NGW-I horiz.	9.5 / 0.375	DCRP	260-270 / 25-26	0.4-0.5 / 10-11	60 / 40	Ar-Co_2
NGW-I horiz.	10-12 / 0.4-0.5	DSRP (120 pulses per second)	220-240 / 24-28	1.1-1.2 / 26-29	18-21 / 12-14	Ar-Co_2
NGW-I flat	9.5 / 0.375	DSRP (120 pulses per second)	280-300 / 29	2.3-2.1 / 57-52	13-15 / 8½-10	Ar-Co_2
NGW-II flat	14-15 / 0.055-0.59	DCRP	450-470 / 28-29	2.3-2.5 / 58-63	20 / 13	CO_2
NGW-II flat	12.5 / 0.5	DCSP	450 / 30-37.5	2.2-2.5 / 54-62	22 / 15	Ar-Co_2
NGW-II flat	12-14 / 0.47-0.55	DCSP	450-550 / 38-42	2.3-2.6 / 57-66	27-31 / 18-21	Ar-Co_2

NGW - I : thin wire (0.9 - 1.5 mm)
NGW - II: large wire (2-3 mm)

The narrow gap welding concept has been extended to include the application of gas tungsten-arc, flux-cored arc, electrogas and electron as well as laser-beam welding processes.

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4. FLUX-CORED ARC WELDING

The flux-cored arc welding (FCAW) process uses a tubular electrode wire filled with flux. The arc established between the electrode wire and the base metal as in all arc welding processes provide the heat for welding. There are two variations: one using auxiliary shielding gas from an external source, the other with self-shielding. The core of the tubular electrode contains fluxing, deoxidizing, and at times alloying additions. The molten slag that floats on the liquid weld metal provides physical protection from atmospheric contamination. Shown in Figures 42 and 43 are the schematic diagrams of the two variations.(7)

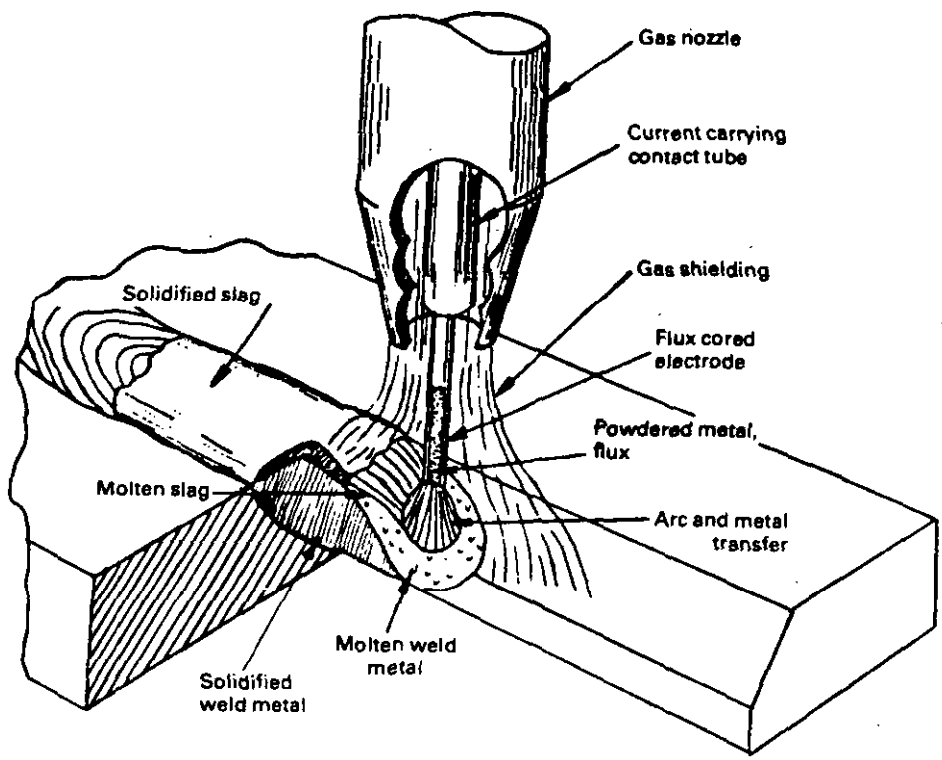


FIGURE 42. Gas-shielded flux-cored arc welding.

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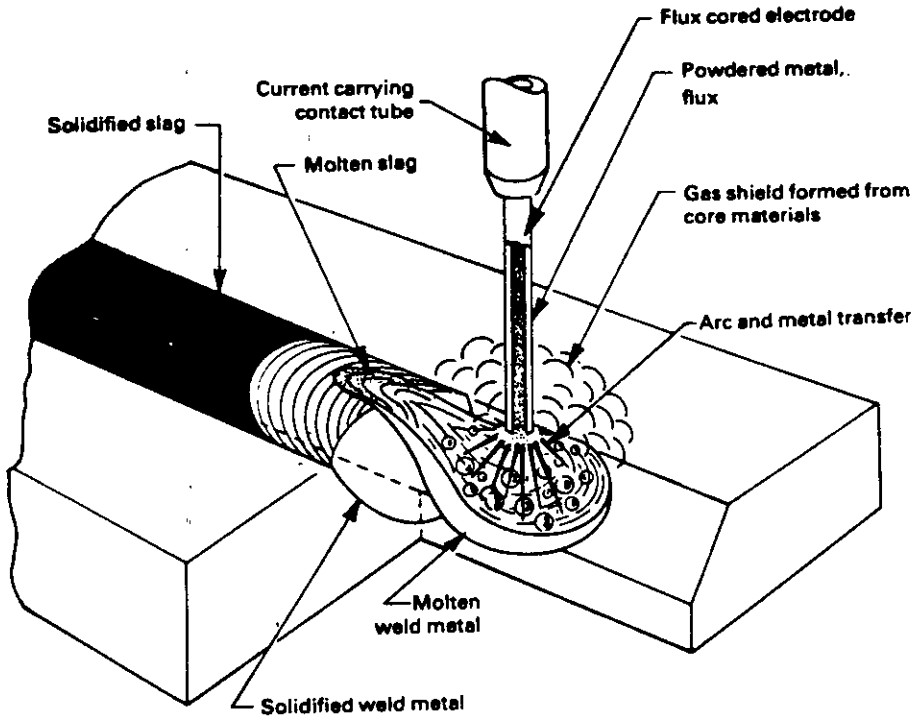


FIGURE 43. Self-shielded flux-cored arc welding.

The nozzle used with auxiliary gas-shielded flux-cored electrode wire is presented in Figure 4.4.

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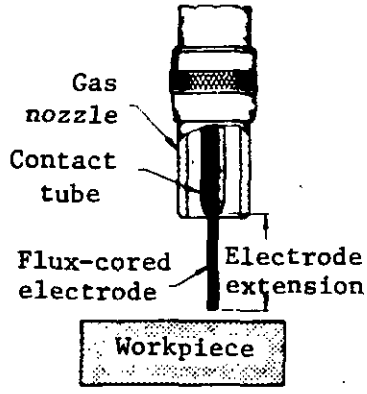


FIGURE 44. Nozzle used in auxiliary gas-shielded FCAW process.

The gas-shielded flux-cored arc welding process can be regarded as a modification of gas metal-arc welding.

The self-shielded electrode nozzle is depicted in Figure 45.

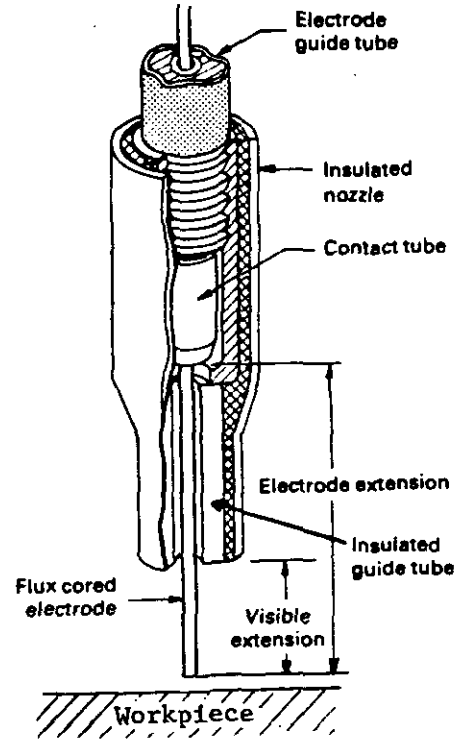


FIGURE 45. Self-shielded electrode nozzle.

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The electrode extension is the length of the electrode wire between the point of electrical contact in the nozzle and the arc. Displayed in Figure 46 is a typical arc welding gun or welding head for semiautomatic self-shielded flux-cored arc welding process.(7)

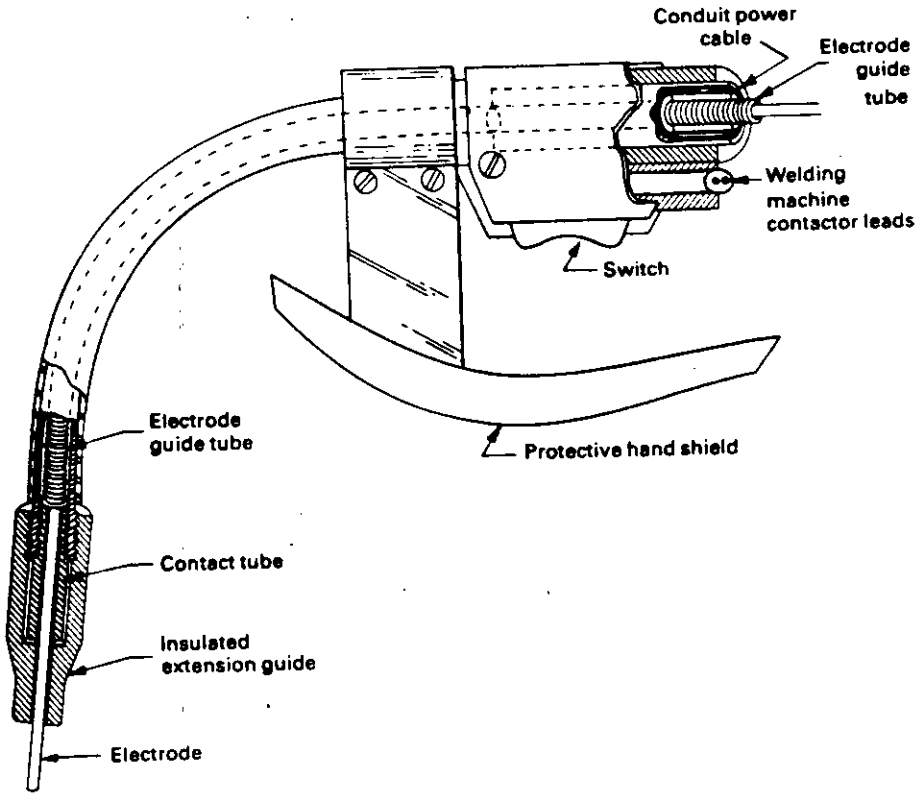


Figure 46. Welding for semiautomatic self-shielded flux-cored arc welding.

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A schematic layout for automatic flux-cored arc welding equipment can be seen in Figure 47.

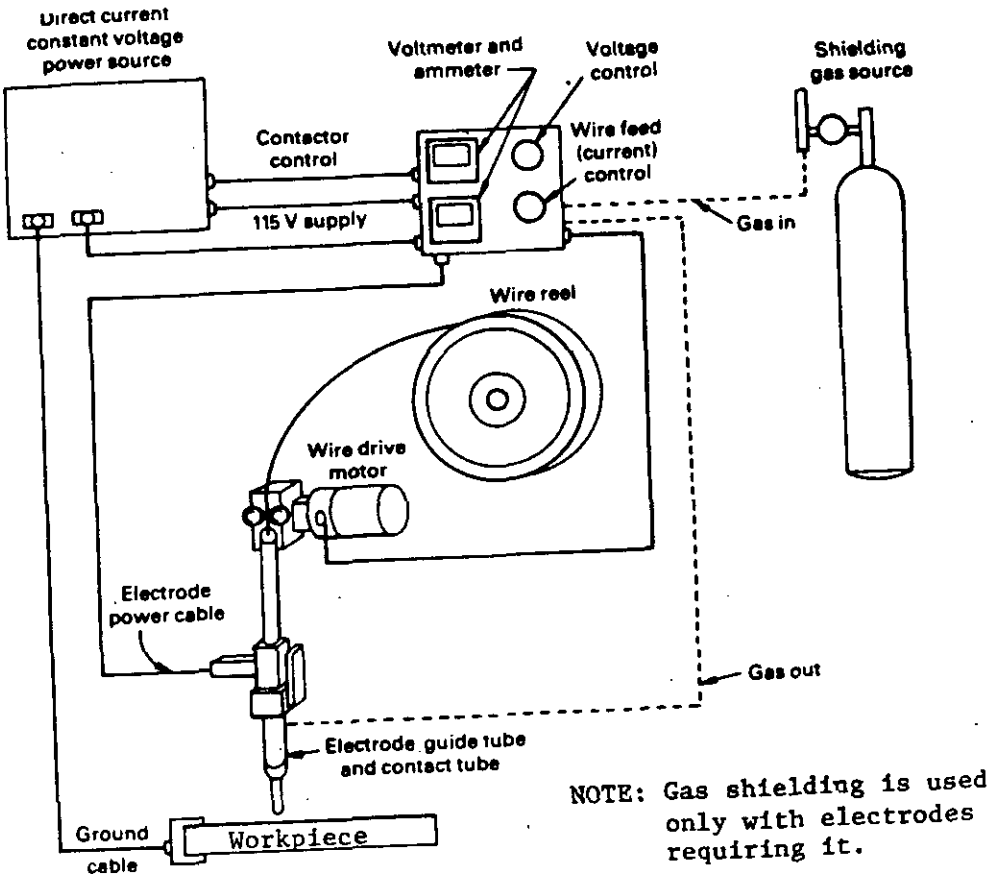


FIGURE 47. Layout for automatic FCAW process.

The major difference between the two versions of FCAW lies in the quality of the resultant weld. The gas-shielded variety can produce weld quality that can satisfy many code requirements, while the self-shielded version may not. The nozzle design is of course different for the two modifications (see Figures 44 and 45). The electrode extension is considerably longer for the self-shielding type, because the electrode can be prevented from touching the nozzle by an insulated wire guide located in the space at the end of the electrode wire holder.

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With the advent of small diameter flux-cored electrode wires of 1.14 mm (0.045 in) or less, the flux-cored arc welding process has become an all-position joining process.

The flux-cored arc welding process generates large amounts of fumes. Due to recent health and safety requirements for controlling air pollution, gas shielded welding guns come with integral fume extractors.

4.1 Electrode Classification

The identification system is similar to other AWS filler metal classifications with minor variation. As an example, E60T-1 designation carries the following meaning. "E" indicates an electrode. The number 6 designates the minimum "as-welded" tensile strength (in 10 KSI units) with a range of 60 - 69 KSI (414-476 MPa). The second digit indicates the recommended weld position. The zero means flat and horizontal positions. The letter "T" means that the electrode is of tubular construction. The number 1 indicates composition of the weld deposit, type of current, polarity, use of auxiliary shielding and a usability factor. The various styles of flux-cored electrode constructions are shown in Figure 48.(5)

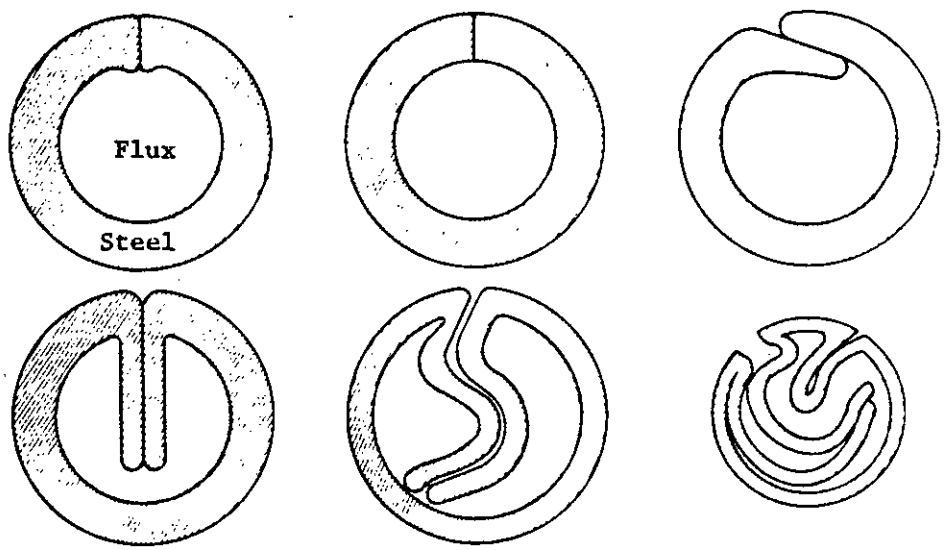


FIGURE 48. Fabrication methods for flux-cored electrodes.

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Compositional requirements for flux-cored arc weld deposits can be seen in table X.

TABLE X.

Chemical Composition Requirements for Flux-Cored Electrodes.

AWS Classification	Chemical Composition, Max. (%)						
	Mn	Si	Ni	Cr†	Mo†	V†	Al
E60T-7	1.50	0.90	0.50	0.20	0.30	0.08	1.8
E60T-8	1.50	0.90	0.50	0.20	0.30	0.08	1.0
E70T-1	1.75	0.90	0.30†	0.20	0.30	0.08	--
E70T-2	No chemical requirements						
E70T-3	No chemical requirements						
E70T-4	1.50	0.90	0.50	0.20	0.30	0.08	1.8
E70T-5	1.50	0.90	0.30†	0.20	0.30	0.08	--
E70T-6	1.50	0.90	0.80	0.20	0.30	0.08	--
E70T-G	No chemical requirements						

† These elements may be present, but are not intentionally added.

Tabulated in table XI are the various test requirements for weld deposits made by FCAW.(10)

TABLE XI.

Required Tests for Mechanical Properties, Soundness, and Usability of Welds Made by Flux-Cored Electrodes.

AWS Classification	Type of Test Required				
	Radio-graphic	All-Weld Metal Tension	Transverse Tension	Longitudinal Guided-Bend	Charpy V-Notch Impact
E60T-7	Yes	Yes	No	No	No
E60T-8	Yes	Yes	No	No	Yes
E70T-1	Yes	Yes	No	No	Yes
E70T-2	No	No	Yes	Yes	No
E70T-3	No	No	Yes	Yes	No
E70T-4	Yes	Yes	No	No	No
E70T-5	Yes	Yes	No	No	Yes
E70T-6	Yes	Yes	No	No	Yes
E70T-G	No	Note 1	Note 1	No	No

1. If electrode is suitable only for single-pass applications, a transverse-tension test is required; if electrode is suitable for multiple-pass work, an all-weld-metal test is required.

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The minimum mechanical property requirements for carbon steel flux-cored electrodes in the as-welded conditions are given in table XII.

TABLE XII

Mechanical Property Requirements for Carbon Steel Flux-Cored Electrodes (AWS A5.20-79).

AWS classification	Tensile strength, ksi	Yield strength, ksi	Elongation in 2 in., %	Charpy V-notch impact
E6XT-1	62	50	22	20 ft·lb at 0 °F
E6XT-4	62	50	22	Not required
E6XT-5	62	50	22	20 ft·lb at −20 °F
E6XT-6	62	50	22	20 ft·lb at −20 °F
E6XT-7	62	50	22	Not required
E6XT-8	62	50	22	20 ft·lb at −20 °F
E6XT-11	62	50	22	Not required
E6XT-G	62	50	22	Not required
E6XT-GS	62	Not required	Not required	Not required
E7XT-1	72	60	22	20 ft·lb at 0 °F
E7XT-4	72	60	22	Not required
E7XT-5	72	60	22	20 ft·lb at −20 °F
E7XT-6	72	60	22	20 ft·lb at −20 °F
E7XT-7	72	60	22	Not required
E7XT-8	72	60	22	20 ft·lb at −20 °F
E7XT-11	72	60	22	Not required
E7XT-G	72	60	22	Not required
E7XT-2, E7XT-3, E7XT-10, E7XT-GS	72	Not required	Not required	Not required

Various low-alloy flux-cored electrodes are available for joining high-strength structural steels and low-alloy steels.(1) (See Table XIII).

TABLE XIII

Mechanical Property Requirements for Low-Alloy Flux-Cored Electrodes (AWS A5.29-80).

AWS classification	Tensile strength, ksi	Yield strength (min), ksi	Elongation in 2 in., min %
E6XTX-X	60-80	50	22
E7XTX-X	70-90	58	20
E8XTX-X	80-100	68	19
E9XTX-X	90-110	78	17
E10XTX-X	100-120	88	16
E11XTX-X	110-130	98	15
E12XTX-X	120-140	108	14
EXXXTX-G	(a)	(a)	(a)

Footnote

(a) properties as agreed upon between supplier and purchaser.

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4.2 Shielding Gas

The auxiliary gas shielding can be similar to that used in gas metal-arc welding. It may be pure carbon dioxide, or 98% argon plus 2% oxygen, or 75% argon mixed with 25% CO_2.

4.3 Welding Variables

In manual or semiautomatic welding methods, the welder can control a number of variables that can affect the resultant weld in several ways. These variables can be classified as primary and secondary. Travel speed, arc voltage and weld current fall in the first category. They influence the morphology of the weld bead, the joint penetration, the deposition rate, arc stability and spatter. The secondary variables consist of electrode extension and angle. The secondary variables have an influence upon the primary ones indirectly . Electrode extension affects the weld current. The greater the extension, the greater the resistance and with constant voltage power source, the weld current is decreased according to Ohm's law. The wire feed rate controls the weld current in the constant voltage power supplies. The relationship between wire feed rate and welding current is affected by electrode extension, shielding media, electrode diameter and polarity.

The nozzle travel angle or electrode angle has an effect on the depth of penetration. Two angles are required to define the angular position of an electrode: (1) work angle and (2) travel angle. The travel angle is further divided into a drag and a push angle.(8) These angles are related to the work and travel planes as illustrated in Figures 49 and 50.

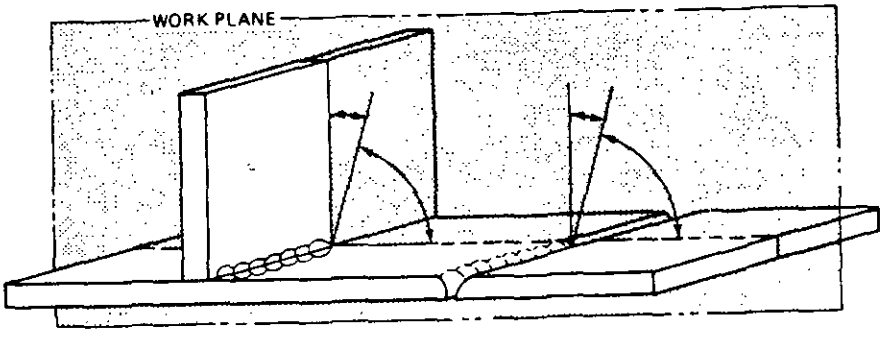


FIGURE 49. Work angle for fillet and groove welds.

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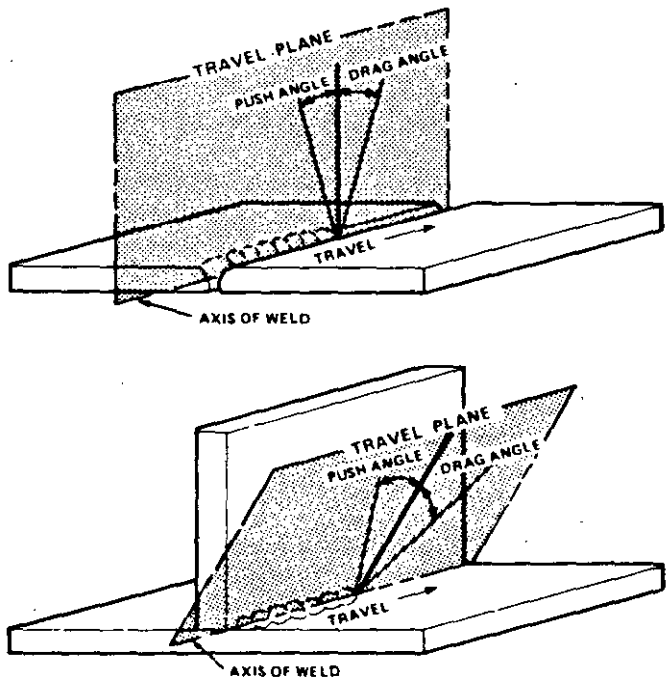


FIGURE 50. Travel angles for groove and fillet welds.

The nozzle-to-work or standoff distance and the electrode extension have an effect on weld performance in terms of spatter, deposition rate and penetration. Typically, the nozzle-to-work distance ranges from 19 mm (0.75 in) to 25.4 mm (1.00 in) which gives an electrode extension of about 25.4 mm to 31.8 mm (1-1.25 in) using a concentric-type nozzle. Too much spatter is generated with an excessively short standoff distance, which plugs up the nozzle and contact tube prematurely.

As with all arc welding processes, the deposition rate is primarily affected by the welding current and the electrode diameter. Electrode extension in this respect is less significant than welding current and electrode size. Plotted in Figure 51 is the deposition rate as a function of welding current for differing diameters of E70T-1 electrode using CO_2 shielding gas. The various welding positions are also indicated on the graph.

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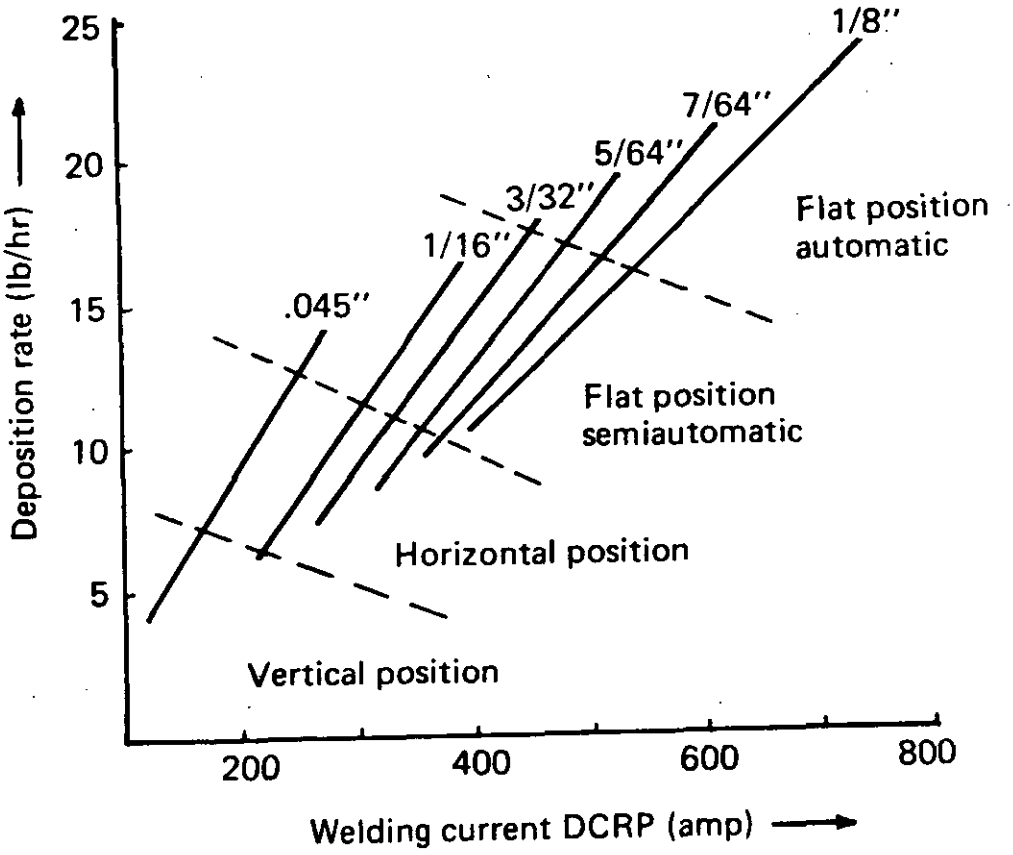


FIGURE 51. Deposition rate as a function of welding current, electrode diameter and welding position.

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5. SUBMERGED-ARC WELDING

A blanket of granular flux is used for shielding the arc and the molten metal in submerged-arc welding (SAW). The arc is struck between a bare electrode wire and the workpiece. The arc is not visible because it is totally covered by the flux. Consequently, there is no flash, spatter, and spark unlike in open-arc welding processes. A block diagram of SAW can be found in Figure 52.

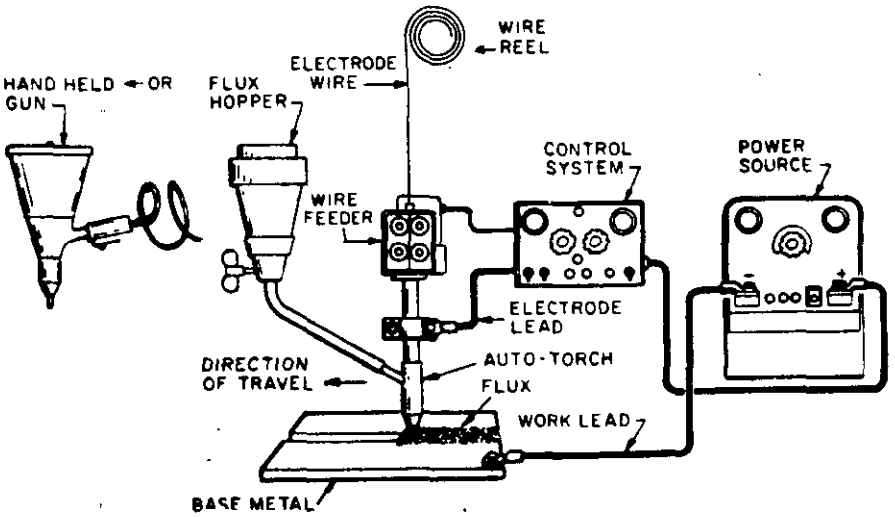


FIGURE 52. Submerged-arc welding.

Represented in Figure 53 is the arc area for the submerged-arc welding process.(8)

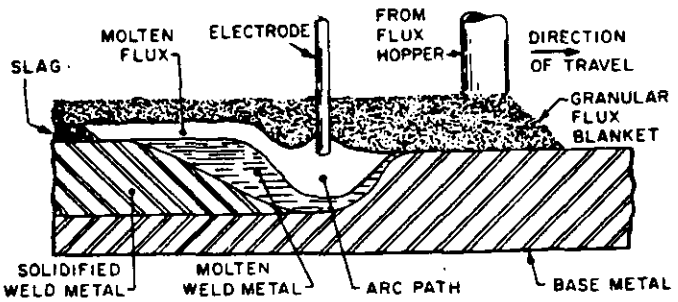


FIGURE 53. Weld region of SAW.

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The heat of the arc melts the inner portion of the flux along with the tip of the filler metal and the base metal. The outer layer of the flux blanket remains unfused, and can be reused. The slag is readily removable from the top of the weld bead.

The high current that characterizes SAW gives rise to high heat input. Since the current is applied to the bare electrode wire a short distance above its tip, high currents can be utilized on relatively small diameter filler metals: hence a very high current density. Welding currents as high as 600 amps may be carried on a 2.0 mm (0.078 in) electrode. This results in current densities of the order of 100,000 amps per square inch. This is 6 - 10 times greater than those carried by SMAW electrodes.

Table XIV contains welding current ranges for various diameter SAW electrodes.(7)

TABLE XIV
Current Ranges for Electrodes Used in SAW.

Electrode Diameter		Current Range
mm	in	AMP
1.6	1/16	150-400
2.0	5.64	200-600
2.4	3/32	250-700
3.2	1/8	300-900
4.0	5/32	400-1000
4.8	3/16	500-1100
5.6	7/32	600-1200
6.4	1/4	700-1600

The high current density produces a relatively high melt-off rate, deep penetration and fast welding speed. The high welding speed minimizes distortion.

A comparison of welding speeds for different welding processes is tabulated in table XV.(10)

TABLE XV.
Comparison of Welding Speeds.

Welding Process	Arc Speed* in/min
Manual shielded metal-arc	4-1/4
Semiautomatic self-shielded flux-cored	10
Automatic submerged-arc	15
Automatic submerged-arc, one wire	18
Twin-arc submerged-arc	21
Tandem submerged-arc, two wires	30
Tandem submerged-arc, three wires	45

*Arc speed for 3/8-in. butt weld

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Small grooves can be used even in thick base metals because of the deep joint penetration capability of the process. The heavy flux blanket, which surmounts the weld, yields a smooth weld appearance.

In single-pass welds, the fused base metal is large in comparison to the quantity of the filler metal employed. Therefore, the chemical and mechanical properties of the weld metal are affected by the base metal. This is known as dilution. It is for this reason that in submerged-arc welding of low-alloy steels, the application of an electrode having similar composition with the workpiece is not necessary. In multi-pass welds, dilution is not nearly as pronounced as in single-pass welds. The composition and mechanical properties of multi-pass welds are more of a function of electrode composition, welding conditions and flux activity. Typical mechanical properties of SAW metal deposits are listed in table XVI.(10)

TABLE XVI
Mechanical Properties for Submerged-Arc Weld Metal Deposits.

AWS Classification Flux-Wire	Tensile Strength (1000 psi)	Yield Strength (1000 psi)	Impact Strength, Charpy V-Notch (ft-lb)		Elongation in 2 in. (%)
			At 0°F	At −20°F	
F60, F70-EH14	74	60	19	–	28
F61, F71-EL12	73	60	30	–	26
F61, F71-EH14	77	61	24	–	28
F62, F72-EL12	71	56	–	61	30
F62, F72-EH14	74	58	–	21	28
F62, F72-EM15K	77	64	–	42	26
F62-EL12	68	57	–	45	28
F70-EL12	78	66	21	18	25
F71-EL12	79	68	25	22	27
F71-EM12K	89	77	30	25	25
F71-EM13K	85	70	22	–	25
F72-EM12K	83	75	–	47	26
F72-EM13K	84	74	–	40	27
F72-EM15K	85	72	–	27	24

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Welding conditions for single electrode, single-pass, machine submerged-arc welding of steel plate with square groove can be seen in table XVII.(7)

TABLE XVII
Weld Conditions for Single-Pass, Single Electrode, Machine Submerged-Arc Welding of Steel Plate.

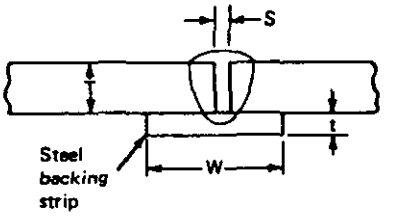


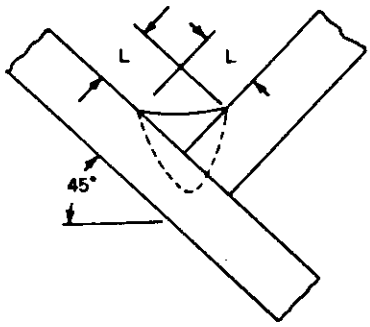
Plate thickness T		Root opening S		Current	Voltage	Travel speed		Electrode diam		Electrode consumption		Backing strip			
												t, min		W, min	
mm	in.	mm	in.	A[a]	V[a]	mm/s	in./min	mm	in.	kg/m	lb/ft	mm	in.	mm	in.
3.6	10 ga.	1.6	0-1/16	650	28	20	48	3.2	1/8	0.104	0.070	3.2	1/8	15.9	5/8
4.8	3/16	1.6	1/16	850	32	15	36	4.8	3/16	0.194	0.13	4.8	3/16	19.0	3/4
6.4	1/4	3.2	1/8	900	33	11	26	4.8	3/16	0.248	0.20	6.4	1/4	25.4	1
9.5	3/8	3.2	1/8	950	33	10	24	5.6	7/32	0.357	0.24	6.4	1/4	25.4	1
12.7	1/2	4.8	3/16	1100	34	8	18	5.6	7/32	0.685	0.46	9.5	3/8	25.4	1

a. DC, reverse polarity (electrode positive)

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Semiautomatic SAW conditions for flat position fillet welds in steel are presented in table XVIII.

TABLE XVIII
Semiautomatic SAW Conditions for Fillet Welds Made in Steel.



Weld leg, L		Current, A	Voltage V	Travel speed		Electrode diam		Electrode comsumption	
mm	in.			mm/s	in./min	mm	in.	kg/m	lb/ft
4.8	3/16	360[a]	30	21	50	1.6	1/16	0.070	0.047
6.4	1/4	425[b]	45	14	33	2.0	5/64	0.209	0.14
7.9	5/16	450[b]	47	10	23	2.0	5/64	0.328	0.22
9.5	3/8	450[b]	47	7	16	2.0	5/64	0.477	0.32
12.7	1/2	450[b]	47	4	9	2.0	5/64	0.849	0.57

(a) DC, reverse polarity (electrode positive).
(b) DC, straight polarity (electrode negative).

To increase weld productivity further, multiple electrodes may be utilized; two side by side or two or more in tandem as demonstrated in Figure 54.

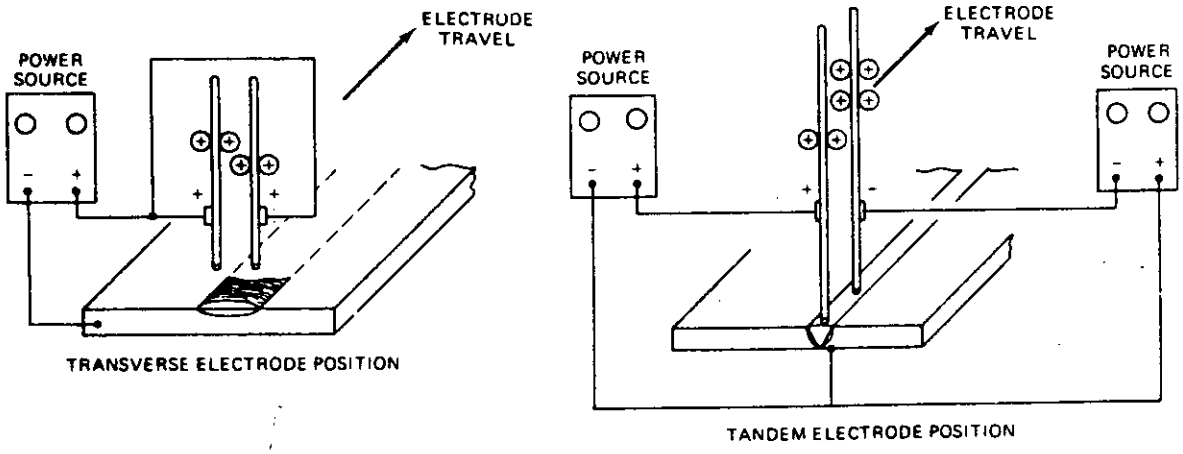


FIGURE 54. Twin-arc SAW system.

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Either direct current or alternating current power sources may be employed in SAW, while recognizing that each has distinct advantages and limitations in specific applications. Dc gives a better control of weld morphology, penetration, speed and arc initiation. Bead shape is best and penetration maximum with DCEP. DCEN produces maximum deposition rate and lowest penetration. Ac is excellent for controlling magnetic arc blow. Penetration is intermediate with alternating current. The twin-arc and tandem-arc systems can have either the same or separate power supplies.

Another variety of SAW is strip electrode welding, seen in Figure 55.

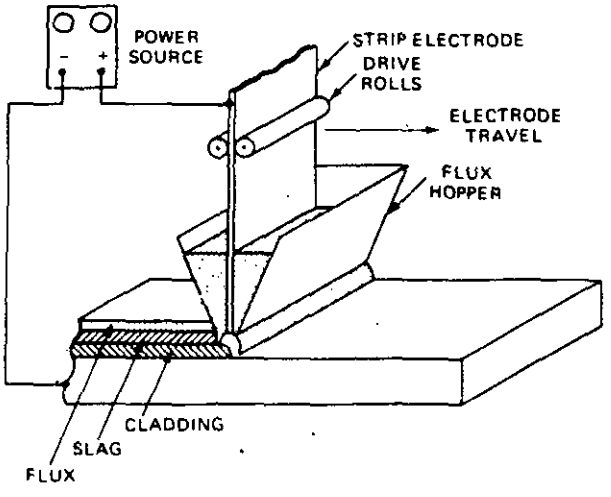


FIGURE 55. Strip welding system.

Strip electrode SAW has the advantage of covering a large surface area uniformly and with minimum penetration.(8) Overlaying mild and alloy steels with stainless steel is extensively used in nuclear pressure vessels.

The deposition rate can also be increased by iron powder additions to the joint under the flux. Special alloys added to the weld metal may be introduced via "cold" filler metals of the solid or flux-cored variety.

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Long electrode extension welding has the advantage of increasing the melting rate and reducing the weld penetration. The electrode beyond the electrical contact point is preheated through resistance heating as the current passes through it. The longer the electrode extension, the greater the heat build-up within the electrode. A voltage drop accompanies the resistance heating of the electrode extension.

Submerged-arc welding is limited to flat position and horizontal fillet position welding. Another limitation of this process has to do with its high heat input and slow cooling when quenched and tempered steels are being welded. Excessively large single-pass weld deposits may show high hardness due to alloy pickup and rapid cooling when welding carbon and low-alloy steels.

SAW is extensively used in heavy plate welded constructions. Butt joints from 18-gage sheet metal to plate thicknesses of several centimeters (inches) are commonly submerged-arc welded.

In sheet metal welding, a copper or steel backup bar is frequently employed to prevent distortion. When a steel backup bar is used, a small gap is maintained between the workpieces. The weld penetrates into the bar and becomes a permanent part of the weldment. If a copper bar is used, it does not fuse to the weld due to its high thermal conductivity. The copper backup bar may be flat or grooved in order to provide the desired shape on the back side of the weld as shown in Figure 56.

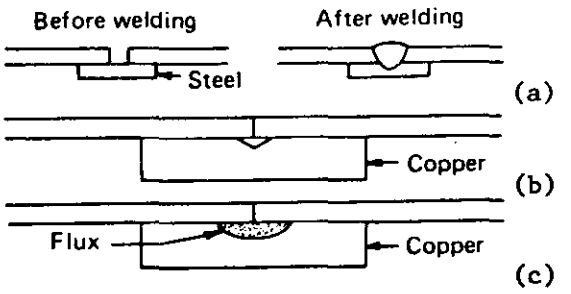


FIGURE 56. Joints with steel or copper backup bars.

(a) steel backup bar becomes permanently attached to weld.
(b) notched copper backup bar does not fuse to the joint.
(c) flux filled copper backup bar yields a smooth, flat weld profile.

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Joint preparation can be costly, if it is unnecessarily complicated for deep-groove welding in thick plates.(10) Figure 57 illustrates the point.

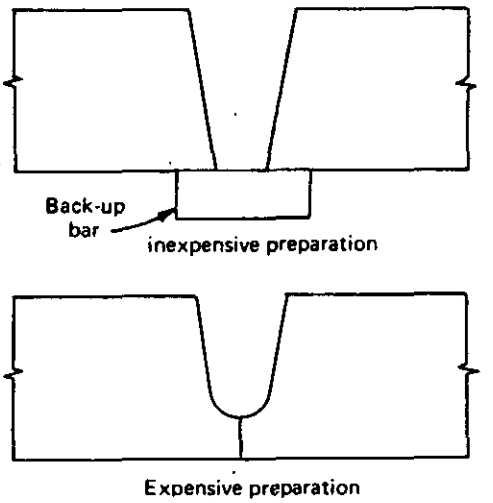


FIGURE 57. Examples of expensive and inexpensive joint preparations.

The largest single-pass fillet weld that can be made in horizontal position with one electrode is a 9.5 mm (0.38 in) leg. Undercut results when attempts are made to produce larger fillet welds as can be seen in Figure 58.

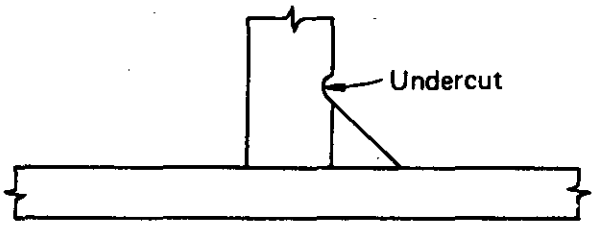


FIGURE 58. Undercut in fillet welds with legs larger than 9.5 mm (0.38 in).

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Horizontal fillet welds up to 15.9 mm (0.63 in) are possible with multiple electrodes. The recommended sequence of weld deposits in multi-pass welding is given in Figure 59.

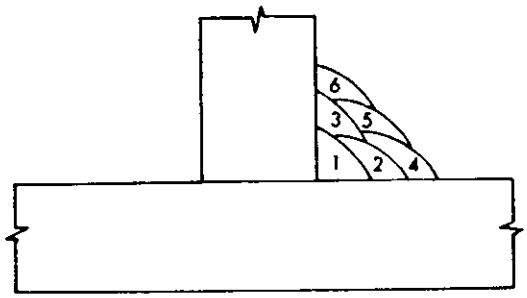


FIGURE 59. Recommended sequence of weld deposits in multiple-pass fillet welds.

Circumferential or girth welds present a problem with flux retention. The liquid weld metal also tends to spill off the work. To prevent run off of the molten weld metal and to minimize bead distortion, the electrode has to be positioned ahead of the centerline opposite the direction of rotation of the cylindrical workpieces as illustrated by Figure 60.

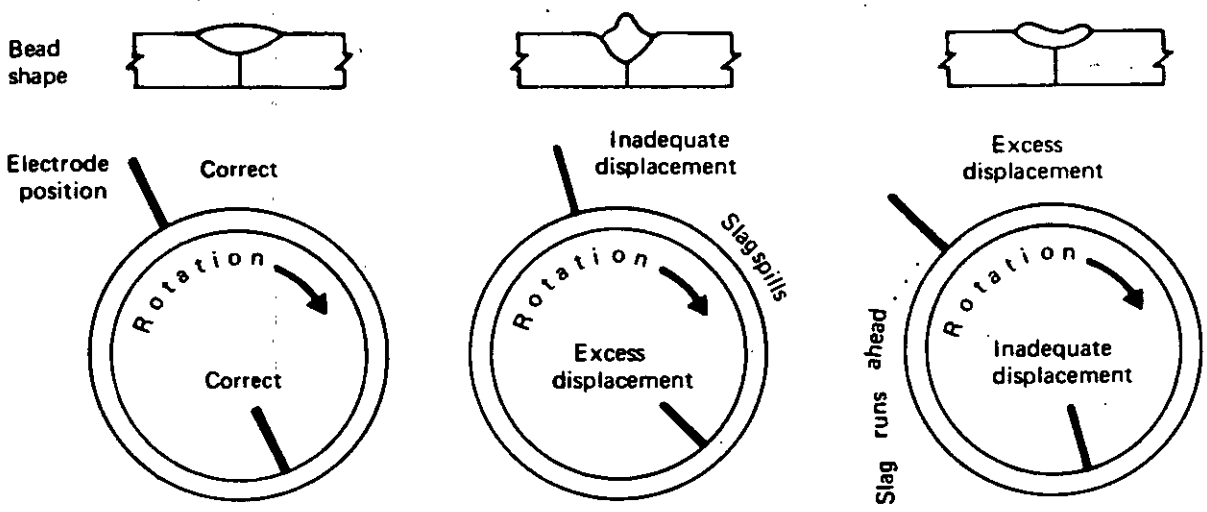


FIGURE 60. Effect of electrode position on weld morphology in girth welds.

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Table XIX lists the suggested electrode displacement for girth welding.(10)

TABLE XIX

Suggested Displacements for the Electrodes Used in Girth Welding.

Workpiece Diameter (in.)	Workpiece Diameter (cm)	Electrode Displacement Ahead of Vertical Center (in.)	Electrode Displacement Ahead of Vertical Center (mm)
1 - 3	2.5 - 7.6	3/8-3/4	9.5 - 19.0
3 - 18	7.6 - 45.7	3/4-1	19.0 - 25.4
18 - 36	45.7 - 91.4	1 1/4-1 1/2	31.8 - 38.1
36 - 42	91.4 -106.7	1 1/2-1 3/4	38.1 - 44.5
42 - 48	106.7 -121.9	1 3/4-2	44.5 - 50.8
48 - 72	121.9 -182.9	2-2 1/2	50.8 - 63.5
72	182.9	3	76.2

Since the flux is granular, it must be supported by a flux dam to retain the flux on the girth and edge welds. The flux supported and unsupported situations are shown schematically in Figure 61.

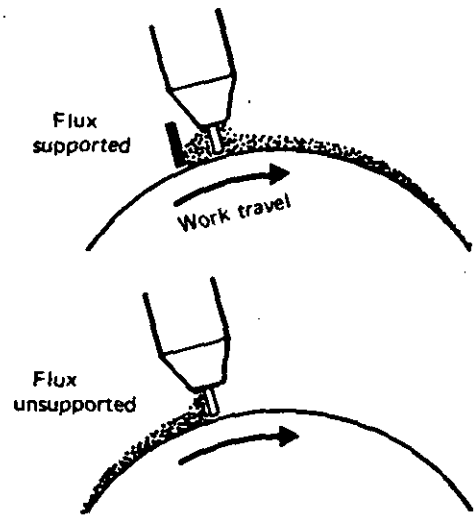


Figure 61. Position of flux support in the upper drawing corrects spillage in girth welds.

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The recommended weld currents for the various workpiece diameters and thickness are plotted in Figure 62.(10)

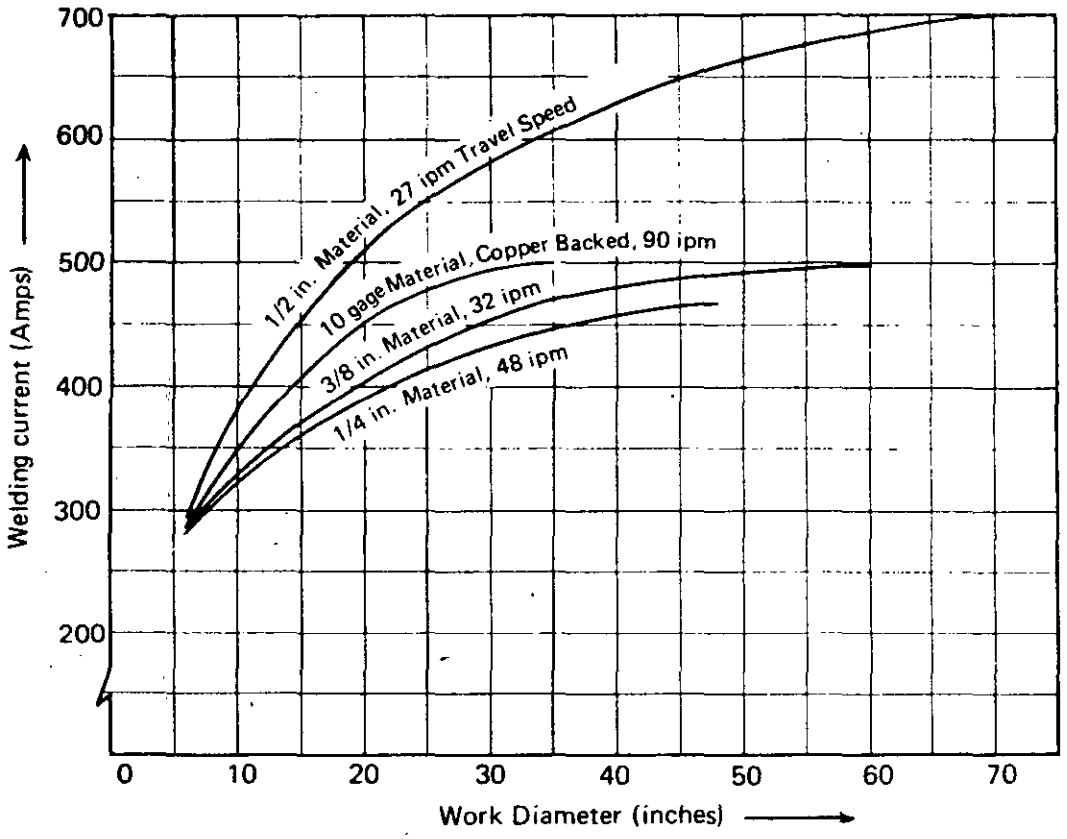


FIGURE 62. Plot of welding current versus work diameter as a function of metal thickness and travel speed.

When a weld starts and ends at abrupt terminal points of the workpieces, it is necessary to provide run-on and run-off tabs to support the weld metal and slag. Otherwise, crater cracking may occur.

In SAW, arc termination can be done by one of three ways. The travel and the electrode feed will stop simultaneously when the "stop" button is pushed. Another method involves travel reversal for a predetermined but brief time period while welding continues. The third technique consists of stopping the electrode travel while filler metal feeding is continued for a controlled length of time. The latter two methods are designed to feed the weld crater.

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SAW is well suited for groove, fillet, lap, plug welding and surfacing. In plug welding, it is difficult to tell when the plug is filled because the weld is surmounted with flux. Consequently, plug welding is usually timed. The current is decreased towards the end of the welding cycle before the arc is broken. This is for minimizing the crater at the top of the plug weld. A schematical arrangement of submerged-arc plug welding is presented in Figure 63.

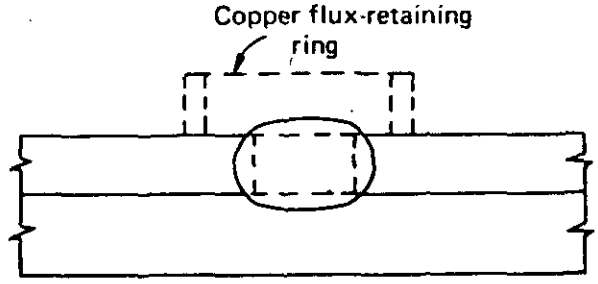


FIGURE 63. Submerged-arc plug weld arrangement.

When plates in excess of 50.8 mm (2 in) are submerged-arc welded, the rate of cooling increases. The larger the cross-section, the greater the heat sink. Also, the restraint increases with plate thickness. Under these conditions, high residual stresses may be induced in the weld. The tendency for cracking is therefore enhanced. There are other factors that promote cracking. They are steel composition, lack of preheat, or too low or too high preheat, undesirable joint configuration, hydrogen, assembly restraint, incorrect weld variables, lack of postheat, improper post weld heat treatment. Figure 64 shows the effect of weld configuration on the susceptibility of weld cracking.(5)

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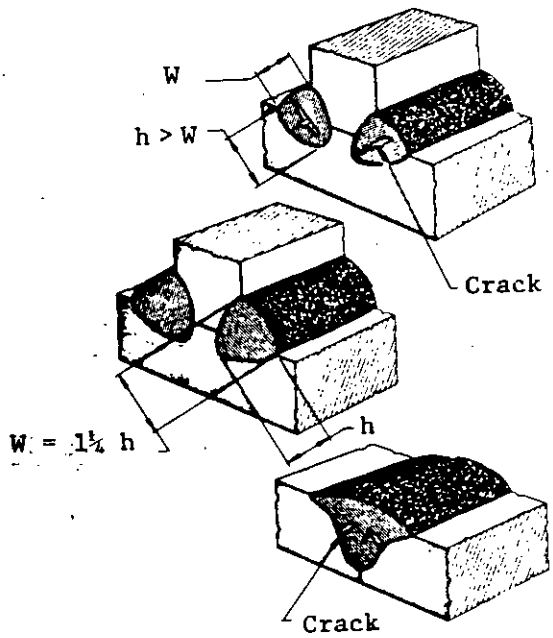


FIGURE 64. Effect of weld shape on tendency of weld cracking.

Every specific case of a cracking problem must be analyzed in order to determine its exact cause.

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6. GAS TUNGSTEN - ARC WELDING

Gas tungsten-arc welding (GTAW) is an arc welding process, which produces coalescence of metals by heating them with an arc between a nonconsumable tungsten electrode and the base metal. The weld pool, the arc, the electrode and the heated section of the workpieces are protected from atmospheric contamination by a gaseous shield. The shielding gas is usually an inert gas such as helium, argon, or a mixture of gases. The flowsheet of GTAW can be seen in Figure 65.

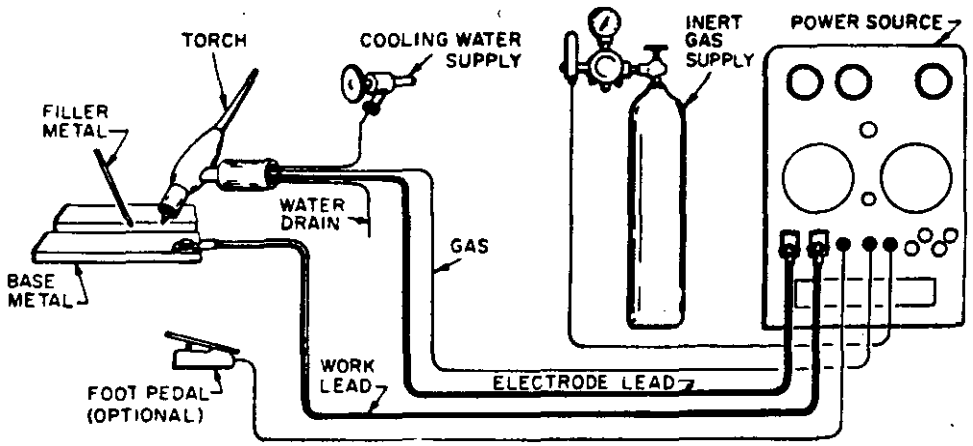


FIGURE 65. Main features of gas tungsten-arc welding.

Tungsten is used for electrodes because other refractory metals would erode too rapidly at the high arc temperature involved. It is also a strong emitter of electrons.

GTAW is an all-position welding process which may or may not utilize a filler metal.(1) Most of the metals can be joined by GTAW. Exceptions are metals with very low melting points such as lead, tin, cadmium and zinc. GTAW is usually employed for joining thin metals up to about 6.4 mm (0.25 in). The concentrated heat generated makes the process well suited for joining metals of thin cross-section including foils.

6.1 Welding Current

Direct current and alternating current can be used with gas tungsten-arc welding. With direct current power supply, both electrode negative (DCEN) and electrode positive (DCEP) polarities are utilized. Selection of polarity with a dc power source is guided by the specifics of the work involved. Most often, DCEN is employed with GTAW because it avoids overheating of the electrode.

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Since 70% of the heat generation is at the positive or anode end of the direct current arc, a DCEN welding produces more heat in the workpiece than in the electrode. This permits the use of smaller diameter electrodes. DCEN produces a deep and narrow penetration in the base metal. DCEP results in more heat in the electrode than in the workpiece. Therefore, larger diameter electrodes are required with DCEP. The net results of DCEP are a shallow penetration and wide bead. In welding aluminum, DCEP is especially cumbersome because the electrode tip attracts the weld pool leading to electrode contamination. Magnesium is repelled by the arc action of DCEP. Figure 66 describes the effects of dc polarity.(8)

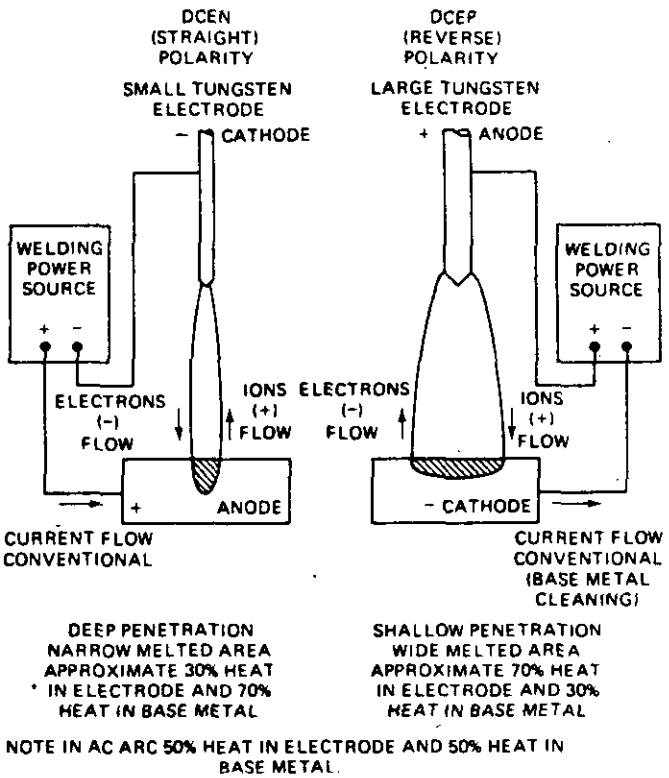


FIGURE 66. Effects of direct current polarities.

The ionic bombardment associated with DCEP results in a cleaning action on the base metal surface. It is most beneficial in gas tungsten-arc welding of aluminum, magnesium and beryllium copper. Metals such as steel and copper do not form the type of oxides which would interfere with GTAW significantly.

Alternating current may be regarded as a series of alternate pulses of direct current electrode negative and direct current electrode positive. In each half-cycle, the voltage goes through zero, at which point the arc must be reignited to provide current flow. During one half-cycle, the base metal is negative (cathode) and is relatively cool and may not emit sufficient quantities of electrons. Therefore, when the electrode is positive, reignition may not take place. This arc rectification (more current flow in one direction than in the other) can be bothersome. It can cause a delayed

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ignition as an example. Arc rectification is the product of unequal emittance of electrons by the electrode and the base metal. It is often referred to as a direct current component. It is influenced by the current density of the arc at the electrode and the workpiece, the arc length and the type of shielding gas used. The superimposition of a high-frequency current or a balanced-wave transformer has solved this problem. The high-frequency stabilization is continuously applied and thereby an ionized medium is maintained for the welding current to flow. Alternating current has a cleaning action every other half-cycle sufficient to remove oxides from metals. The joint penetration obtained by ac is intermediate between DCEN and DCEP. Typical weld current ranges are presented in table XX.(7)

TABLE XX.

Typical Ranges of Weld Current for Tungsten Electrodes.[a]

Electrode diameter		Direct Current, Amp		Alternating current, Amp					
		Straight polarity	Reverse polarity	Unbalanced wave			Balanced wave		
		EWP EWTh-1 EWTh-2 EWTh-3	EWP EWTh-1 EWTh-2 EWTh-3	EWP	EWTh-1 EWTh-2 EWZr	EWTh-3	EWP	EWTh-1 EWTh-2 EWZr	EWTh-3
mm	in.								
0.26	0.010	up to 15	b	up to 15	up to 15	b	up to 15	up to 15	b
0.51	0.020	5-20	b	5-15	5-20	b	10-20	5-20	10-20
1.02	0.040	15-80	b	10-60	15-80	10-80	20-30	20-60	20-60
1.59	1/16	70-150	10-20	50-100	70-150	50-150	30-80	60-120	30-120
2.38	3/32	150-250	15-30	100-160	140-235	100-235	60-130	100-180	60-180
3.18	1/8	250-400	25-40	150-210	225-325	150-325	100-180	160-250	100-250
3.97	5/32	400-500	40-55	200-275	300-400	200-400	160-240	200-320	160-320
4.76	3/16	500-750	55-80	250-350	400-500	250-500	190-300	290-390	190-390
6.35	1/4	750-1000	80-125	325-450	500-630	325-630	250-400	340-525	250-525

a. All values are based on the use of argon as the shielding gas. Other current values may be employed depending on the shielding gas, type of equipment, and application.
b. These combinations are not commonly used.

In pulsed-current gas tungsten-arc welding, the current is varied between a low background level and a high pulse level. Between pulses, the weld metal is allowed to solidify. While the background current level and its duration are not critical, the magnitude and duration of the pulse or peak current are important indeed. Usually, the background current is about 15% of the peak current. The magnitude and duration of the peak current depend on the type and the thickness of the material to be welded. Normally, a simple rectangular wave current pulse is satisfactory. After pulsing, a reduction or "slope out" can be beneficial in filling the weld crater and in minimizing weld cracking. The main application of pulsed gas tungsten-arc welding is for joining dissimilar metals and thicknesses. It is also useful in girth welding pipes where continuous current would result in an excessive heat build up.

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Gas tungsten-arc spot welding is a modification of the basic GTAW using a timing device and a modified torch to allow escape of the shielding gas during spot welding. The process is utilized to join overlapping sheets of similar or dissimilar thicknesses. The workpieces are firmly held together. The torch nozzle is pressed against the upper sheet. Then a DCEN arc is initiated for a predefined time. The time of current flow is usually in the range of 0.5 - 1.5 seconds. The result is a nugget melted in the upper sheet which penetrates into the lower sheet, thus fusing the two workpieces together. Displayed in Figure 67 is the gas tungsten-arc spot welding process.(7)

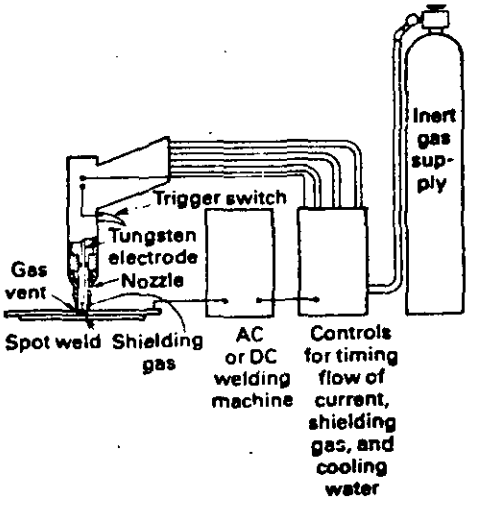


FIGURE 67. Gas tungsten-arc spot welding.

Two electrodes are used in twin gas tungsten-arc welding in applications where a smooth and wide weld is required.

The gas tungsten-arc hot wire system exhibited in Figure 68 is analogous to the cold wire method except the filler metal is resistance heated to a desired temperature before being fed into the weld pool. The preheated filler metal must be protected from oxidation and is usually done by an argon/helium mixture of 75/25 percent. The deposition rate with the hot wire system is considerably greater than that with the cold wire method.

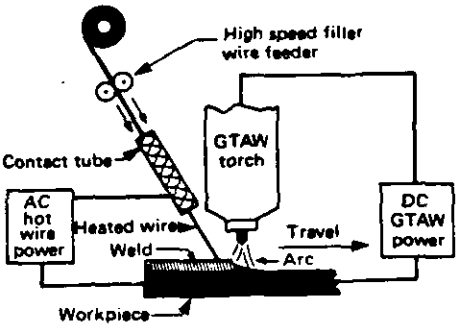


FIGURE 68. Hot wire addition to GTAW.

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6.2 Electrodes

Pure tungsten or tungsten alloyed with either thoria or zirconia have proved to be the best electrode for gas tungsten-arc welding. Pure tungsten electrode is less expensive than the tungsten alloys. However, the tungsten alloys are superior in terms of electron emissivity, endurance, resistance to contamination and current carrying capacity. Table XXI shows the compositon of the various types of tungsten electrodes.

TABLE XXI

AWS Classifications and Compositions of Tungsten Electrodes

AWS Classification	Tungsten % min (a)	Thoria, %	Zirconia, %	Other % max (b)
EWP	99.5	-	-	0.5
EWTh-1	98.5	0.8-1.2	-	0.5
EWTh-2	97.5	1.7-2.2	-	0.5
EWTh-3(c)	98.95	0.35-0.55	-	0.5
EWZr	99.2	-	0.15-0.40	0.5

(a) By difference
(b) Total
(c) EWTh-3 is a tungsten electrode with an integral lateral segment throughout its length that contains 1.0 to 2.0% thoria: average thoria content of the electrode is as shown in this table.

The size of the electrode should be so selected that its operation is near the maximum current-carrying capacity. The heat input is more concentrated this way. The current-carrying capacity of every type of tungsten electrode is affected by the type of weld current used, polarity, electrode extension, shielding gas, weld position, and electrode holder involved.

The shape of the electrode tip may be pointed or tapered, partially or completely hemispherical and bulbous as shown in Figure 69.

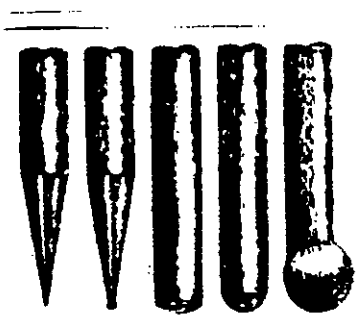


Figure 69. Shapes of tungsten electrode tips.

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Conditioning of the electrode tip is an important process variable in GTAW. Elements that vaporize more readily at any given temperature will condense on the tip and can pose problems with arc ignition, spatter and contamination. Therefore, the tip of the electrode has to be redressed occasionally. Chemical cleaning and grinding methods are used for this purpose. Table XXII gives the current ranges recommended for the various electrode sizes and tip profiles.

TABLE XXII

Tungsten Electrode Sizes, Tip Profiles and Weld Currents.

Electrode diameter, mm	Electrode diameter, in.	Diameter at tip, mm	Diameter at tip, in.	Included angle, degrees	DCEN Constant current range, A	DCEN Pulsed current range, A
1.02	0.040	0.125	0.005	12	2-15	2-25
1.02	0.040	0.25	0.010	20	5-30	5-60
1.59	0.062	0.5	0.020	25	8-50	8-100
1.59	0.062	0.8	0.030	30	10-70	10-140
2.38	0.093	0.8	0.030	35	12-90	12-180
2.38	0.093	1.1	0.045	45	15-150	15-250
3.18	0.125	1.1	0.045	60	20-200	20-300
3.18	0.125	1.5	0.060	90	25-250	25-350

6.3 Filler Metals

The filler metal composition is in general adjusted so that its properties in the as welded condition will at least match or exceed those of the base metal. Deoxidizers are often included in order to help assure satisfactory quality in the welded joint. Allowance in the chemical composition of the filler metal is also made for losses due to vaporization induced by the high arc temperature. Furthermore, filler metal selection must take into consideration metallurgical compatibility, in-service requirements, and costs

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6.4 Shielding Gases

Argon and helium and their mixtures are the type of shielding gases most often used in GTAW. Sometimes, hydrogen and nitrogen are added to either argon or helium depending on the type of metal to be welded and on whether the process is mechanized or not. Tabulated in table XXIII are the shielding gases, materials, thicknesses and type of power supplies utilized in GTAW.

Table XXIII

Shielding Gases and Power Sources for GTAW of Materials Indicated.

Material	Thickness[c]	Shielding gas[a] and power[b] used	
		Manual	Machine
Aluminum and its alloys	Under 3.2 mm	Ar (achf)	Ar (achf) or He (dcsp)
	Over 3.2 mm	Ar (achf)	Ar-He (achf) or He (dcsp)
Carbon steel	Under 3.2 mm	Ar (dcsp)	Ar (dcsp)
	Over 3.2 mm	Ar (dcsp)	Ar-He (dcsp) or He (dcsp)
Stainless steel	Under 3.2 mm	Ar (dcsp)	Ar-He (dcsp) or Ar-H_2 (dcsp)
	Over 3.2 mm	Ar-He (dcsp)	He (dcsp)
Nickel alloys	Under 3.2 mm	Ar (dcsp)	Ar-He (dcsp) or He (dcsp)
	Over 3.2 mm	Ar-He (dcsp)	He (dcsp)
Copper	Under 3.2 mm	Ar-He (dcsp)	Ar-He (dcsp)
	Over 3.2 mm	He (dcsp)	He (dcsp)
Titanium and its alloys	Under 3.2 mm	Ar (dcsp)	Ar (dcsp) or Ar-He (dcsp)
	Over 3.2 mm	Ar-He (dcsp)	He (dcsp)

a. Ar-He contains up to 75% helium; Ar-H_2 contains up to 15% hydrogen.
b. Abbreviations: achf = alternating current, high frequency; dcsp = direct current, straight polarity (electrode negative).
c. 3.2mm = 1/8 in.

6.5 Flow Rate

The adequacy of gas shielding depends on several factors as follows:

- type of shielding gas,
- size and shape of gas nozzle,
- metal being welded,
- nozzle-to-work distance,
- joint geometry,
- intensity of draft,
- welding speed,
- weld position,
- angular position of the torch.

The exact flow rate is usually determined by trial and error. Insufficient gas flow results in unacceptable weld quality. Excessive flow of gas, on the other hand, is unnecessarily costly. Once the optimum amount of flow rate has been established, gas consumption is controlled by a flowmeter calibrated in liters per minute (cubic feet per hour).

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6.6 Arc Initiation

Emission of electrons and ionization of the shielding gas by some means are necessary to start the arc. This can be done by, (a) touching the energized electrode to the workpiece and rapidly withdrawing the electrode to the appropriate arc length, (b) the use of a pilot circuit, or (c) the application of a high frequency auxiliary apparatus. The pilot arc starting method can be used with manual and automatic GTAW involving direct current power supplies only. The high frequency technique is applicable to manual gas tungsten-arc welding only but with both alternating and direct currents. It is advisable to strike the arc on a piece of scrap metal to "warm up" the electrode, then re-strike on the work once the electrode has reached the operating temperature. This approach minimizes the loss of tungsten from the electrode to form inclusions in the resultant weld.

The positions of the torch and filler metal, if used, in manual GTAW are illustrated in Figure 70.

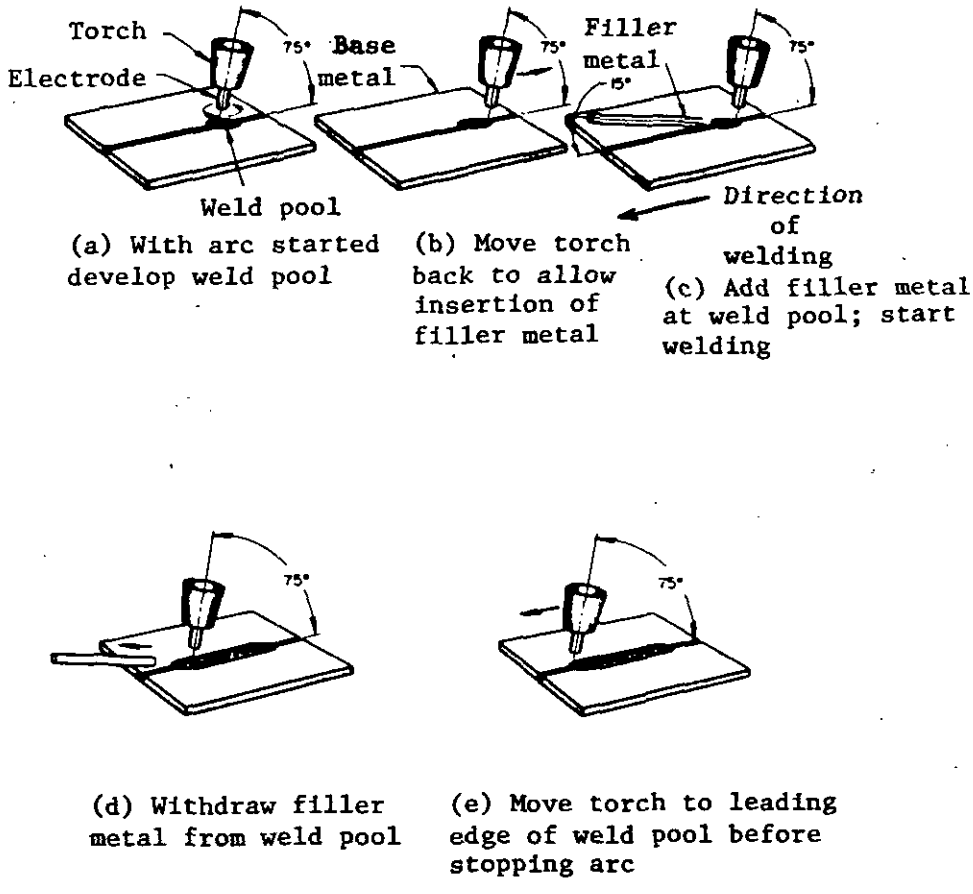


FIGURE 70. Torch and filler metal positions in GTAW done manually.

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Welding begins with moving the arc in circles until a pool of base metal is formed. Welding then proceeds as described in Figure 70. In most applications, the arc length is about 1.5 times the diameter of the electrode. The introduction of filler metal to the weld pool can disturb the gas shielding and air may be entrained. Therefore, it is beneficial to keep the end of the filler metal within the gas shielding zone at all times. This also prevents oxidation of the filler metal.

The size of the filler metal is important. If it is too small, it will melt rapidly forming a globule at its end. If it is too large, the filler metal may cause arc disturbance which can give rise to oxide inclusions in the weld. The filler metal and torch must be moved smoothly in order to avoid exposure of the liquid weld metal, the hot filler metal end and the hot solidified weld bead to air. In mechanized welding, the filler metal in the form of wire is continuously fed into the leading edge of the weld puddle.

6.7 Arc Termination

When welding is about to be terminated, the current is cut off by a foot- or a hand-controlled switch, while the flow of shielding gas is maintained to prevent excessive oxidation of the weld. In most GTAW, the flow of gas is controlled by a solenoid valve. It is timed to open before the arc is struck and to close after adequate time has passed to compensate for cooling of the electrode and the weld metal.

The arc can also be extinguished by abruptly withdrawing the electrode from the workpiece. This practice is not recommended because it may lead to crater cracking and undesirable oxidation.

6.8 Base Metal Cleaning

Since GTAW process is quite susceptible to contamination during welding, the workpieces must be carefully cleaned. Contaminants such as grease, oil, markings, paints, dirt, oxides, and fingerprints have to be removed from the weld area. The methodology of cleaning may consist of chemical or mechanical processes. Ordinary degreasing is sufficient to remove contaminants other than oxides. Grinding or abrasive blasting is necessary for dislodging of tenaceous oxides.

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7. PLASMA-ARC WELDING

The plasma-arc welding process produces coalescence of metals by heating them with a constricted arc between an electrode and the base metal or the electrode and the constricting nozzle. The density of the arc energy is increased when the arc is constricted resulting in much higher arc temperatures. The higher temperatures and electrical changes caused by constricting an electric arc are indicated in Figure 71.

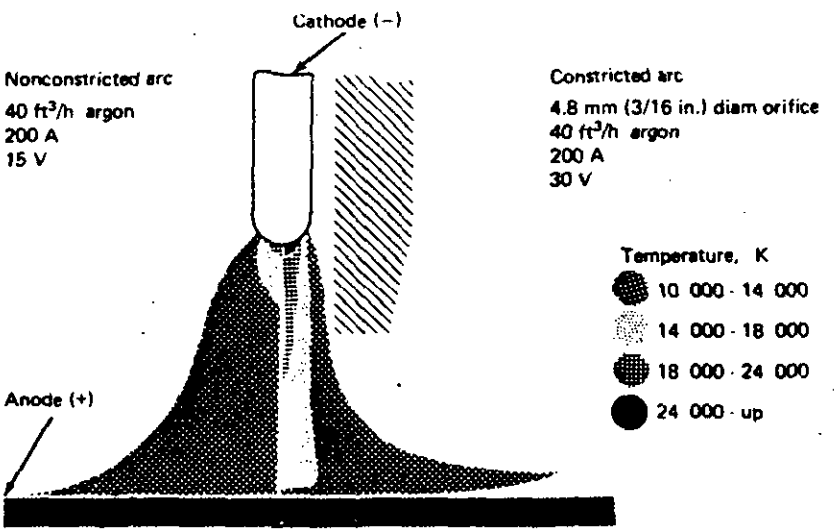


FIGURE 71. Effect of constricting an electric arc on changes of temperature and voltage.

There are two types of the plasma-arc heat sources: the non-transferred and the transferred or constricted.(7) When the arc is truck between the electrode and the constricting orifice it is known as the non-transferred plasma-arc. This is used for metal spraying and cutting as well as joining non-conductive materials. There is no electrical connection to the base met in the non-transferred arc process. When the constricted arc is struck between the electrode and the workpiece, it is called a transferred arc.

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Transferred and non-transferred plasma arc modes are illustrated in Figure 72.

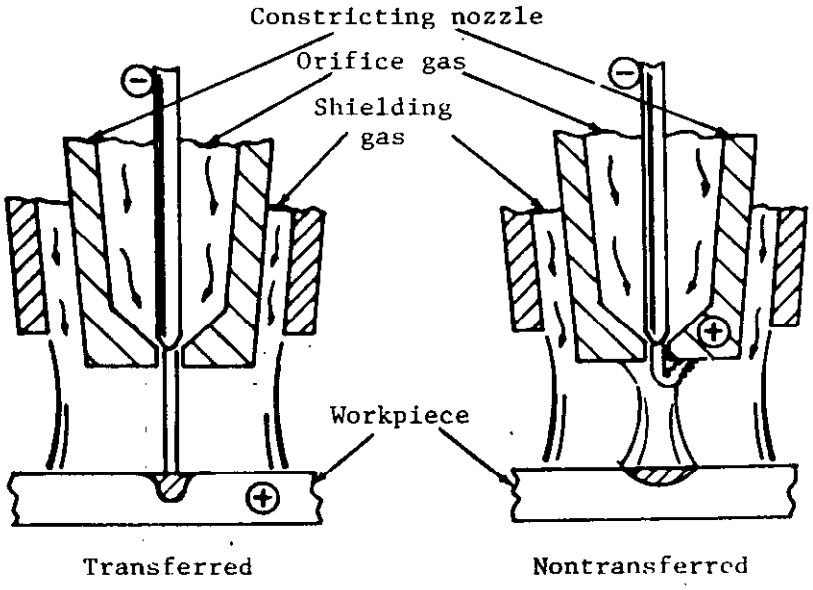


FIGURE 72. Modes of arc transfer in PAW.

When the plasma-arc is employed for welding purposes, auxiliary inert-gas shielding is required to protect the weld metal from atmospheric contamination.

Two different techniques of plasma-arc welding (PAW) are known: low currents for sheet metals under 1.6 mm (0.06 in) and high currents up to 400 amperes for thick metals using the deep penetration keyhole technique.

Plasma-arc welding is essentially a modification of the gas tungsten-arc welding. A comparison of the two processes are schematically illustrated in Figure 73. Flowsheet of PAW is given in Figure 74.

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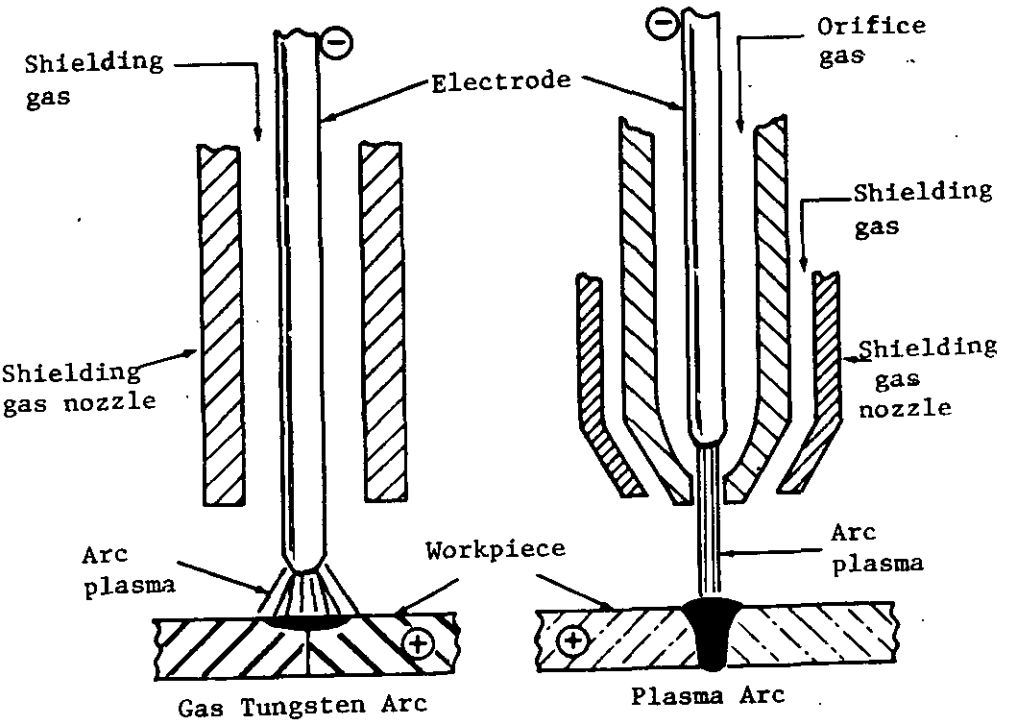


FIGURE 73. Schematic representation of GTAW and PAW.

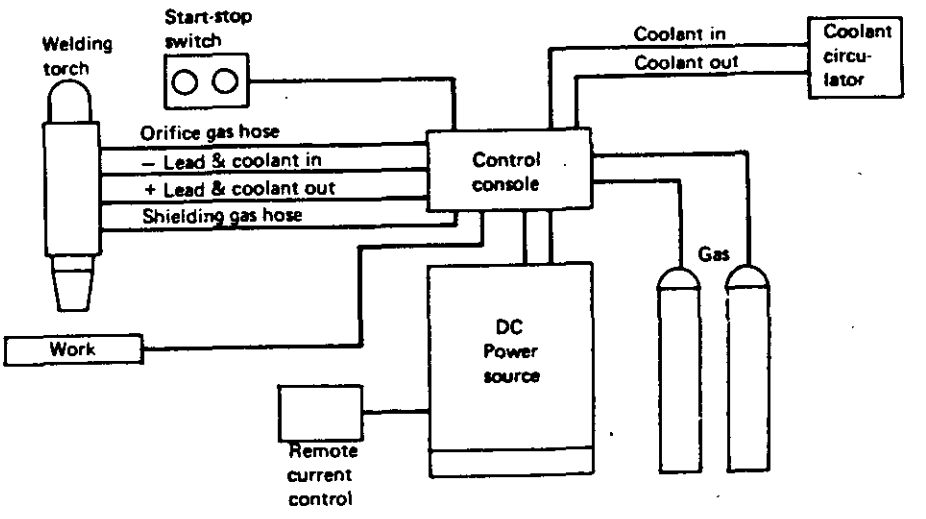


FIGURE 74. Characteristic components of PAW.

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A schematic presentation of the keyhole obtained by the high current keyhole technique can be seen in Figure 75.

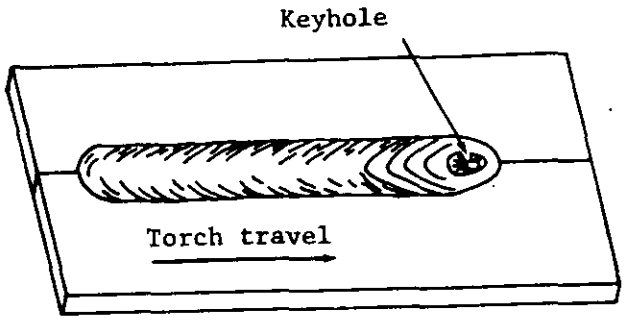


FIGURE 75. Schematic drawing of the keyhole obtained by high current PAW.

The high current keyhole welding process operates just below the gas velocity necessary for cutting.(7) This means that the orifice flow rate of the plasma gas must be carefully controlled. A flow rate variation of no more than 0.12 liter per minute (0.25 ft^3/h) is usually suggested. When the so called melt-in technique is utilized, the plasma gas flow rate in the orifice is decreased from that of the keyhole welding method. Thus, the melt-in technique of PAW is identical with that of GTAW.

7.1 Power Source

Plasma-arc welding uses direct current with a drooping voltage with a current output ranging from 0.1 - 400 amperes. For the low current version of PAW a separate power supply is used which has a current range of 0.1 - 10 amperes. The pilot arc power supply is set at 5 amperes which is delivered continuously during the process. The preferred polarity of PAW is straight (electrode negative).

Power supplies of the rectifier type are used with an open-circuit voltage range of about 65 - 80 volts when the shielding gas is either pure argon or a mixture of argon plus up to 7% hydrogen. If the shielding gas iseither pure helium or argon mixed with more than 7% hydrogen, a higher open-circuit voltage is needed to ensure a reliable arc starting.

Besides steady direct current power sources, pulsed direct current power supplies can also be employed for PAW similar to those utilized in pulsed GTAW.

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7.2 Arc Initiation

Since the electrode is within a constricted chamber, the arc cannot be started by touching the workpiece. A separate low power supply is necessary to energize the electrode. The arc is initiated by touching the electrode to the constricting nozzle and rapidly withdrawing it to the desired distance. The electrode is advanced or retracted by means of an adjusting screw. After the pilot arc has been established, the main transferred-arc current is energized by closing a contactor in the work lead (ground) attached to the base metal. The hot ionized plasma gas serves as a conductive path between the electrode and the base metal provided the torch is held close enough to the surface of the workpieces. This conductive path gives rise to arc initiation between the electrode and the workpieces. In the high-current PAW, the arc is initiated by a high-frequency current superimposed on the welding circuit as illustrated in Figure 76.

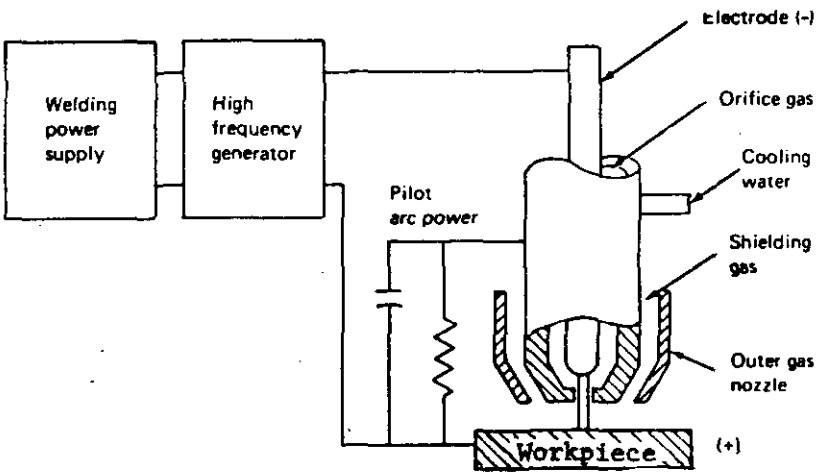


FIGURE 76. PAW with high-frequency pilot arc initiation.

7.3 Electrodes

Both PAW and GTAW use the same type of nonconsumable electrode. The electrode tip is ground to an included angle of about 60°, and has a small flattened end. The diameter of the flattened tip surface is 0.8 mm (0.03 in) for 3.2 mm (0.13 in) and 4 mm (0.16 in) diameter electrodes.

7.4 Filler Metals

Filler metals when used in PAW are the same as those employed with GTAW. They come in the form of cut lengths for manual welding and in spools of wire for semi-automatic and fully-automatic processes.

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7.5 Orifice and Shielding Gases

It is customary to use the same type of gas for the orifice gas and the shielding gas. This eliminates variation in arc effluence that usually occurs when dissimilar gases are utilized for the orifice (or plasma) gas and the shielding gas. The choice of gases is governed by such things as the type of metal to be welded, and requirements on arc temperature, travel speed, weld quality, electrode life, cost and base metal thickness. Gas selection for various metals to be high-current plasma-arc welded is presented in table XXIV.

TABLE XXIV
Recommended Gases for High-Current PAW.[a]

Metal		Thickness mm	Thickness in.	Welding technique: Keyhole	Welding technique: Melt-in
Carbon steel (aluminum killed)	under	3.2	1/8	Ar	Ar
	over	3.2	1/8	Ar	75% He-25% Ar
Low alloy steel	under	3.2	1/8	Ar	Ar
	over	3.2	1/8	Ar	75% He-25% Ar
Stainless steel	under	3.2	1/8	Ar, 92.5% Ar-7.5% H_2	Ar
	over	3.2	1/8	Ar, 95% Ar-5% H_2	75% He-25% Ar
Copper	under	2.4	3/32	Ar	75% He-25% Ar, He
	over	2.4	3/32	Not recommended[b]	He
Nickel alloys	under	3.2	1/8	Ar, 92.5% Ar-7.5% H_2	Ar
	over	3.2	1/8	Ar, 95% Ar-5% H_2	75% He-25% Ar
Reactive metals	under	6.4	1/4	Ar	Ar
	over	6.4	1/4	Ar-He (50 to 75% He)	75% He-25% Ar

a. Gas selections are for both orifice and shielding gases.
b. The underbead will not form correctly. The technique can be used for copper-zinc alloys only.

7.6 Joint Design

Joint preparation for thicknesses of up to 6.4 mm (0.25 in) consists of square-groove butt joints. Since the plasma arc is "stiffer" and less affected by voltage changes than the conventional gas tungsten-arc, the PAW process is less critical to variations in root opening and misalignments. Metal thicknesses in excess of 6.4 mm (0.25 in) require a V- or U-groove preparation. The first pass is normally done by the keyhole technique. The fill-in passes are carried out with the melt-in methodology. Figure 77 shows a comparison between a V-groove joint for GTAW and PAW in 10 mm (0.38 in) thick metal.

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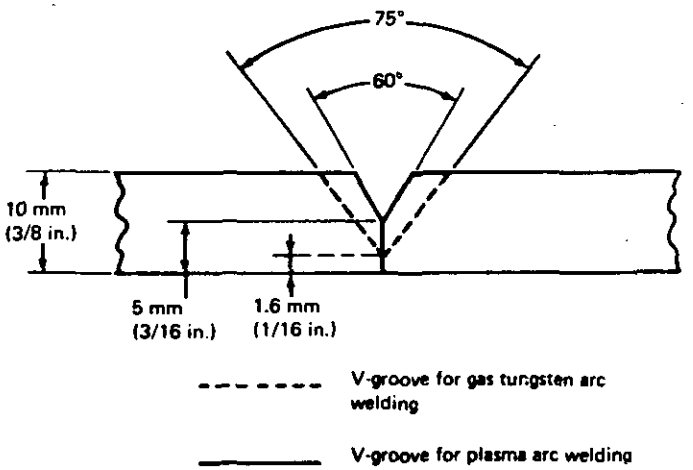


FIGURE 77. V-groove weld joint for PAW and GTAW.

7.7 Weld Backing

A grooved copper bar is adequate for backing a weld made with the fill-in technique. The shape of the groove should be such that it forms an acceptable root profile. Metals which have a high affinity for oxygen necessitate the application of shielding gas to protect the back side of the joint from atmospheric contamination.

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The keyhole technique penetrates through the base metal and forms a keyhole. As the weld progresses, the molten weld metal flows in behind the the hole. The weld metal is supported by the surface tension of the weld puddle. Therefore, a closely fitted backing bar is not necessary. The function of the backing bar is simply to direct the underbead shielding gas to the weld root, to align and support the workpieces and to vent the plasma jet. A typical backing bar for keyhole welding is displayed in Figure 78.

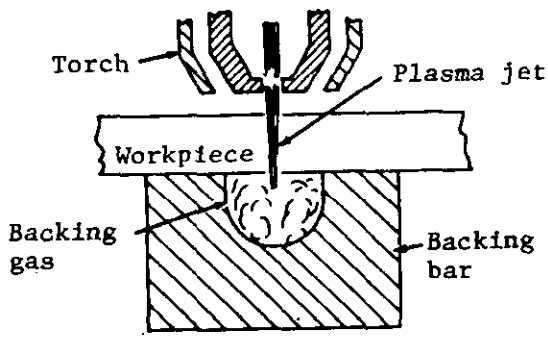


FIGURE 78. Backing bar for keyhole welding.

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8. ELECTROGAS WELDING

Electrogas Welding (EGW) is a modification of the gas metal-arc (GMAW-EG) or flux-cored arc welding (FCAW-EG) processes designed for single-pass, vertical position welding of butt, corner and T-joints. Water-cooled molding shoes confine the molten weld metal between the workpieces having a thickness range of about 13 mm (0.5 in) to 76 mm (3.0 in). The schematic diagrams of the two varieties of EGW are exhibited in Figures 79 and 80, respectively.

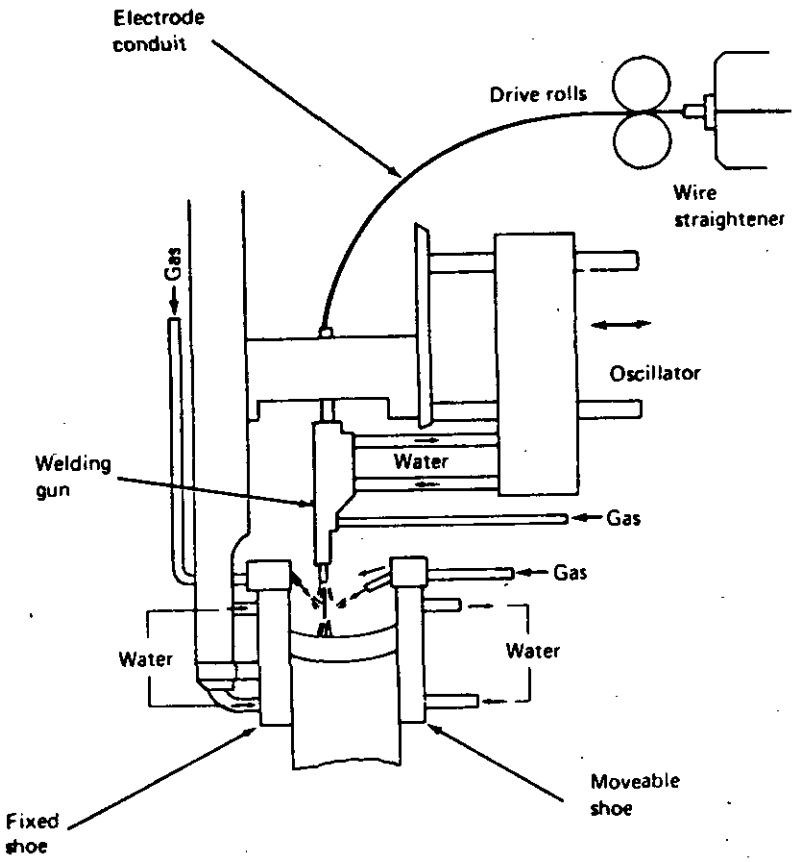


FIGURE 79. Electrogas welding of the GMAW-EG variety (using a solid wire electrode).

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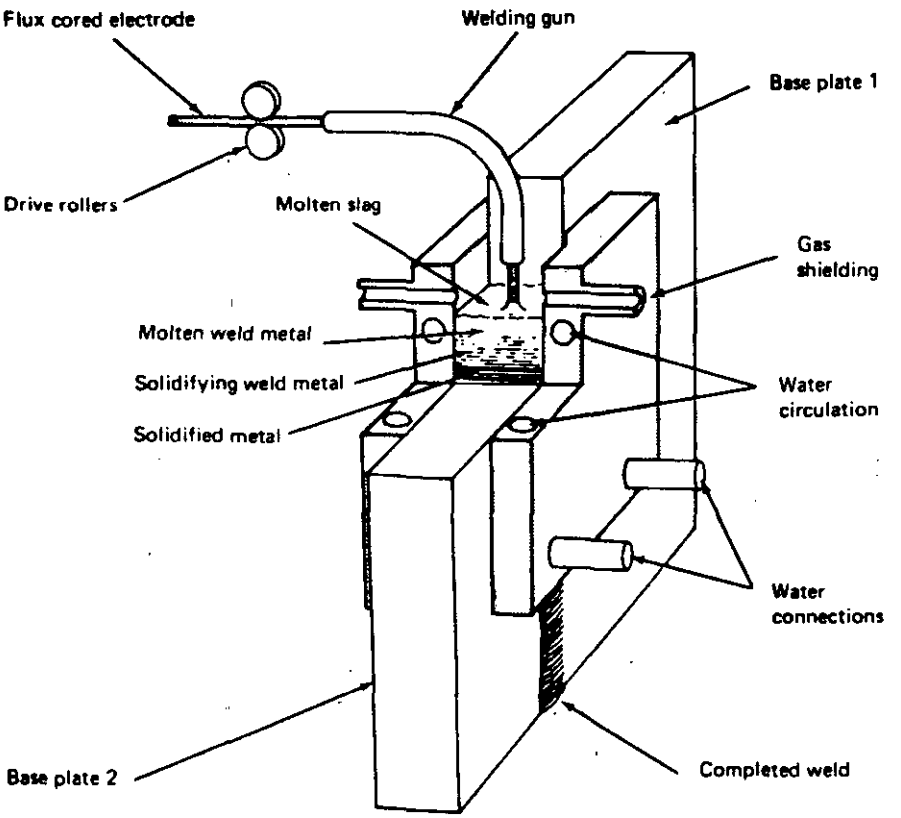


FIGURE 80. Electrogas welding of the FCAW-EG version (using flux-cored wire electrode).

Deposition rates of as much as 13 kg/h (30 lb/h) can be achieved with single electrode EGW. The essential components of this process consists of a DC power source, water-cooled copper retaining shoes, a welding torch, an electrode wire feeding device and shielding gas and water supply unit.

An electric arc is struck between the electrode and a starter plate at the bottom of the vertical joint. The consumable electrode is oscillated horizontally in welding thick cross-sections in order to distribute the heat and weld metal uniformly across the joint.(1) Oscillation helps prevent the formation of incomplete fusion, which is one of the more common weld discontinuities that may occur in EGW.

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The wire feeder is an integral part of the welding head, which moves in a vertical-up direction. Electrogas welding is operated on direct current reverse polarity power, that may be of either the constant voltage or constant current type. The rated capacity of an EGW power source can vary from 750 - 1000 amperes with 100% duty cycle. The mechanical properties of EGW are affected by the form factor, which is a ratio of width to depth of the weld pool.(7) A high form factor is desirable from a point of view of good mechanical properties. EGW produces a coarse-grained microstructure. The grain structure is coarser at the fusion line than in the center of the weld. Electrode composition, type and thickness of the base metal, and the welding conditions have a marked effect on the fracture toughness of the welded joint. Presented in table XXV are typical mechanical properties of different grades of structural steels welded by FCAW-EG.

TABLE XXV

Mechanical Properties of Structural Steels Electrogas Welded with Flux-Cored Electrode Wire.

Base metal, ASTM	Thickness mm	Thickness in	AWS electrode class.	Shielding gas[2]	Yield strength MPa	Yield strength ksi	Tensile strength MPa	Tensile strength ksi	Elong. % in 51 mm (2 in.)	Reduction of area %	Charpy V-notch impact strength[1] Test temp. °C	Test temp. °F	WHCL J	WHCL ft-lb	HAZ J	HAZ ft-lb
A36	19	3/4	EG70T1	None	390	56.6	550	79.8	24	-	-16	4	49	36	-	-
A131-C	38	1-1/2	EG70T1	None	415	60.2	542	78.6	27	31.6	-18	0	19-23	14-17	23-37	17-27
A441	51	2	EG70T1	None	442	64.1	563	81.7	24	-	-18	0	58	43	-	-
A441	51	2	EG72T4	CO_2	472	68.4	637	92.4	24	61	-18	0	30	22	-	-
A516-70	38	1-1/2	EG70T1	None	423	61.4	558	80.9	29	35	-18	0	61	45	39	29
A516-70	38	1-1/2	EG72T4	AG20	538	78	619	89.8	29	67	- 7	20	42	30	38-91	28-67
A537-A	25	1	EG72T4	CO_2	430	62.3	572	83	29	70	- 7	20	34	25	39	29
A537-A	29	1-1/8	EG72T4	AG20	510	74	690	100	26	68	- 8	18	46	34	-	-
A537-A	38	1-1/2	EG70T1	None	385	55.9	549	79.6	30	40	-18	0	23	17	23	17

1. Test specimen notch locations: WHCL - weld metal center line, HAZ - heat-affected zone in the base metal.
2. AG20 is a mixture of 80% argon-20% CO_2.

The mechanical properties and microstructure of electrogas welded joints are similar to those produced by submerged-arc welding.

A long electrode extension of about 38 mm (1.5 in) used in EGW provides electrical resistance preheating of the wire and thereby increasing the electrode melting rate. The most widely used shielding gas is a mixture of 80% argon and 20% carbon dioxide.

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9. ELECTROSLAG WELDING

The conventional multiple-pass arc welding processes become progressively uneconomical as the base metal thickness increases over 76 mm (3.0 in). Electroslag welding (ESW) overcomes this problem. ESW is a metal joining process in which the heat obtained from a molten slag melts a consumable electrode and the surfaces of the base metal being welded.

The arc is struck between the energized electrode and a run-on plate with a steel wool ball located at the bottom of the joint. The run-on plate made usually from the same material and thickness as the parent metal, forms a sump which is discarded after welding. A water-cooled copper sump can also be used. The arc is not struck on the copper sump because it would be melted. Instead, small blocks of the base metal are placed on the copper sump for arc initiation purposes.

As the energized electrode contacts the steel wool violent arcing occurs. A small quantity of flux is added immediately or even before the arc is struck. The violent action of the arc ceases as soon as the flux melts. Then, more flux is added to extinguish the arc and welding commences. The objective is to have a steady welding by the time the flux level reaches the bottom of the joint so that a sound weld is obtained in the weldment. The power supply and the welding conditions are deliberately chosen to avoid arc maintenance. Flux is added manually during the welding process to keep the molten flux depth at about 38 - 51 mm (1.5 - 2.0 in). Welding should proceed smoothly until completion. Electroslag welding is analogous to continuous casting in vertical position. Graphical representation of the process in two versions can be seen in Figures 81 and 82.

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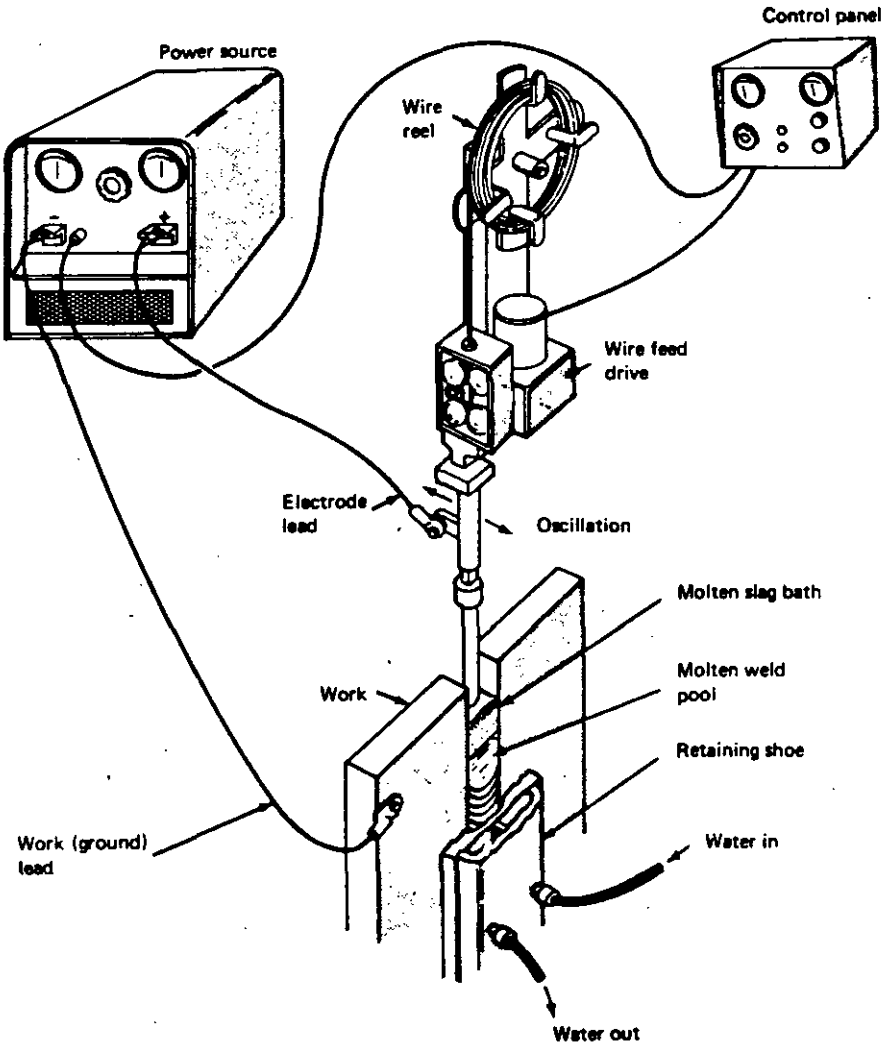


FIGURE 81. The consumable guide version of ESW.

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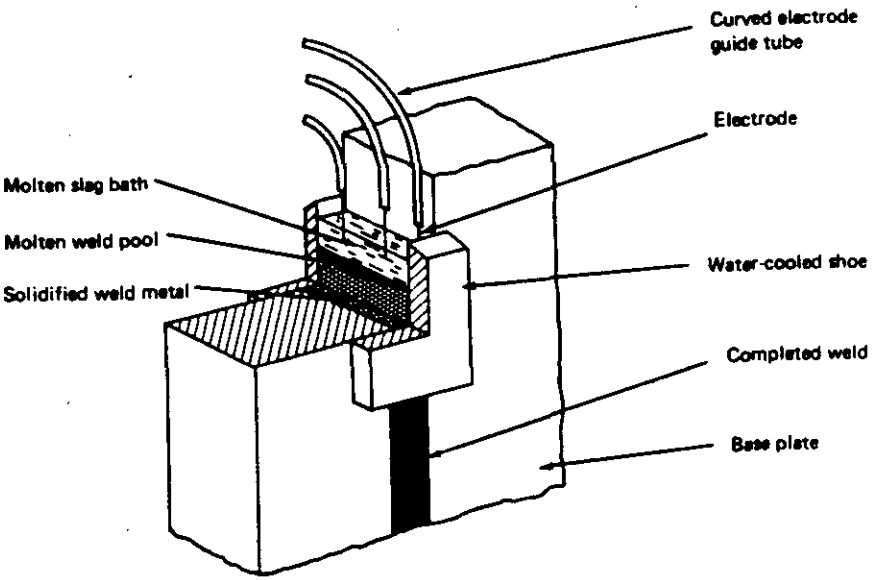


FIGURE 82. Non-consumable guide version of ESW with three electrode wires.

The consumable electrode may come in the form of a wire or plate. With metals over 100 mm (4 in) thick, multiple electrodes may be utilized. As the electrodes are melted, the weld pool rises and the water-cooled molding shoes are raised at the proper rate. Speeds of one meter per hour or more can be achieved with ESW.

The molten slag floating on top of the liquid weld metal provides the heat source and the shielding cover.(1) Once the right amount of molten slag has been established, the electroslag welding process progresses by resistance heating of the conductive slag. Hence, electroslag welding is not an arc welding process. In terms of basic equipments, ESW is similar to EGW. There is no spatter in ESW, since no arc exists. Solidification is progressive upward as welding continues. The weld is terminated on a run-off tab so that the solidified slag and weld metal extend beyond the top of the joint. The run-on and run-off tabs are usually cut off the weldment.

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The electrode is oscillated when the metal thickness is in the range of 51 mm (2.0 in) to 127 mm (5.0 in). Two oscillating wires are utilized for thicknesses of 127 mm (5.0 in) - 305 mm (12.0 in). Three oscillating electrodes are employed when the thickness is over 305 mm (12.0 in). The electrode diameter most often used varies from 2.4 mm (0.09 in) - 4.0 mm (0.16 in).

Electrodes must be guided from the wire feed rolls into the molten slag. The nonconsumable guide tube is normally made of beryllium copper and is insulated from the work to avoid short circuiting. The diameter of the guide tube must be small enough to fit into the root opening of the joint.

The consumable guide tube is made of steel with a composition that is complementary to the parent metal to be welded. The outside and inside diameters of the consumable guide tube are 16 mm (0.63 in) and 3.2 to 4.8 (0.13 - 0.19 in), respectively. The smaller inside diameter tube is used for welding thicknesses under 19 mm (0.75 in).

Electroslag welding is a very high deposition rate process as can be seen in Figure 83.

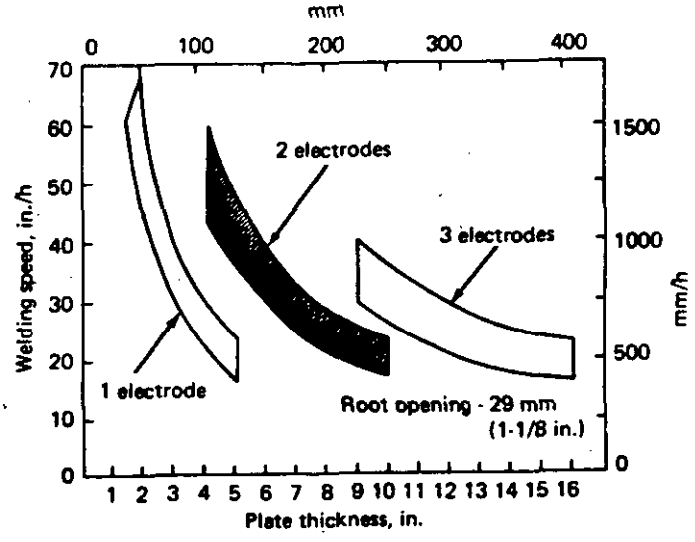


FIGURE 83. ESW speeds as a function of plate thickness.

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9.1 Fluxes

Fluxes used for electroslag welding are in some respects similar to those used in submerged-arc welding. The flux in the molten state must have adequate conductivity to carry the weld current from the electrode to the weld pool without arcing. The flux needs to have a lower melting point than the base metal. Electroslag fluxes contain more fluorides than submerged-arc fluxes in order to reduce viscosity and improve electrical conductivity. Low viscosity is beneficial in that it allows stirring of the slag to distribute heat uniformly across the joint.

The flux should have a low density to float on top of the weld pool and not adhere to the weld after solidification has taken place. A basic flux is preferred for welding steels. The boiling points of the flux components must be high enough to prevent evaporation losses at the temperatures involved.

9.2 Electrode Composition

The carbon content of the electrodes designed for welding carbon and low-alloy steels is less than that of the base metal. The filler metal composition for higher alloy steels matches that of the base metal. While electroslag welded joints involving mild and low-alloy steels do not normally require preheat or postweld heat treatment due to the inherent slow cooling, postweld heat treatment may, however, be advisable in higher-alloy steels to develop optimum properties in the weld and the heat-affected zone.

9.3 Dilution

The solidified weld deposit obtained by ESW is a combination of filler metal and base metal in a ratio of about 7 to 3. Consequently, dilution must be considered in developing the optimum weld procedure to satisfy the chemical and mechanical property requirements imposed on the weldment.

9.4 Metallurgical Considerations

The microstructure of electroslag welds is markedly different from that of conventional, multiple-pass arc welds. Large grains dominate the microstructure which result in inferior fracture toughness properties for the weld and the heat-affected zone (HAZ). This can create a cracking problem at low temperatures. Electroslag welds are characterized by a rather wide HAZ. The slow cooling rate produces a microstructure dominated by high-temperature transformation products such as ferrite and pearlite in the case of welding low-carbon steels.

For applications where the coarse-grained microstructure may be a problem, a normalizing heat-treatment can bring about sufficient improvement in the fracture toughness. Electroslag weldments are known to have relatively low residual stresses. Furthermore, the rotational distortion is virtually absent. Typical mechanical properties of electroslag welded steels can be found in table XXVI.

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TABLE XXVI

Mechanical Properties of Electroslag Welds (Consumable-guide)
As-Welded Coondition.

Structural Steels

Base metal, ASTM	Thickness mm	Thickness in.	Electrode Type (AWS)	Electrode No.	Yield strength MPa	Yield strength ksi	Tensile strength MPa	Tensile strength ksi	Elong. % in 51 mm (2 in.)	Reduction of area %	Impact strength[a] J	Impact strength[a] ft-lb
A441	25	1	EM13K-EW	1	344	49.9	523	75.8	28.0	59.0	23	17
A441	64	2-1/2	EM13K-EW	1	316	45.8	505	73.3	26.5	66.0	37	27
A36	152	6	EM13K-EW	2	317	46.0	548	79.5	28.5	52.8	-	-
A36	305	12	EM13K-EW	2	255	37.0	464	67.3	33.5	71.0	-	-
A572 Gr.42	203	8	EM13K-EW	2	400	58.2	585	84.8	25.0	67.6	38	28
A572 Gr.60	57	2-1/4	EH10Mo-EW	1	423	61.5	680	98.5	18.0	35.6	-	-

a. Impacts at - 17.8°C (0°F).

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9.5 Applications

ESW is primarily a joining process for heavy cross-sections in the vertical position. Its main advantage lies in high weld productivity and little or no distortion. The most common application areas of ESW are in bulky components of machinery, ships, pressure vessels, castings and large constructions.(1) Electroslag welding has a limited application in materials joining due to the high heat imput generated and its undesirable consequences.

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10. OXYFUEL GAS WELDING

Oxyfuel gas welding (OFW) is a manual welding process in which the base metal surfaces are fused progressively by heat from a gas flame, with or without the use of filler metal and pressure. One of the most common sources of heat is the flame produced at the tip of a welding torch. The major components of the oxyfuel gas welding system include compressed gas cylinders - one for oxygen and one for fuel gas - gas regulators, hoses and a welding torch. The rate of gas flow from the respective cylinders is controlled by the gas regulators. The gas flows through the rubber hose to a gland connected to the inlet on the welding torch. The gases are mixed in a mixing chamber. The mixed gases in turn exit through the tip of the welding torch. The gas is lit by a spark lighter.

The filler metal, if required, comes in the form of a welding rod. The welding rod is placed in the flame at an angle of about 90° to the axis of the flame. The welder controls the rate of heat input into the base metal, the temperature of the weld zone and the flame conditions (neutral, oxidizing, reducing, and carburizing).

One of the most important aspects of the flame is its temperature, which to a great extent determines the welding rate. Heat is transferred from the torch to the workpiece by forced convection and radiation. Flame temperature is dependent on the net calorific value of the fuel gas and the volume as well as the specific heat of the products of combustion. The combustion ratio is the volume of oxygen required to complete the combustion of a unit volume of fuel gas. The flame temperature decreases when there is an excess of either the fuel gas or oxygen in the gas mixture. Combustion in air does not produce hot enough flame for welding most metals. The combustion temperature is also decreased by the presence of nitrogen in the flame.

The oxyfuel gas welding equipment is used for a variety of related functions, namely preheating, postheating, bending, straightening, forming, hardening, surfacing, cutting, braze welding, torch brazing and soldering.

Metals that are normally oxyfuel gas welded include steels and most of the nonferrous metals. The reactive and refractory metals are not suitable for oxyfuel gas welding. Different metals and thicknesses require different torch nozzle sizes, gas pressures and flame temperatures.

The gas velocity can be regulated by the pressure and the orifice diameter. When the gas pressure is too high for a given nozzle size, the flame becomes too harsh and unsteady. Such a flame will disturb the weld pool. If the gas pressure is too low, the flame will not have an adequate force necessary for welding and cutting. The nozzles for oxyacetylene welding are usually graded and numbered to indicate their heating power. The grading is an index of the volume of acetylene consumed per hour at the optimal setting.

A spark lighter is usually used for starting the flame. Cigarette lighters and matches should not be used to light the oxyacetylene torch because the hand may be too close to the flame, which may result in skin injury.

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The velocity of the gas must be properly adjusted to the flame velocity, otherwise one of two situations may arise. If the gas velocity is excessive, the flame will separate from the nozzle. If, on the other hand, the gas velocity is less than that of the flame, a flame back-fire will occur. Each fuel gas when mixed either with air or oxygen does have an optimal velocity to maintain the desired flame. This characteristic flame velocity for the gas mixtures is lowest at the two extreme limits of flammability and reaches the maximum between these two extreme values. The limits of flammability is defined by the range of mixtures of the gases involved wherein combustion occurs. Hydrogen and acetylene are characterized by wide limits of flammability and high flame speeds, while propane is flammable within a very narrow range of mixtures and has a low flame velocity.

In order to determine the suitability of commercially available fuel gases for welding purposes, the following attributes must be assessed;

(a) flame temperature,
(b) flame speed,
(c) heat content,
(d) extent of chemical reaction of the flame with both the base and the weld metal.

Of the commercial fuel gases, acetylene approaches the above requirements. Gases like propylene, propane, natural gas and various other proprietary gases may have a high flame temperature but lack sufficient flame speed. These gases have an unacceptable oxidizing characteristic at oxygen-to-fuel gas ratios high enough to yield acceptable heat transfer rates. However, these commercial gases are good enough for cutting operations, torch brazing, and soldering where the flame and heat transfer characteristics are not as rigorous as in gas welding.

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The chemical characteristics of the flame can be changed to meet the needs of the welding procedure by adjusting the acetylene to oxygen ratio. The neutral flame is utilized in most applications. Since it is less desirable to have a slightly oxidizing flame than one which is reducing, a slight excess of acetylene is therefore preferred. The type of flame obtained is judged by the size, shape and color of the combustion zones in the flame.(1) (See Figure 84).

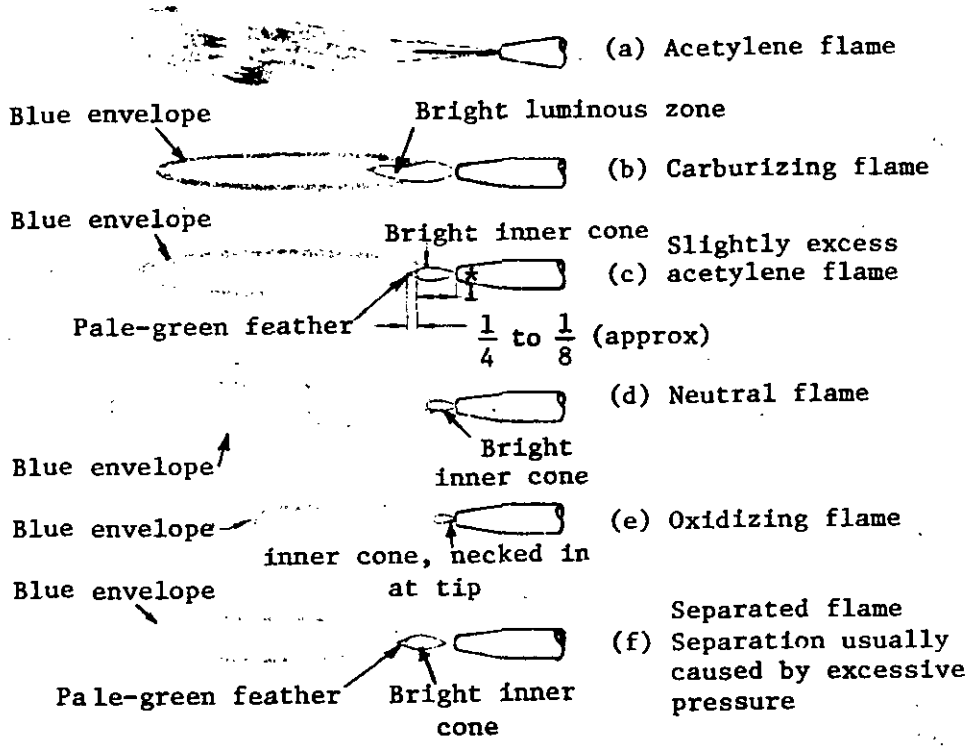
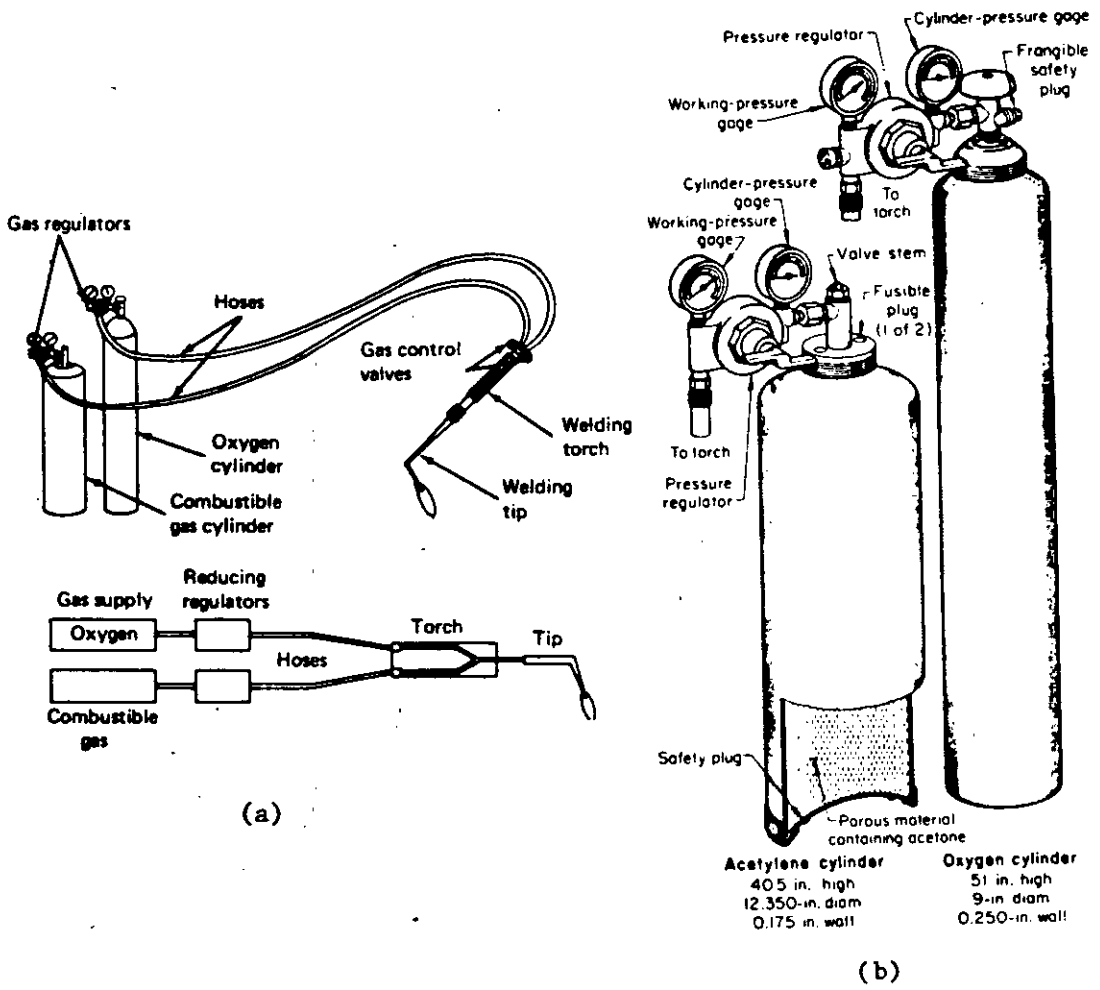


FIGURE 84. Visible changes in the flame as a result of adjustments.

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The principal components of gas welding equipment are shown in Figure 85.



(a) (b)

FIGURE 85 Major components of oxyfuel gas welding.

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Neutral flame is obtained upon the disappearance of the feather extension of the inner zone (or cone). When an oxidizing flame is needed, the welder customarily begins with the neutral flame and either decreases the acetylene or increases the oxygen until the length of the inner cone is reduced sufficiently.

The chemical reactions which describe the complete combustion of the more common fuel gases are given in table XXVII.

TABLE XXVII

Chemical Reactions for Complete Combustion of Common Fuel Gases.

Fuel Gas	Reaction with Oxygen
Acetylene	$C_2H_2 + 2.5O_2 \rightarrow 2CO_2 + H_2O$
Methylacetylene-propadiene (MSP)	$C_3H_4 + 4O_2 \rightarrow 3CO_2 + 2H_2O$
Propylene	$C_3H_6 + 4.5O_2 \rightarrow 3CO_2 + 3H_2O$
Propane	$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$
Natural gas (methane)	$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O$
Hydrogen	$H_2 + 0.5O_2 \rightarrow H_2O$

There are different temperature zones within the flame. The maximum flame temperature is near the end of the inner cone beyond which the temperature decreases towards the end of the flame envelope. The flame temperature is an inherent property of a fuel gas. It can, however, be altered by changing the oxygen-to-fuel ratio. So, an oxidizing flame - while undesirable for welding purposes - has a higher temperature than the neutral flame.

Another physical property of a fuel gas is its combustion velocity, otherwise known as rate of flame propagation. It is the speed with which the front of the flame travels through the unburnt gas. This property affects the temperature and size of the primary flame. It also affects back-fire and flame separation from the tip of the torch.

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A third characteristic property of fuel gases is the so called combustion intensity. The combustion intensity is a function of the heating value of the oxygen-and-fuel gas mixture, the area of the flame cone emitted from the torch tip and the speed of combustion.

An expression of combustion intensity is as follows;

$$C_i = C_v \times C_h$$

where,

C_i = combustion intensity ($J/m^2.s$) or ($Btu/ft^2.s$)
C_v = normal combustion velocity of the flame (m/s) or (ft/s)
C_h = heating value of the gas mixture (J/m^3) or (Btu/ft^3)

There are four varieties of OFW, namely oxyacetylene, airacetylene, oxyhydrogen and gas pressure welding. For all intents and purposes, all but oxyacetylene gas welding became outmoded as a result of arc welding processes and other more modern joining methodologies. Therefore, this discussion is confined to that of oxyacetylene gas welding.

The welder has almost complete control over the entire gas welding process and the resultant weld quality. The welder controls the temperature of the base metal, the filler metal deposition rate and the weld profile. OFW is a simple, and low cost welding process. In today's climate of high weld productivity needs, oxyfuel gas welding has a limited application. Like other manual welding processes, OFW has been rapidly replaced by the introduction of high weld productivity technologies such as automatic welding and robotics.

The oxyacetylene flame is the result of combustion of nearly equal volumes of oxygen and acetylene at the tip of a torch. The combustion can be described chemically as follows,

$$C_2H_2 + O_2 \qquad 2CO + H_2 + 106{,}500 \text{ cal.}$$

The structure of this flame is illustrated in Figure 86.

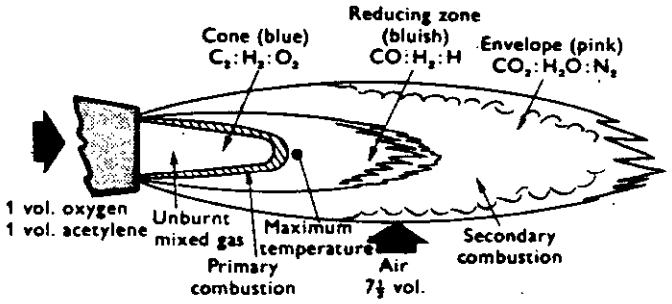


FIGURE 86. The Structure of the oxyacetylene flame.

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The most frequently used fuel gases produce maximum flame temperatures over 2760°C (5000°F) when mixed with oxygen. At the maximum temperature, the flame has a strongly oxidizing nature. Therefore, they are unsuitable for welding because of oxide formation in the weld. Flame temperatures for a number of oxyfuel gas mixtures are given in table XXVIII.

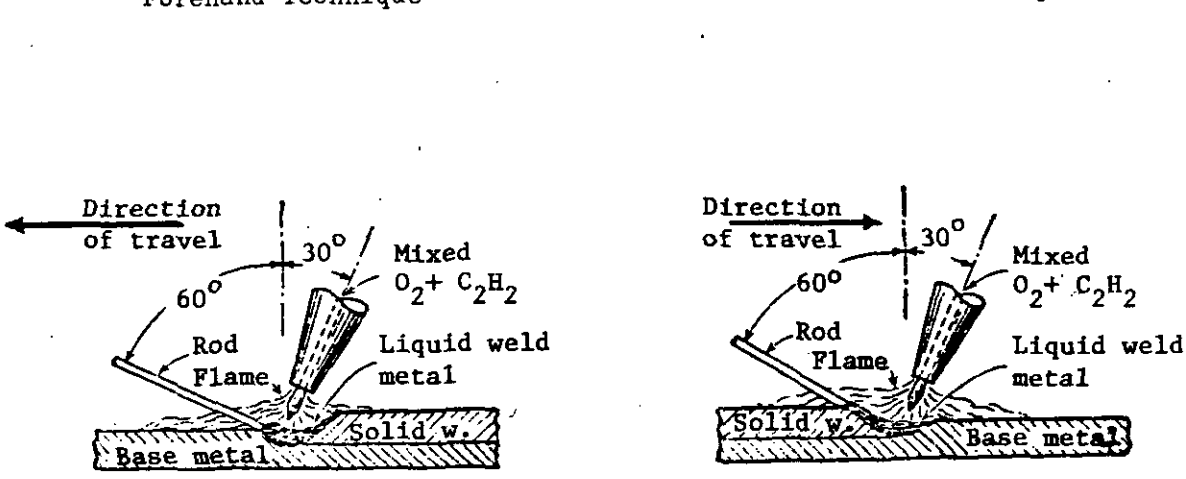
TABLE XXVIII

Flame Temperatures of Oxyfuel Gases.

	Max temp.		Neutral flame temp.	
	°C	°F	°C	°F
Acetylene	3102	5615	3100	5612
MAPP (a)	2902	5255	2600	4712
Propylene	2857	5174	2500	4532
Hydrogen	2871	5200	2390	4334
Propane	2777	5030	2450	4442
Natural gas/methane	2742	4967	2350	4260

(a) Methylacetylene-propadiene (stabilized)

The distortion and the heat-affected zone obtained in welds made by OFW are substantial due to the wide spread of the flame. There are two types of OFW techniques: the forehand and backhand techniques graphically shown in Figure 87.



w. = weld

FIGURE 87. Forehand and backhand OFW techniques.

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In forehand welding, the flame is pointed in the welding direction so that the hot inner cone of the flame lies above the surface of the base metal and preheats the area ahead of the weld pool. The filler metal should be held below the surface of the molten weld to minimize atmospheric contamination.

The same relative positions of the torch and filler metal are used in backhand welding. The outer region of the flame postheats the weld, when the backhand technique is employed. Forehand technique is more suitable for sheet metal joining. The backhand technique is preferred for oxyfuel gas welding heavier cross-sections.

In general, gas welding is limited to the joining of relatively thin sections due to the limited heat content of gas flames compared to those of electric arc and radiant energy welding processes.

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11. STUD ARC WELDING

Stud arc welding (SW) is an arc welding process for joining a stud or a fastener to the surface of a base plate. Joining can be accomplished by means of arc, resistance, percussion and friction. The two basic methods include stud arc welding and capacitor-discharge stud welding.

11.1 Stud Arc Welding

In many ways, this process resembles the manual shielded metal-arc welding. Fusion between the surfaces of a stud and base plate occurs as a result of a direct current arc drawn between the two pieces of metal and pressure. The electrical power is supplied by either a transformer-rectifier or a motor-generator. The welding time and current as well as the plunging of the stud are controlled automatically. The stud with a ceramic ferrule around its tip is held in a gun or electrode holder and positioned over the spot where it is to be attached by plunging rapidly into the liquid metal created on the surface of the base plate. Flux is used for stud arc welding of ferrous metals. The flux provides a cleaning action and protection during welding. Arc time is in milliseconds. The short arc time and the relatively low heat input result in minimal distortion, residual stresses and heat-affected zone. Figure 88 illustrates the circuit diagram of stud arc welding.

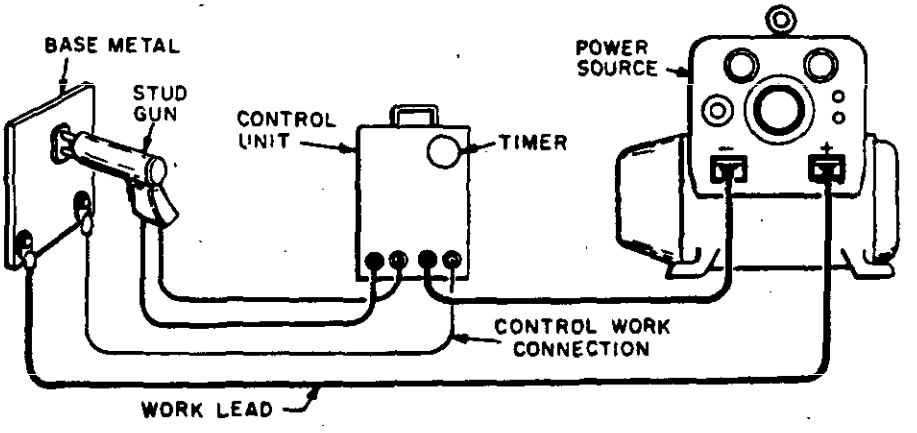


FIGURE 88. Setup for stud arc welding.

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Studs come in many different sizes and shapes. In designing the stud, it must be recognized that the length of the stud will be reduced due to material loss caused by the welding action. Molten metal will be expelled from the interface. A part of this metal loss will manifest itself in a flash around the welded joint. The ferrule fulfills several functions, specifically heat concentration, retention of weld pool, minimization of air entrapment in the weld and operator protection.(7) The sequence of arc stud welding is presented in Figure 89.

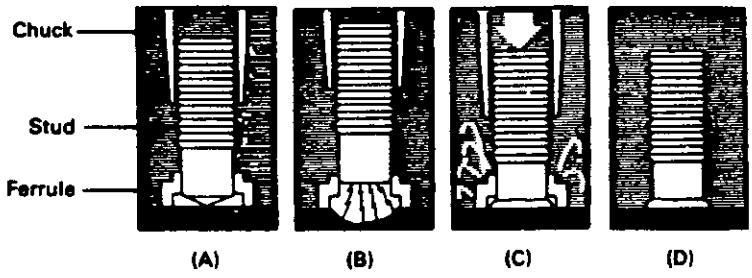


FIGURE 89. Operational sequence in stud arc welding. A - gun positioned, B - arc established C - stud plunged, D - ferrule removed.

The welding gun can either be hand-held for portable applications or mounted on a device for stationary use under production run conditions.

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Stud arc welding of nonferrous metals require the use of inert gas shielding. Stud arc welding conditions for aluminum can be found in table XXIX.

TABLE XXIX
SW conditions for Aluminum Alloys[a].

Stud weld base diameter		Weld time, cycles (60Hz)	Welding current,[b] Amp	Shielding gas flow[c]	
mm	in.			Liter/min	ft^3/h
6.4	1/4	20	250	7.1	15
7.9	5/16	30	325	7.1	15
9.5	3/8	40	400	9.4	20
11.1	7/16	50	430	9.4	20
12.7	1/2	55	475	9.4	20

a. Settings should be adjusted to suit job conditions.
b. The currents shown are actual welding current and do not necessarily correspond to power source dial settings.
c. Shield gas - 99.95% pure argon.

Stud arc welding of ferrous metals involves direct current straight polarity (stud negative, base plate positive). Reverse polarity is used for nonferrous metals.

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To develop the maximum fastener strength, the base plate thickness should be at least one third of the diameter of the weld base. Stud welding should not cause a melt-through of the base plate. Therefore, there is a minimum base plate thickness for every stud size as illustrated in table XXX.

TABLE XXX
Minimum Steel and Aluminum Plate Thicknesses Recommended for Stud Arc Welding.

Stud base diameter		Steel without backup		Aluminum			
				Without backup		With backup[a]	
mm	in.	mm	in.	mm	in.	mm	in.
4.8	3/16	0.9	0.04	3.2	0.13	3.2	0.13
6.4	1/4	1.2	0.05	3.2	0.13	3.2	0.13
7.9	5/16	1.5	0.06	4.7	0.19	3.2	0.13
9.5	3/8	1.9	0.08	4.7	0.19	4.7	0.19
11.1	7/16	2.3	0.09	6.4	0.25	4.7	0.19
12.7	1/2	3.0	0.12	6.4	0.25	6.4	0.25
15.9	5/8	3.8	0.15				
19.1	3/4	4.7	0.19				
22.2	7/8	6.4	0.25				
25.4	1	9.5	0.38				

a. A metal backup to prevent melt-through of the plate.

11.2 Capacitor-Discharge Stud Welding

Capacitor-discharge stud welding generates heat from an arc produced by a rapid discharge of stored electrical energy with pressure applied either during or right after the electrical discharge. The energy used for welding is stored in a capacitor. Ferrule or flux is not required for steel. Argon shielding is utilized for aluminum welded by the drawn-arc method.

Three modifications of the capacitor-discharge stud welding process are known: (a) initial-contact, (b) initial-gap and (c) drawn-arc.

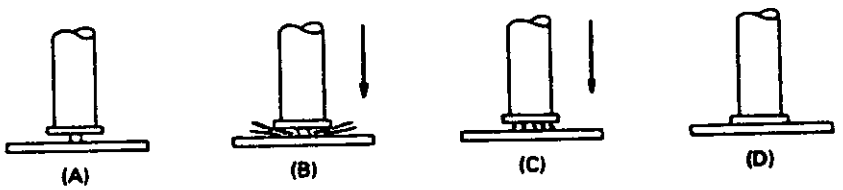
(a) Initial-contact stud welding involves a stud with a tip which concentrates the weld current, which in turn rapidly melts the interfacial area of both the stud and the base metal. A compressive force of about 2.25 - 3.15 kg (5 - 7 lb) plunges the stud into the liquid metal. The arc is then extinguished and the weld cycle completed. Arc duration is about 3 - 6 milliseconds resulting in practically no marking on the reverse side of the base metal.

(b) Initial-gap stud welding is characterized by a gap between the stud and the workpiece. When the gun is triggered the stud is released and plunges toward the parent metal. At the same time, a voltage is impressed on the stud and the workpiece. Upon contact, an arc is formed producing fusion between the stud and the base metal.

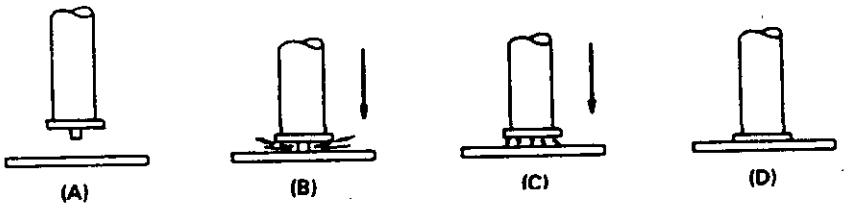
(c) Drawn-arc stud welding resembles the arc stud welding process. The stud does not have a tip and is initially in contact with the workpiece. A solenoid in the gun body lifts the stud while the welding circuit is energized thereby drawing a pilot arc between the stud and the base plate. The welding capacitors are discharged as the lifting coil is de-energized. The high current melts the surfaces and the stud is forced into the weld puddle by the spring action of the welding gun to complete the weld.

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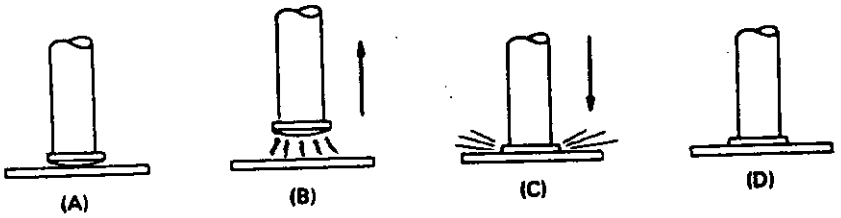
The operational steps of the three different capacitor-discharge stud welding processes are prescribed graphically in Figure 90.



initial-contact



initial-gap



drawn-arc

FIGURE 90. Operational sequences - from start to finish - in the three varieties of capacitor-discharge stud welding.

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The plot of welding current and time for the three different types of capacitor-discharge stud welding technique is exhibited in Figure 91.

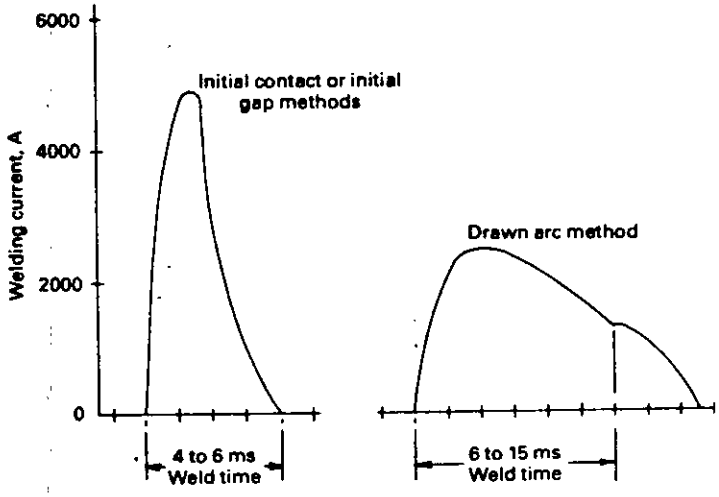


FIGURE 91. Welding current versus weld time for capacitor-discharge stud welding.

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The principal difference between stud arc and capacitor-discharge stud welding can be found in the quantity of molten metal produced.(5) A relatively large amount of stud and base metal is melted in SW as compared to capacitor-discharge stud welding. Consequently, there is a deeper penetration into the base metal and a noticeable heat tint on the back side of the base plate welded by the stud arc process. These process attributes are illustrated in Figure 92.

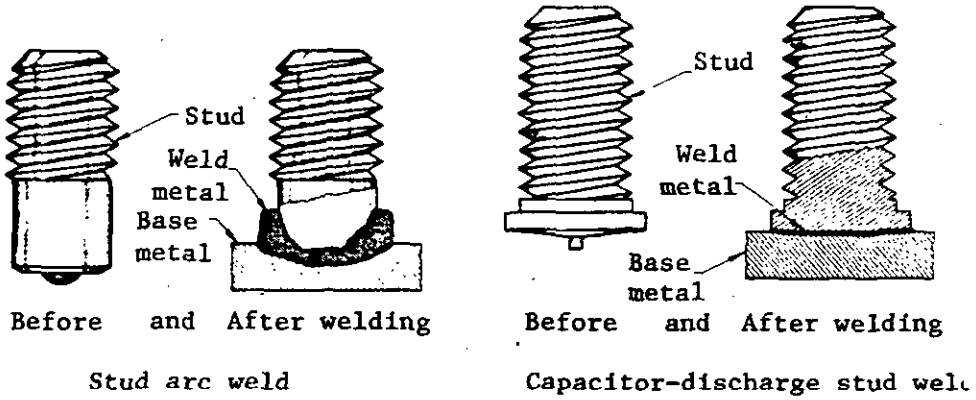


FIGURE 92. Comparison of the two basic stud welding methods.

11.3 Quality Control

A satisfactory weld quality requires that attention be paid to the maintenance of optimum welding current and other variables developed for the particular stud being welded. The work surface should be cleaned so that a sound weld is obtained. The stud must be positioned correctly in the gun which in turn must be held at the proper angle to the work surface.

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Visual examination is the most common nondestructive examination (NDE) method applied on evaluating the quality of stud welds.(5) Mechanical testing can involve either a torque test or a bend test. For the torque test, the steel may be twisted until failure occurs. The bend test may consist of either striking the steel with a hammer or bending it with a piece of tubing. The acceptable angle of bending varies from application to application. The degree to which the stud may bend or fail depends on the composition of the stud and base metal, the welding conditions and the design of the stud. Presented in Figure 93 are some examples of acceptable and unacceptable stud welding results.

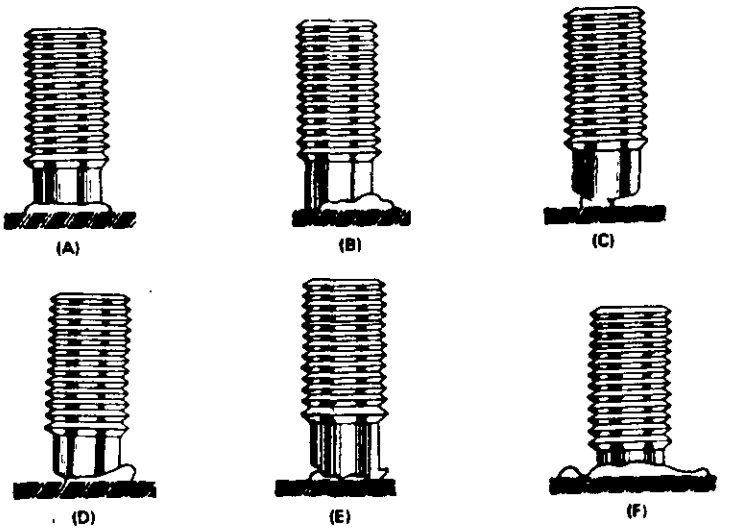


FIGURE 93. A - satisfactory stud weld, B - plunge too short C - hanging, D - misalignment, E - insufficient weld current, F - excessive weld current.

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12. RESISTANCE WELDING

Resistance welding (RW) produces coalescence at the faying surfaces by the heat obtained from resistance to electric current through the workpieces held together under pressure by the electrodes. Modifications of the process include resistance spot welding, seam welding, projection welding, percussion welding, upset welding, flash welding and high-frequency resistance welding.

The welding current, time, pressure and the characteristics of the electrode constitute the essential variables of resistance welding.

12.1 Resistance Spot Welding

Sheet metal workpieces are clamped between an upper and a lower electrode by pressure applied through the electrodes by pneumatic means. The welding current is passed through the electrodes, which concentrate the heat in one spot to form a nugget. Typical ranges of current and voltage values employed in resistance spot welding of steels are 5,000-20,000 amperes and 5-20 volts. This requires a high kilovolt-ampere (kva) demand. This kva demand is even higher for resistance spot welding metals of good conductivity such as aluminum and copper. Important features of electrical spot welding are shown in Figure 94.

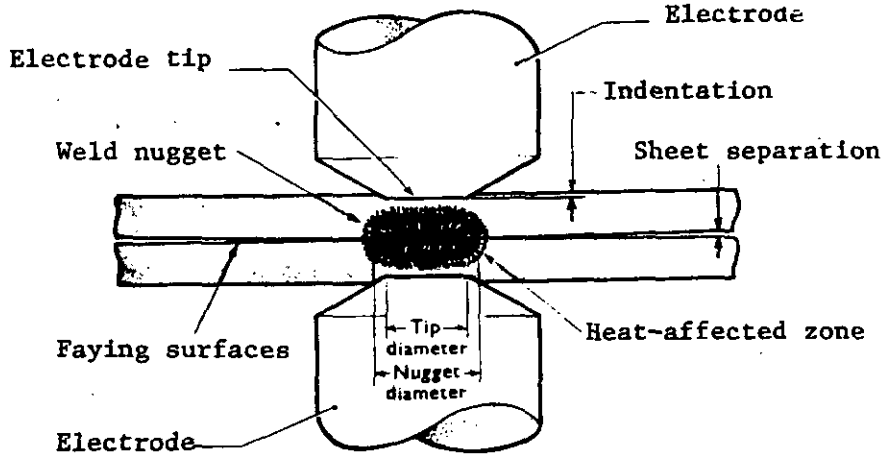


FIGURE 94. Highlights of resistance spot welding.

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The concentration of the welding current is determined by the contact area between the electrodes and the work, which in turn is a function of the size and shape of the electrode tip. The welding time is very brief, usually a fraction of a second. Therefore, the most important advantage of RSW is the high weld productivity.

The electrode pressure is maintained during solidification of the nugget after the welding current is cut off. The water-cooled electrodes are retracted from the workpieces after each spot weld. The welding current, time and pressure are controlled automatically.

RSW machines take their energy from 220-volt or 440-volt in-plant power lines of either single-phase or three-phase 60-cycle alternating current. The in-plant power source is stepped down to about 5-20 volts and supplied directly to the electrodes when a weld is made.

The sequence of operation consists of four steps as follows,

1. Squeeze time
2. Weld time
3. Hold time and
4. Off time.

Contact between the electrodes and the workpieces is established during the squeeze time. Initiation and maintenance of current flow take place in step two. During hold time, the current is cut off while the electrode pressure is maintained to consolidate the nugget. This is the cooling period of the weld metal under pressure. The electrodes are retracted to complete the spot welding cycle.

There are slope controls for increasing and decreasing the value of the welding current. The upslope control regulates the current from a low value to the full current value needed for the base metal involved. A downslope control is utilized to decrease the full welding current at the end of the welding step to a lower value gradually instead of cutting off the supply of welding current abruptly. These upslope and downslope features of resistance spot welding machines serve to control both the heating and the cooling rates of the base materials.

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A rocker-arm type spot welding machine is shown schematically in Figure 95.

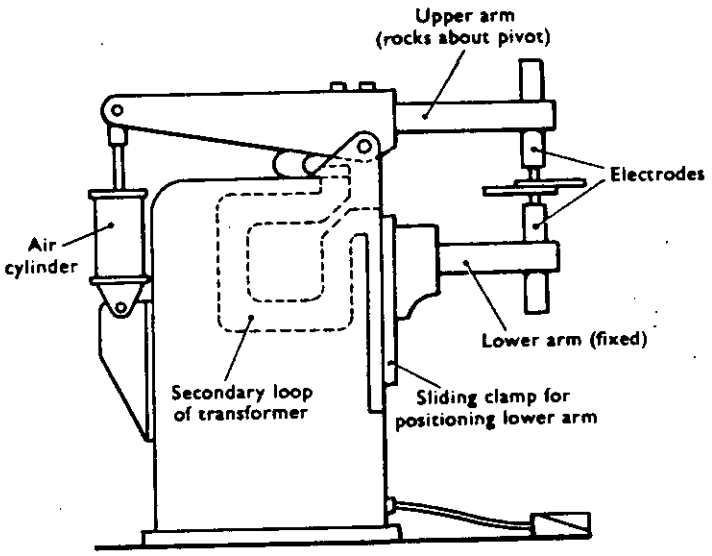


FIGURE 95 Features of rocker-arm spot welding machine.

Electrode materials must have sufficient electrical and thermal conductivities, low contact resistance and adequate strength. High electrical conductivity and low contact resistance are necessary to reduce the heating of the electrodes. The dissipation of heat from the interface between the electrodes and workpieces necessitates a good thermal conductivity. The higher the hardness of the electrodes, the greater the resistance to heat and current flow as well as to permanent deformation. Electrodes have been divided into copper-base alloys or group A and refractory-metal compositions or group B.

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The electrodes in group A have been further subdivided into three classes and those in group B into five classes. Some of the properties of resistance spot welding electrodes are presented in tables XXXI and XXXII.

Table XXXI

Minimum Values of Properties Given for Copper-Base Electrodes (Group A).(a)

Diameter or thickness of electrode material, in.	Proportional limit, ksi			Hardness, HRB			Electrical conductivity(b) % IACS			Tensile strength ksi			Elongation (c) in 2 in., %		
	Class 1	Class 2	Class 3	Class 1	Class 2	Class 3	Class 1	Class 2	Class 3	Class 1	Class 2	Class 3	Class 1	Class 1	Class 3
Round rod stock															
Up to 1	17.5	35	50	65	75	90	80	75	45	60	65	100	13	13	9
1 - 2	15	30	50	60	70	90	80	75	45	55	59	100	14	13	9
2 - 3	15	25	50	55	65	90	80	75	45	50	55	95	15	13	9
Square, rectangular, and hexagonal bar stock															
Up to 1	20	35	50	55	70	90	80	75	45	60	65	100	13	13	9
Over 1	15	25	50	50	65	90	80	75	45	50	55	100	14	13	9
Forgings															
Up to 1	20	22(d)	50	55	65	90	80	75	45	45	55	94	12	13	9
1 - 2	15	21(d)	50	50	65	90	80	75	45	40	55	94	13	13	9
Over 2	15	20(d)	50	50	65	90	80	75	45	40	55	94	13	13	9
Castings															
All	-	20	45	-	55	90	-	70	45	-	45	85	-	13	5

(a) Nominal compositions: Class 1, 1% Cd, rem Cu; Class 2, 0.8% Cr, rem Cu; Class 3, 0.5% Be, 1.0% Ni and/or 1.0% Co, rem Cu. (b) International Annealed Copper Standard. (c) Or in a length equal to four times the diameter of the test specimen. (d) For hot worked and heat treated (but not cold worked) electrode material.
Source: "Resistance Welding Equipment Standards," American National Standard C-88.2.

TABLE XXXII

Minimum Values of Properties Indicated for Group B Electrode Materials

Class(a)	Rockwell hardness	Electrical conductivity(b), % IACS	Compressive strength ksi
10	72 HRB	35	135
11	94 HRB	28	160
12	98 HRB	27	170
13	69 HRA	30	200
14	85 HRB	30	-

(a) Classes 10, 11 and 12 are copper-tungsten mixtures; class 13 is unalloyed tungsten; class 14 is unalloyed molybdenum. (b) International Annealed Copper Standard.
Source: "Resistance Welding Equipment Standards," American National Standard C-88.2.

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The high temperature and pressure are the two most important factors that must be taken into account in designing resistance spot welding electrodes. Pickup or alloying and plastic deformation of the electrode tip are a direct result of elevated temperature and pressure used during welding. Consequently, the face of the electrodes must be redressed occasionally in order to meet weld quality control requirements. The geometrical shapes of standardized electrodes can be seen in Figure 96.

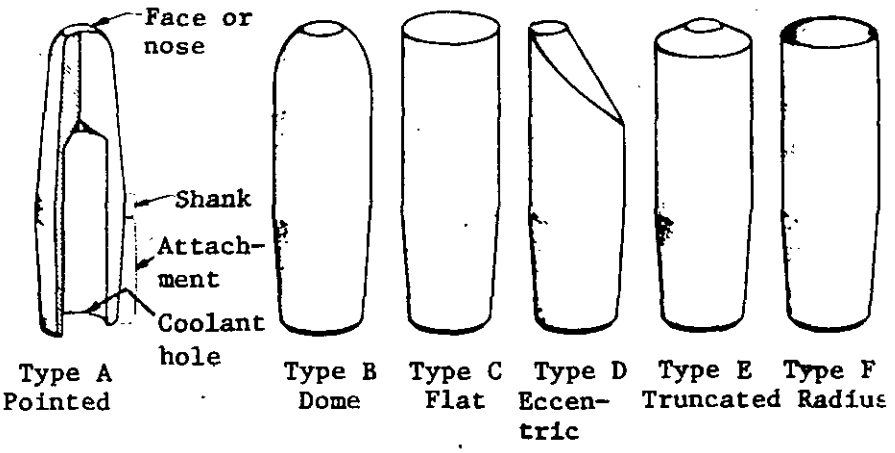


FIGURE 96. Different types of standardized electrodes.

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Dimensions of the electrode face are selected on the basis of base metal thickness, required nugget size and the geometry of the weld assembly. The minimum face diameter -d- for electrode types A, B, D and E is determined by the following empirical formula:

$$d = 0.10 + 2t \text{ (inch)}$$

where,

t = base metal thickness (mm or inch)

Too small of a face diameter produces a weak weld because of the resultant small nugget size. The high current density of an excessively small electrode face diameter may cause an unacceptable indentation or heat tint on the work surface. Voids, blow holes or undesirable surface appearances may be the consequence of overly large electrode face diameters.

Direct or series (indirect) welding implies different arrangements of the electrodes and workpieces in the secondary circuit. When all of the secondary current flows through the weld nugget(s) without shunting, it is referred to as a direct spot welding. If, on the other hand, a part of the current through a shunt path bypasses the nugget, this is referred to as series or indirect welding.

The welding current flows through the secondary circuit which is made up of the secondary coil of the transformer, the flexible bands connecting the coil to the horns, the horns, the electrodes and the base metal. The secondary circuit, therefore, consists of a series of resistances affecting the flow of the welding current. The heat generated at any point is directly proportional to the resistance at that point. There are seven resistances connected in series in the case of a direct single-spot welding of a two-thickness joint, as shown in Figure 97.(1)

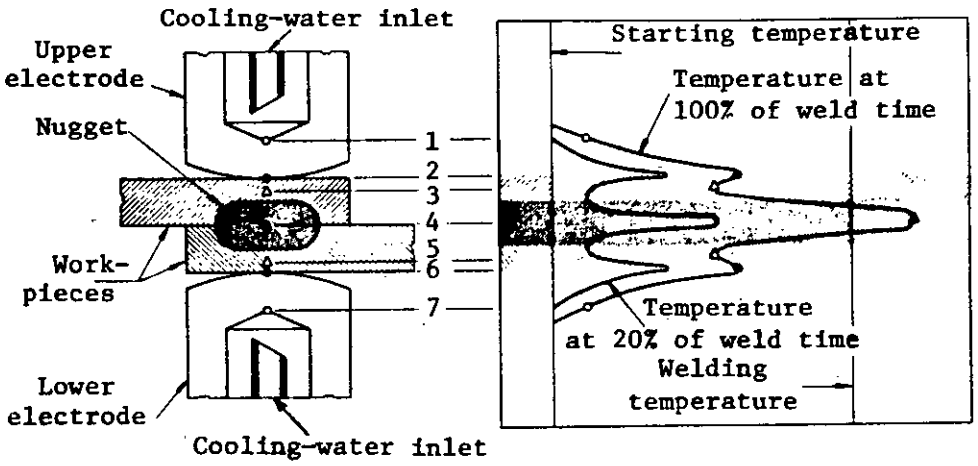


FIGURE 97. Points of resistance and attendant heat generations.

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The heat generated at any point in the secondary circuit is expressed as follows,

$$H = kI^2Rt$$

where,

H = heat in watt-seconds or joules
I = welding current in amperes
R = total resistance in ohms
t = time of current flow in seconds
k = heat losses or efficiency factor

The electrode pressure has an influence on heating because the contact resistance decreases with an increase in pressure. The electrode pressure must be adjusted so as to avoid excessive indentation but to ensure sufficient physical contact at the spot weld area.

Attention must be paid to the surface preparation of the workpieces in order to produce satisfactory weld quality. Scale, dirt, grease, oil, paint, oxide film, rust and other forms of contaminants on the base metal surface must be removed prior to welding.

Resistance spot welding is applicable to joining ferrous, non-ferrous and dissimilar metals and thicknesses. The welding parameters must be developed to satisfy the properties and thicknesses of the base metals involved.

12.2 Weld Spacing

If the spot welds are too close to one another, the welding current may be shunted. The current flow is inversely proportional to the resistance of the paths available. The current can flow either through the metal at the first spot weld or via the base metal at the point of the second weld to be made. If the distance separating the two spot welds is great enough so that the resistance of the path through the first spot weld is high relative to that of the second spot weld to be made, shunting will be negligible.

The resistance increases with a rise in metal temperature during welding. This can create an additional shunting effect. Base metals of high electrical resistivity are not affected by shunting as much as metals of low electrical resistivity. Spot spacing is also influenced by base metal thickness, surface cleanliness and nugget diameter.

When the spot weld is placed too close to the edge of the base metal, hot metal may be ejected from the weld because there is insufficient amount of metal to withstand the electrode force and heat.

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12.3 Quality Control

The quality of spot welds is determined mainly by either visual means or by destructive testing such as peel, tension-shear, tension, impact, twist and hardness. Peel testing is the most widely used destructive test method for spot welds. Peel testing can be done manually or automatically. Most specifications require that a full nugget be torn from one sheet. A schematic illustration of the test coupons used in two of these destructive testing methods is displayed in Figure 98.

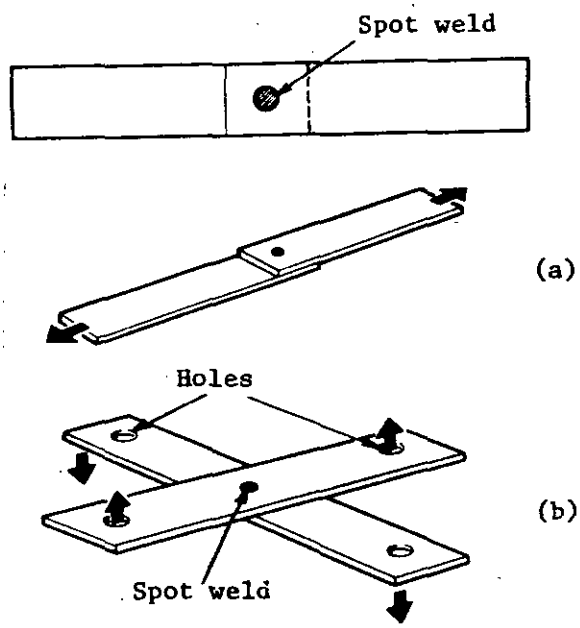


FIGURE 98. Methods of destructive testing resistance spot welds.
(a) tension-shear
(b) tension

12.4 Resistance Seam Welding

Resistance seam welding (RSEW) works on the same principle as resistance spot welding, except for the rotating wheel electrodes. Resistance seam welding produces a series of overlapping spot welds - sometimes known as stitch welding which are leak-tight.[13] The overlapping spot welds along a seam are the result of current pulses either from two roller electrodes or from one circular electrode and one bar-type electrode. The work moves between the rollers without retracting the wheel electrodes. The amount of overlap is a function of the peripheral speed of the rollers and the time lapse between current impulses. The overlapping spot welds can be made either by a continuous or by an intermittent motion of the electrode wheel(s) and the work. The welding speed, weld spacing and base-metal thickness govern the type of motion employed. Continuous motion is utilized for base metal thickness under 4.8 mm (0.19 in), while intermittent motion is used above that thickness. In intermittent motion or step-by-step seam welding, the work and the electrode wheels are stationary while the current flows to make a weld. The current is then switched off before the rollers move the work to the next weld position.

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Roll spot welding is a modification of RSEW in that spot spacing is increased to the point where a series of individual welds are made without overlapping. The resultant weldment is no longer leak-tight. A schematic representation of the two types of resistance seam welding method can be found in Figure 99.

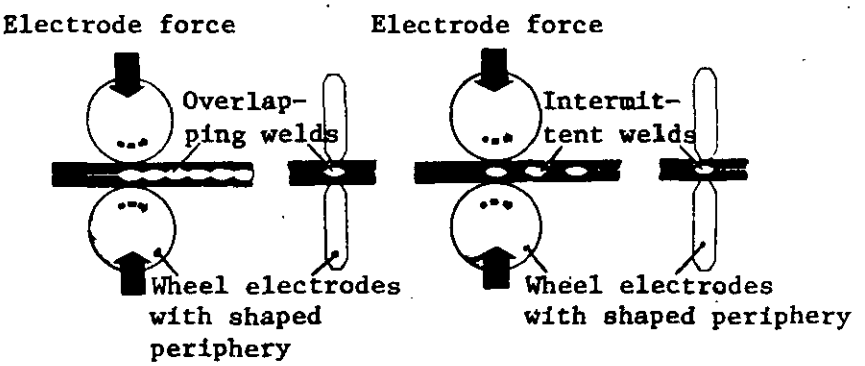


FIGURE 99. Overlapping spot and roll spot welding processes.

Since internal cooling of the roller electrodes is not possible, the electrodes and the work are flooded with water externally. Shunting of the welding current takes place through the previous welds so that higher currents are necessary for resistance seam welding than for spot welding.

The size of the weld is determined by the contact area between the rollers and the work. The power source is similar in design to that used for spot welding.

A simple lap joint is the most common type of joint in resistance seam welding. Other forms of resistance seam welding inclue flange-joint lap seam welding, mash-seam welding, butt-seam welding and foil butt-seam welding. If the overlap in seam welding is decreased to no more than 1.5 times the base metal thickness, the overlap can be "mashed" down by applying a high electrode force and a high continuous welding current as the work moves through the rollers. Low-carbon steels of up to 1.5 mm (0.06 in) thick can be mash-seam welded satisfactorily.

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Butt-seam welding involves heating and forging together under a slight pressure two abutting edges. High-quality butt-seam welds can be achieved by resistance heating the edges of tubing roll-formed continuously from strip stock.(13) The diagram of this process is displayed in Figure 100.

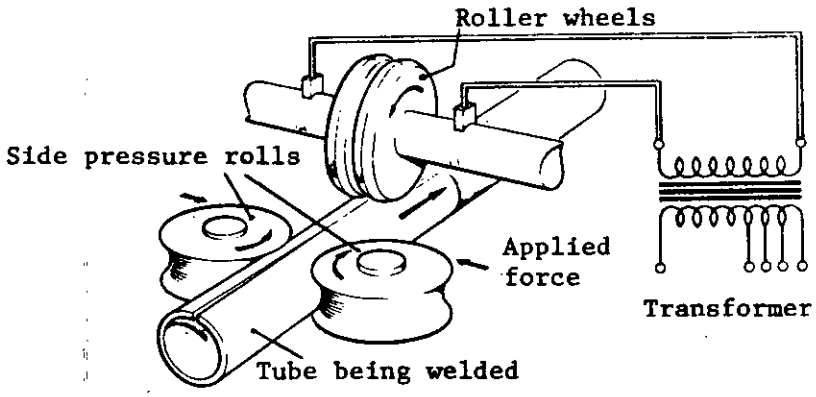


FIGURE 100. Resistance butt-seam welding of tubing made from strip stock.

The welded seam has a small upset on each side of the joint, which is removed by scarfing tools.

Figure 101 illustrates the cross-sections of a foil butt-seam weld.

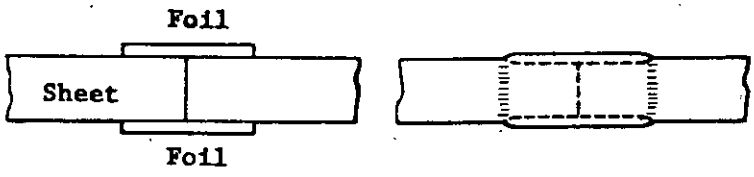


FIGURE 101 Cross-section of foil butt-seam weld (a) before, and (b) after welding.

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A thin and narrow foil of about 0.2 mm (0.01 in) thick and 3.2 mm (0.13 in) wide is applied between the seam welding rollers and the work, which has a close butt joint. The workpieces and the foil are fused together to give a weld slightly thicker than the base metal.

The high-frequency resistance welding makes use of ac power sources of frequencies of up to 450 kHz and voltages of up to 100 V. The current is introduced through two probes on either side of the joint as shown in Figure 102.

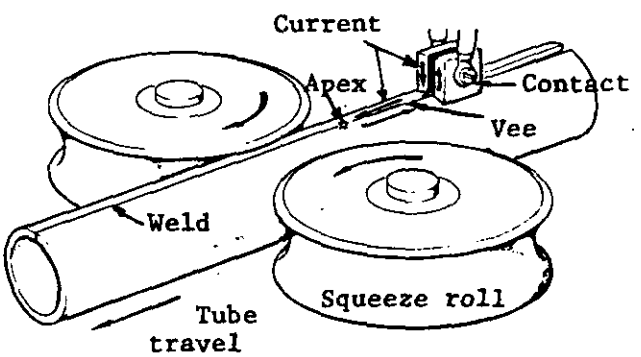


FIGURE 102. High-frequency resistance butt welding of tubing.

The contact with the probes is made a short distance before the heated edges of the joint are pressed together. The current path between probes lies alongthe edges of the seam through the apex of the Vee that is formed as the base metal is formed into a tubing.

Superficial melting may occur during welding, which is squeezed out as the surfaces of the joint meet. The melting on the surface makes oxide film removal easier thereby enabling the welding of non-ferrous metals and others which readily form refractory oxide films on their surfaces. Metals of this type are most difficult to weld by low-frequency resistance welding processes.

High-frequency induction welding, like high-frequency resistance welding, makes use of the skin effect by which the welding current flows on the surface of the work. The difference between the two processes is that the induction coil wrapped around the work does not make a direct contact with the formed tubing. The joint is forged together after induction heating.

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12.5 Projection Welding

Projection welding is closely related to resistance spot welding except for the means of achieving localization of current flow. Current concentration is accomplished by shaping the base metal so that when the two pieces to be welded are pressed together between flat-nosed electrodes or platens, the welding current passes through limited points of contact.(13) The process is shown schematically in Figure 103.

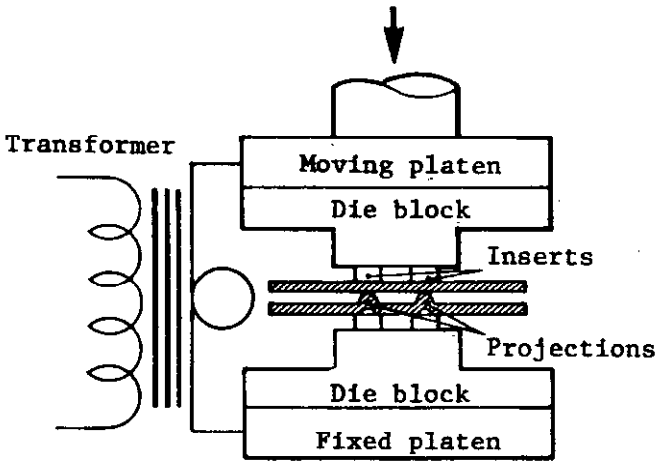


FIGURE 103. Basic principle of projection welding.

In sheet metal work, the projection of the desired shape and size is performed in one of the two pieces overlapped. Localized heating and collapse of the projection occur as a result of current flow and electrode force applied.

The use of inserts extends the working life of the flat-faced electrodes, platens or dies. Single or multiple projection welds can be made at one time. The electrode diameter and contact resistance are not as important in projection welding as in RSW. The formation of a projection weld is governed by the geometry of the projection, the mechanical and thermal properties of the base metal, welding current, electrode force and time of current flow. All metals except for those which cannot support the projection are suitable for this joining process. Dissimilar metals and thicknesses can also be projection welded. Allowance must be made for the differences in properties and thicknesses. This may be accomplished by forming the projection in the metal of higher conductivity or by using a low-conductivity electrode in contact with the higher-conductivity base metal. Surface preparation requirements are just as important in this process as in resistance spot welding. The five basic projections include spherical, pyramidal, annular, elongated and cross-wire.

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12.6 Flash Welding

Flash welding is a resistance butt welding process. The workpieces are abutted and heated by the contact resistance to the flow of current and subsequently coalesced by pressure. Flashing and upsetting are characteristic features of this process. Current flows through descrete points of contact on the abutted surfaces initially. This causes localized melting. When these molten bridges of contact are broken by the applied pressure, flashing occurs. Upsetting is a result of metal expulsion from the joint. The joint surface of one of the workpieces may be raised in the center slightly to facilitate the initiation of flashing. A schematic representation of flash welding can be seem in Figure 104(8)

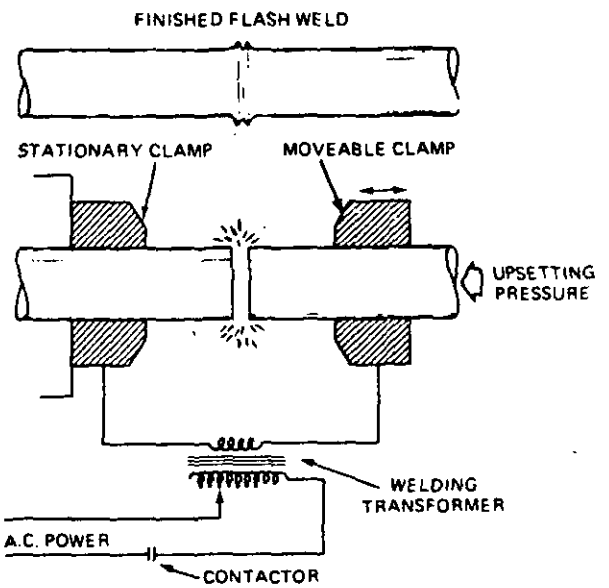


FIGURE 104. Main features of flash welding.

One of the two workpieces is held stationary while the other is gripped in a moveable clamp. Upon heating the faying surfaces of the base metal above the melting point, the two pieces are upset by a pressure sufficient to cause coalescence. Fixtures either in place of or in addition to clamping dies are used to support and align the workpieces that are large in size and somewhat complicated in shape. The process is most suitable for joining rods and bar stock.

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If the welding conditions of current, time and pressure are not optimized for the workpieces involved, an unsatisfactory weld will result.(1) The salient features of acceptable and unacceptable flash welds are described in Figure 105.

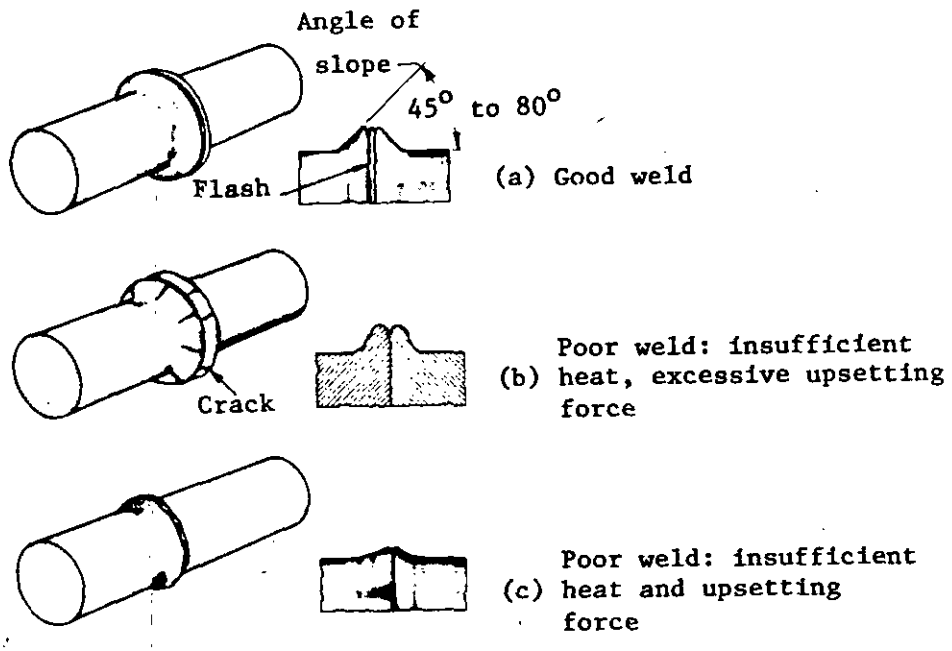


FIGURE 105. Features of acceptable and unacceptable flash welds.

Preheating of the base metal, if required, is done by resistance heating. It is done by bringing the two pieces together and applying a higher current and pressure than necessary for flashing. The pressure is adjusted so that the contact area is larger than that for sparking but less than what is required for upsetting. Preheating offers several advantages, namely, heat balance between dissimilar masses of workpieces, reduction in power demand, decrease in metal expulsion and shorter weld time.

Proper alignment of the workpieces is always important to obtain good weld results. With thin-wall tubing and thin sheets with miter joints especially, proper alignment is much more critical and correspondingly more important in order to produce a satisfactory weld quality. The microstructure of the flash weld varies with the composition of the base metal. A hardness traverse across the weld shows little change in mild steel. There is a pronounced difference in hardness and microstructure across flash-welded joints in alloy and higher carbon steels.(5) Figure 106 illustrates examples to this effect.

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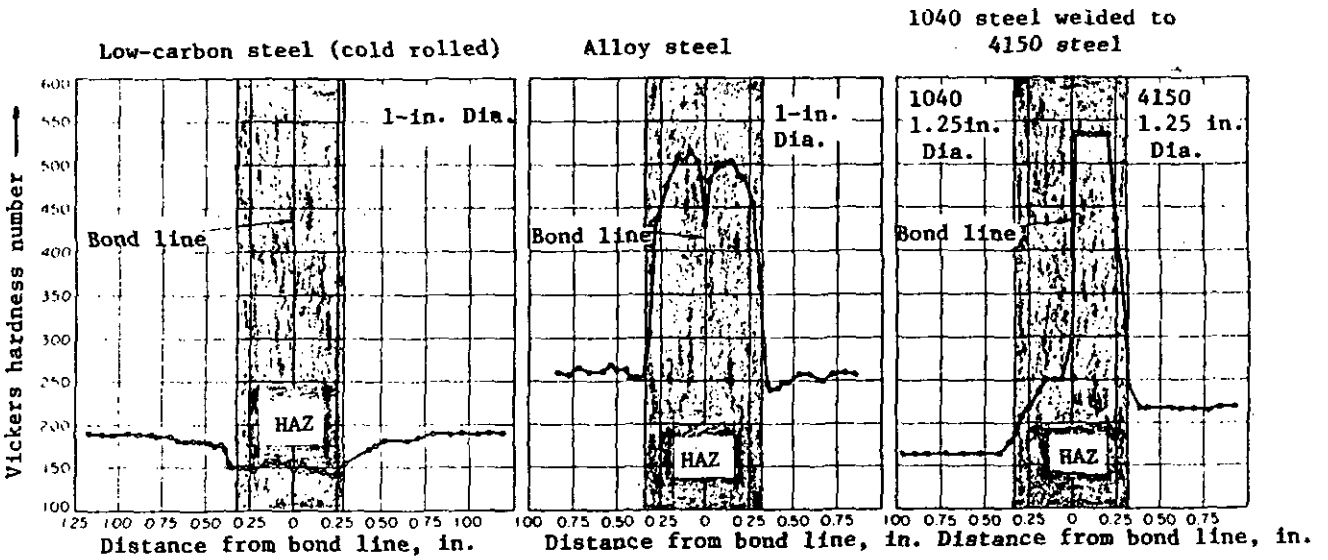


FIGURE 106. Hardness traverse across flash-welded joints in steels indicated.

12.7 Percussion Welding

In percussion welding, a short-time high-intensity arc is produced by a sudden discharge of electrical energy stored in capacitors. Pressure is applied percussively either during or immediately after the electrical discharge. A thin layer of molten metal on the contact surfaces of the workpieces is formed by the intense heat input. One of the two pieces, usually the smaller of the two, is impacted into the surface of the larger component. The arc is extinguished and the molten metal is expelled as a result of the percussion. The percussive force can be applied by electromagnetic, electromechanical devices and other means. Not only a bank of capacitors but a welding transformer may be used to supply electrical energy.

The primary application of percussion welding is in the area of joining small diameter wires to large metallic surfaces.

The short arc time produces a very shallow heat-affected zone. The welding force needed to make a good weld is determined empirically and in conjunction with the welding current. Because there is a possibility of springback of the component which is impacted into the surface of the other workpiece, a method of damping must be provided.

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In capacitor-discharge percussion welding, a direct current with either a low or a high voltage is supplied by a capacitor or a bank of capacitors.

In low-voltage welding, arc starting is done with either a nib or a high-frequency pulse of alternating current. A nib is a projection on the end of one of the base metals serving as an arc initiator. Auxiliary arc initiation is not necessary in high-voltage welding, because the direct current voltage is high enough to overcome the resistance of the air in the joint. The high voltage is capable of ionizing the air and thus provide a path for the passage of current. The high-frequency alternating current superimposed on the low-voltage direct current ionizes the air and the arc is then maintained by the low-voltage direct current. The third method of arc initiation consists of making a starter nib on the end of one of the workpieces to increase the current density high enough to melt the nib. The molten particles ejected from the nib help form the arc in the joint.

In magnetic-force percussion, a separate transformer is employed for developing and controlling the welding force of an electromagnet. The welding current at a voltage of about 10-35 volts is provided by a resistance welding transformer. The system is similar to a low-voltage direct current percussion welding process without a supplementary device for energy storage.

Typical applications of magnetic-force percussion welding include copper to copper, copper to silver-tungsten, silver-cadmium oxide to brass, tungsten tips to copper shanks and other dissimilar metals.

12.8 Upset Welding

Upset welding is a modified version of flash welding in that sufficient pressure is applied throughout the welding cycle not just after the termination of current flow as is the case in flash welding. There is no arc of flashing in upset welding. Figure 107 shows the salient features of upset welding.(8)

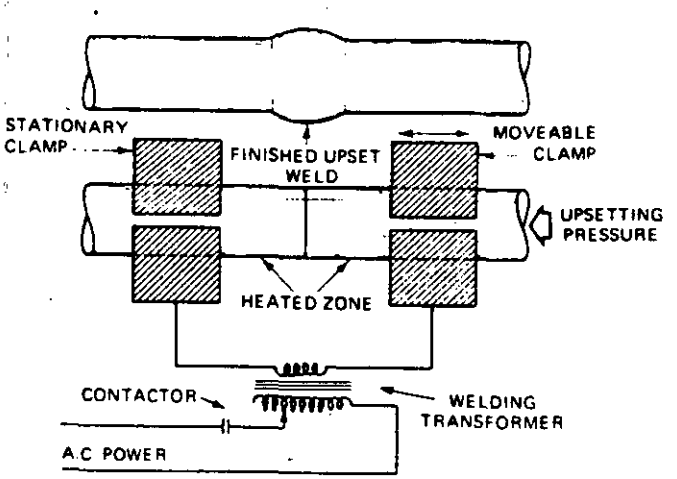


FIGURE 107. Features of upset welding.

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The welding current flows through the abutted joint surfaces and the heat for melting is obtained from the contact resistance to the electric current flowing for a predetermined time. The greatest application of upset welding is in butt welding ends of wires to make continuous wire drawing possible. Metals that are most commonly upset welded include carbon and stainless steels of different grades, aluminum-base, copper-base, nickel-base alloys and refractory metals.

The sequence of operation is as follows:

a. Clamp workpieces
b. Apply welding force
c. Apply welding current
d. Apply upset force
e. Stop current flow
f. Release pressure
g. Remove workpiece.

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13. THERMIT WELDING

A number of powdered metallic oxides such as iron oxide (Fe_2O_3 or Fe_3O_4) can be reduced with metallic aluminum (Al) powder according to the following chemical reaction:

$$(1)\quad Fe_2O_3 + 2Al \longrightarrow Al_2O_3 + 2Fe + heat$$

$$(2)\quad 3Fe_3O_4 + 8Al \longrightarrow 4Al_2O_3 + 9Fe + heat$$

These are exothermic reactions. The heat of these reactions are used for welding.(8) The theoretical temperature generated by the reducing reaction of iron oxide with pure aluminum is nearly 3090°C (5600°F). The ferric oxide (Fe_2O_3) reacted with aluminum produces a maximum temperature of 2450°C (4440°F), and heat of about 850 kJ, while the ferrous oxide (Fe_3O_4) and aluminum reaction liberates about 3350 kJ (800 kcal) heat. Considering heat losses, the reaction (2) generates a pouring temperature of approximately 2480°C (4500°F). The excess heat in the reaction results in superheated iron. The superheated steel is held in a crucible located above the weld joint. Other materials are usually added to the thermit charge for purposes of controlling the temperature of the reaction, the chemical analysis and the mechanical properties of the weld produced. Both the quality and the properties of the weld can be modified by special additions to the charge, namely scrap steel or alloys.

In order to ignite the thermit charge, the temperature of 1300°C (2370°F) has to be exceeded. The reaction is set off with a special mixture and an ignitor. The actual reaction lasts between 30 - 60 seconds regardless of the size of the charge.

The superheated steel is poured into a mold, which is built around the joint to be thermit welded. The end sections of the workpieces and the interior of the mold are preheated by a torch through a preheating gate to a red heat prior to letting the liquid steel from the thermit reaction enter the mold. Thermit welding is to a large extent analogous to the foundry practice of pouring a casting. The most important advantage of the process is the low residual stress developed in the weldment as a result of an inherent slow cooling rate.

Thermit welding is most typically used for joining rails. The chemical composition of thermit welded railroad joints is about C = 0.6 wt%, Mn = 1.4 wt %, Si = 0.13 wt%, Al = 0.4 wt% and Ti = 0.04 wt%.

A typical analysis of thermit utilized to weld mild and medium carbon steels is given below in weight percent.

C = 0.20 - 0.30
Mn = 0.50 - 0.60
Si = 0.25 - 0.50
Al = 0.07 - 0.18
Fe = remainder

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During the exothermic reaction, the molten steel settles to the bottom of the crucible. The aluminum oxide rises to the top in the form of slag and protects the liquid weld metal from atmospheric contamination. The liquid steel is tapped and flows into the mold through the pouring gate and in-gate into the cavity thereby making the weld. Complete filling of the cavity is ensured through providing gates, risers and vents similar to foundry practice.

13.1 Process

The workpieces are prepared with a square groove joint. While the root opening varies according to the cross-section of the base metal, for railroad rails, as an example, it ranges from 19 - 37 mm (3/4 to 1-1/2 in), as an example,

An empirical relation between the joint gap and the cross-sectional area has been developed and is as follows;

$$\text{Gap} = \sqrt[3]{A/0.14} \quad \text{(mm)}$$

or

$$\text{Gap} = \sqrt[3]{A/3.6} \quad \text{(in)}$$

where,

A = cross-sectional area of the base metal (mm or in)

The parts to be joined are cut off square and spaced apart to the required gap. The joint surfaces are cleaned of all materials such as scale, grease, oil, moisture, etc., which would cause an unsound weld. The parts are then properly aligned, braced and the mold made around the butt joint.

The mold cavity for complex shapes is formed by the lost wax technique. The weld joint is filled with wax prior to making the mold. The wax is melted out after the mold has been rammed. A conical crucible is positioned over the mold. The thermit is ignited in the crucible. The slag floats on top of the molten metal. The liquid weld metal is tapped into the mold to complete the weld. The thermit welding process and the details of the crucible and the mold are shown in Figure 13.1-13.2.

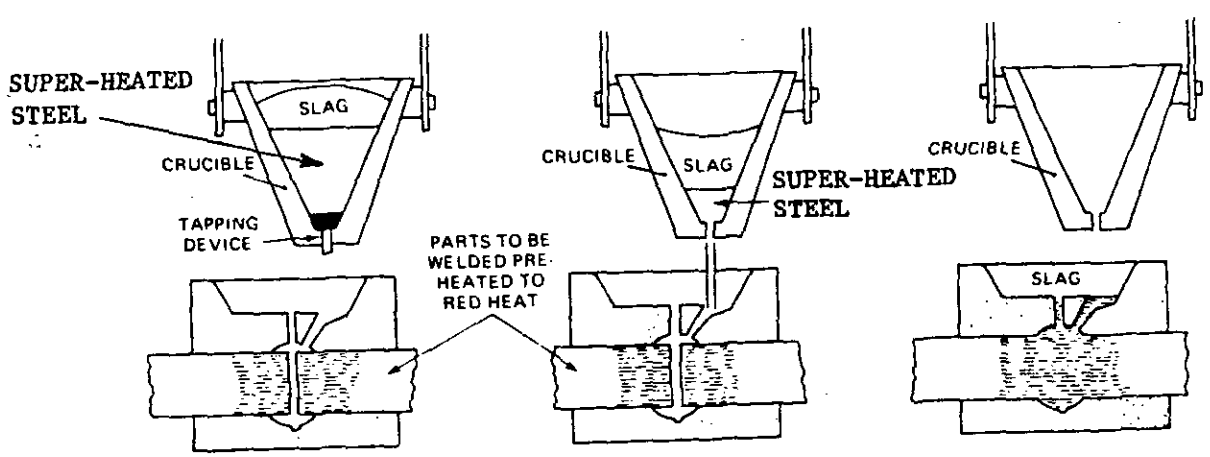


FIGURE 108. Steps in making a thermit weld.

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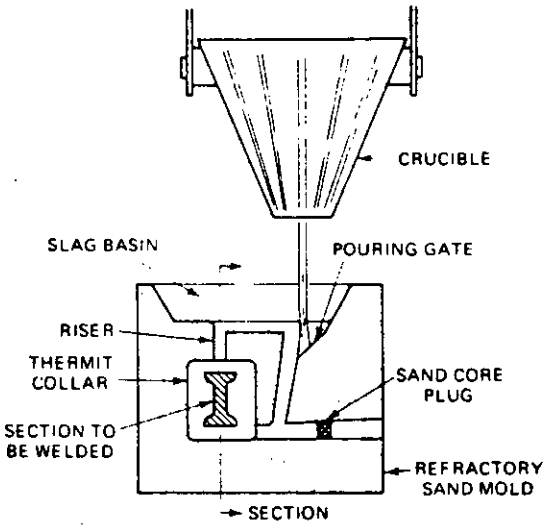


FIGURE 109. Details of the crucible and mold.

After solidification and cooling of the weld, the mold is broken away and discarded. The gates and risers are cut off by an oxyacetylene flame. The surface and the contour of the weld are such that no additional finishing work is required.

Permanent molds are used for repetitive jobs such as joining rails or reinforcing bars. These molds are made in halves that are clamped around the ends of the rails or bars to be thermit welded. Thermit welding has been used for many special applications involving large cross-sections, crank shafts, I-beams, stern frames for ships, craneway rails and for making repairs to large structures. Thermit welding is not limited to ferrous metals. Copper and aluminum cables are joined together by using a non-ferrous aluminothermic mixture. The reduction of copper oxide by aluminum produces a superheated liquid copper. The molten copper flows into the mold and fuses the ends of the fitted cables together. Tapping is automatic because the charge develops a high enough temperature to melt the consumable steel disc which replaces the tapping pin. (See Figure 110) The mold for thermit welding electrical cables can be made out of graphite, and can be used repeatedly.

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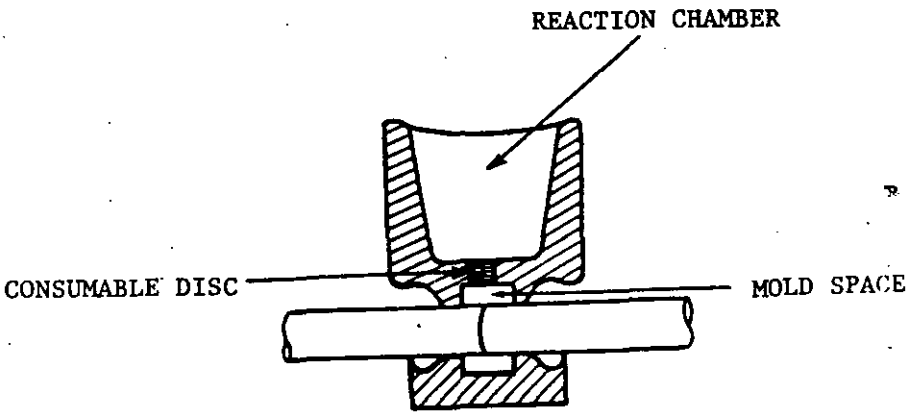


FIGURE 110. Permanent mold for repetitive thermit welding.

Thermit welding is a simple and inexpensive metal joining process.

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14. ELECTRON-BEAM WELDING

Electron-beam welding (EBW) is a welding process which produces coalescence of metals with the heat obtained from a focused beam of high velocity electrons impinging on the surfaces to be joined. The essential features of EB welding equipment are the electron gun, the beam control system and the welding chamber. The electron gun producing the electron beam is composed of a cathode, a grid and an anode. The cathode, made of tungsten or tantalum, is heated to above 2540°C (4600°F) to emit electrons. The control system gathers, accelerates and focuses the electrons. An example of an EB welding machine is exhibited in Figure 111.

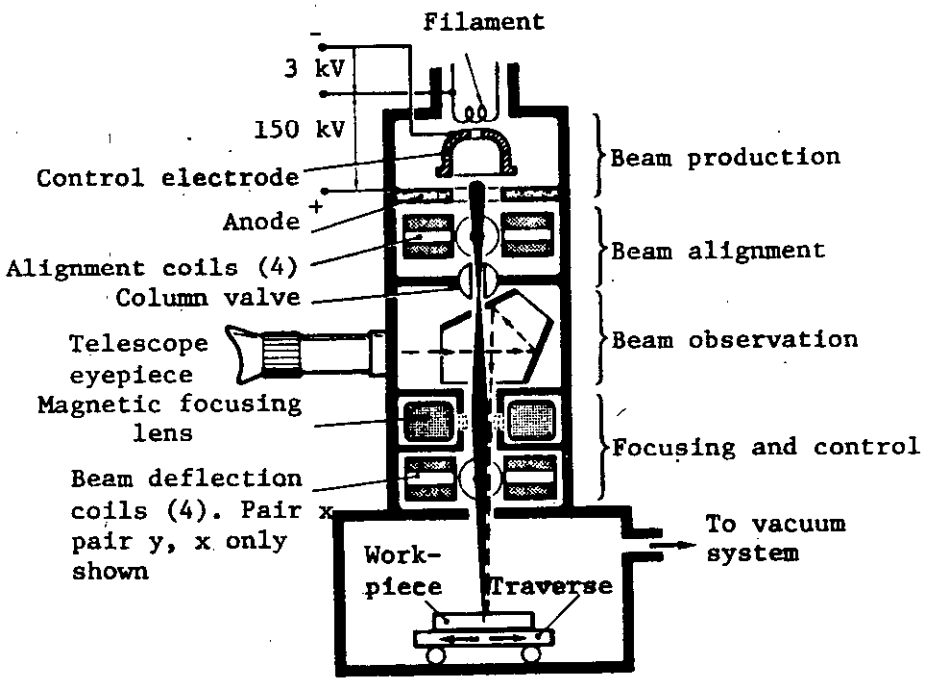


FIGURE 111. High-voltage triode EB welding machine with co-axial optical viewing.

The gun is usually isolated from the welding chamber and held at a high vacuum (10^{-5} torr or less) to prevent oxidation of the cathode and to avoid short-circuiting by metal vapors emanating from the welding chamber.

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The focused beam diameter may range from 0.3 - 0.8 mm (0.01 - 0.03 in) and contains about 10^6 watts of power per unit area, which is sufficent to vaporize any metal. The beam power is a function of the kinetic energy of all the electrons produced in unit time. The kinetic energy, E_k, is expressed by the formula

$$E_k = 1/2\ mv^2$$

where,

v = velocity
m = mass of a body

The velocity, however, is proportional to the square root of the accelerating potential. Therefore, the energy per electron is directly related to the accelerating voltage. The equation for power density, P, is given as follows,(3)

$$P = \frac{neE}{A} = \frac{EI}{A}$$

where,
n = total number of electrons present in the beam per second
e = electronic charge in coulombs (1.602×10^{-19})
E = accelerating voltage on the gun in volts
I = beam current in amperes
A = area of the focused beam on the workpiece.

The depth of joint penetration is dependent on beam current, accelerating voltage and welding speed. Energy concentration is voltage dependent. EB welding is customarily used in a voltage range of 20 - 150 kV. The primary advantage of EBW is a high ratio of depth to width (as much as 30:1), which makes the process suitable for welding thick cross-sections. The concentrated heat input produces little distortion and a narrow heat-affected zone. Other attributes of EBW include no flux or shielding gas requirements, welding speeds as high as 200 mm/s (500 in/min) and clean working environments. Since the beam can be projected over a long distance, welds can be made in inaccessible working places.

The essential variables of EBW are accelerating voltage, beam current, beam focusing, travel speed and gun-to-work distance.

A comparison of weld penetration obtained by GMAW and EBW is shown in Figure 112.

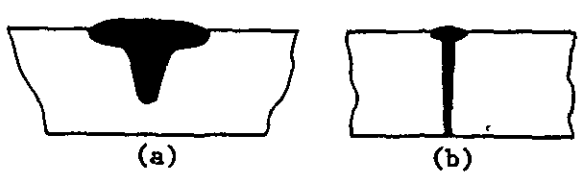


FIGURE 112. Schematic representation of weld penetration profile obtained by (a) GMAW and (b) EBW.

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The high power density vaporizes the metal which in turn creates a cavity or keyhole within the finger of the penetration similar to plasma-arc welding. In EBW, the high-intensity beam first makes a crater or hole in the base metal. As the beam moves along the joint, the metal on the leading side of the hole vaporizes, then, condenses and flows to the trailing side. The molten metal is held in place by surface tension and vapor pressure. The metal solidifies as the beam advances. The keyhole is the most likely site for weld discontinuities to occur.

Joint preparation for EBW involves closely abutted square groove faces so that the beam fuses the base metal on both sides of the joint without a filler metal. Therefore, a good fit-up is extremely important to avoid incomplete fusion, porosity, underfill and undercut. Surface cleaning must be carefully implemented since contamination will increase pumpdown time and can cause weld defects. Fixturing is not as important in EBW as in conventional electric arc welding processes because of the lower and more concentrated heat input. There are three variations to EBW Welding: high vacuum operating at a maximum pressure of 10^{-4} torr (0.013 Pa) or less, soft vacuum working at 10^{-1} torr (13.0 Pa) and nonvacuum or out-of-chamber (1 atm).

14.1 High Vacuum EBW

Scattering of the beam is practically negligible in a high vacuum chamber [10^{-4} torr (0.013 Pa) or less]. Consequently, the electron beam can be held in a sharp focus over a working distance of several centimeters (several feet). Better weld quality, deeper penetration, narrower weld and HAZ, and broader range of metals and thicknesses to be welded are the attributes of high vacuum electron-beam welding. Limitations of this process are the size of the workpiece that can be placed in the vacuum chamber, and the high cost of the hardware needed for the process. Because of the high vacuum, auxiliary shielding of the weld pool is not needed to prevent contamination. The high vacuum has a degassing influence on the weld pool as well. For these reasons, high vacuum EBW is well suited for the joining of metals that have a great affinity for oxygen, such as titanium.

Refractory metals can be electron beam welded successfully. However, attention has to be paid to vaporization losses of alloying elements that are present in refractory metal alloys. Most of the refractory metals and their alloys are quite brittle and are therefore sensitive to thermal shock. A decrease in travel speed helps to avoid this problem.

Electron beam welding is characterized by high cooling rates, which is an added benefit in joining austenitic stainless steels. The high cooling rate through the sensitizing-temperature range inhibits carbide precipitation.

During EBW, the workpieces are not exposed to high temperatures very long due to the high travel speed inherent in the process. Consequently, the base metal properties are hardly affected. This is of great value in joining alloys which obtain their properties through heat-treatment, or work-hardening.

14.2 Soft Vacuum EBW

The soft vacuum or medium vacuum process does not require as elaborate pumping system as does the high vacuum electron-beam welding process. A

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simple mechanical pump is sufficient to achieve the typical working pressure of about 10^{-1} torr (13.0 Pa).

Pumpdown time is relatively short. This process is therefore suitable for large production runs of workpieces that are simple in shape and small in size. The vacuum pressure has an effect on weld penetration and geometry as illustrated in Figure 113.

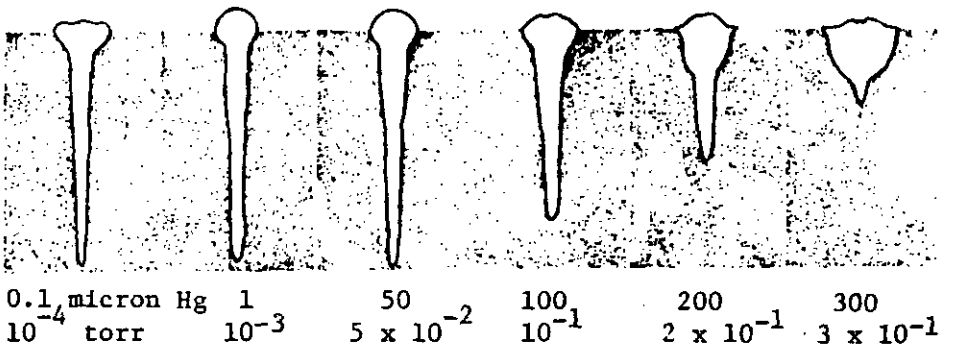


FIGURE 113. Effect of vacuum pressure on weld penetration and profile obtained in 304 stainless steel.

The welding conditions pertaining to Figure 14.3 involved 40.6 cm (16 in) beam length without changing focus, beam voltage of 150 kV, beam current of 30 mA and welding speed of 152 cm/min (60 ipm).(1)

Contamination level in soft vacuum may not be satisfactory for highly reactive metals. The weld penetration and the depth-to-width ratios are less in medium vacuum than in high vacuum EBW. The application of vacuum chambers permits beam deflection and oscillation. EBW has the capability of reaching into deep and narrow openings due to the small diameter and long working distance of the focused electron beam. The beam can also be deflected or projected at an angle desired to reach areas which otherwise would be inaccessible.

The weld area has to be in line-of-sight from the beam source. The joint has to have adequate clearance to prevent beam interference. The beam path must also be devoid of magnetic fields.

The principal purpose of beam oscillation is to weld joints with large root openings. Beam oscillation also decreases the cooling rate because of

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the larger weld pool produced and makes beam alignment as well as joint tracking less exacting. Joint penetration and general weld metal behavior can be controlled not only by oscillation but by beam pulsation, travel speed modification and beam deflection. Pulsed beams result in lower peak temperatures and are therefore applicable for repair procedures.

14.3 Nonvacuum EBW

This process is often referred to as inert atmosphere electron-beam welding. Scattering of the beam is significant due to collision of the electrons with air molecules. For this reason, the power loss is substantial and the working distance is markedly less than in vacuum EBW. Reactive metals cannot be welded satisfactorily with nonvacuum EBW.

A schematic diagram of nonvacuum or out-of-chamber EBW process can be found in Figure 114.

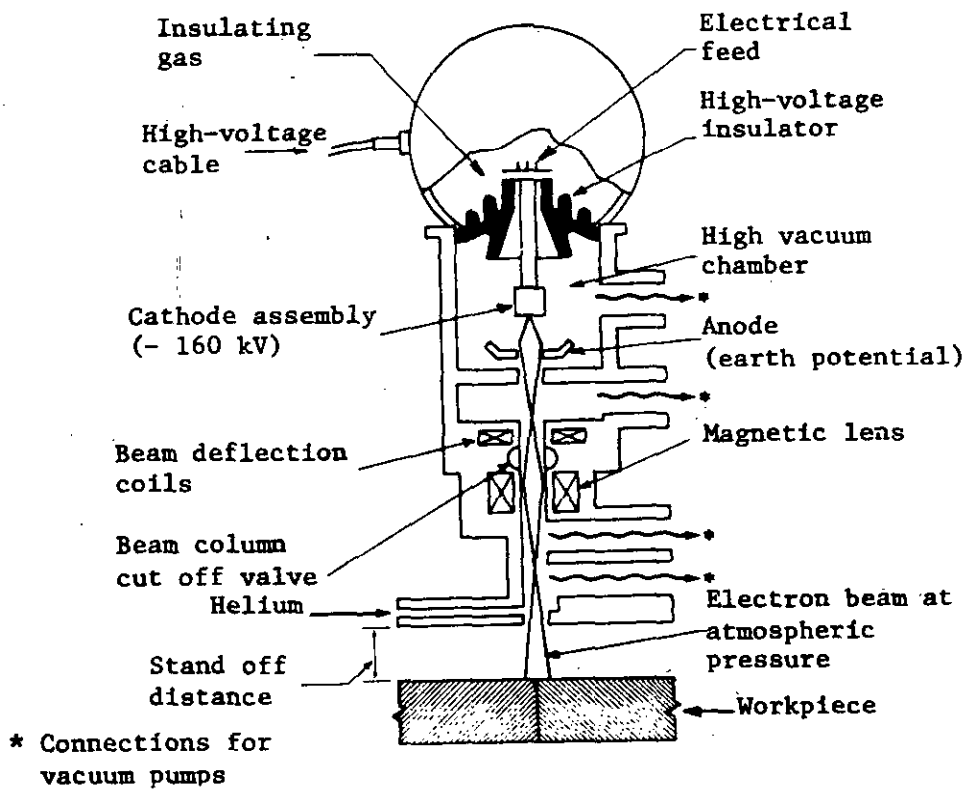


FIGURE 114. Features of nonvacuum electron-beam welding process.

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The electron beam is generated in a vacuum chamber just as in high-or soft-vacuum EBW processes. However, the actual welding is carried out at atmospheric pressure. The beam is transmitted via a transfer column, which consists of a series of chambers with progressively higher pressures before the beam is finally emitted into the atmosphere. Obvious advantages of out-of-chamber welding are the higher production rates possible due to the elimination of pumpdown time and the lower investment cost. While there is no limitation on the overall size of the base metal because of the absence of a vacuum chamber for the work, nonvacuum electron-beam welding is limited in terms of base metal thickness, depth of weld penetration, working distance and shape of weld profile.

The effect of variation in working distance on weld profile in 6.4 mm (0.25 in) thick aluminum alloy 2219 is shown in Figure 115.

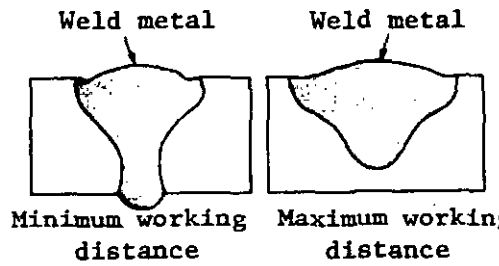


FIGURE 115. Effect of changing the working distance on weld profile.

Inert gas shielding of the molten weld metal must be provided as well as lead shielding of the work area from the x-rays that are generated when the beam strikes the workpiece.

Since beam dispersion is substantial, the maintenance of sufficient electron velocity necessitates a reduction in the working distance and an increase in accelerating voltage. Typical values include 6.4 - 19.0 mm (0.25 - 0.75 in) for gun-to-work distance and 100 - 175 kV for beam voltage. The influence of welding speed variation on penetration in aluminum and steel is plotted in Figure 116.

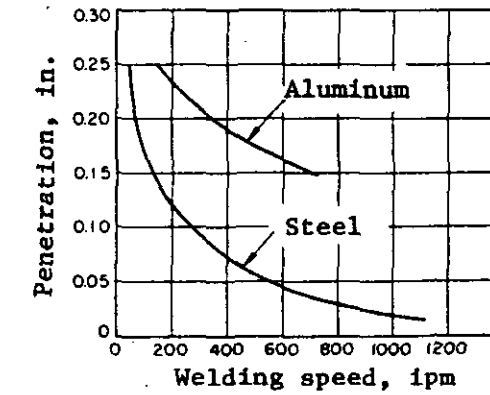


FIGURE 116. Influence of welding speed on weld penetration.

The corresponding beam power was 9 kW using helium shielding gas for weld metal protection.

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The bulkiness of the welding head and the short working distance may to some extent be overcome by the application of specially designed nozzles to reach into restricted areas of complex shapes. Normally, the EBW process involves no filler metal because of the close butt joints used. However, when dissimilar metals are electron beam welded, filler metals may have to be added to compensate for the compositional difference between the workpieces in order to produce acceptable weld properties and qualities. The use of filler metal may also be required in case the weld has to be built up for reasons such as cross-sectional changes, large gaps, or extra stock needed. Deoxidizing filler metals are added to the joint to eliminate severe porosity in electron-beam welding of rimmed steel and certain copper-base alloys. Filler metals are available in the form of a thin foil, or wire.

Multiple-tier welding, which entails the welding of two or more stacks of joints simultaneously, can be carried out with EBW process provided the air space between joints can be evacuated.(13) The joints of each tier must be carefully aligned with the electron beam so that the metal flows in behind the traversing beam as in keyhole welding. Examples of joints suitable for electron-beam welding can be seen in Figure 117.

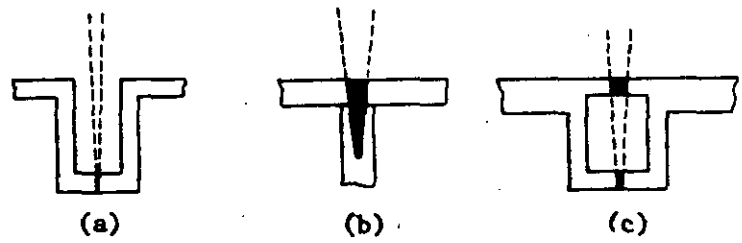


FIGURE 117. Types of EBW joints: (a) Welding in a recess, (b) Spike welding, (c) Multiple-tier welding.

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15. LASER-BEAM WELDING

Light amplification by stimulated emission of radiation, or laser for short, can be used for welding or cutting of materials. Laser-beam welding (LBW) works on a principle similar to EBW in that a focussed, high intensity monochromatic light beam vaporizes the metal at the point where it strikes the surface. In doing so, the beam penetrates the base metal through a column of vapor which is surrounded by liquid metal. The depth-to-width ratio of laser-beam welds is generally four to one.

The beam diameter emitted from a laser source ranges between 1-10 mm (0.04 - 0.4 in), which has to be focussed for usefulness in welding and cutting.

In general there are two types of lasers: Ruby, Nd-Glass (neodymium doped glass) and Nd-YAG (neodymium doped yttrium-aluminum-garnet) constitute the former group, whereas the CO_2 laser is the most useful gas laser.

Ruby and Nd-Glass emit energy in intermittent pulses. The operating principle of these lasers involves a rod of ruby 10 mm (0.4 in) diameter and about 100 mm (4.0 in) long having accurately polished ends. One end is fully silver coated, while the other end is only half silver coated. Usually, the active medium in the ruby crystal is chromium ion with a concentration level of about 0.05%.

The crystal is irradiated by the light from a powerful xenon flash tube. The chromium ions emit radiation which travels to and fro between the ends of the crystal. As the intensity of radiation builds up, the laser beam leaves from the half-silvered end. The input energy has to exceed a certain threshold value in order to induce laser beam emission of sufficient energy and frequency for welding or cutting. The active ions and the input energy determine the amount of output energy and pulse frequency. The intermittent nature of these types of lasers is due to the discontinuous discharging of a band of capacitors through the xenon flash tube and the necessary, periodic cooling of the crystal in which heat builds up during the process. Owing to the low pulsing characteristics of these lasers, they are usually used for cutting and drilling holes. The pulsed energy of a laser beam striking a metallic surface is absorbed in a localized area. The depth to which this energy is felt depends on the thermal diffusivity of the base metal. For effective melting to occur, the pulse duration must be long enough to compensate for thermal conduction. Welding requires a continuous heat input or a much higher frequency of pulsation than 1-50 pulses per second, which is normal for ruby and Nd-glass lasers.

In contrast, the CO_2 and Nd-YAG lasers have capacities for generating high enough energy continuously or in pulses at rates as high as several thousand pulses per second. The most powerful CO_2 laser is capable of producing power densities of $10 kW/mm^2$ ($6.45 MW/in^2$) or greater, which can make welds with deep penetration in 13 mm (0.5 in) thick steel at a travel speed of 25 mm/s (60 in/min).

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A CO_2 laser comprises a glass tube with two electrodes at each end through which a lasing medium or a gas mixture (1:1:10) of carbon dioxide, nitrogen and helium flows. The gas pressure ranges from 20 - 50 torr. A high-voltage discharge (10-30,000 volts) is set up between the electrodes. A fully reflecting water-cooled, gold-plated mirror is placed at one end of the tube. At the other end of the glass tube, a partially reflecting mirror is attached. The space between the two mirrors is called the laser cavity. Since most of the discharge energy goes to increasing the temperature of the lasing medium and the glass wall, the output of this laser is small. The glass tube must also be water-cooled. The gas can be wasted or recycled through a heat exchanger and reconditioning unit. This type of gas laser is referred to as a slow axial-flow laser with an output of up to 1 kW.[14] A schematic diagram of a slow-flow CO_2 laser is presented in Figure 118.

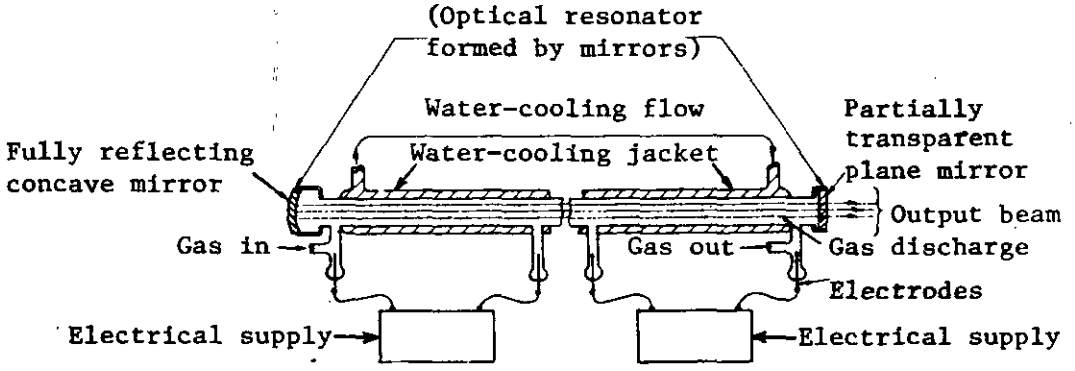


FIGURE 118. Diagram of a slow-flow CO_2 laser.

If the gas is forced through the laser tube at high speed by a blower in a direction either parallel or transverse to the optical axis, the heat losses to the wall is then negligible. There is then no need for water cooling. These types of high power lasers are called fast axial flow (FAF) and fast transverse flow (FTF), which are schematically illustrated in Figures 119 - 120.

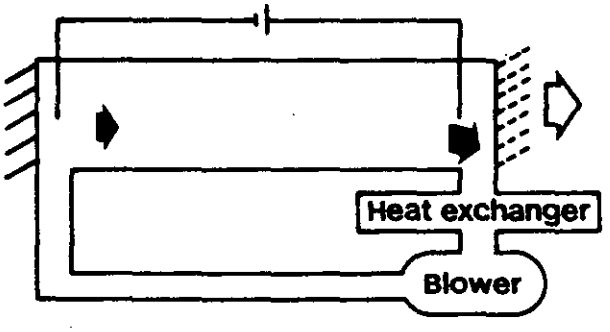


FIGURE 119. Schematic representation of FAF CO_2 laser system.

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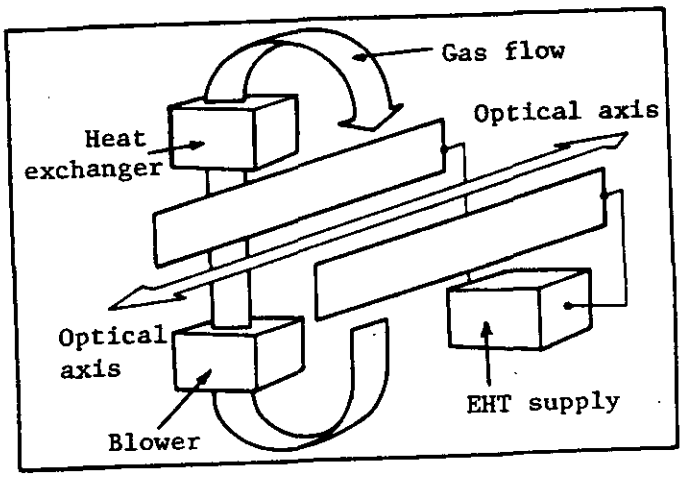


FIGURE 120. Highlight of FTF CO_2 laser system.

The main features of FAF CO_2 laser capable of delivering 2kW power are displayed in Figure 121.

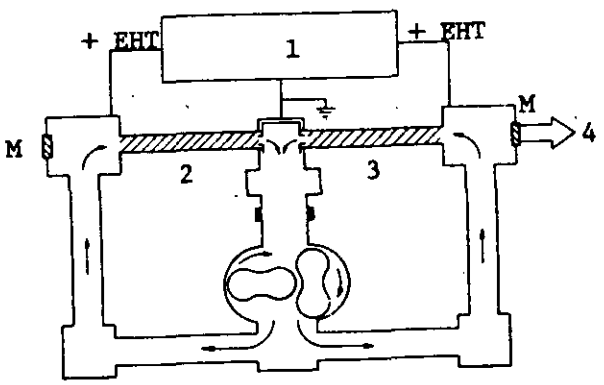


FIGURE 121. FAF CO_2 laser: (1) power supply, (2-3) laser tubes, (4) laser output, (M) lasermirrors. Arrows indicate gas flow. Shaded region is the discharge area.

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A heat exchanger is used to ensure that the lasing gas temperature does not exceed 200°C (390°F). The FTF CO_2 laser can generate an energy output of about 15 kW which is adequate to weld steel thickness up to 19 mm (0.75 in) in one pass. The FAF CO_2 laser with 2 kW power is sufficient to weld steel thickness up to 5 mm (0.2 in) in one pass. The beam of FAF CO_2 laser can be focussed to a smaller diameter than that of FTF CO_2 laser.

The focussed beam of a high power laser in butt joints of thick cross-sections produces a "keyhole" weld at relatively slow travel speeds.(13) (See Figure 122).

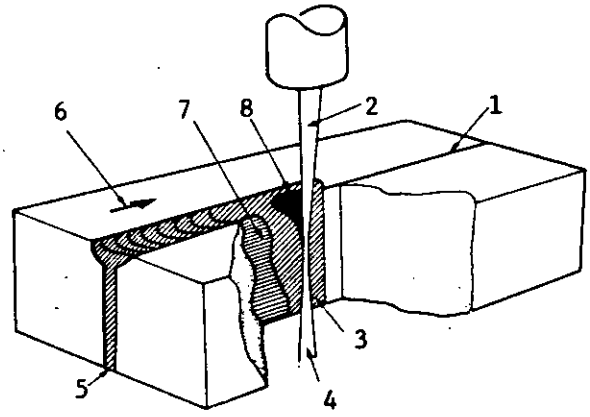


FIGURE 122. Perspective diagram of keyhole butt welding with laser. (1) close butt joint, (2) focussed beam, (3) molten metal, (4) energy lost through keyhole, (5) full penetration weld, (6) welding direction, (7) solidified weld, (8) cavity.

The factors that affect the depth of keyhole penetration include beam power, travel speed, size of the focussed spot, shielding gas employed and the thermal properties of the base metal.

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The depth of penetration as a function of beam power is plotted in Figure 123.

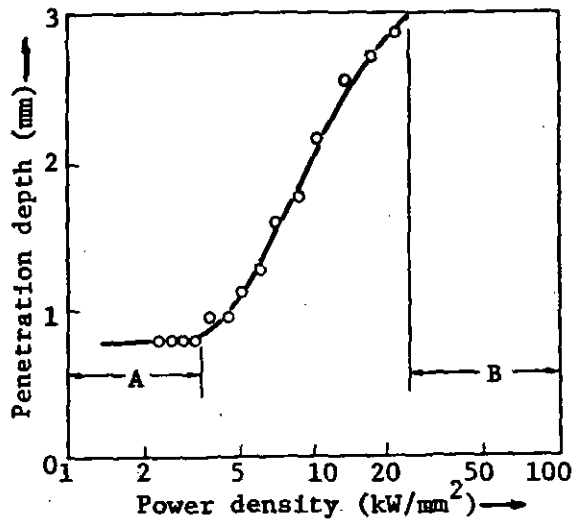


FIGURE 123. Plot of weld penetration versus power density for 3 mm (1/8 in) thick stainless steel. Power 1.75 kW, speed 10 mm/s.

Regions A and B in Figure 123 represent surface melting and full keyhole penetration.(13) During welding, a plasma is generated at the top of the keyhole as a result of interaction amongst the laser beam, shielding gas (usually helium) and metal vapor. This plasma serves as a secondary heat source and increases the amount of weld pool on the surface of the base metal. This gives rise to a decrease in weld penetration, a greater distortion, and a weld geometry shaped like a keyhole. The plasma can be controlled by blowing it away from the joint with a stream of inert gas or mixtures thereof (helium or helium-argon).

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16. SOLID-STATE WELDING

Solid-state welding (SSW) represents a group of welding processes in which coalascence of the base metals occurs below their respective melting points without the addition of a filler material. Pressure may or may not be used.

The application of pressure causes plastic deformation of the faying surfaces. The magnitude of the pressure varies according to the specific solid-state welding process used. The condition of the abutting surfaces has an important effect on the mechanical properties of the weld. Therefore, surface contaminants must be removed. Oxide layers on the faying surfaces can be dispersed by the applied pressure as is the case in forge welding.

Solid-state welding is customarily done without a control of the surrounding atmosphere at ambient and moderately low temperatures. When solid state welding is carried out at an elevated temperature, a reducing atmosphere or a vacuum might be utilized so as to help remove oxide films. Solid-state welding of reactive metals requires inert gas shielding.

Solid-state welding includes the following materials joining processes:

- cold welding
- diffusion welding
- explosion welding
- forge welding
- friction welding
- ultrasonic welding
- roll welding
- hot pressure welding
- electromagnetic impact welding

While all these processes have their unique characteristics in some respects, diffusion of atoms is an essential feature of all solid-state welding methods.

16.1 Cold Welding

Cold welding is accomplished by means of an applied pressure across the interface between metals held at room temperature. Ductile materials such as aluminum and copper are especially suitable for cold welding. The harder the base material and the higher the melting point, the less weldable the workpiece becomes. Cold welding involves either a lap or butt joint and requires a good surface preparation technique to produce a satisfactory weld.

In lap welding, the surface preparation consists of wire brushing usually. When butt joints are involved between bar stocks or tubes, the workpieces are only freshly cut or sheared prior to welding. The presence of thin oxide films particularly on aluminum is not necessarily an impediment to cold welding. The oxide film is broken up by the applied pressure. This results in a fresh metal surface to aid the welding process. When the workpieces are lapped, cold welding becomes essentially a spot welding methodology. The applied pressure causes a reduction in thickness at the point of indentation.

In butt welding, the workpieces are aligned and held securely in dies. The applied pressure forces the pieces together producing a flash around the joint. The success of cold welding depends in part upon the design of the tools and dies involved. For lap welding sheet metals, an indenter with a flat surface has been found satisfactory. As the indenter is forced into the workpieces, the sheets tend to separate outside the die. Consequently, the size of the weld is slightly smaller than the actual cross-section area of the indenter face.

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The strength of the welded joint is affected by the size of the weld area and the degree of indentation at the spot weld. It has been shown through experimentation that the size of the indenter should be between one and three times as wide as the individual sheet thickness and about six times as long. If the indentation is excessive, the joint strength is markedly decreased.

An increase in spot weld can be achieved through the use of double-acting dies. When a collar surrounding the indenter grips the sheets before the weld is made, separation and distortion of the workpieces are eliminated. Then, under a sufficiently high pressure, the welded area becomes larger than the size of the indenter through what is known as corona welding as illustrated in Figure 124.

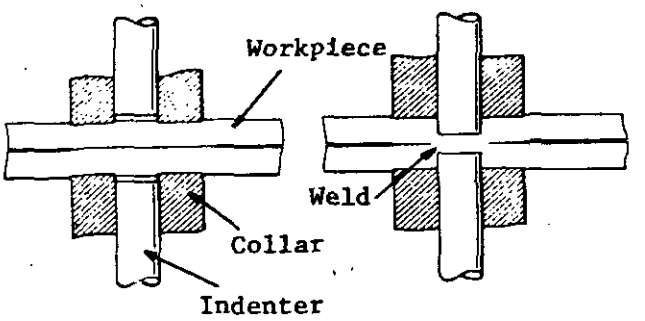


FIGURE 124. Schematic of double-acting die used in cold welding.
(a) collar clamped in position,
(b) applied pressure produces a corona weld.

16.1.1 Process

The method of surface preparation usually consists of wire brushing or degreasing to remove the oxide film, oil, grease, paint, finger print and other surface contaminants. Cold welding should take place as soon as possible after cleaning before the surfaces begin to oxidize again.

Cold welding of heavy sections requires powered presses to exert sufficient pressure. Cold welding of wires and spot welding of foils need only hand tools.

The alignment of the workpieces to be cold welded is done by dies. The dies for butt welding have either flat or conical faces as illustrated in Figure 125.

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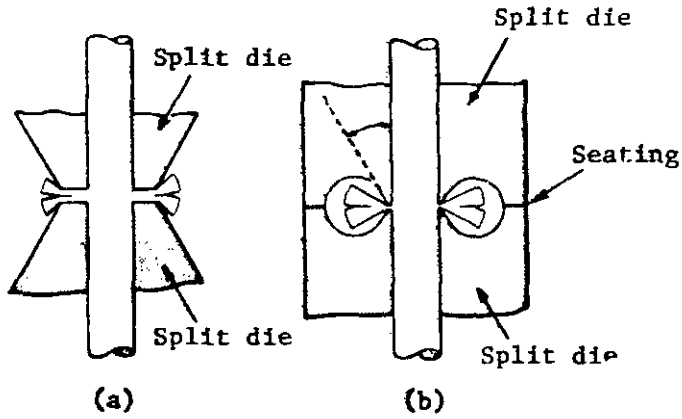


FIGURE 125. Cold-pressure butt welding: (a) flat-faced, (b) conical-faced dies.

Dies with conical faces have two specific advantages over dies with flat surfaces: (1) welding pressure is greatly reduced and (2) the flash is automatically trimmed off upon closing of the dies.(13) A conical angle of 120°is used most frequently.

The amount of deformation of the workpieces is dependent on the geometry of the face of the dies and the projection of the parts beyond the face of the dies. Round bars require a projection of 40% of the diameter on both sides of the interface. With square bars the projection is 40%. With parts of a rectangular cross-section, the amount of projection is based on the smaller dimension When metals with different hardness values are cold-welded, the softer metal deforms first. The projection of the softer metal must therefore be greater than that of the harder material. Cold welding results in the formation of a flash around the joint. Cavities in the die have to be provided for the flash such as shown in Figures 126 and 127 for butt welding of wires and tubes, respectively.

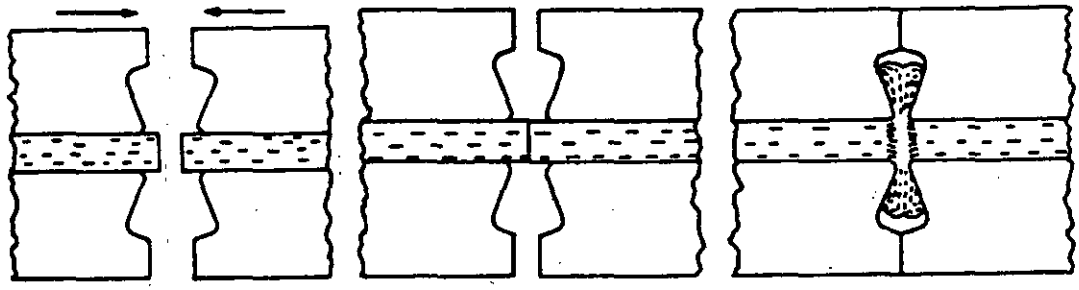


FIGURE 126. Die action in cold butt welding.

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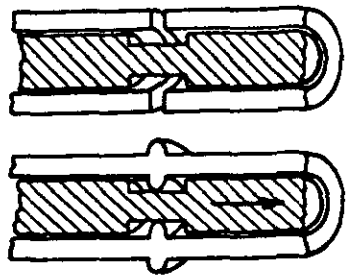


FIGURE 127. Use of mandrel in cold butt welding tube.

16.2 Diffusion Welding

Diffusion welding is a modification of the pressure-welding process in that higher temperatures and longer times are involved. The higher temperature allows a reduction in the pressure to a level well below that needed for cold welding. However, the higher temperature and longer exposure time necessitate the use of a protective atmosphere, or a vacuum to avoid excessive oxidation and contamination of the faying surfaces with some base metals.

Diffusion welding allows the joining of materials with markedly dissimilar properties, which would not be possible by fusion welding processes. For example, non-metals, namely ceramics and powder metal products can be joined to metals by diffusion welding. Complex shapes, multiple joints and intricate configurations are possible to bond in one operation by the diffusion welding process.

Two types of diffusion welding are known: solid-state and liquid phase.(13) Solid-state form of diffusion welding is normally carried out at about 0.7 T_m (where T_m is the melting point of the base metal), and pressures of 5-15 N/mm^2. Welding time ranges from 2-3 minutes.

When dissimilar materials are diffusion welded, usually a filler metal in the form of an insert or an interlayer such as electroplating may be utilized. The purpose of the filler metal is either to decrease the required temperature, pressure and time for welding, or to permit welding in a less costly atmosphere. When an interlayer is introduced, it is possible for this filler metal to liquify at the welding temperature, if its melting point is significantly below that of the base metal. The liquid-phase diffusion welding produces a cleaner joint surface. The molten filler metal fills the surface cavities that may be present. Liquid-phase diffusion welding is considered to be diffusion brazing and is extensively used in the aircraft, aerospace and nuclear industries. Diffusion welding in the aerospace industries is employed for joining titanium components. The titanium parts are placed in a low-carbon steel container. Stainless steel spacers and

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positioners are used to hold the parts of the assembly together in the correct position. The container is then seal welded and evacuated prior to heating to the welding temperature. The container is then pressed in an open press and opened after cooling.

16.3 Forge Welding

Forge welding is a solid state welding process wherein coalescence is produced by heating and by applying pressure sufficient to cause permanent deformation at the interface.

In forge welding, the parts are usually preheated to over 1000°C (1835°F) and then hammered together. Pressure can be applied manually and mechanically. Machine forge welding can be done by rolling, hammering and pressing within a die. Forge welding is believed to be the oldest welding method known. The workpieces are preheated to the forging temperature in a fuel-fired furnace.

Most of the pieces require some edge preparation to insure that the initial point of contact during forge welding occurs in the interior of the joint. By forging from the center outward, oxides and foreign particles are forced out of the joint during the welding process. Oxidation can be inhibited by adjustment of the atmosphere during heating and forging to create a neutral or a reducing atmosphere. In addition, fluxing materials may be added to dissolve the more tenacious oxides present on the joint surface. Silica sand, fluorspar, and borax are used for this purpose.

The control of dimensions and shapes of forge-welded components to close tolerances is not possible, especially with the manual hammering method. Dimensional and shape controls are improved by the use of dies. Roll bonding used for cladding purposes can be mentioned as a form of forge welding.

Drop hammers, hydraulic hammers or presses of up to 50,000 tons were built for joining large masses of metals. Forge welding has in many areas been taken over by the more modern and high productivity welding processes. As a result, the forge welding process is of small industrial consequence today.

16.4 Explosion Welding

Explosion welding is a solid-state welding process in which coalescence of the parts to be joined is effected by high-velocity movement produced by a controlled detonation.(3)

One of the components called the flyer collides with the stationary part in explosion welding. Upon collision, the kinetic energy of the flyer plate is converted into a compressive stress wave on the surface of the two workpieces. To protect the flyer plate from surface damage such as erosion and to control the explosive force a buffer or attenuator layer is placed between it and the explosive charge. Welding is done in microseconds without a noticeable effect on prior work-hardening and heat treatment of the workpieces.

The parts can be arranged with the faying surfaces parallel to each other or at an angle. With an angular interface a so called "bonding front" is established that moves across the interface between the two colliding components. A thin layer on the interfaces is melted and jetted out leaving behind clean surfaces.

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Explosion welding yields no heat-affected zone, but the interface invariably has a wavy pattern as can be seen in Figure 128.(13)



FIGURE 128. Photomicrograph of weld interface between explosion welded stainless steel cladding (top) and a carbon steel substrate (bottom). x 150

The amplitude and the wavelengths while depending on the welding conditions vary from 0.1 - 4.0 mm and 0.25 - 5.0 mm, respectively. The mechanism of ripples and jet formation is that under the condition of high pressure, temperature and velocity the metal in the region of the welding front is plastic to the point of behaving like a liquid. The flyer or the cladding plate resembles a jet traversing across the base or target plate. A ripple is formed in the plastic region of the target or backer plate ahead of the jet which passes over the ripple periodically.

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When the two pieces to be explosion welded differ in thickness, the heavier piece may be placed on an anvil. If one of the pieces is thick enough, it may serve as its own anvil. Explosion welding with an angled interface with and without a separate anvil is shown schematically in Figure 129.

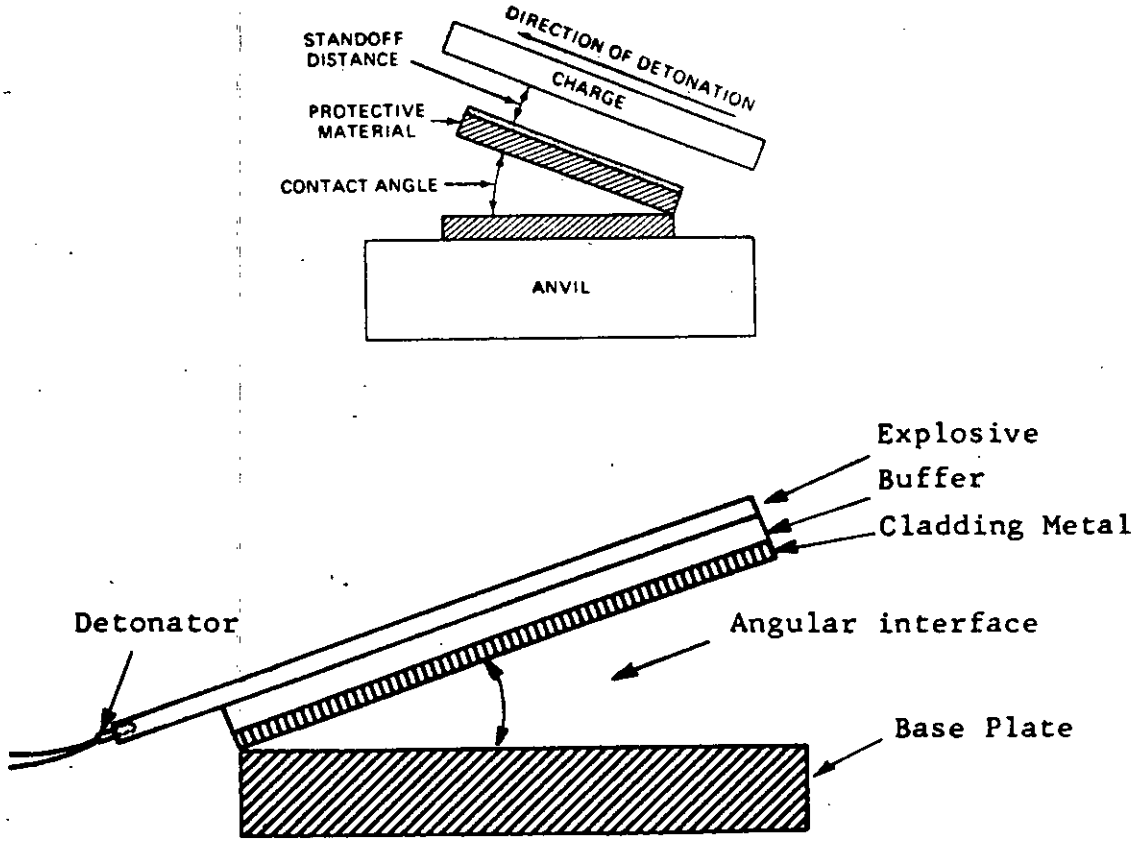


FIGURE 129. Explosion welding:angled interface.

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When the arrangement of the two pieces of metal is parallel, a small gap is maintained between the workpieces as exhibited in Figure 130.

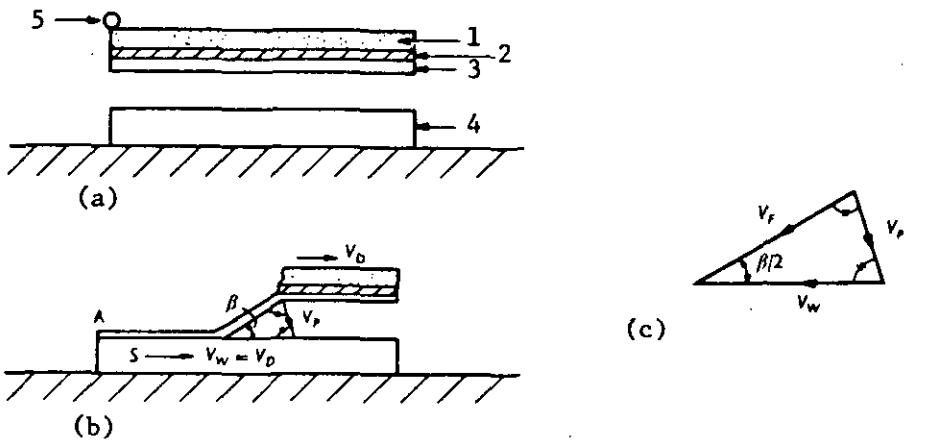


FIGURE 130. Arrangement for explosion cladding. 1, explosive; 2, buffer; 3, flyer-plate; 4, base metal; 5, detonator; V_D; detonation velocity; V_w, welding velocity; V_f, V_P, components of flyer plate velocity. (a) before welding, (b) during welding, (c) velocity diagram.

The angular interface or gap clearance is a function of the velocity of the explosive applied. In terms of angles, satisfactory welding requires that the two surfaces be inclined between 5-25° provided all other welding parameters are optimized. When the two surfaces initially are parallel with one another, this angle can only develop if the standoff distance is from half to two times the thickness of the cladding plate depending on the detonation velocity. The detonation velocity must be within limits to generate the necessary impact velocity to make the weld and not exceed the highest sonic velocity within the materials being welded. There is a certain minimum velocity for each material type to be explosion welded satisfactorily. For example, copper and aluminum alloys require detonation velocities in excess of 120 m/s and 255 m/s, respectively. The minimum required velocity may be determined by the point at which the flyer material becomes sufficiently plastic upon impact to produce a divided jet. Explosives should have a detonation velocity of about 2400-3600 m/s (8000 - 12000 ft/s). When the detonation velocity is below the sonic velocity of the material being welded, the pressure generated by the expanding gases travels faster than the detonation itself and spreads out ahead of the detonation front. Therefore, a shock wave is not generated. If the

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detonation speed is slightly greater than the sonic speed in the metal, the shock wave may be detached and move in front of the detonation line. Jetting action in explosion welding is illustrated in Figure 131.(15)

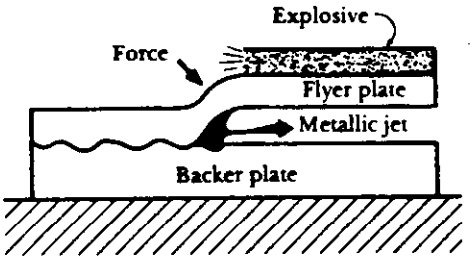


FIGURE 131. Jetting action in explosion welding.

Explosives used may be in the form of a flexible plastic sheet, cord and granulated, pressed, cast as well as liquid shapes. Explosives vary in detonation velocity. Changes in detonation velocity is accomplished by altering the composition, thickness and density of the explosive sheet. The most common explosives are based on ammonium nitrate. The actual detonation is set off by a commercial blasting cap.

Low velocity explosives can be placed in direct contact with the flyer plate. Thick buffers are necessary with the use of high velocity explosives to prevent shock wave damage in the form of microcracking at the weld interface and spalling along the edges. Thus, low detonation velocity explosives are preferred wherever possible to use for easier control of the process.

The speed of the flyer plate can be changed by changing the explosive charge per unit area. If the flyer plate velocity is too low, welding will not take place between the two pieces of metal. If on the other hand, the velocity of the flyer or cladding plate is too high, the brittle intermetallic compound formation will be excessive. Larger waves will also be produced. Moreover, the angle of incidence at which waviness begins will increase and the range of angles within which waves are generated will widen. The maximum impact velocity is determined by the sonic velocity in the base or backer plate because at supersonic velocities the wave in the backer plate is unable to propagate in front of the welding line. Due to the pressure being reduced along the edges of a plate which in turn results in a lowering of impact velocities, explosion welding along plate edges tends to be incomplete.

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The most extensive application of explosion welding has been in cladding of metals: flat surfaces, pipes and tubings. Explosion welding is unsuitable for brittle materials and metals of low melting point because of the high impact energies and temperatures involved. Dissimilar metals that cannot be welded by the more conventional fusion welding techniques are often joined by explosion welding in open spaces. Explosion welding of smaller pieces is done in a vacuum, which not only confines the explosion but also dampens the noise.

Surface smoothness rather than surface cleanliness is the most important aspect of surface preparation. A surface finish of better than 0.5 um is desirable for satisfactory explosion welding.

16.5 Friction Welding

In friction welding (FRW), heat is generated by a conversion of mechanical energy to thermal energy. Welding is obtained between a stationary and a rotating workpiece as a result of frictional heat developed at the interface by subjecting them to a constant or a gradually increasing compressive loading. This welding process requires that the rotating member be essentially symmetrical about the axis of rotation. The stationary member can be of any geometric shape.

The maximum temperature at the interface is below the melting point of the base metal. When the appropriate level of temperature or plasticity has been reached at the interface, rotation is stopped and the compressive force may be increased to induce welding. When dissimilar metals are friction welded, that which has the greater deformation characteristics will show a greater weld upset.

Friction welding can be divided into two categories: continuous-drive FRW and inertia-drive FRW.[1] In the continuous-drive friction welding method, the rotating workpiece is pressed against a stationary member. When sufficient temperature is generated at the interface, rotation is stopped suddenly while the compressive force is increased and held for a predetermined time to complete the weld. In inertia-drive welding, the rotating piece is held in collet chuck-flywheel assembly. Welding is done by thrusting the rotating part against the stationary piece (or vice versa), at which point the flywheel is disconnected from the power supply. The pieces are welded together under a steadily decreasing rotational speed. The schematic diagrams showing the stages of the two friction welding processes are displayed in Figure 132.

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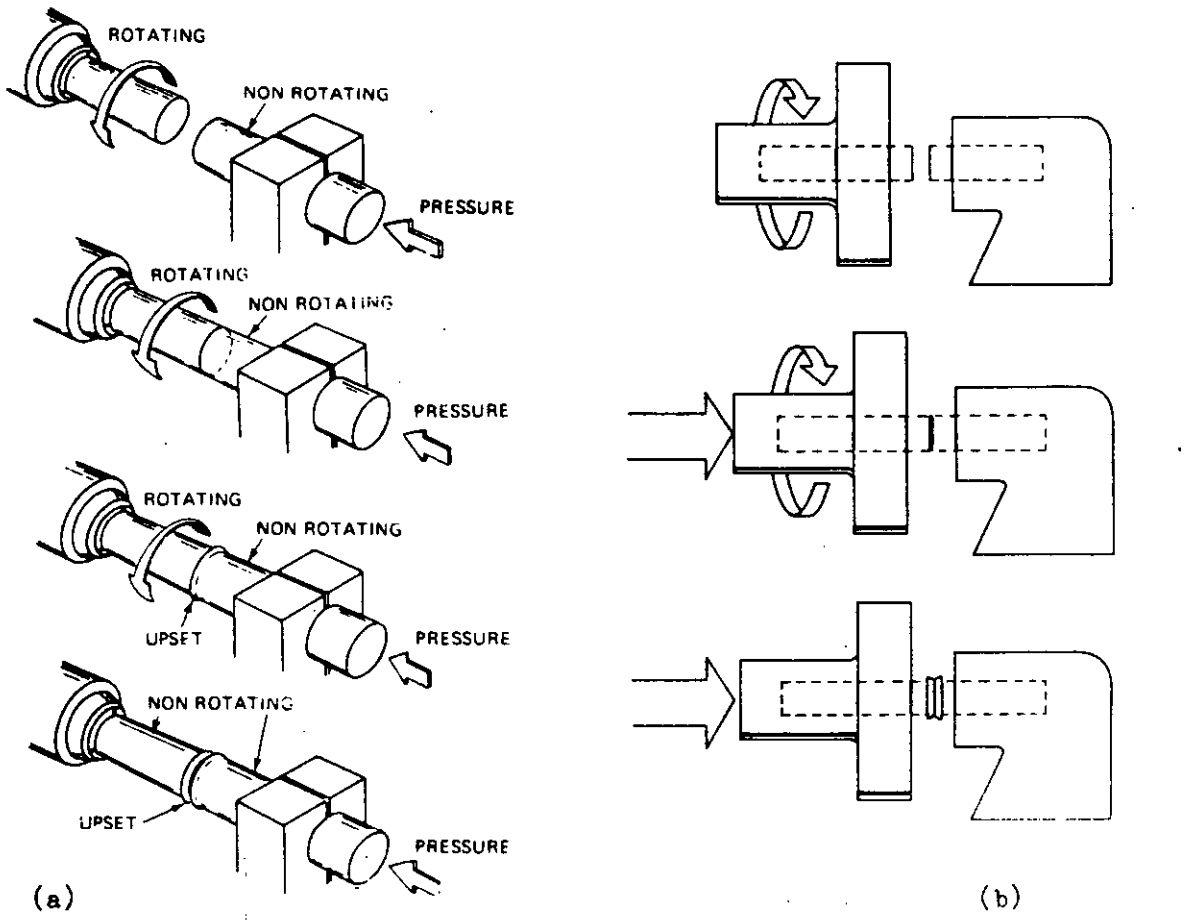


FIGURE 132. Schematic presentation of the steps involved in (a) continuous-drive friction welding and (b) inertia-drive FRW processes.

16.5.1 Process Characteristics

Surface preparation of the workpieces is not critical because the rubbing action removes the oxide film and irregularities leaving a clean, smooth interface which is heated sufficiently high to give a metallic bond between the two pieces. When friction welding large-diameter bars, a small projection at the center of one of the two pieces is used to make sure that heating and forging occur and to eliminate weld defects in the center of the weld joint.

This process is suitable for welding both ferrous and non-ferrous alloys and metals of dissimilar properties. Friction welding does not require filler metals, protective atmospheres or fluxes to obtain a sound weld. Other characteristics of the process are that the heat-affected zone is very thin, the grain size is frequently smaller than that of the parent metal and little fumes and spatter are generated. The total energy necessary to make friction welds is a fraction required for many other welding processes.

There are a number of limitations to the friction welding method which are as follows:

1. Workpieces need be nearly round at the weld interface.

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2. The parts must have physical dimensions that can fit into the chuck of the machine.
3. The workpieces and the holding devices have to have adequate strength to withstand the torque and axial compressive forces applied.
4. The process is restricted to flat and angular butt welds which are normal to and concentric with the axis of rotation.

Machines for friction welding are similar to lathes that are provided with a laterally fixed rotating chuck and a laterally movable stationary chuck. The lateral motion of the chuck which holds the stationary part during welding is powered either hydraulically or pneumatically. The machine is provided with controls to change the welding variables which include the rotational speed, the weld-cycle timer and the axial compressive force for heating and forging. Friction welding machines must have a feature of sudden stop. Braking is effected by a spindle-braking system and reverse electrical fields in the driving motor.

16.5.2 Continuous-Drive Friction Welding Procedure

The part to be rotated is clamped in the spindle chuck and brought to a predetermined speed. The stationary workpiece is clamped in another chuck. The two pieces are then brought into contact under a constant or gradually increasing axial pressure. Once the welding temperature is reached, the brake is applied to stop the rotation as rapidly as possible. The tailstock pressure is simultaneously raised to complete the weld.

The welding variables are adjusted to meet the needs of a given cross-section, and physical and mechanical properties of the base metals being welded.

16.5.3 Inertia-Drive Welding Procedure

Inertia-drive welding takes advantage of the kinetic energy of a freely rotating flywheel. A major difference between the continous-drive and the friction-drive welding processes is that while the former uses a constant spindle speed, the latter starts out with a high speed which steadily decreases due to friction. The torque at the weld interface also changes continuously in inertia-drive FRW.

The moment of inertia of the flywheel can be adjusted by adding or removing disks. Hot working and rapid cooling which characterize inertia-drive FRW welding produce a fine grain structure in the weld zone. The grain size can be changed by post weld heat treatment.

The energy input can be controlled by modifying the flywheel size and/or the peripheral velocity of the rotating workpiece. The optimum combination of moment of inertia and rotational velocity to produce the best weld properties varies with the type of base metal being welded. In steel to steel welding, the initial velocity is usually between 500 and 1500 sfm. Low velocities (under 300 sfm) result in low center heating and uneven weld upset.

The effect of varying the axial pressure is similar but opposite to the influence of changing the rotational velocity. For instance, excessive pressure gives rise to poor quality weld in the center of the interface and produces a large weld upset.

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The moment of inertia of the flywheel is so selected as to give the right amount of forging. Forging comes from an increase in torque which occurs at the weld interface as the flywheel slows down. This torque in conjunction with the axial pressure produces forging. Large, low-speed flywheels induce greater forging than small, high-speed flywheels in spite of having the same kinetic energy in both.

Inertia-drive FRW is suitable for welding practically any metal that is forgeable, be it similar or dissimilar in terms of properties.

16.6 Ultrasonic Welding

Ultrasonic welding (USW) is a type of solid state bonding in which coalescence of the workpieces is accomplished by the local application of high-frequency vibration while they are held together under a low compressive force. Although the process is primarily used to join thin gages of aluminum alloys, it has also been utilized for welding a whole variety of other metals including dissimilar ones. The method is also ideally suited for joining thin to thick pieces.

Ultrasonic welding does not normally require an extensive surface preparation because the lateral vibratory frequencies at 15 to 75kHz disperses the oxide film and removes other surface contaminants that may be present at the interface. The low pressure produces minimum surface deformation which is of particular advantage from a hydrodynamic and aerodynamic point of view.

16.6.1 Procedure

The procedure involves placing the workpieces between an anvil and a vibrating probe called a sonotrode or an electrode or a welding tip.(1) The sonotrode induces lateral vibration and slip locally between the faying surfaces, thereby disrupting surface oxides, raising the temperature locally and permitting metal-to-metal contact. The low applied pressure helps ensure a metallurgical bond between similar and dissimilar metals. Under normal vibratory power and compressive loading, the maximum local temperature reaches about 35-50% of the melting points of the metals involved. The major components of an ultrasonic welding machine are shown schematically in Figure 133.

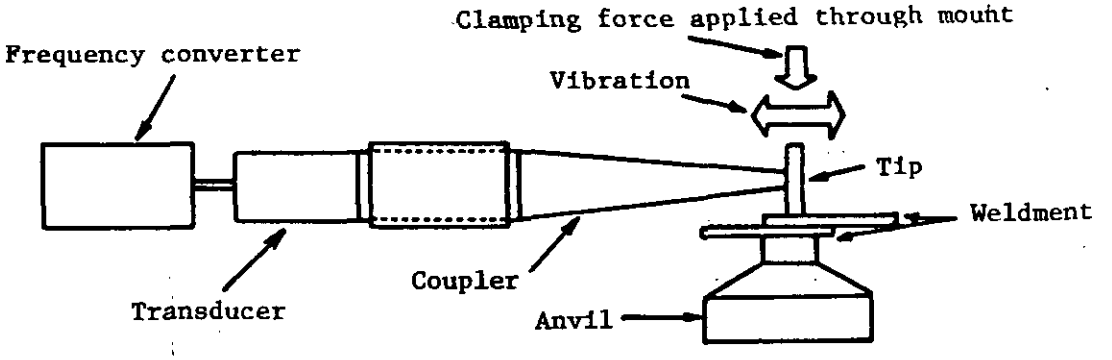


FIGURE 133. Schematic of ultrasonic welding machine.

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The vibratory energy is produced by a transducer, which convert the electrical energy into mechanical energy. Transducer materials undergo a dimensional change when exposed to an electromagnetic field. Materials with magnetostrictive and piezoelectric properties such as barium titanate, nickel or quartz crystals are used for transducers. Ultrasonic vibrators consist of the transducer, which is a resonant laminated magnetostrictor, and a velocity transformer. The latter is made of a low-loss, high-strength metal such as titanium. The oscillator converts 60 Hz line power into ultrasonic power, and transmits it to the transducer. The sampler transmits the vibrations of the transducer to the electrode without subjecting the transducer itself to the clamping force. The electrode applies the clamping force or welding pressure to the workpieces and the anvil. The welding tip also transmits the vibrating energy to the joint. The ultrasonic welding head assembly is shown in Figure 134.

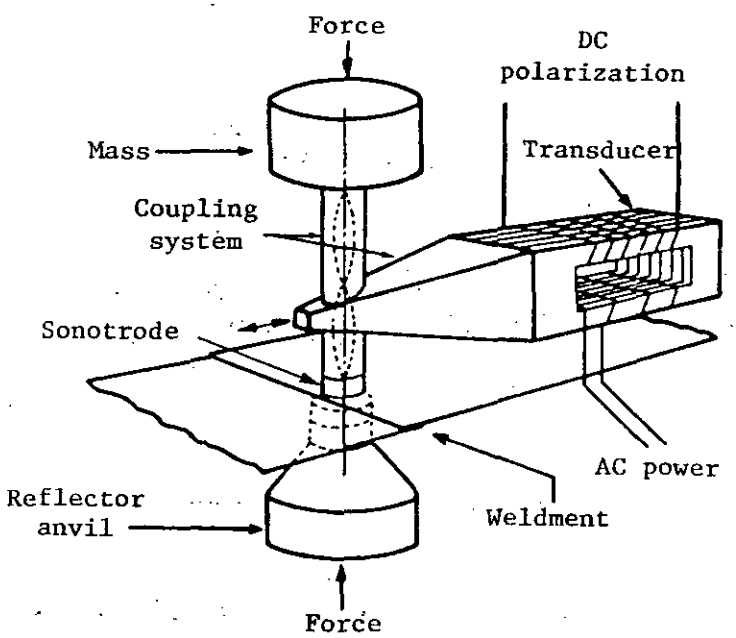


FIGURE 134. Ultrasonic welding head assembly.

For reasons of ruggedness, most of the transducers used for ultrasonic welding are magnetostrictive. A rod of nickel wrapped with wire through which an alternating electric current flows, does expand and contract longitudinally at the same frequency as the alternating current, under the influence of the magnetic field associated with the current. The strongest vibrations are reached when the system is in resonance, i.e., when the dimensions and mass of the transducer element are matched with the frequency of the imposed field. Therefore, ultrasonic welding machines are designed to operate only at one particular frequency.

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The welding pressure is provided to the welding tip and anvil either hydraulically or mechanically. The welding tips are generally made from hardened high-speed steels or other specialty alloys. Tips come in various forms depending on the type of joint being made. For spot welds, the welding tips are similar to those used in resistance spot welding. The surface finish on the tip may be smooth or cross-hatched. The latter is claimed to produce a spot weld with improved fatigue strength. A shim stock can be placed between the electrode and the workpiece to eliminate surface indentation. Rotating disc tips either with a counter-rotating roller anvil, or with a table-type anvil are utilized for continuous ultrasonic seam welding. These various welding tips and anvils can be seen in Figures 135 and 136.

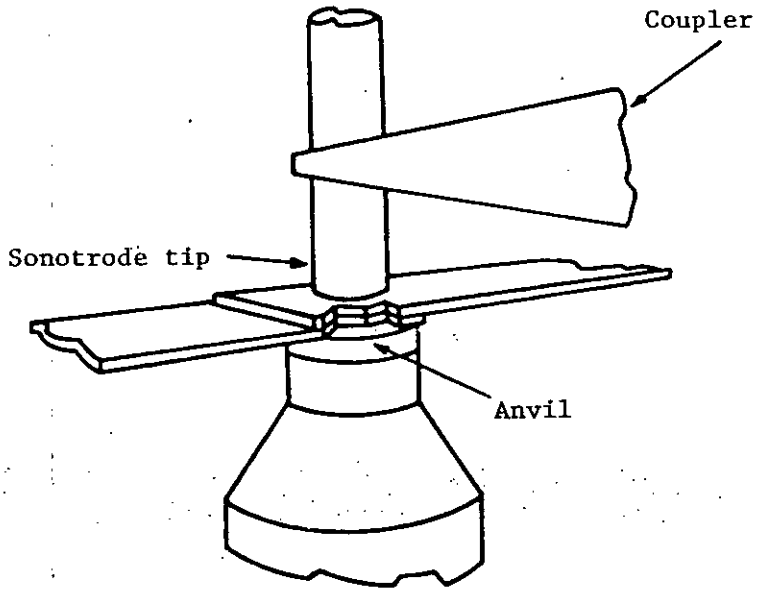


FIGURE 135 A typical flat-faced welding tip and anvil.

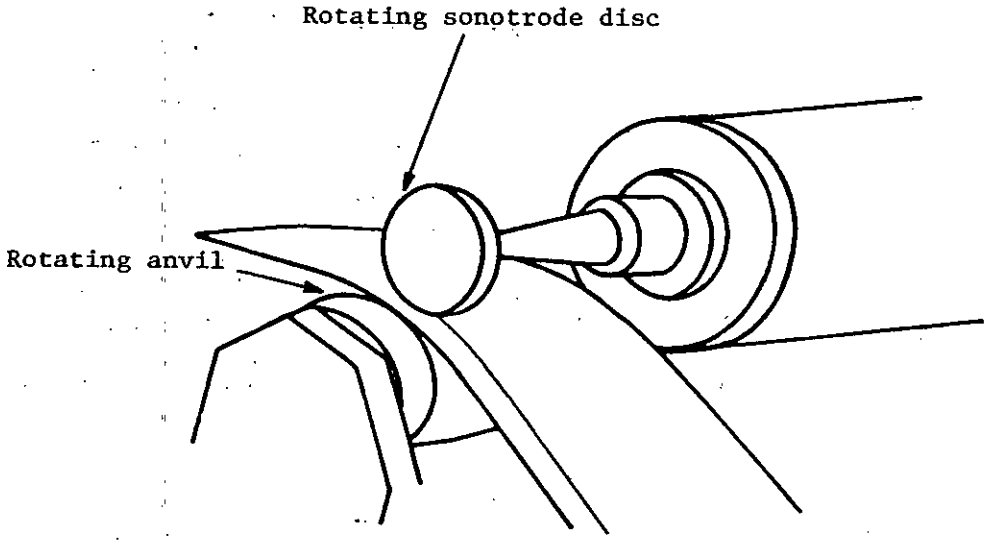


FIGURE 136. Rotating disc electrode with rotating anvil.

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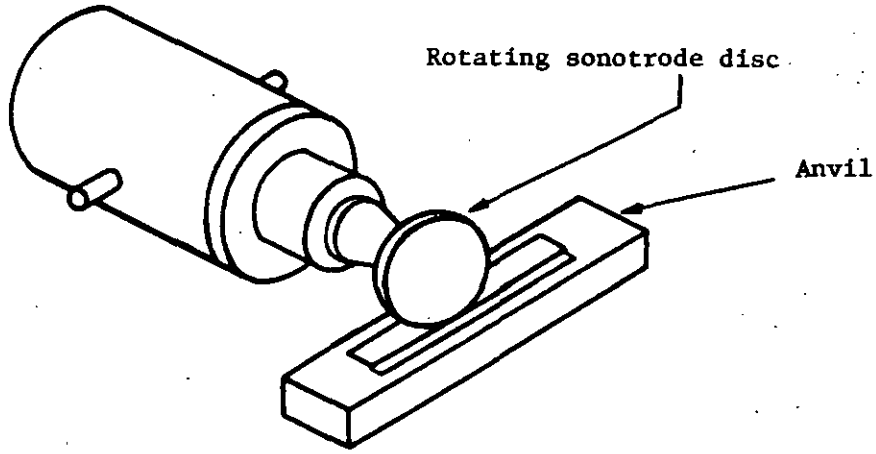


FIGURE 137. Rotating disc with flat anvil.

The ultrasonic welding process has its broadest application in the aerospace, electronics, packaging and instrument industries.

Unlike in resistance spot welding, there is no electric current flowing through the parts being ultrasonically welded. Consequently, there is no minimum space requirement between weld spots to avoid short circuiting the equipment. The advantages of USW include low heat generation, absence of flux and dispersion of surface oxides.

At present, commercially available ultrasonic welding machines have power ratings ranging from 25 watts to 8 kilowatts. This range of power rating is adequate for welding for example aluminum alloys in thickness range of 0.0043 mm (0.00017 in) to 3.0 mm (0.12 in) and dissimilar metals in various thicknesses such as copper/nickel to carbon steel of 2.54 mm (0.1 in) and 19.0 mm (0.75 in) thick, respectively.

The essential variables of ultrasonic welding are weld power, weld time and clamping force. In general, as the thickness of the base material increases, all of these variables have to be correspondingly increased to obtain an acceptable weld. As an example, aluminum foil of 0.0254 mm (0.001 in) thick can be welded with 100 watts of weld power and 22.5 kg (50 lbs) to 33.8 kg (75 lbs) of clamping force in 0.05 second. Aluminum alloy 2024 in thickness of 2.0 mm (0.08 in) requires 4 kW of weld power and 315 kg (700 lbs) of compressive force for about 1.0 second. Usually, experimentation is necessary to optimize the weld variables that will produce satisfactory welds in a specific type of material and thickness.

16.7 Hot Pressure Welding

Hot pressure welding often called thermocompression welding is a form of pressure welding carried out at some elevated temperatures. It has been developed for joining miniature components in the electronics industry. Notable examples include fine wires of 0.025 mm diameter (0.001 in) or less

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and metal films on glass or ceramic.(13) The weld temperature may be 200°C (390°F) or higher. The specific techniques of thermocompression welding were developed to accommodate the geometries of the microcircuit joints

Induction welding and resistance butt welding constitute modifications of the hot pressure welding process. A low-frequency induction heating of the joint is used as the energy source in induction welding. Pressure may or may not be used.

The resistance to the flow of electric current across the interface of the butt joint provides the heat input in upset welding (also known as resistance butt welding). The clamps that hold the workpieces together not only provide the pressure but conduct the current as well. Upset welding is customarily used for joining wire and rod. For welds that require a post weld heat treatment, such as high-carbon steels, a second pulse of current to modify the cooling rate may be used in order to change the resultant micro- structure.

16.8 Electromagnetic Impact Welding

The forces that impact the workpieces together in this welding process are induced by electromagnetic means. The basic features of the process comprise a coil of wire wrapped around the joint line. The discharge of a considerable amount of electric current through the coil causes a vaporization of the coil which in turn breaks the circuit in about one microsecond. The rapid generation and collapse of the magnetic field in the coil induces a heavy current in the workpieces, and one of them collapses with an appreciable impact. The action of vaporizing the coil produces a mechanical shock wave which serves as a secondary impact force.

Another version of electromagnetic impact welding consists of a permanent coil and a field shaping piece which concentrates the magnetic field at the proper point on the part being welded. Electromagnetic impact welding can be done on cold and hot workpieces.

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17. MAGNETICALLY IMPELLED-ARC WELDING

Magnetically impelled-arc welding (MIAB) is in essence a forge butt welding process for tubes in which heat is developed before forging by causing a rapid movement of an arc around the ends of the workpieces through the application of an external magnetic field.

The workpieces are held in clamps. Two solenoids are placed around the base metal, one on each side of the joint.(13) The solenoids are energized by a direct current power source so that some poles are on each side of the joint as shown in Figure 138.

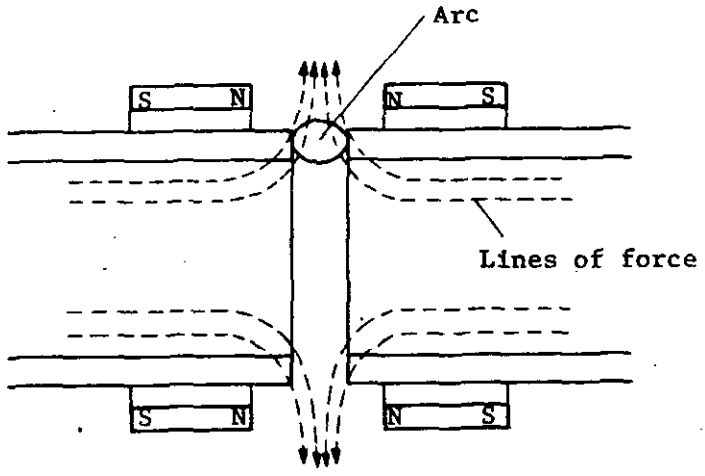


FIGURE 138. Principle of MIAB welding.

The workpieces are in contact initially and a dc welding supply with 28-48 V range is connected across the joint. Upon withdrawing the workpieces from contact, an arc is initiated across the joint, which is opened to 1-2 mm (0.04 - 0.08 in). The magnetic field induced by the solenoids is radial with respect to the axis of the work. This causes the arc to move around the outer edges of the base metal, thereby melting the ends. The workpieces are subsequently forged together by a pressure of about 69 MPa (5 tons/in^2) to give a solid state weld. The molten metal is squeezed out similar to flash welding.

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The major advantages of this process consist of high welding speed, no shielding gas requirement, low electrical power consumption, cleanliness, proper alignment of the workpieces and a relative insensitivity to upset forces. The MIAB process is limited to wall thicknesses under 5 mm (0.2 in) and cylindrical shapes.

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18. THERMAL SPRAYING

Thermal spraying is a process in which metallic or non-metallic materials are heated and then deposited in finely divided particles in molten or semimolten condition on a substrate. The process may be subdivided on the basis of heat-generating source as flame, electric arc and plasma spraying methods.(16,17,18) The spray material may come in the form of powder, wire and rod. Because the sprayed particles flatten out upon impacting the substrate, the sprayed deposit has a lamellar structure. The surface of the substrate is intentionally roughened to provide anchorages for the deposited particles. The anchor pattern serves to mechanically bond the deposited layer to the substrate. While there are some points at which metallurgical bonds develop between some of the particles and the substrate, the primary bonding mechanism is of the mechanical type.

In thermal spraying, oxide formation is extensive because of the high surface to volume ratio and the high-velocity blast of air used to propel the heated particles. Consequently, oxide-to-oxide bonding is also extensive. Therefore, thermal sprayed deposits are characteristically quite brittle. Furthermore, the deposits are porous and have a lower density and ductility than the original material. Thus, sprayed deposits resemble sintered metals. The physical and mechanical properties of the sprayed metal deposits differ significantly from those of the wrought metal. Alloys deposited by thermal spraying processes may not have the same chemical composition as that of the original material. In the plasma and electric arc processes, a noticeable loss due to vaporization of the lower melting point constituents can occur because of the high temperatures involved. Ceramics of the oxide type retain their chemical composition after deposition, though the crystal structure may in some cases change. The reactive ceramics, namely carbides, borides and silicides undergo a chemical change when deposited with flame or plasma in air. The change in chemical composition is the result of the formation of carbon dioxide, boron trioxide and silicon dioxide. So that, for instance, when silicon carbide is thermal sprayed, the deposited layer obtained is silicon dioxide only. Therefore, with reactive ceramics special environmental chambers or special techniques must be employed in order to mitigate compositional changes.

Plastics have also been deposited by thermal spraying. When metal powder is utilized, a carrier gas is necessary to pick up the powder from the hopper and propel it through the heat source. The velocity has to be so selected that the temperature of the powder to be deposited will be raised to the molten or semimolten condition.

Thermal spraying is often substituted by terms such as metallizing, metal spraying, or flame spraying. The thermal spraying process can be applied to all materials that do not volatilize upon reaching their respective melting points. Parts to be sprayed must first be formed to the final shape, because the as-sprayed deposit does not have the necessary ductility and tensile strength to allow extensive plastic deformation. There is no size limitation to thermal spraying. The more simple the shape, the more suitable the workpiece for spraying application.

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18.1 Surface Preparation

The success or failure of thermal spraying processes depends on the surface preparation of the base metal. A good surface preparation implies not only cleanliness but an anchor pattern as well. Generally accepted methods of surface preparation include shot blasting, rough threading, undercutting, grinding, studding, etching, sanding, grooving, dovetailing, use of chemical solvents and self-bonding materials as well as electric bonding.(17)

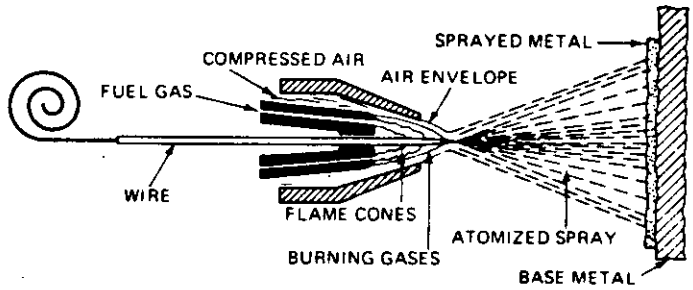
It is a good practice to clean the surface beyond the area to be thermal sprayed. If the prepared surface is not immediately coated, means of surface protection should be provided. Porous base materials having been in service with a lubricant should be baked at a temperature high enough to evaporate the lubricant.

The choice of surface preparation method depends on the type of base and spray material involved and the service conditions. Preheating may be necessary for certain parts to prevent condensation of moisture and to minimize distortion of the part or cracking of the coating.

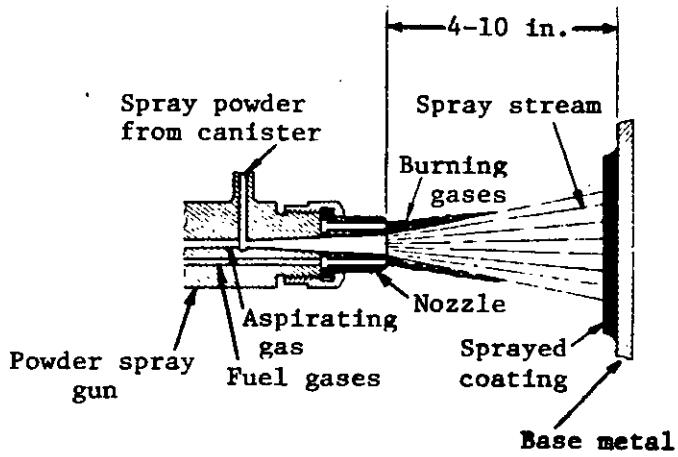
18.2 Flame Spraying

This process uses oxyfuel gas as the heat source for melting the coating material which is atomized and propelled onto a substrate by a stream of compressed air or other gas. Flame spraying has two variations: one utilizes metal in wire form, the other employs material in powder form.(16) These two versions are shown schematically in Figure 139.

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(a)



(b)

FIGURE 139. Schematic drawing of two flame spraying techniques: (a) wire gun, (b) powder gun.

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Using acetylene as the fuel gas, the flame temperature is approximately 3130°C (5660°F), which is used for oxides and alloys with higher melting points. Propane gas is used for spraying aluminum, zinc, tin, lead and babbitt metal.

The ratio of oxygen to fuel gas should be balanced properly to prevent excessive oxide film formation on the metal particles which would interfere with fusion of the deposited particles. The quality of the compressed air is important. Too much moisture, oil or sulfur in the air will cause poor atomization and decrease the bond strength.

Thermal spraying of ceramic rods is similar in concept to wire and powder flame spraying.[17] The ceramic rod is fed into a multiple flame source where it is melted and atomized by a stream of compressed air. The high melting points and low conductivities of ceramics require a more efficient heat transfer than metal spraying. The gas nozzles for ceramic spraying is therefore different and the rod feed system is slower than those for metallizing. A cross-sectional view of a ceramic spray gun is exhibited in Figure 140.

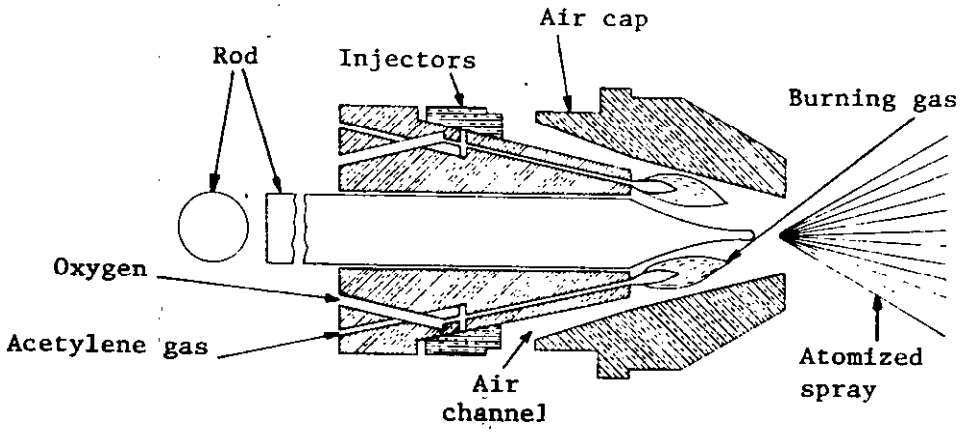


FIGURE 140. Cross-sectional view of a ceramic spray gun.

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18.3 Electric Arc Spraying

Two consumable wire electrodes pass through electrical wire guides, become energized and contact each other.[8] This creates a high current density at the contact point. An arc forms and melts the ends of the two wires. A jet of compressed air carries the liquid metal away onto the part being surfaced. The arc is interrupted momentarily. The wire feed system continues to feed the wire causing them to contact again and the cycle is repeated.

The power source is a direct-current, constant-voltage welding machine. The wire feed mechanism is analogous to that used in gas metal-arc welding. The magnitude of the electric current and voltage employed in electric arc spraying ranges from 300 to 500 amps and from 25 to 35 volts, respectively. The pressure and amount of air is about 9 kg/mm^2 (80 psi) and 0.9-2.3 m^3/min (30-80 ft^3/min), respectively. Salient features of the process is illustrated in Figure 141.

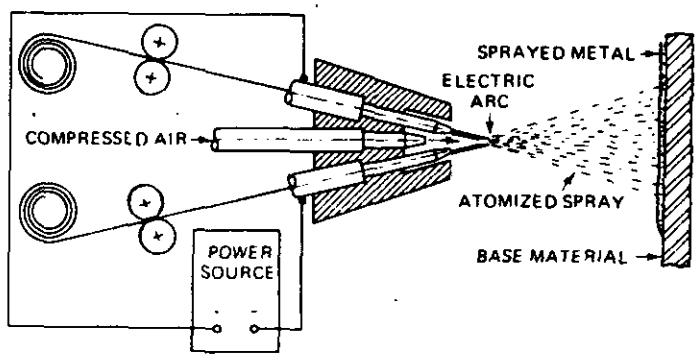


FIGURE 141. A schematic drawing of electric arc spraying.

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Surface preparation of the workpieces is just as important with the electric arc spraying process as with the other thermal spraying methods. Electric arc spraying produces more atomized metal and the resultant particles are hotter than those produced by flame spraying. Therefore, the electric arc gun must be manipulated faster than the flame spray gun. Otherwise, the substrate will be overheated. Aluminum, copper, nickel, tin, zinc, molybdenum, carbon steel, stainless steel, brass and other materials have been metallized by electric arc spraying.

18.4 Plasma Spraying

Plasma spraying is a process in which a suitable gas is injected into an electric arc confined in the nozzle of a water-cooled jacket to produce an ionized gas called plasma as a heat source for melting and ejecting the surfacing material onto the workpiece. The temperature of the plasma may be as high as 16,650°C (30,000°F). Consequently, any material which does not volatilize upon melting can be plasma sprayed.

The spray material is usually supplied in powder form and is suspended in a carrier gas as it enters into the plasma spray gun. The gases used in plasma spray systems have three functions: (1) as plasma generating gas, (2) as powder-carrier gas and (3) as auxiliary gas. The plasma generating gas also helps sustain the arc. Small volumes of the auxiliary gas may be mixed with the plasma generating gas for certain reasons. The source of the plasma generating and powder-carrier gas is the same.

In plasma spraying, monatomic or diatomic gases may be utilized. Of the monatomic gases, argon or helium are most commonly used and produce high enough temperatures for most applications. Because argon and helium are inert gases, the resultant metallized surface has a lower oxide content than either flame sprayed or electric arc sprayed coatings: hence improved ductility in the deposited layer.

Nitrogen is the most widely used diatomic gas. The type of gas employed determines not only the temperature evolved but the overall quality of the plasma, which in turn has a bearing on life of the torch (or gun). Hydrogen in amounts of 5 to 25% is used as an auxiliary gas mixed in with the plasma generating gas such as argon, helium or nitrogen so as to either raise the arc voltage or permit the application of higher levels of power. Nitrogen is the least expensive of all the gases mentioned above: hence its broad application in plasma spraying. The diagram seen in Figure 142 describes the main features of the plasma spray process.(8)

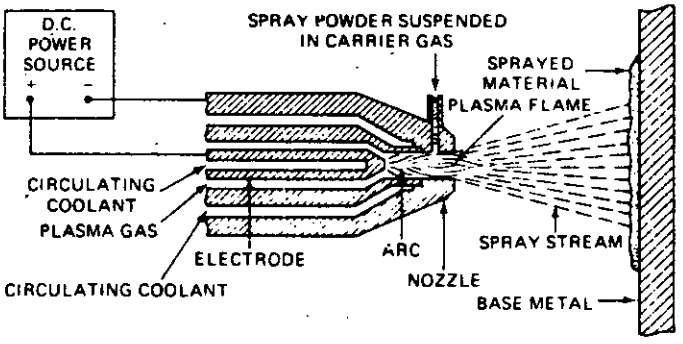


Figure 142. A sketch of the plasma spray process.

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Surface preparation is important and the techniques are similar to those mentioned in connection with the other thermal spray processes. The torch-to-work distance as in all thermal spray processes has to be adjusted as the particular requirements of a specific case dictate. Once the optimum distance has been determined, it should be maintained constant for that application. The traverse speed should be adjusted to maintain the proper thickness in each pass. The specific thickness is dependent upon the particular thermal process used, the type of material deposited, the substrate involved and the total thickness of the surface layer to be built up.

Thermal spraying in general has two major application areas: one in building up worn-out surfaces due to abrasion, erosion, cavitation or general wear and tear; the other in providing protection from environmental effects such as corrosion and oxidation.

18.5 Post-Spray Operations

There are several operations which may follow thermal spraying to meet various but specific requirements.[16] Included in the post-spray operation are fusing, controlled cooling, grinding, machining, sealing, diffusing, wire brushing, burnishing, polishing, honing and lapping. Some of the more important finishing operations are discussed briefly.

18.5.1 Fusing

The materials for fused coating comes in powder form because they are usually too hard and brittle to be drawn into wire. Fusing should be started right after spraying to take advantage of the residual heat in the workpiece.

The most common tool used for fusing is the oxyacetylene torch. The flame is adjusted to be either neutral or reducing. With cylindrical parts, the technique involves revolving the part in a lathe or some rotating fixture at a speed so that the operator can observe the surface visually. The base metal adjacent to the sprayed area is heated first to a dim red heat. The torch is gradually moved over towards the surfacing area. The temperature of the hardfacing area is raised to about 1010-1300°C (1850-2370°F) depending on the type of coating used. At these elevated temperatures, the coating fuses to a denser and more uniform surface layer. Moreover, the bond strength between the coating and the substrate increases. The coating thickness changes as a result of fusion. The shrinkage ranges from 15-20% of the as-sprayed coating thickness. Therefore, allowance must be made when the part is sprayed for the decrease in coating thickness that results during fusion. Overheating must be avoided since it causes the surfacing material to flow and can produce other undesirable effects such as excessive diffusion of the coating into the substrate, a decrease in corrosion resistance, distortion and change in hardness.

18.5.2 Controlled Cooling

The rate of cooling from the fusion temperature is often critical. To prevent cracking of the coating may necessitate furnace cooling at a predetermined rate. This is particularly important in situations where the ductility and the coefficient of thermal expansion of the coating and the substrate differ significantly.

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18.5.3 Grinding

Sprayed coatings are usually deposited on components for hardfacing purposes. Therefore, grinding is perhaps the most cost-effective method of finishing. Grinding may be carried out either wet or dry. Excessive pressure exerted during either method of grinding can result in surface checking or smearing. Thus, proper selection of grinding pressure is important to produce a satisfactory finish. The finish on a thermal sprayed coating is generally considered acceptable when the finished surface is dull and finely porous. If the surface is bright or shiny, it is indicative of smearing, caused by the use of a wrong grinding wheel or improper dressing. Usually, grinding is done by a wheel of low bond strength and coarse structure. Grinding wheels are composed of particles of oxides or carbides of a particular type depending on the coating to be ground.

18.5.4 Machining

While even the harder sprayed coatings can be finished by the proper selection of carbide tools, close tolerance work is very difficult to achieve due to the large amount of heat evolved during machining. The wear on the cutting tool is also excessive. Consequently, machining is applied to softer sprayed coatings namely copper and aluminum alloys and low-carbon steel.

18.5.5 Sealing

Sealing is intended to increase the life and corrosion resistance of the coating. Sealing is also applied in order to help in the subsequent finishing operation. Waxes or liquid phenolics are used for coatings on machine components. Waxes should be applied immediately after spraying to make use of the residual heat left in the part for melting and diffusing the wax into the coating. Phenolics can be utilized at any time before finishing. Both waxes and phenolics aid in minimizing the attack of corrosive elements and penetration of oil.

Corrosion-resistant sealers may consist of various vinyls and resin compounds. Some of them may contain pigments of aluminum flakes or other coloring agents. Sealers of this type are used on aluminum or zinc sprayed coating. Aluminum and zinc coatings are extensively used in corrosion prevention of iron and steel. As a general rule, sealers should penetrate the coating completely to obtain their maximum benefit.

18.5.6 Diffusing

Aluminum coating can be diffused into the substrate at an elevated temperature. Diffusing aluminum into steel can be accomplished at about 790°C (1450°F) for 1 hour per inch (1 hr. per 2.54 cm) of thickness. The diffused aluminum coating provides protection against damage by hot gases up to about 870°C (1600°F). After spraying, the workpiece is coated with a bituminous aluminum sealer. The purpose of the seal is to eliminate high temperature oxidation of the aluminum coating at the diffusion temperature.

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18.5.7 Burnishing and Polishing

Soft coatings such as tin, zinc and babbitt can be finished by burnishing. The result is a dense and smooth surface. Burnishing is typically used on surfaces that are subsequently painted or plated.

Every type of sprayed coating can be polished or given a superfinish by honing, lapping or buffing. Superfinishing usually involves stones of increasingly finer grit sizes or laps made of different materials in combination with diamond paste in mesh sizes up to 50,000 (0.2 microns) and an oscillating unit. The resultant finish is a function of the fineness of the abrasive, the speed of the workpiece as well as the oscillation and the applied pressure. The surface finish will improve with increasing speed of the workpiece, decreasing grit size of the abrasive, decreasing pressure and oscillation speed.

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19. BRAZING

Brazing is a joining process which occurs above the liquidus of the filler metal and below the solidus of the base metal. Brazing involves the use of a nonferrous filler metal with a melting point above 450°C (840°F) but always below that of the base metal.(3) The filler metal is distributed in closely fitted joints by capillary action. The filler metal may be preinserted in the joint in the form of a solid strip, washer, ring, powder, shim, wire and subsequently heated above its melting point in order to fill the joint by capillary flow and metallurgically bond the workpieces together upon solidification.

Typically, brazing filler metals are pure copper, copper alloys, silver alloys and aluminum alloys. While pure copper is an excellent brazing material for ferrous metals and alloys, its high melting point may preclude its use due to grain coarsening in the base metal, which may be brought on by the high brazing temperature involved. Copper-, silver- and aluminum-base alloys have lower melting temperatures than pure copper. However, the bond strength of the brazement is less than that obtained with pure copper as the brazing filler metal. Phosphorus-containing filler metals are widely used for brazing copper-base alloys because they are self-fluxing. Aluminum-base alloys can be brazed with other aluminum alloys which have a melting range below the solidus of the base metal. Since the differences between the melting points of the base and filler metals are not as great as those of brazing steels with copper alloys, the brazing of aluminum requires a closer temperature control.

The brazing filler metals classified by the American Welding Society (AWS) are listed in Table XXXIII.

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Table XXXIII
AWS Classification of Brazing Filler Metals.

AWS Classification	Brazing Temperature Range F°	Brazing Temperature Range C°	Nominal Composition (%)
Aluminum-silicon alloys			
BAlSi-2	1110-1150	599-621	92.5 Al, 7.5 Si
BAlSi-3	1060-1120	571-601	86 Al, 10 Si, 4 Cu
BAlSi-4	1080-1120	582-604	88 Al, 12 Si
BAlSi-5	1090-1120	588-604	90 Al, 10 Si
Magnesium alloys			
BMg-1	1120-1160	604-627	89 Mg, 2 Zn, 9 Al
BMg-2a	1080-1130	582-610	83 Mg, 5 Zn, 12 Al
Copper-phosphorus alloys			
BCuP-1	1450-1700	788-927	95 Cu, 5P
BCuP-2	1350-1550	732-843	93 Cu, 7 P
BCuP-3	1300-1500	704-816	89 Cu, 5 Ag, 6 P
BCuP-4	1300-1450	704-788	87 Cu, 6 Ag, 7 P
BCuP-5	1300-1500	704-816	80 Cu, 15 Ag, 5 P
Copper-copper zinc alloys			
BCu-1	2000-2100	1093-1149	99.9 Cu (min)
BCu-1a	2000-2100	1093-1149	99.0 Cu (min)
BCu-2	2000-2100	1093-1149	86.5 Cu (min)
RBCuZn-A	1670-1750	910-954	57 Cu, 42 Zn, 1 Sn
RBCuZn-D	1720-1800	938-982	47 Cu, 11 Ni, 42 Zn
Silver alloys			
BAg-1	1145-1400	618-760	45 Ag, 15 Cu, 16 Zn, 24 Cd
BAg-1a	1175-1400	635-760	45 Ag, 15 Cu, 16 Zn 24 Cd
BAg-2	1295-1550	700-843	45 Ag, 26 Cu, 21 Zn, 18 Cd
BAg-2A	1310-1550	710-843	30 Ag, 27 Cu. 23 Zn, 20 Cd
BAg-3	1270-1500	688-816	52 Ag, 15 Cu, 15 Zn, 15 Cd, 3 Ni
BAg-4	1435-1650	780-899	40 Ag, 30 Cu, 23 Zn, 2 Ni
BAg-5	1370-1550	743-843	45 Ag, 30 Cu, 25 Zn
BAg-6	1425-1600	774-871	53 Ag, 31 Cu, 16 Zn
BAg-7	1205-1400	651-760	56 Ag, 22 Cu, 17 Zn, 5 Sn
BAg-8	1435-1650	780-899	77 Ag, 23 Cu
BAg-8a	1410-1600	766-871	77 Ag, 23 Cu
BAg-13	1575-1775	857-635	54 Ag, 40 Cu, 5 Zn, 1 Ni
BAg-13a	1600-1800	871-982	56 Ag, 42 Cu, 2 Ni
BAg-18	1325-1550	718-843	60 Ag, 40 Cu
BAg-19	1610-1800	877-982	92 Ag, 8 Cu
Precious metals			
BAu-1	1860-2000	1016-1093	37 Au, 63 Cn
BAu-2	1635-1850	890-1010	79.5 Au, 20.5 Cn
BAu-3	1885-1995	1030-1090	34 Au, 62 Cn, 4 Ni
BAu-4	1740-1840	949-1004	82 Au, 18 Ni
Nickel alloys			
BNi-1	1950-2200	1066-1204	14 Cr, 3 Br, 4 Si, 4 Fe, 75 Ni
BNi-2	1850-2150	1010-1177	7 Cr, 3 Br, 4 Si, 3 Fe, 83 Ni
BNi-3	1850-2150	1010-1177	3 Br, 4 Si, 2 Fe, 91 Ni
BNi-4	1850-2150	1010-1177	1 Br, 3 Si, 2 Fe, 94 Ni
BNi-5	2100-2200	1149-1204	19 Cr, 10 Si, 71 Ni
BNi-6	1700-1875	927-1025	11 Br, 89 Ni
BNi-7	1700-1900	927-1038	13 Cr, 10 Br, 77 Ni

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Good bond strength requires that the surfaces of the joint be adequately cleaned prior to brazing and protected during heating.(1,8)

Protection from contamination is accomplished by the application of flux, special atmosphere, or vacuum. Capillary flow, as a condition of good brazing, necessitates that the surfaces of the joint be wet by the liquid brazing filler metal.

The brazing procedure consists of the following major steps:

- cleaning the joint surfaces
- applying flux on the surfaces of the joint
- preheating the joint area to melt the flux
- applying and heating the brazing filler metal at points on the surfaces of the joint
- removing flux residues.

As the molten filler metal is drawn into the joint, the flux is displaced by the brazing metal. Since the braze joint has a small clearance ranging from 0.025-0.64 mm (0.001 - 0.025 in), the filler metal must have a good fluidity to fill the gap properly.

Joint clearance is an important consideration in brazing. Too small a clearance can prevent the flow by capillary action of the molten filler metal throughout the joint. If the clearance is excessive, the filler metal may not fill the joint uniformly. Thus, joint clearance must be assessed in terms of mechanical restraints to metal flow, amount of filler metal needed, possibility of slag entrapment, porosity and the relationship between clearance and capillary force for braze metal distribution. The selection of joint clearance is also affected by the depth of the joint, surface finish and the method of surface protection. Fluxes require a larger clearance than atmospheres. When dissimilar metals are brazed, allowance must be made for the difference between the coefficients of thermal expansion. This is to avoid the development of deleterious tensile stresses during cooling.

Surface finish should be between 30 - 80 microinches. If the finish is too smooth, wetting is inhibited and filler metal distribution by capillary action is not favored. Excessively rough surfaces are on the other hand brazed at high points only resulting in a weak bond.

A shallow joint depth or a small joint area needs a tighter joint clearance than a large joint area. Some of the more typical joint details are illustrated in Figure 143.

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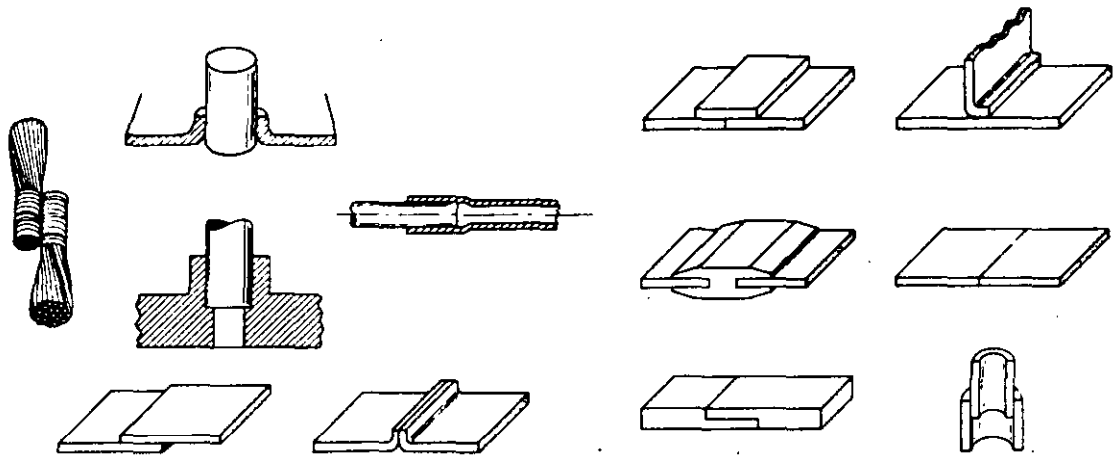


FIGURE 143. Joint details for brazing.

Joints for brazing can be divided into two categories: lap and butt joints. The scarf joint is a modification of the butt joint. Lap joints are easy to fabricate and have an excellent joint efficiency. Their shortcoming is in creating a stress concentration and greater thickness at the joint. These limitations are overcome by the use of butt joints. If the brazed joint is free of significant discontinuities, the joint strength will then be a function of the extent of bond between the filler and base metals, the strength of the filler metal used and the area of the joint. Dry hydrogen, dissociated ammonia and argon constitute the protective atmospheres most commonly employed. Vacuum furnaces are usually reserved for the brazing of reactive metals which otherwise may not be brazed satisfactorily.

19.1 Fluxes and Atmospheres

Protection of surfaces from contaminants both at ambient and elevated temperatures must be ensured to obtain optimum joint strength. Fluxes of various chemical compositions fulfill such a function in brazing processes. Fluxes not only minimize the formation of oxides on metal surfaces but dissolve oxides that are present. Mixtures of fused borax and boric acid are most widely used for brazing steel with copper alloys. Refractory oxides of chromium, beryllium, or aluminum require more active fluxing agents, namely fluorides and chlorides of the alkaline and alkaline-earth metals. In general, flux ingredients include, chlorides, fluorides, borates, borax, boric acid, fluoborates, wetting agents, and water. There is no universal flux for all brazing applications. Classifications of fluxes are done in accordance with their effectiveness on a particular base metal used and the brazing temperature range involved, as indicated in Table XXXIV.

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Table XXXIV
Brazing Fluxes.

AWS Brazing Flux Type No.	Base Metals Being Brazed	Recommended Filler Metals	Recommended Useful Temp. Range F°	Recommended Useful Temp. Range C°	Major Flux Ingredients	Forms Available
1	All brazeable aluminum alloys	BAlSi	700-1190	371-643	Chlorides Fluorides	Powder
2	All brazeable magnesium alloys	BMg	900-1200	482-649	Chlorides Fluorides	Powder
3A	Copper & copper-base alloys (except those with aluminum) iron base alloys; cast iron; carbon & alloy steel; nickel & nickel base alloys; stainless steels; precious metals	BCuP Bag	1050-1600	566-871	Boric Acid Borates Fluorides Fluoborates Wetting Agent	Powder paste liquid
3B	Copper & copper-base alloys (except those with aluminum); iron base alloys; cast iron; carbon & alloy steel; nickel & nickel base alloys; stainless steel; precious metals	BCu BCuP BAg BAu RBCu Zn BNi	1350-2100	732-1149	Boric Acid Borates Fluorides Fluoborates Wetting Agent	Powder paste liquid
4	Aluminum bronze, aluminum brass & iron or nickel base alloys containing minor amounts of Al &/or Ti	BAg (all) BCuP (copper base alloys only)	1050-1600	566-871	Chlorides Fluorides Borates Wetting Agent	Powder paste
5	Same as 3A & B above	Same as 3B (excluding BAg through -7)	1400-2200	760-1204	Borax Boric acid Borates Wetting Agent	Powder paste liquid

Dry hydrogen is capable of reducing oxides of iron and copper. Therefore, a hydrogen atmosphere serves as a flux for steel and copper. It should be recognized that the use of atmospheres which contain hydrogen, oxygen, or nitrogen may cause an embrittlement in titanium, tantalum, columbium (niobium) and zirconium. Non-deoxidized copper is also embrittled by hydrogen.

Some brazing filler metals contain deoxidizers which make them self-fluxing. Phosphorus and lithium are two examples of such deoxidizers. Fluxes may be available as pastes, powders and gases. Regardless of the form at room temperature, a flux must be liquid or gaseous at the brazing temperature in order to have a complete contact between the flux and the base metal so as to remove the oxide and protect the surface of the joint.

When brazing is completed, the residual flux should be removed since it would induce corrosion in the joint. Washing in hot water followed by air drying is the most common practice for removing residual flux. Chemical and mechanical cleaning may be necessary on tenacious flux residues.

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Inert atmospheres are used in induction brazing of reactive metals such as titanium, zirconium and beryllium, and refractory metals. A large variety of controlled atmospheres are available for brazing of different metals. These controlled atmospheres are the result of combustion of gas mixtures. They require a moisture or dew point control because of the undesirable consequence of moisture that may be present during brazing. The dew point must be carefully controlled in dry hydrogen brazing.

Vacuum chambers are very effective methods for protecting metals from contamination during brazing. Vacuum is the most reliable environment for brazing highly reactive metals since gases entrapped in close-fitting joints can be removed, which could cause severe embrittlement or even disintegration at the brazing temperature.

19.2 Brazing Techniques

Brazing processes are categorized according to the method of heating. Thus, there are

- furnace brazing,
- torch brazing,
- induction brazing,
- resistance brazing,
- dip brazing,
- exothermic brazing.

Furnace brazing is used in brazing small assemblies under mass-production conditions. The parts to be joined are cleaned, fluxed, assembled and the filler metal is preplaced. The assembly is heated in the furnace using a suitable protective atmosphere, which is base material dependent. The flux cleans the surfaces of the joint and the filler metal is attracted into the joint by capillary forces and makes a metallurgical bond with the base metal. The proper brazing atmosphere helps the wetting of the joint surfaces by the molten filler metal.

Low-carbon and low-alloy steels are most often brazed with pure copper and occasionally with silver alloys. The furnace atmosphere used must satisfy two basic requirements: one, to prevent oxidation and two, to promote wetting. Both requirements necessitate the application of a reducing atmosphere. Atmospheres most often employed in brazing steels involve exothermic-based, endothermic-based and nitrogen-based atmospheres. They can be prepared with any carbon potential in the range of 0.2% to in excess of 1.0% C to match the carbon content of the workpiece so as to avoid carburizing or decarburizing of the base metal. Air atmosphere furnaces can be utilized for workpieces that are protected with a paste flux. While copper is the preferred filler metal for furnace brazing of carbon and low-alloy steels without flux in a reducing atmosphere, the silver-base filler metals are preferred in brazing dissimilar metals (e.g., carbon steel to stainless or to nonferrous metal) and in applications where the high brazing temperature of pure copper [1090-1150°C (2000-2100°F)] would have undesirable consequences on base metal properties.

Unless a flux is used, furnace brazing of stainless steel is carried out in a protective atmosphere of dry hydrogen or dissociated ammonia. These atmospheres are reducing in character and capable of effectively preventing oxidation at the brazing temperature and promote wetting by the filler metal. Argon as an inert gas, or a vacuum are also appropriate media for brazing stainless steels. Exothermic atmospheres cannot be used for brazing stainless

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steels. The type of furnace atmosphere employed is dependent on the extent of protection of the base metal involved, flow characteristics of the filler metal and cost. Air atmosphere may be utilized with a suitable flux and filler metals of lower melting points.

Heat-resistant alloys are typically furnace brazed in controlled atmospheres or in a vacuum. Pure dry hydrogen and inert gases can also be used. Hydrogen with a dew point of -51°C (-60°F) dissociates the oxides of most metallic elements present in heat-resistant alloys.

Furnace brazing of aluminum alloys is implemented in ambient air or inert gases with a dew point of about -40°C (-40°F). Flux slurry is used to provide additional protection to aluminum alloys, which may be applied by dipping, brushing or spraying. Vacuum brazing of aluminum eliminates the necessity of using flux.

Various protective atmospheres may be used for furnace brazing of copper and copper alloys as long as the deleterious effect of hydrogen at the brazing temperature is eliminated. This implies dew point control. The application of a flux is not necessary in dry hydrogen and prepared nitrogen-based atmospheres.

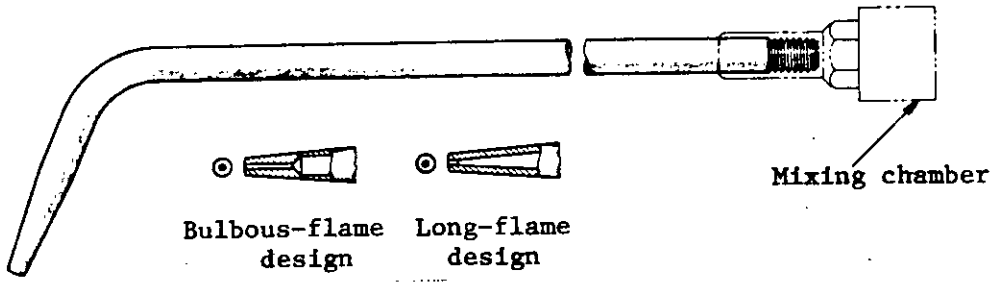
Reactive and refractory metals, owing to their high reactivity with oxygen, hydrogen, nitrogen and carbon, require vacuum or dry inert gases for brazing.

Stop-offs may be required to prevent the flow of filler metal beyond the joint. Stop-off materials may be milk of magnesia, or various proprietary chemical formulations of oxides, thinners, gelled vehicles, water solution of chromic acid and graphite. The specific type of stop-off material used is governed by the base material and atmosphere involved in the brazing process.

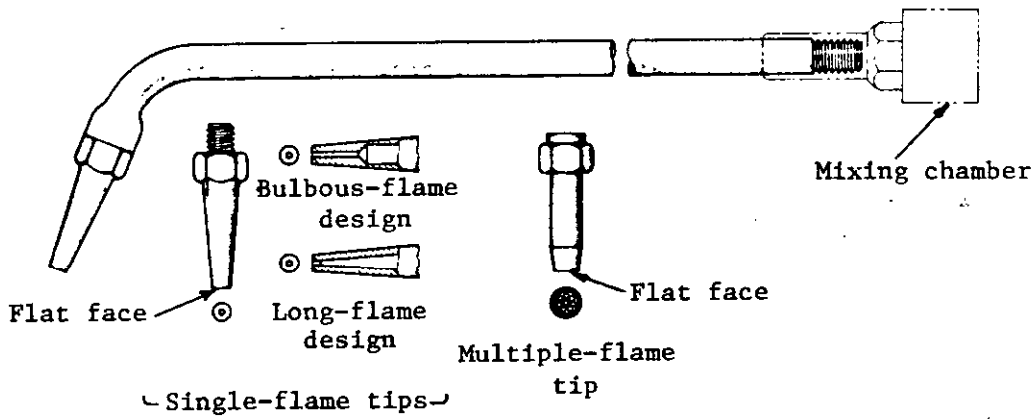
In torch brazing, heat is provided by a gas flame or flames to raise the temperature of the joint so that the filler metal flows by capillary action in the close-fitting joint. While torch brazing is the most versatile brazing method, the quality of the brazement is dependent on operator skill. It is easy to overheat thin sections and localized areas, which can produce an appreciable amount of distortion. Filler metal can be applied manually from a wire or rod during brazing. Filler metal may also be preplaced in the form of a ring, washer, shim or insert shaped to fit the joint geometry. Multiple flames can be provided to heat the joint uniformly in order to minimize distortion. The principle of torch brazing is analogous to furnace brazing. Only the technique and the method of applying heat differ. Manual torch brazing is customarily used with small quantities of production.

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The flame is either neutral or reducing and is obtained from oxyfuel or air-fuel gas. The types of torches used depend on the gas mixtures: air-natural gas, oxyacetylene, air-acetylene, oxy-hydrogen and others. Figure 144 shows a one-piece and a two-piece tip and tube assembly employed in oxyacetylene brazing.(5)



(a)



(b)

FIGURE 144. Oxyacetylene torch for brazing.
(a) one-piece tip and tub assembly,
(b) two-piece tip and tub assembly.

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Air-natural gas gives the lowest flame temperature. Oxygen mixed with natural gas, or propane, or butane yield a higher flame temperature. The tip size and the number of tips used will affect the brazing heat input.

Torch heating requires the application of flux or filler metals with self-fluxing characteristics such as copper-phosphorus. The inner cone of the flame should not come in contact with the joint (except for preheating) since the melting of the base metal, dilution of the filler metal, overheating of the flux and evaporation of low melting point constituents would have an adverse effect on the quality of the brazement.

In induction brazing, a high-frequency electric current is passed through a coil which induces heat by means of I^2R losses or eddy current in the workpieces placed within the coil.(1) The fluxed workpieces with the preformed filler metal placed in the proper location are heated by an inductor to the specified brazing temperature. Prepared atmospheres are not used in induction brazing. The depth of heating is a function of the frequency of the alternating current source. The heat penetration is inversely proportional to the frequency. Typically, frequencies of 60 Hz to 450 kHz are involved in induction brazing. In addition, factors that affect heating include number of turns, space between turns, shape of the inductor, air gap between inductor and workpiece, presence of sharp corners of the workpiece within the magnetic field, presence of metallic shields within the inductor and ac power input. Heat distribution occurs by conduction.

Inductors are constructed with different contours to match the geometry of the workpieces. Some are exhibited in Figure 145.

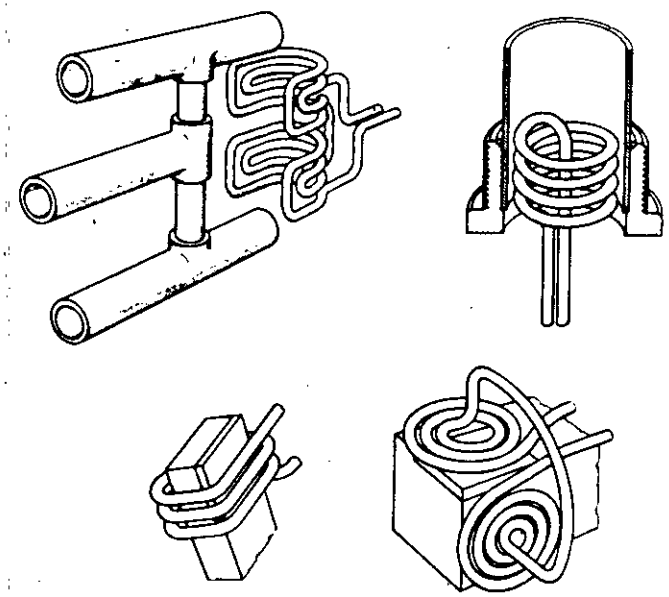


FIGURE 145. Various shapes of inductors used for brazing.

The chief advantage of induction brazing is rapid heating which in turn decreases oxidation during brazing and minimizes cleaning after brazing. The controlled localized heating in induction brazing produces very little warpage. The components of induction brazements are held together by fixturing. The

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design of fixturing must satisfy the size, shape, support and holding requirements of the assembly. Induction brazing is well suited for joining dissimilar metals. The inductor for such an application is so designed as to admit the component in the weaker part of the magnetic field that heats more rapidly.

Resistance brazing consists of the same equipment that is utilized in resistance welding. The application of flux and preplacement of filler metal have the same purpose as in other brazing processes. When flux is not used, protection is provided by means of reducing or inert atmospheres and vacuum. Resistance brazing in air is possible when brazing is carried out immediately after cleaning.

The choice of electrodes is governed by the conductivity of the workpieces, the geometry of the joint, the appearance requirement of the brazement in terms of markings, amount of brazing to be done and cost.

Conductors of high electrical resistance are used for resistance brazing, namely carbon-graphite, electrographite, refractory metal electrodes, and chromium copper.

Dip brazing in molten salt generally known as salt-bath dip brazing makes use of a fused salt bath to heat the joint and melt the filler metal. The salt composition is so selected as to behave as a flux as well.

A variety of salts and fluxing agents can be utilized in salt-bath brazing of carbon and low-alloy steels. They are neutral chloride salts with or without fluxing agents, carburizing and cyaniding salts. When brazing is carried out in a neutral chloride salt bath, a flux is normally preapplied to the assembly before dipping. Hardening and carburizing can be combined with dip brazing. Preheating the assembly has the advantage of driving off the moisture from the preapplied flux and the fixture. This practice avoids spattering upon immersion of the assembly in the molten salt bath. Cyanide baths have their own fluxing agents and allow treatments such as austenitizing, carburizing and cyaniding during brazing. The selection of proper quenching media is important in order to prevent an explosion hazard as would be the case in a nitrate-nitrate quenching bath. The filler metal must solidify before quenching may be permitted. Typically, this is done in the following fashion. The brazement is transferred from the cyanide salt bath to a neutral chloride salt held below the solidus of the filler metal but within the austenitizing temperature regime. Then quenching is carried out in the nitrate-nitrate bath. The quenched brazement is air cooled, rinsed and tempered as necessary. Fixtures must be thoroughly rinsed and dried before reuse in the cyanide and neutral chloride baths to avoid explosion.

Salt-bath furnaces are metal or ceramic pots that are either internally or externally heated with gas, electrical resistance, or oil.(5) (See Figure 146.)

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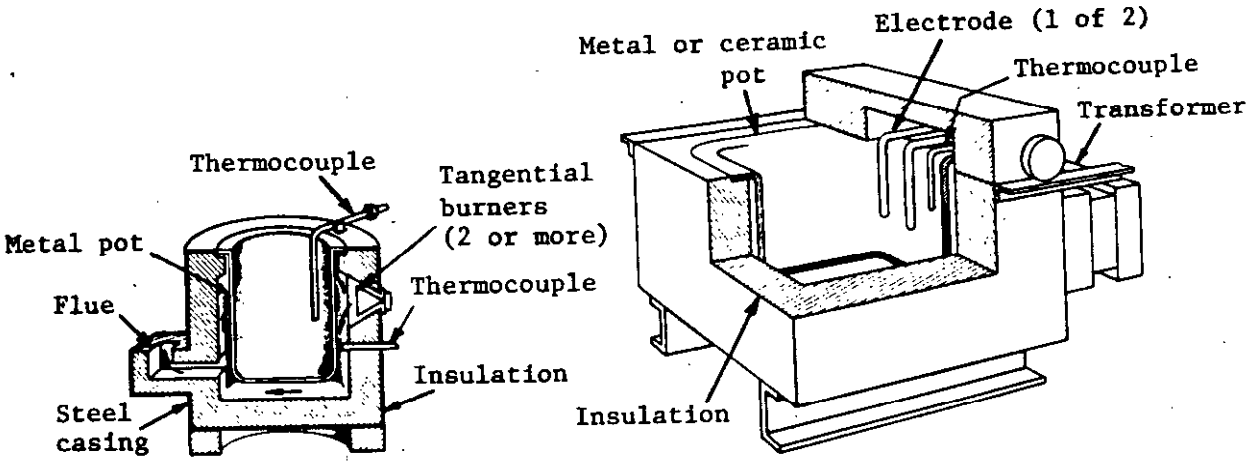


FIGURE 146. Dip brazing furnaces: (a) externally heated (gas or oil) (b) internally heated (electrical resistance)

A liquid metal bath may also serve as a heating medium for brazing. The metal-bath brazing can fulfill the role of providing filler metal and heat source. The flux floats on top of the metal-bath.

The molten flux bath method consists of a container for holding the flux, which is heated to the brazing temperature. Preforms of filler metal are placed in the joint prior to brazing. The workpieces are held together in fixtures. The whole assembly is usually preheated to evaporate entrapped moisture. The preheating temperature is close to the melting point of the flux.

Exothermic brazing takes advantages of the heat of reaction between chemical reactions. Metal-metal oxide reactions are most appropriate for exothermic brazing processes in which the filler metal is preplaced.

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20. SOLDERING

Soldering is conducted at a temperature below 450°C (840°F). The filler metal is heated above its liquidus, while the base metal remains well below its solidus. As in brazing, the filler metal is distributed in the close-fitted joint by capillary action. Soldering is analogous to brazing in many ways except for the temperature limitation. The functions of soldering fluxes are identical with brazing fluxes in that they protect the surface from contamination and enhance wetting of the base metal by the solder.

The interaction between the solder and the base metal depends on their chemical compatibility and soldering temperature involved. A metallurgical bond can develop and an intermetallic compound may form at the interface depending upon the type of filler and base metals involved. The most common solders are the lead-tin alloys.(7) Compositions of 50% lead-tin or 60-40% lead-tin are the most universal.

The melting temperature and the spread between solidus and liquidus of lead-tin solder alloys vary with composition. The melting point can range from 182°C (360°F) to 315°C (600°F). The phase diagram of lead-tin alloy system is presented in Figure 147.

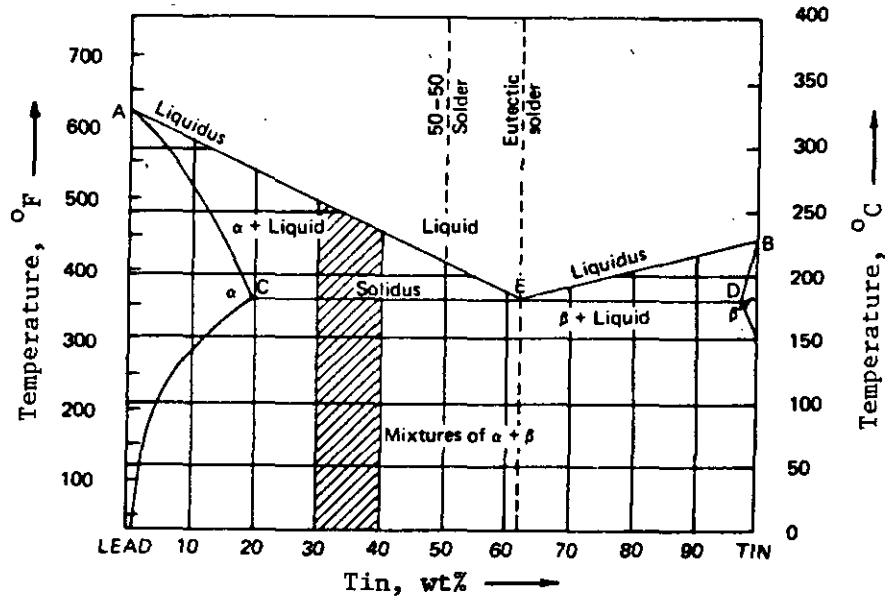


FIGURE 147. Phase diagram of Pb-Sn system.

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Customarily, soldering is used in low-strength joint applications as in tin-plated containers, radiators, electrical and electronic components. In sheet-metal work, lock seams, spot welded joints can be sealed with solder.

During solidification, the joint must remain undisturbed to prevent fracture of the joint. After soldering, the flux residue must be removed in order to avoid corrosion damage in the joint. Scrubbing the part while it is immersed in hot water is a common method to remove flux residue. If non-corrosive flux is employed, no flux removal is necessary. Impurities of aluminum, antimony, arsenic, bismuth, cadmium and copper present in Pb-Sn solders are harmful in various ways. Aluminum in excess of 0.005% causes lack of adhesion, grittiness and surface oxidation of the solder. Dewetting, which is a retraction of the molten solder after it has first spread over a given surface area is promoted by small amounts of arsenic. Cadmium decreases wetting. Excessive copper content in the form of an intermetallic compound settles to the bottom of a solder bath. Antimony influences the spread of a solder. Grittiness and dewetting are also adversely affected by sulfur and phosphorus. Bismuth causes discoloration. Table XXXV contains the nominal compositions and temperature ranges of lead-tin solders.

TABLE XXXV

Lead-Tin Solders

ASTM solder classification[a]	Composition, wt. %		Solidus		Liquidus		Melting range	
	Tin	Lead	°C	°F	°C	°F	°C	°F
5	5	95	300	572	314	596	14	24
10	10	90	268	514	301	573	33	59
15	15	85	225	437	290	553	65	116
20	20	80	183	361	280	535	97	174
25	25	75	183	361	267	511	84	150
30	30	70	183	361	255	491	72	130
35	35	65	183	361	247	477	64	116
40	40	60	183	361	235	455	52	94
45	45	55	183	361	228	441	45	80
50	50	50	183	361	217	421	34	60
60	60	40	183	361	190	374	7	13
70	70	30	183	361	192	378	9	17

a. See ASTM Specification B32, *Solder Metal*.

Besides lead-tin, other low melting temperature alloys are available for soldering. These are tin-antimony, tin-silver, tin-zinc, tin-lead-silver, tin-lead-antimony, cadmium-silver, cadmium-zinc, zinc-aluminum, indium-based alloys, lead-silver, and other binary as well as ternary systems. The various solder alloys including the precious metal solders were developed to meet specific requirements and applications.

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Solderability of metals is determined by a simple vertical dip test, which is visually examined for wetting characteristics. Most often, base metals used in soldering are selected for electrical and thermal conductivity. Then, their solderability is evaluated. Precious metals tend to have excellent solderability. This is because of either the absence of surface oxides in some cases, or because of the rapid displacement of surface oxides by lead-tin solders. Fluxes of low chemical activity are sufficient to promote wetting on noble metals.

Copper, brass and bronze are easy to solder. Copper forms an intermetallic alloy with tin. Since copper oxidizes readily, the application of flux is needed to ensure good wetting. Soldering of carbon and low-alloy steels requires a greater attention than that of copper-base alloys in terms of choice of flux needed. Aluminum, stainless steel and the reactive as well as the refractory metals are quite difficult to solder. This problem can be circumvented by preplating or precoating these base metals with a solderable metal. Precoating also makes ceramics solderable. Tin, copper, silver, cadmium and binary alloys of tin-lead, tin-zinc and tin-copper are utilized for this purpose. Precoating increases the rate of soldering and reduces the need to use aggressive fluxes. The solderability of coated metals depends on the coating involved. The judgement is based upon melting, or solubility of the coating at the soldering temperature. Metallic coatings are applied by either hot dipping or electroplating. The nonmetallic coatings which include lacquers hinder solderability.

Soldered joints are used in low-strength applications.(1) Examples of soldered joints are displayed in Figure 148.

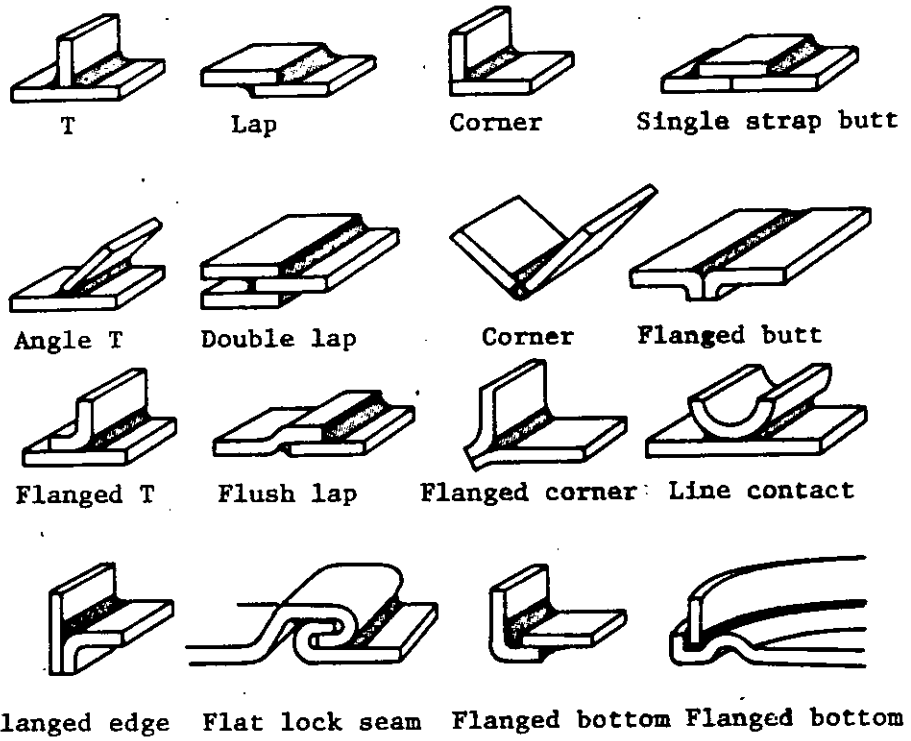


FIGURE 148. Examples of soldered joints.

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Precleaning of joint surfaces to remove oil, grease, paint, dirt and markings are important prerequisites of a satisfactory soldered joint. Degreasing, acid cleaning and mechanical methods constitute the surface preparation techniques. Fluxes can be inorganic or organic in nature mixed with vehicles of one kind or another. The flux residue must be removed after soldering to minimize subsequent corrosion in the joint. This is especially true of the chemically more active fluxes.

Soldering processes are classified on the basis of heating methods employed. Accordingly, there are torch, hot dip, induction, resistance, furnace, infrared soldering and the traditional soldering iron. Wave and ultrasonic soldering techniques are modifications of the hot dip soldering process.

If the soldered joint is determined to be unsatisfactory to meet the intended service requirements, the joint is repaired. The repair procedure consists of desoldering. The solder is remelted and removed under a negative pressure (vacuum) or by the use of a copper braid to wick the liquid filler metal from the joint. Resoldering involves recleaning and application of new flux and solder.

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21. UNDERWATER WELDING AND CUTTING

Underwater welding has several modifications in terms of work implementation in an underwater environment.[1] Underwater welding may be divided into the following categories:

- wet or open-water welding
- dry-spot welding
- dry-box welding
- habitat welding
- chamber welding and
- remote underwater welding.

These classifications reflect the specific underwater environment of the workpiece, the arc and the diver or welder. Conventional welding processes are used in underwater environments. The primary difference has been in the selection of electrodes and electrode coverings so as to minimize the effect of the moist environment. Shielded metal arc welding is most often used in wet welding. GMAW, GTAW, PAW and FCAW need a dry environment which are provided by means of boxes and chambers. The effects of underwater welding can be manifested in quenching, hydrogen embrittlement and weld quality problems. Porosity and slag inclusions tend to be more of a problem in underwater welding due to the rapid solidification of the weld. The aqueous environment has a number of other effects on welding processes. The arc is more stabilized than it is under atmospheric conditions. Penetration is deeper and the metal transfer is faster. The hydrostatic pressure constricts the arc and increases its temperature. In general, underwater welding requires a higher weld current and an increasing pressure of the shielding gas with increasing water depth. The hydrostatic pressure may lead to decomposition of some flue gases and liquefication of others.

Underwater cutting uses the same methodology as surface cutting. The heat losses are however greater in an underwater environment, which must be compensated for. Given the proper care and equipment, underwater welding can result in welds similar to those made in air. However, there are a number of differences such as visibility, buoyancy effects (or work instability), accessibility, impediment of the diving suit, restriction of movement and more rigorous safety procedures in general.

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22. FITNESS-FOR-PURPOSE

Nature is not perfect. It logically follows that neither construction materials nor welded structures are free from discontinuities or flaws. Weld repair is no exception. Fortunately, not all discontinuities are harmful. Therefore, one of the engineering tasks must be directed towards defining the real significance of weld discontinuities, when they occur. Obviously, the only time weld acceptance standards of any type are needed is when a discontinuity has been clearly located, identified, sized, its orientation determined and its structural significance questioned by an engineering criterion. Engineering critical assessment of weld discontinuities is done for the purpose of defining the acceptability of discontinuities present in a weldment without a sacrifice in reliability. Various fracture-mechanics concepts and principles provide a quantitative framework for such an assessment.

A good failure analysis requires involvement of people from a variety of disciplines, production people inclusive. Broad consultation is the very essence of valid conclusions to be derived especially from highly complex situations. A given expert tends to see all failures in the light of his or her specialized training. All possible factors must be evaluated in a balanced way.

The significance of any given discontinuity should be viewed in the context of fitness-for-purpose of the welded construction. Fitness-for- purpose is a fracture-mechanics based philosophy, which attempts to bring a balance among three essential elements of welded structures: (1) Quality, (2) Reliability and (3) Economy.(1) Fitness-for-purpose is not a constant. It can vary depending on the service requirements of the structure and on the properties of the weld and base metals.

A distinction must be made between the words discontinuity or flaw, and defect due to their implied meaning. Discontinuity or flaw does not have a compelling negative implication. The word defect connotes that something is wrong with the material if it contains a defect. This suggests that a corrective procedure or rejection should be imposed. A definition of discontinuity and defect by the American Society of Mechanical Engineers is as follows. Discontinuity or flaw is an interruption in the normal physical structure of a material or a product. It implies neither a lack of conformance with specification requirement nor a lack of fitness-for-purpose. Discontinuities may or may not be rejectable. A defect is a condition which must be removed or corrected. Defects are always rejectable either because of a lack of conformance to specification requirements, or a lack of fitness-for-purpose, or both.

22.1 Weld Discontinuities

The general characterization of discontinuities may be divided into three categories:(1)

1. Design
2. Welding processes
3. Metallurgy (or Material)

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1. Design Related Discontinuities

- design or structural details
- type of weld joint
- changes in cross-section

2. Welding Process Related Discontinuities

- undercut
- slag inclusion
- porosity
- misalignment
- overlap
- tungsten inclusion
- backing left on
- shrinkage
- geometric
- oxide films
- surface roughness
- incomplete fusion
- inadequate joint penetration
- crater
- burn through
- concavity
- convexity
- sharp reinforcement angle
- spatter
- arc-strikes (arc-burns)
- underfill (lack of fill-in)

3. Metallurgical Discontinuities

- cracks or fissures
- fish-eyes
- microstructural changes and segregation
- lamellar tear or base plate delamination

In decreasing order of harmfulness, weld discontinuities may be classified as

(a) crack or crack-like
(b) geometric
(c) incomplete fusion (IF) and inadequate joint penetration (IJP)
(d) slag inclusion
(e) porosity

There are two other ways of categorizing weld discontinuities from a fracture mechanics point of view, specifically:

1.1 Planar or
1.2 Nonplanar (or volumetric)

and

2.1 Surface
2.2 Buried (or internal)
2.3 Through-thickness

The type of weld discontinuities that can occur in welds depends on

1. welding process used
2. inspection method applied
3. type of weld made
4. joint designed and fit-up obtained
5. material utilized
6. working and environmental conditions

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The most frequent weld discontinuities found during manufacture, when categorized on the basis of welding processes most commonly used are shown below:(19)

Welding Processes	Ranking	Weld Discontinuity
Shielded Metal-Arc Welding	1	Slag
	2	Porosity
	3	IF/IJP
	4	Undercut
	5	Others
Submerged-Arc Welding	1	IF/IJP
	2	Slag
	3	Porosity
	4	Others
Fluxed-Cored Arc Welding	1	Slag
	2	Porosity
	3	IF/IJP
	4	Others
Gas Metal-Arc Welding	1	Porosity
	2	IF/IJP
	3	Others
Gas Tungsten-Arc Welding	1	Porosity
	2	Tungsten inclusion

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Ranking of discontinuities when established on the basis of NDE methods and manual versus automatic welding used are as follows.

Detection by RT

Manual Welding		Automatic Welding	
Ranking	Discontinuity	Ranking	Discontinuity
1	Slag	1	IF/IJP
2	Porosity	2	Cracks
3	IF/IJP	3	Slag
4	Cracks	4	Porosity

Detection by UT

Manual Welding		Automatic Welding	
1.	Slag	1.	IF/IJP
2.	IF/IJP	2.	Slag
3.	Porosity	3.	Porosity

Detection by Visual Means

Manual Welding

1. Undercut
2. Surface porosity
3. Slag
4. Undesirable weld profile
5. IF/IJP
6. Cracks at craters

The overall ranking of discontinuity types based on observation and detection is given below.

1. slag
2. IF/IJP
3. porosity
4. undercut
5. crack

22.2 Causes and Cures of Major Weld Discontinuities

There are several factors responsible for causing a discontinuity to occur in a weld. Equally, different remedies exist for the elimination of specific weld defects. The flow charts presented in Figures 149 through 154 contain the causes and cures for the major weld discontinuities.

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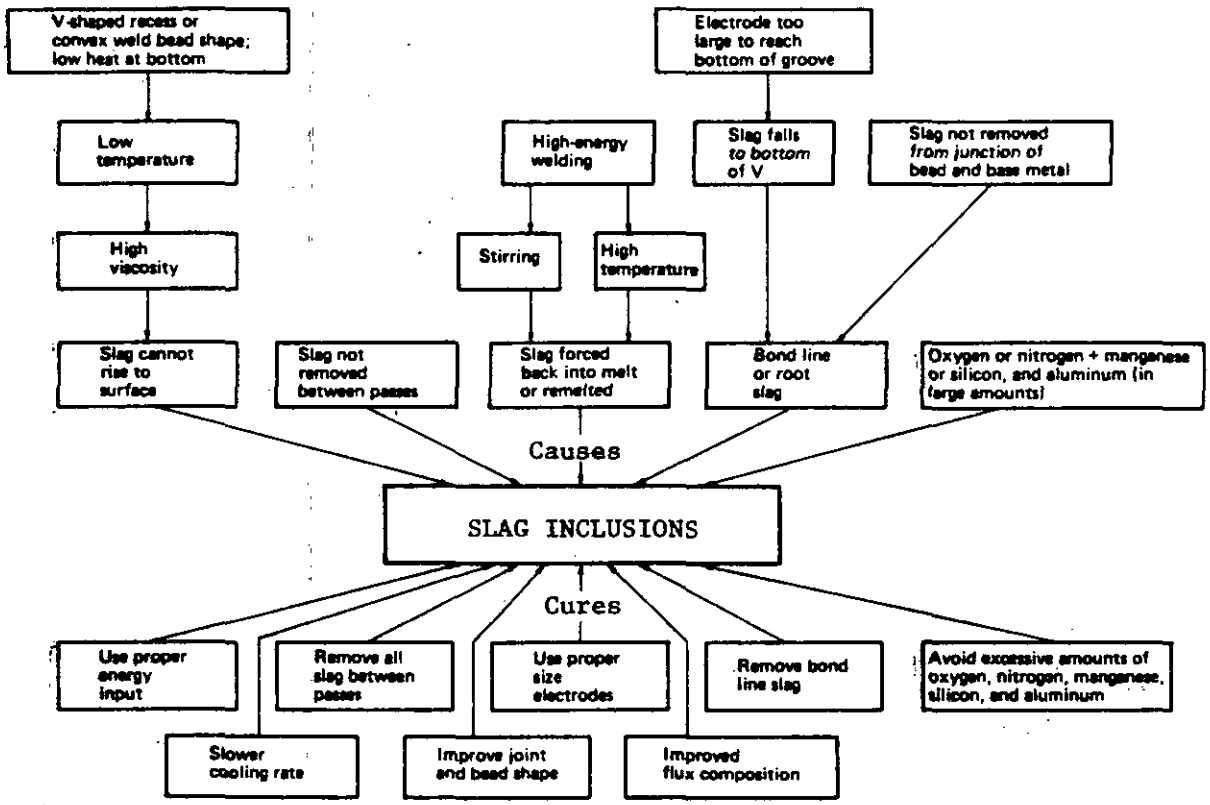


FIGURE 149. Causes and cures for slag inclusions.

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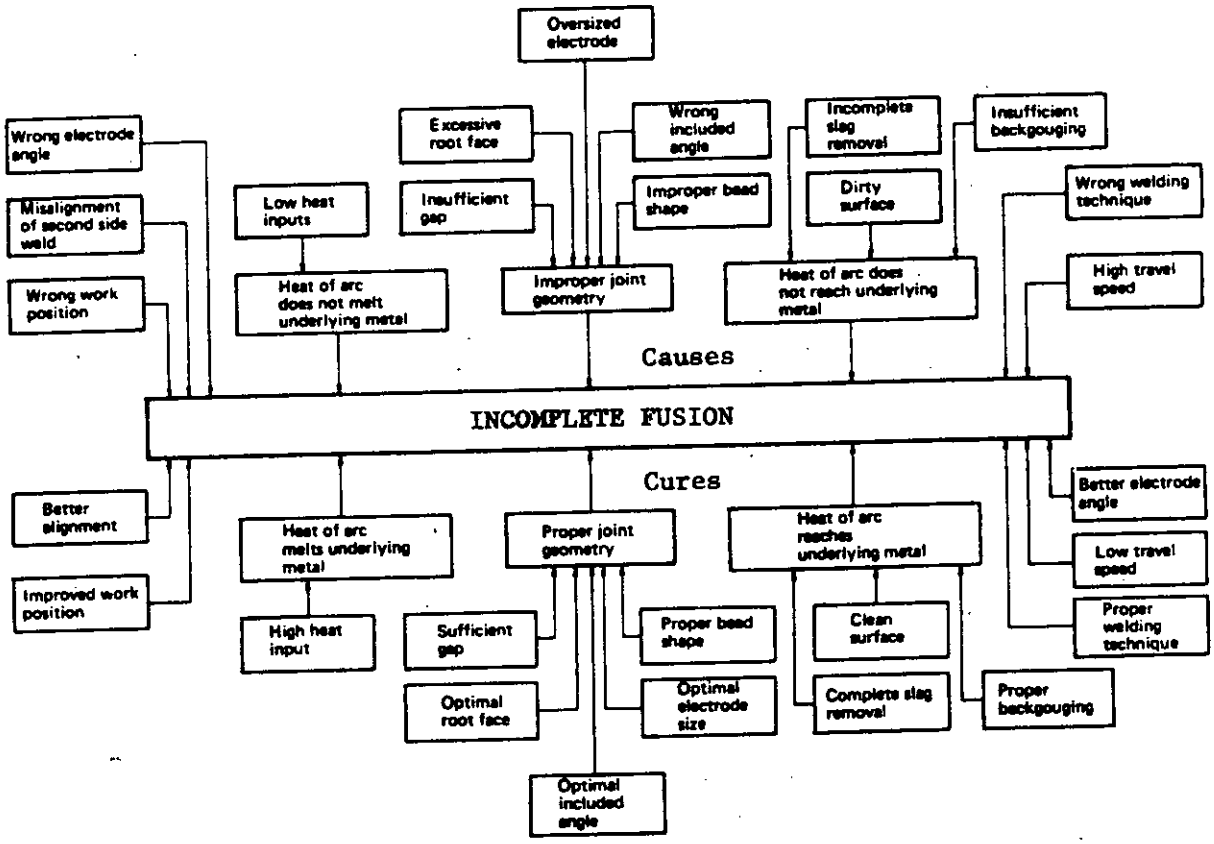


FIGURE 150. Causes and cures for incomplete fusion.

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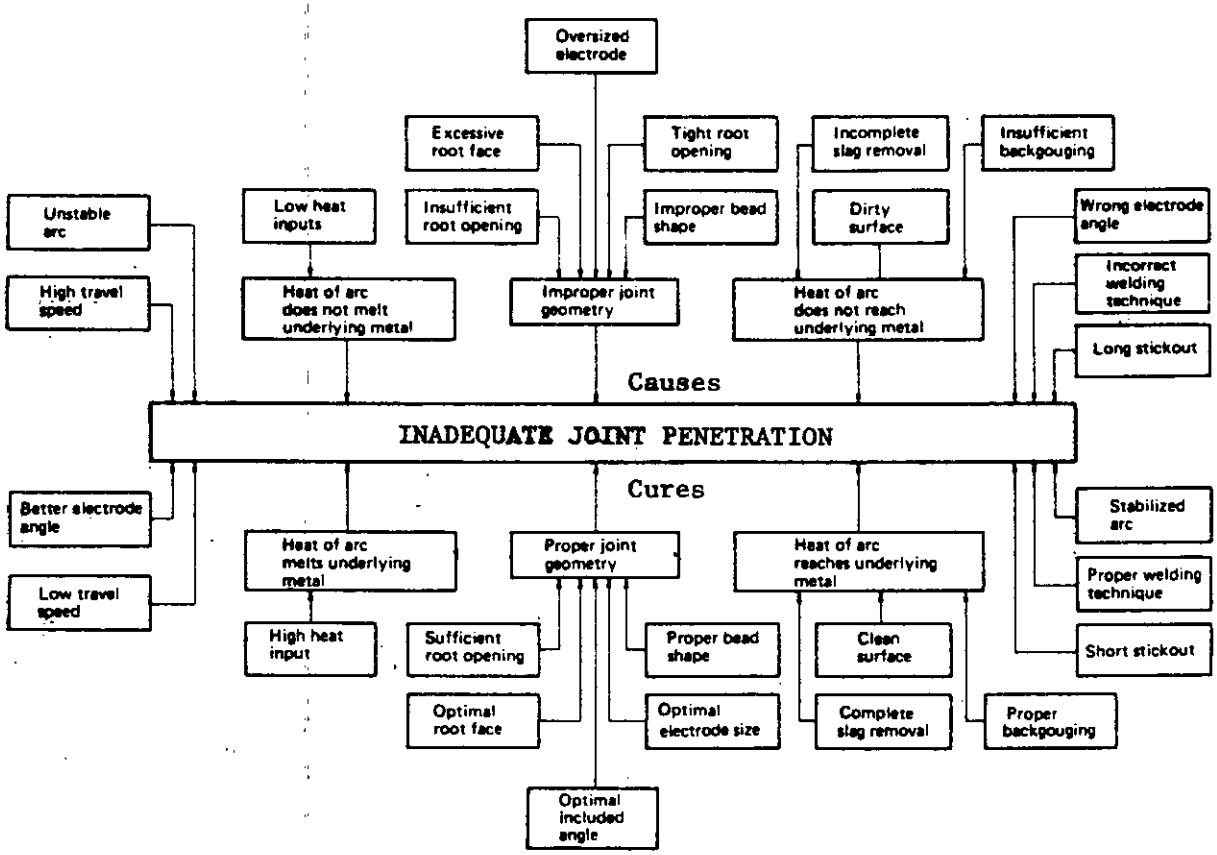


FIGURE 151. Causes and cures for inadequate joint penetration.

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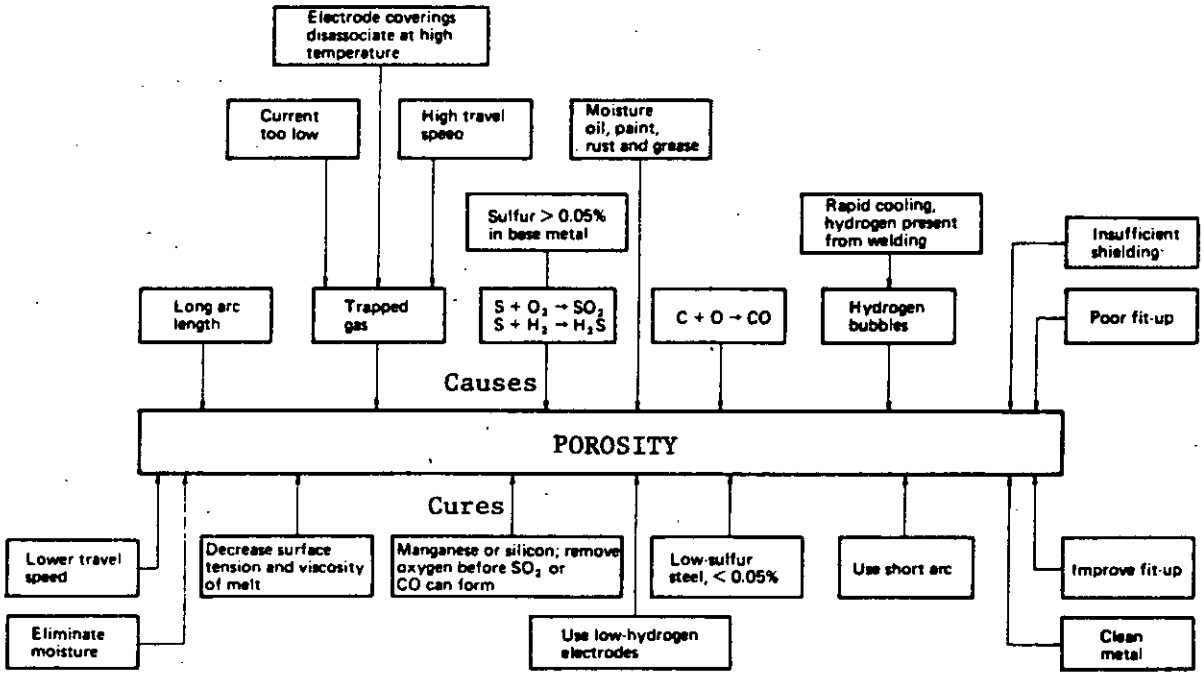
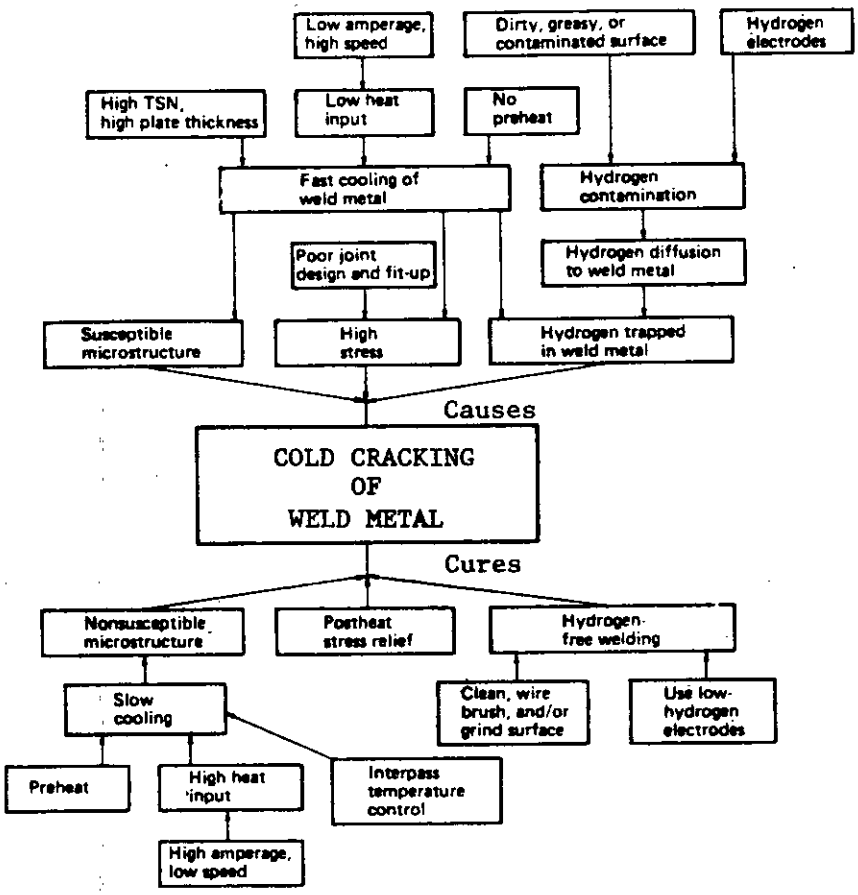


FIGURE 152. Causes and cures for porosity.

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(TSN: Thermal Severity Number, which is four times the total plate thickness capable of removing heat from the joint and thus a measure of the member's ability to serve as a heat sink.)

FIGURE 153. Causes and cures for cold cracking.

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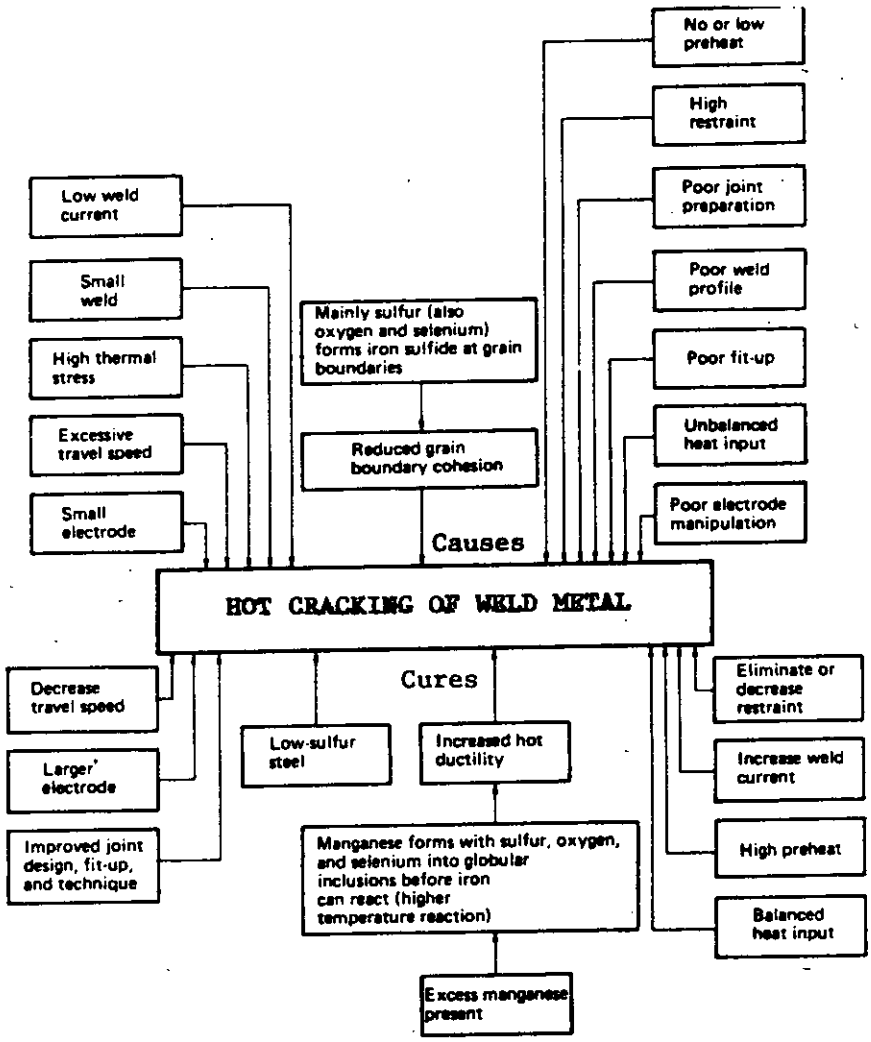


FIGURE 154. Causes and cures for weld metal hot cracking.

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Geometric discontinuities are in general associated with the weld profile, although misalignment, weld spatter, arc-strikes, overlap, excessive reinforcement, underfill and burn-through also fall in this category. These discontinuities are almost always caused by poor welding techniques. The morphology of these geometric discontinuities varies from type to type. One way of controlling the morphology of fillet welds in particular is by selecting electrodes with good wetting characteristics.

Weld undercut is a gap or furrow located at the toe of a weld and occurs when the weld melt does not completely fill the gap at the surface of the groove to form a smooth junction at the weld toe.

To minimize undercut involves the use of one or more of the following steps;

- decrease in current, travel speed and electrode diameter. These allow managing the weld pool size
- change in electrode angle so that the arc force holds the weld metal in the corners
- avoid too much weaving
- maintain constant travel speed
- proper backing
- change the surface tension and viscosity of the molten weld metal.

Undercut is the least desirable type among the geometric discontinuities due to its harmful effect on fatigue strength. Large reinforcement angles at toes of fillet welds are quite deleterious.

22.3 Significance of Weld Discontinuities

The significance of weld discontinuities is evaluated on the basis of their effect on the life and the integrity of a welded structure. The full spectrum of this effect can range from harmful to innocuous. Therefore, weld discontinuities are ranked according to the nature of their influence upon the behavior of the structure, which is a function of many factors. Namely,

Type	Environmental conditions
Size	Design conditions
Geometric shape	Structural discontinuities
Quantity	Distribution
Location	Material thickness
Orientation	Material chemistry
Acuity of the extremities	Microstructure in general
Type and magnitude of operating stresses	Size of weldment
	Rate of loading
Transients	Welding processes and procedures

The fracture behavior of a given metallic material from which a structure is fabricated will, in general, be a function of material properties, presence of discontinuities, level of stress acting at the tip of a crack-like discontinuity and the mechanism of crack extension. Fracture mechanics is a tool to assess the tolerable size of a given discontinuity present in a weldment. Fracture mechanics involves a thorough analysis of stress, discontinuity, material and environment.(1)

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Stress analysis consists of measuring, or calculating all the stresses that operate at or near the tip of a given discontinuity. Non-destructive examination is involved in discontinuity analysis; namely, detection, identification, location, sizing and orientation of the discontinuity. Material analysis entails destructive testing of the base metal and the weld metal in order to determine their mechanical properties, including fracture toughness. Fracture toughness is a measure of the materials resistance to crack propagation.

A discontinuity that does not affect the strength of a statically loaded member can, however, significantly reduce the fatigue strength of the member as a result of a concentration of stress that occurs at the discontinuity. In a statically loaded member the increased stress is relieved by plastic strain of the material near the discontinuity. However, in a member subjected to a fluctuating load, the range of stress at the discontinuity remains much higher than in the surrounding material and causes a reduction in fatigue strength.

The significance of a discontinuity cannot fully be established until it is determined what modes of failure would be affected by the presence of a given type of discontinuity.

In general, the following failure modes are recognized:

- brittle fracture
- fatigue
- elastic-plastic
- stress corrosion
- leakage
- instability (buckling)
- creep
- general yielding due to overload
- corrosion, erosion-corrosion
- corrosion fatigue

Overall, weld discontinuities are not considered influential in elastic-plastic, general yielding, creep and general corrosion failure situations.

The effect of IF and IJP can be masked by weld metal over-matching and reinforcement. Cracks should not in general be tolerated in welds especially during manufacture. Porosity is regarded as the least harmful of all discontinuities. The importance of slag inclusions is treated in terms of their size, shape, amount, distribution and location. The tensile strength of the material is decreased in proportion to the projected area of the slag. The vertical height (through-thickness dimension), the length and distance from the surface or other discontinuity constitute the most important dimensions.

22.4 Weld Repair

Weld repair is one of the corrective measures that are used for repairing a defective weld. Weld repair, however, does not mean an automatic improvement in weldment reliability. Restraint is inherently more enhanced. The undesirable consequences of an unsatisfactory weld repair may be an increase in residual stress, greater distortion, introduction of new discontinuities, degradation of microstructure, sensitization (of stainless steel) and extension of pre-existing discontinuities which were not detected originally.

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22.5 Fatigue and Fracture Control

The approach to fatigue and fracture control of welded structures invariably entails methodologies that reduce stress concentrations (which is design related for the most part), tensile residual stresses (fabrication, installation or service related), introduce compressive stresses, avoid critical levels of applied stress (operation control related), eliminate critical size discontinuities (welding process and quality control related) and apply materials of adequate toughness for the service intended (design related). This approach serves the basis of the fitness-for-purpose concept.

22.5.1 Quality Control Systems Loop (QCSL)

In most cases, failures are the results of the combination of more than one cause, during manufacture, installation, or in service. Therefore, to derive the best possible lessons from failures requires a quality control systems approach in the context of fitness-for-purpose. A good quality control system is extended beyond construction to include installation and in-service monitoring of the structure, information feedback, data analysis, establishment of cause-and-effect relationship and dissemination of conclusions and recommendations to those concerned. Such an approach is called Quality Control Systems Loop (QCSL), schematically shown in Figure 155.(20)

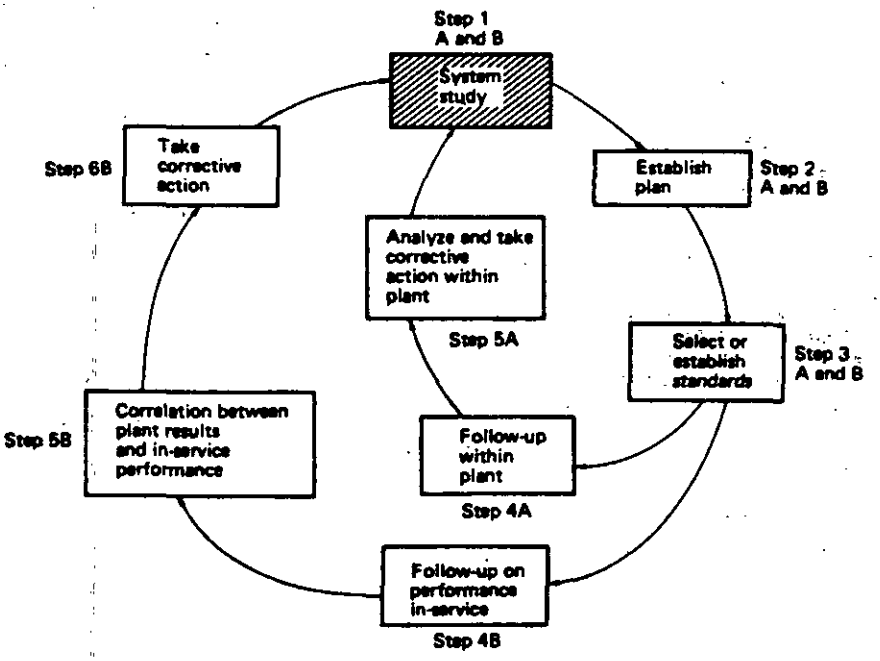


FIGURE 155. Quality Control Systems Loop (QCSL).

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The conventional practice of quality control is open ended, isolated and tends to be reactive to crisis situations that may arise. Usually, it has no follow-through provisions to check the installation and field performance of the inspected or noninspected weld(s). Consequently, there is no reliable way to validate the action taken at the time of original inspection in order to satisfy code compliance and owner/operator requirements. Typically, the only time feedback from the field is obtained if there was a major failure which might have been averted by a systematic and comprehensive approach which is implicit in QCSL. It should be remembered that properties and conditions of the material or the construction can change - at times unexpectedly - during fabrication, installation or service. Obviously, there are several different corrective actions that may be taken to solve a given problem. Corrective actions can include the following:

- stress relieving (post weld heat treatment)
- spot heating
- controlling the weld profile
- peening
- GTA or plasma dressing
- griding
- quenching
- cold working
- improving procedures
- application of ductile materials in selected areas of a welded structure (so called "crack arresters")
- plastic coating
- local pressing
- drilling "crack arrester" holes
- prior overloading
- bolted splice plates
- decreasing the applied stress (or elimination of in-service overload)
- improving the joint design
- weld repair
- minimization of residual stress
- environmental control
- process and procedure control (in the plant and in the field).

The material aspects of failure - in general - consist of

- inappropriate selection of the material
- unexpected alterations of properties during fabrication and/or service caused by a variety of factors
- high residual stresses
- unanticipated overload
- presence of weld defects.

In terms of design considerations, poor design details are almost always one of the principal causes of failure. The others are an inadequate control of processes and procedures, and sometimes a judgement based on wrong economic considerations.

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23. QUALITY ASSURANCE

Quality assurance (Q.A.) is a term used to describe a formal system within a company for managing quality control activities. It is a system designed to meet customer satisfaction with the quality of products manufactured.

Inspection alone does not ensure quality or conformance to contract requirements. There are several essential elements to an effective Q.A. system. These include

- independence of Q.A. departments from other departments
- quality standards that reflect the customer's needs and the attributes of the process
- written procedures for all phases of design, production inspection, installation and service with a program of periodic or continuous review and update of relevant documents as well as controls thereof
- traceability of parts
- detection and segregation of nonconforming materials and parts
- schedules for periodic calibration of equipment
- training and qualification of personnel
- system of controlling specifications, quality and inspection of purchased goods
- system of controlling manufacturing, assembly and packaging processes
- system for periodic audit.

The development, operation and monitoring of an effective Q.A. program is normally the responsibility of the Q.A. manager. There is no single quality level that should be universally applied. Codes and standards cover broad requirements which often have to be interpreted in terms of purchase order for each product or class of products. Written procedures should be as specific and clear as possible to avoid misinterpretation.

All paperwork should reach the work station on time. Identity and traceability of materials and parts should encompass not only markings but also locations and count. Nonconforming materials should be evaluated on the basis of fitness-for-purpose.

Continued use of equipment can have an adverse effect on accuracy. Therefore, periodic calibration requirements should be written into the quality assurance procedures. Calibration standards should be traceable to recognized industry or national standards of measurements. Calibration records should be kept in a central file.

Q.A. records should be retained for a specified minimum time period. Some of the more important records may be maintained for as much as 25-40 years depending on product and industry requirements. There are codes that state the minimum qualification requirement for personnel with special skills. Certification requirements can involve classroom and on-the-job training.

Inspection of incoming material that may affect product quality should be included in the overall Q.A. program. The purchaser should periodically examine the Q.A. system of major suppliers to assure that the expected level of quality is satisfied.

Audit of Q.A. system performance is advisable to determine actual compliance or violations of established quality control procedures.

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24. QUALITY CONTROL

Quality control involves the measurement of properties, soundness and dimensions of manufactured products and comparing the obtained results with predetermined standards. Measurements of these results can be done by direct or indirect techniques. Direct measurements often consist of destructive test methods. While this method yields more accurate and reliable results, the parts cannot be used after destructive testing.

Nondestructive examination (NDE) is in general less accurate and less reliable than the destructive technique. However, the parts can be used after NDE. There are a large variety of NDE methods that have been developed to characterize materials or components. Their accuracy, sensitivity and reliability vary from method to method. Accuracy is the attribute of an examination method which describes the correctness of the technique within the limits of the technique's precision. In other words, the technique is highly accurate if the indications it provides are correct. This does not mean that the technique is able to detect all discontinuities present. Sensitivity refers to the capability of a NDE method to detect discontinuities that are small in size or have properties only slightly different from the material in which they reside. Reliability is a combination of both of these characteristics. The factors that influence the reliability include

a) examination procedure,
b) instrumentation,
c) human factors,
d) data analysis.

Uncalibrated equipment, improper application of the method, poor equipment quality and incorrect procedure can affect quality control results.

Reliability of results is a strong function of such human factors as level of training or qualification and concentration.

Nondestructive examination methods include visual, dye penetrant, magnetic particle, radiography, eddy-current, ultrasonic, and acoustics. It is generally recognized that present-day NDE methods are not accurate enough for fracture mechanics evaluation. The relationship between NDE and product quality is not necessarily proportional.

A number of factors affect the selection of the most appropriate NDE method for a specific job. They can be grouped into three basic categories

1) discontinuity characteristics,
2) fracture mechanics requirements,
3) application constraints.

The properties of a given discontinuity may be chemical, electrical, magnetic, mechanical and thermal. The most important ones are those that are different from those of the material in which it lies. Fracture mechanics requires that the discontinuity be detected, identified, located, sized and its orientation with respect to the applied and residual stresses be established. Application constraint may force the choice of an NDE method different from the ideal.

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Nondestructive examination has two functions to fulfill. One relates to quality control, the other involves the acceptance or the rejection of a welded, or brazed, or soldered joint on the basis of fitness-for-purpose.(1)

Visual and dye penetrant examinations are limited to the detection of surface discontinuities without limitation on material type. Magnetic particle and eddy-current techniques can be used for detecting surface or near surface discontinuities in ferromagnetic materials only.

Radiography provides a qualitative picture of the internal soundness of a joint, which can be interpreted in terms of workmanship criteria. Nonplanar or blunt discontinuities which are relatively innocuous can best be detected by radiography. It is difficult and in most cases impossible to determine the through-thickness dimension of a planar discontinuity by means of x-ray technique.

Ultrasonic examination is more sensitive to the detection of planar discontinuities than radiography. Principle limitations of ultrasonic technique are the subjective nature of the interpretation, need of highly skilled operator, absence of permanent record, difficulty in differentiating between true and spurious signals, and smooth surface finish requirement. Its reliability is also dependent on calibration, type of equipment, weldment geometry and the metallurgy of the joint.

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25. QUALIFICATION TESTING

Codes and specifications require qualification tests for procedures, welders and inspectors to satify certain minimum prescribed standards of acceptance. There are a number of code-writing bodies in the United States. Foremost among them are the American Welding Society, the American Society of Mechanical Engineers and American Petroleum Institute. Qualifying under one particular code does not necessarily qualify under another.

The primary objective of issuing codes, standards, or guideline documents is the public safety of welded structures. Most of the codes have similar requirements for welding, testing and inspecting welded samples. Most of the specifications state that the safety of the welded structure is the responsibility of the contractor, manufacturer, owner, or user. A part of this responsibility is the replacement of any work which does not meet inspection or test requirements stated by a given specification.

25.1 Procedure Qualification

The welding procedure for a structure to be constructed in accordance with a relevant code must be qualified before fabrication can begin. In instances where a prequalified and valid procedure exists for a specific job, no procedure qualification test may be necessary. However, if a contractor wants to deviate from these requirements, he must qualify the procedure.

A welding procedure is a detailed method and practice involved in the production of a particular welded joint. Typically, it consists of a written description, drawings of the joint, tables of the welding schedule and data sheet showing the test results. Codes have established standard procedures that each qualifying welder must follow so that consistent quality can be maintained. The reason for standardized procedures is that there are several essential variables that can have an effect on the quality of a weld. If a welder who already demonstrated an ability to produce sound welds, has followed the written procedure, his welds would have a predictable quality.

Codes require that the welder prepare the test samples and follow the written welding procedure in detail. Small test plates with the same chemical composition as the production weldments are welded by the welding procedure proposed for use on the actual weldment. These test plates are subsequently tested to determine whether the weld has adequate soundness and mechanical properties to satisfy relevant standards of acceptance.

The procedure qualification plates are examined nondestructively by the same method of NDE as required for the production weld. Destructive tests used on procedure qualification include tensile and bend tests. Additional tests required for some welds are nick-break tests, impact tests, and metallographic examination. The procedure qualification test is useful for demonstrating the quality and properties of the weld before actual production. It is, however, made on a small test plate under simulated production welding conditions. It is practically impossible to duplicate production-welding conditions when the procedure plate is being made. Situations will arise that could not have been anticipated when the procedure

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was being qualified. Close supervision must therefore be exercised to recognize factors occurring in the production phase that were not considered during procedure qualification. If these factors could alter the quality of the weld, they should be evaluated.

25.2. Personnel Qualification

The personnel qualification test requires the welder or welding operator to demonstrate that he can produce sound welds by the qualified welding procedure.

In personnel qualification tests, as in procedure qualification tests, the welder is required to weld a small test plate in a material which is the same or similar to that of the actual production weldment using the procedure qualified for that weldment. This test plate is then examined nondestructively and destructively to assure that the weld soundness meets the minimum acceptance standards required by the welding code governing the particular application.

The testing methods used for personnel qualification tests usually include the nondestructive examinations applicable to the job. Nondestructive tests are used exclusively for some applications. Where destruction tests are required, the primary test is normally the bend test. Nick-break tests may also be used. These test methods are selected because of their emphasis on weld soundness, rather than weld properties. Thus, tensile tests are not customarily required for personnel qualification. Destructive tests are categorized as chemical, metallographic, hardness, and mechanical in nature.

Chemical tests are generally used to validate the chemical composition or the corrosion resistance of the base metal and weld metal. Chemical analysis can show that the metals being used conform to specifications. Chemical analysis of the weld metal can show that the result of the joining process is as anticipated, or may indicate whether undesirable constituents have been introduced into the weld metal as a result of the welding process. Corrosion tests are used to demonstrate that a weldment intended for use in a corrosive environment will actually stand up satisfactorily when placed in that environment. Cost and time considerations do not usually permit testing for corrosion resistance by actual use under service conditions. Therefore, tests that can be conducted under accelerated laboratory conditions have been developed to provide the necessary information.

Hardness tests measure the resistance of materials to permanent deformation. For some metals, ductility and corrosion resistance decrease with increasing hardness. Since each operation during welding has metallurgical effects, some specifications will call for an upper limit to the acceptable hardness of various areas of the weld. Basically, hardness testing machines operate by forcing a small hardened steel ball or diamond point into the surface of the metal, under a specific load. The depth of penetration is either measured directly by the machine, or inferred from the dimensions of the impression. By associating a number with each possible impression depth, a hardness scale can be estimated. This principle for measuring hardness is used by the three most common hardness measuring devices: the Brinell, the Rockwell, and the Vickers hardness tests. Because of the differences in the formulas used to define hardness numbers, differences in the material and

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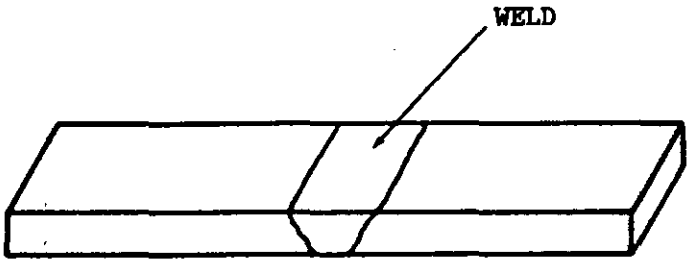
shape used to make the impression and differences in the imposed load, hardness numbers obtained by any of these methods will not necessarily be the same as hardness numbers obtained by the other methods. However, tables of approximately equivalent hardness numbers among these three methods have been constructed for some materials, particularly steels.

Metallographic tests are specified infrequently for weldments, but are especially invaluable in procedure qualification testing and in welding development work. These tests are used to determine the visually observable characteristics of the welds. Metallographic test samples consist of sections of welds cut through the weld in any desired plane, then polished and etched to reveal the structure. Examination may be made visually or microscopically. Among the characteristics that can be examined through this method are the soundness, location, depth of penetration of the weld, the metallurgical structure of the weld, fusion zone, and heat-affected zone. The extent and distribution of inclusions in the weld, and the number of weld passes can also be determined.

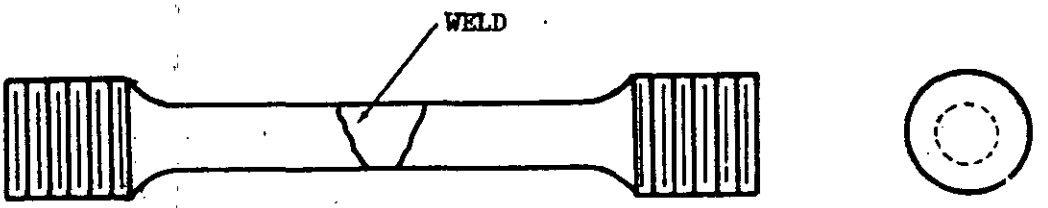
Mechanical tests (exclusive of hardness) have also been designed to determine a number of weld properties. Almost all of the standard tests can be categorized as tensile, bending, or shear tests.

Tensile tests are conducted using specimens cut from a test weld and machined to one of several standard sizes for mounting in a testing machine. Specimens are usually taken perpendicular to the weld, with the weld centered in the specimen, as shown in Figure 156. However, specimens are sometimes taken parallel to the direction of weld and consist entirely of weld metal. Specimens may have round or rectangular cross-sections, depending on the requirements of the appplicable welding code. The testing machine appplies a tensile force on the specimen until it ruptures. From readings on the machine and measurements on the specimen before and after the test, properties such as yield strength, tensile strength and ductility can be calculated. The primary purpose of the test is to demonstrate that the weld metal deposited by the selected procedure meets or exceeds the minimum values specified in the applicable code.

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(a)



(b)

FIGURE 156. Tensile test specimens: (a) rectangular and (b) round.

Guided bend test determines the ductility of a weld. Bends are described as root-bend, face-bend, or side-bend tests, referring to the surface that is stretched in bending. Rectangular test specimens, similar to those prepared for tensile tests are machined or ground to remove any weld reinforcement. A test jig such as shown in Figure 157 is used to make the bend. With the plunger removed, the specimen is placed across the shoulders of the jig, with the weld centered. The plunger is then forced down until the specimen is bent into a U-shape. The elongation of the tension surface is determined by the

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relationship between the thickness of the specimen and the radius of the die. Failure of the specimen is determined by the presence of cracks or other open defects greater than a specified number and size, or by fracture of the weld.

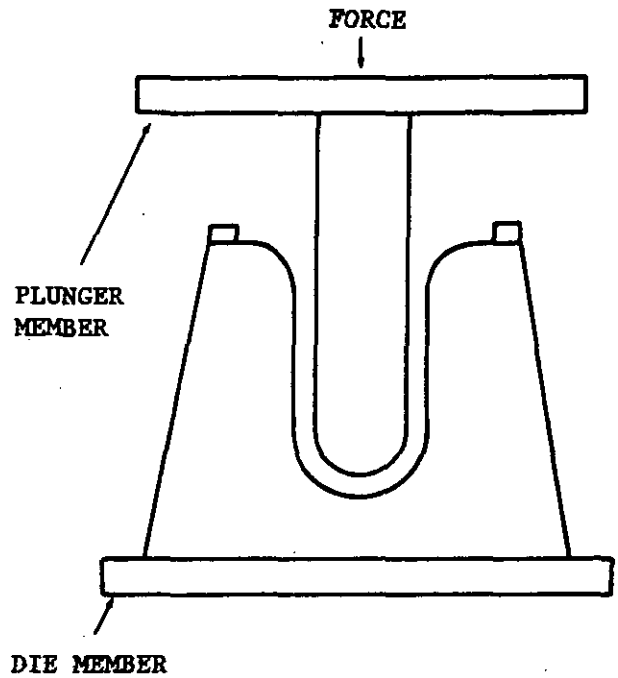
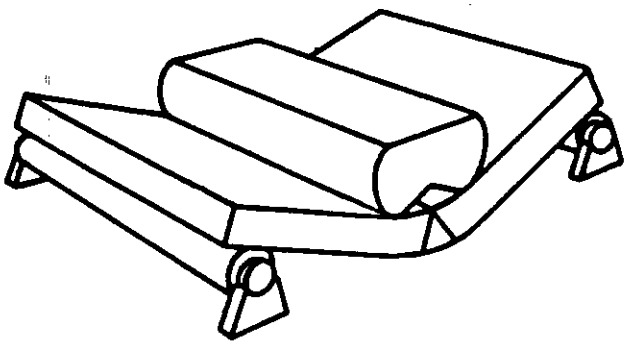


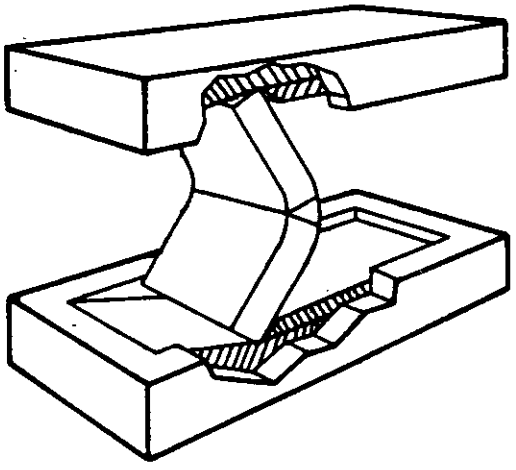
FIGURE 157. Jig used for guided bend testing of welded specimens.

Free bend tests, also for ductility, use specimens similar to those employed in guided bend tests. Gage lines are inscribed across the width of the sample before the test, marking off a distance on the deposited weld metal, about 3.2 mm (1/8 in.) less than the width of the weld. The sample is given an initial bend as shown in Figure 158a, by supporting the ends or shoulders on rollers, then forcing the center down until the specimen takes a permanent set. Then, using a testing machine or vise, as in Figure 158b, the sample is compressed longitudinally until a crack or depression appears on the convex face of the specimen, or until the specimen is bent double. The load is removed immediately if a defect appears before the specimen is bent double. The percent of elongation is calculated from the initial and final distances between the gage marks.

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(a)



(b)

Figure 158. Free bend test: (a) initial set and (b) specimen compression.

Shear tests of fillet welds are conducted by pulling specimens apart in a tensile machine. The dimensions of the specimens are such that shear strength in kilograms per linear millimeter (pounds per linear inch) of weld is easily obtained from the test results. Nick-break tests are required in some welding codes to detect weld discontinuities. The nick-break test specimen is similar to a rectangular tensile test specimen, except that notches are cut at the center of the weld. The specimen is then broken either by pulling it in tension or by bending it. The specimen is forced to break in the weld metal because of the notches. After the specimen has been broken, the fracture surfaces are examined for presence of weld discontinuities. If the discontinuites exceed the allowable size, the weld is rejected.

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Various types of impact tests are sometimes used to determine the fracture toughness of the weld metal. These tests are designed to measure the ability of the weld metal to resist crack propagation under conditions of low temperature. Pull tests and shear tests are used to measure the strength of resistance welds and brazed joints.

In many instances, a proof test is used to check the rated capacity of a welded assembly. The part is loaded to its rated capacity to see if it remains intact. If the part is never abused or used to support loads higher than the proof test, it is safe to assume that the part will perform properly. Leak testing is similar to proof testing in that a container being tested is filled with a liquid or gas and is pressurized. The rate at which the liquid or gas leaks from inside the vessel is measured. Leak detection can involve the use of soap film, immersion in a liquid, pressure gages, dyes, and listening to the sound of the escaping gas.

Virtually every nondestructive examination method available has been used for the examination of welds, including visual, magnetic-particle, dye-penetrant, eddy-current, ultrasonic, and radiography using X-rays and gamma rays. The method and the extent of the NDE will vary with the nature of the work and the criticality of certain joints. Some factors that should be considered in selecting NDE methods for weldments are:

1. material to be tested
2. joining process
3. joint geometry
4. type of weld discontinuities
5. attributes of NDE technique involved.

The welding inspector examines the welded joints for visible flaws, dimensions and conformance with other specification requirements. AWS established a procedure for certifying welding inspectors. The procedure requires a certain amount of welding experience, training, and technical knowledge of welding and its related processes that inspectors must have.

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26. NONDESTRUCTIVE EXAMINATION

Care in design, materials selection, materials processing and service conditions help prevent failures of one type or another. But, how is it determined that the engineering process has been successful? Nondestructive examinations and destructive tests can be performed in an effort to answer this question. (See Section 23.1 for details of destructive testing.)

The primary objective of all NDE methods is to detect weld discontinuities and defects in a given joint and base material. There are a large variety of NDE techniques available today. The most common ones are visual, liquid penetrant, magnetic particle, radiography, ultrasonic, acoustic emissions and eddy-current examination. Each of these methods have certain advantages and limitations. It is generally recognized that present-day NDE methods do not fully meet the fracture mechanics-based fitness-for-purpose criteria. Simply, it means that all discontinuities or flaws that may be present in a structure cannot be characterized completely by current NDE methods in order to provide full assurance. There have been numerous case histories reported where the structure had been inspected 100%, and in spite of it, significant defects were discovered later on in service.

26.1 Visual Examination

Visual examination is probably the most widely used NDE method. It is simple and inexpensive. The only instrument that may be involved is a magnifying glass (10 x or less). Its primary disadvantage is its limitation to the detection of surface discontinuities or flaws that may be present in any type of material.

Visual examination is the first stage in the inspection of a finished weld. The following quality factors can usually be determined by this technique:

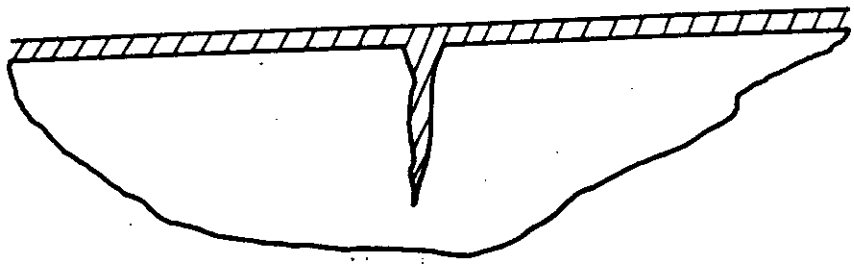
1. Dimensional accuracy of the weldment
2. Conformity to specification requirements regarding the extent, distribution, size, contour and continuity of welds
3. Surface discontinuities
4. Surface finish.

Although visual examination is an invaluable method for detecting surface flaws, it cannot be used for determining the presence of sub-surface discontinuities.

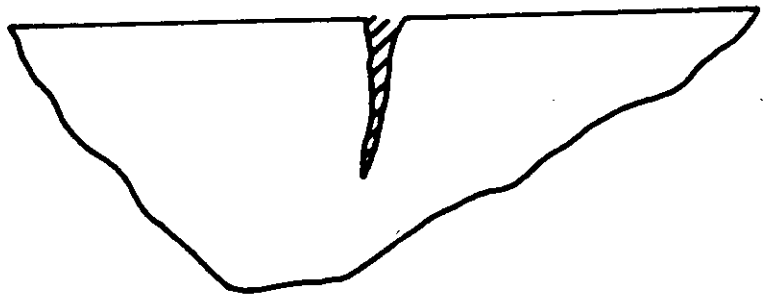
26.2. Liquid Penetrant Examination

Liquid penetrant examination is capable of detecting surface discontinuities that are invisible to the unaided eye. Liquid penetrant procedures usually require ground surfaces, (improper griding can smear the surface and close surface openings of the defects) although some "as welded" surfaces can be inspected by this technique. The basic steps in penetrant inspection are shown in Figure 159.

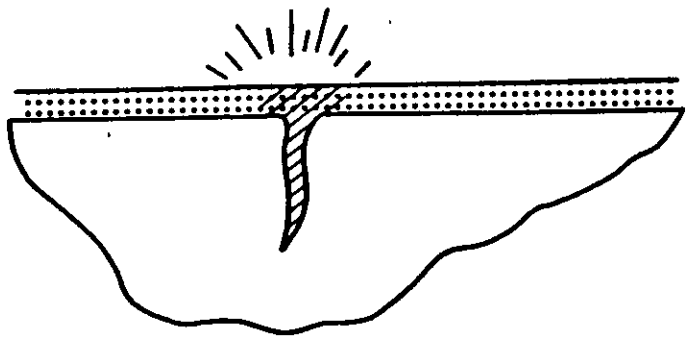
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(a)



(b)



(c)

FIGURE 159. Steps in liquid-penetrant examination:
(a) Pentrant is drawn into a flaw by capillary action
(b) Excess penetrant removed from surface
(c) Developer draws the penetrant from the flaw and produces an indication.

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Liquid-penetrant examination is most effective when the temperature of the materials is in the range of 20-30°C (68-86°F). The part is sprayed with a cleaner to remove oil and grease. It is then sprayed with the liquid-penetrant, which is drawn into surface discontinuities by capillary action. After the penetrating period, the excess penetrant remaining on the surface is removed. The penetrant is colored by a red dye. For this reason, this inspection is often referred to as a dye-penetrant examination. The part is sprayed with the developer, a white chalky substance that dries on contact. The developer is stained by the red dye, which is drawn from the flaw. The fluorescent-penetrant examination uses a highly fluorescent liquid. It is similar to the dye-penetrant examination, except for the application of ultra-violet lights for inspection purposes. White light is utilized for dye-penetrant examination.

26.3. Magnetic-Particle Examination

Magnetic-particle examination is limited to detecting surface or near-surface discontinuities in ferromagnetic materials only. The technique involves the establishment of a suitable magnetic field in the material being examined, the application of magnetic particles to the surface, and the examination of the surface for accumulation of particles. Accumulation of particles is indicative of the presence of a discontinuity. A schematic presentation of this process is illustrated in Figure 160.

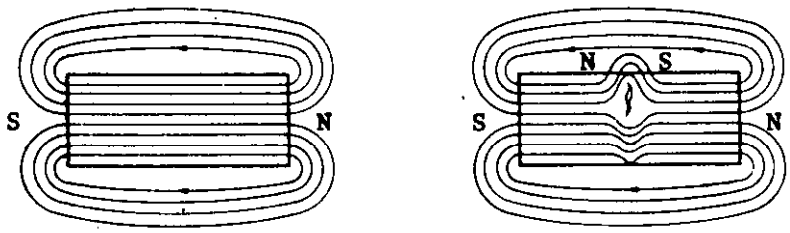


FIGURE 160. Magnetic-particle examination.

The present of a flaw disrupts the normal lines of magnetic flux causing lines of flux leak from the surface. Magnetic particles are attracted to the flux leakage through the creation of local north and south poles. This is the location of the discontinuity. The particles can be added dry or in a liquid such as water or light oil for better movement. The particles may be dyed with a fluorescent material for better detection.

Detectability of a discontinuity requires that

a. the discontinuity be perpendicular to the lines of the flux
b. the discontinuity must be near the surface
c. the discontinuity must have a lower magnetic permeability than the metal
d. the material must be ferromagnetic.

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The electromagnetic process is a modification of the basic magnetic-particle method and is used for sorting materials and for analyzing microstructures and heat treatments.

26.4. Radiographic Examination

Radiographic examination can detect the presence of discontinuities in the interior of materials provided the discontinuity is volumetric or in the case of planar flaws, it is favorably oriented with respect to the penetrating rays.

The method makes use of the ability of short-wavelength radiations, such as X-rays or gamma rays, to penetate objects opaque to ordinary light. In general, the shorter the wavelength, the greater the penetrating power. The procedure for radiographic examination is displayed in Figure 161.

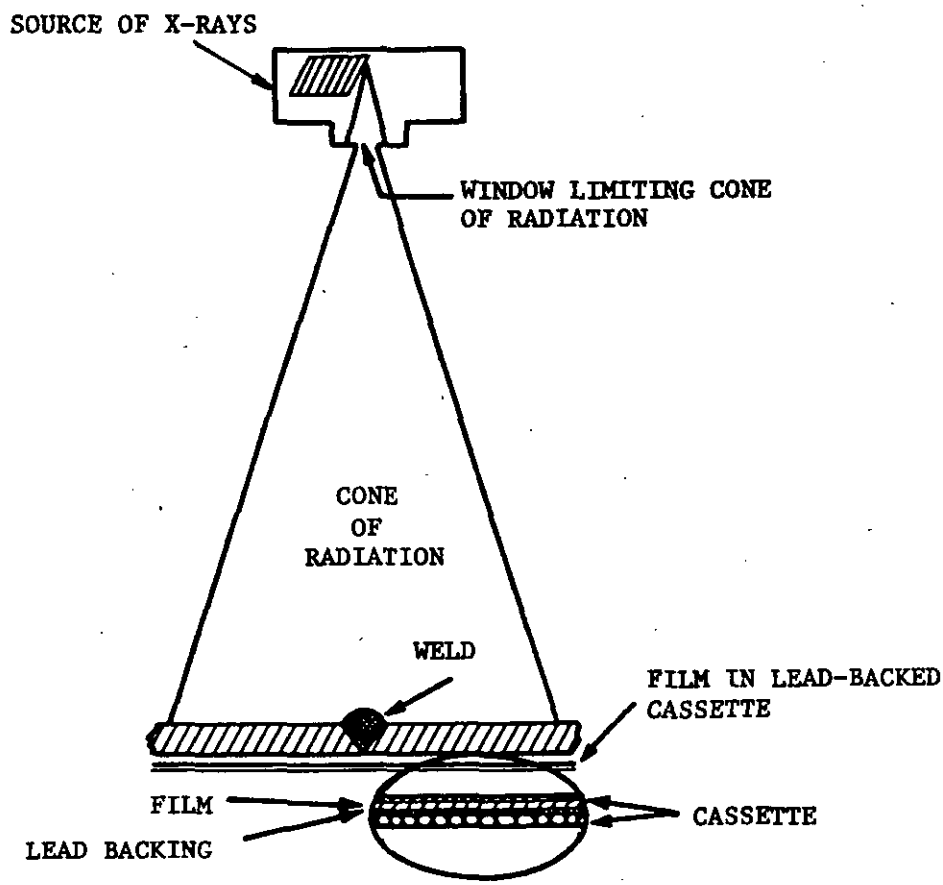


FIGURE 161. Elements of an X-ray radiographic setup.

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In radiographic examination, the transmission and absorption characteristics of a material are used to produce a visual image of a given type of flaw present in a material. These are several components in a radiographic technique.

1. Source of a penetrating radiation. X-rays emitted from a tungsten target are the most common, but occasionally gamma rays or neutrons are required.

2. Detection system. Normally, special films are used to detect the amount of radiation that is transmitted through the material. Other detectors involve fluorescent screens, Geiger counters, television screens, or xerographs.

3. Discontinuity must have absorption characteristics different from that of the material in which the discontinuity lies.

The emitted x-rays are directed to the material to be examined. A small fraction of the x-rays is transmitted through the material to expose the film. Obtaining a radiograph that can sensitively detect a discontinuity requires optimization of the film, accelerating voltage, exposure and various geometric factors. Sensitivity is defined as a ratio of the smallest change in the thickness of the material that is visible on the film to the total material thickness. The sensitivity tells the minimum size of a discontinuity which can be detected. If the sensitivity is very small, it is possible to detect very small flaws.

The geometry and physical properties of the flaw have a significant effect on detection. Porosity can be detected easily, if the size of the voids falls within the sensitivity of the radiographic technique. Since voids are three-dimensional, orientation of the material with respect to the direction of x-rays is not important. The density of a pore is different from that of the adjacent sound metal which will result in different absorption coefficients.

Nonmetallic inclusions in rolled products may be flattened during plastic deformation. If the x-rays are perpendicular to the flattened inclusions, they may not cause a noticeable change in intensity. In this case, the geometric arrangement between the x-ray source and the thin inclusion must be changed in order to detect such a flaw. Alternatively, some other NDE method may have to be sought.

Detection of cracks is dependent on their geometry. If a crack is perpendicular to the incident radiation, it may escape detection. However, if the crack is parallel to the x-rays, the crack may be easily observed.

The image quality of a radiograph can be shown by using a penetrameter. This is a piece of metal similar to the one being examined. It is placed near the metal during radiography. The penetrameter thickness is about 2% of the thickness of the base metal. Holes of various diameters are drilled in the penetrameter to provide an assessment of the resolution available in the radiograph.

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Gamma ray radiography employs an intense radiation of a single wavelength produced by nuclear disintegration of radioactive materials. The intensity of the radiation depends on the type and size of the radioactive source. Normally, cobalt 60 is used for thick, absorptive materials and iridium 192 or cesium 137 is used when less intense radiation is needed.

Neutron radiography can be used in materials that have similar absorption coefficients for x-rays or gamma rays for sorting purposes.

26.5. Ultrasonic Examination

Ultrasonic examination employs mechanical vibrations similar to sound waves, but of a higher frequency. A material may both transmit and reflect elastic waves. An ultrasonic transducer utilizing the piezoelectric effect introduces a series of elastic pulses into the material at a high frequency, usually greater than 100,000 Hz. The elastic wave, or phonon, is transmitted through the material at a rate that depends on the modulus of elasticity and the density of the material.

Three common techniques are applied to the ultrasonic inspection of a material.(15)

1. In the pulse-echo or reflection method, an ultrasonic pulse is generated and transmitted through the material (Figure 162). When the elastic wave strikes an interface, a portion is reflected and returns to the transducer. Both the initial and the reflected pulse can be displayed on an oscilloscope.

   From the oscilloscope display, the time required for the pulse to travel from the transducer to the reflecting interface back to the transducer is measured. If the velocity at which the pulse travels in the material is known, it is possible to determine how far the elastic wave traveled and can calculate the distance below the surface at which the reflecting interface is located. If there are no flaws in the metal, the beam reflects from the opposite side of the metal and our measured distance corresponds to twice the wall thickness.

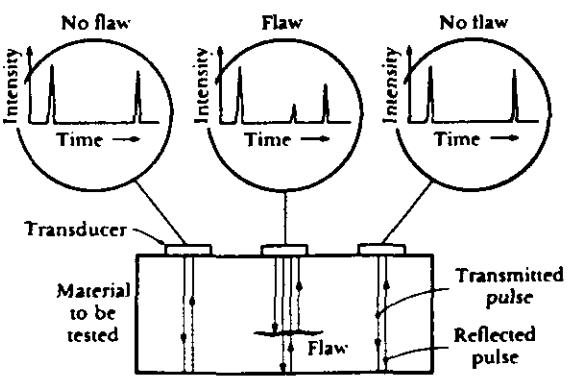


FIGURE 162. Pulse-echo ultrasonic examination.

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If a discontinuity is present and properly oriented beneth the transducer, at least a portion of the pulse reflects from the discontinuity and registers at the transducer in a shorter period of time. This indicates that a discontinuity is in the material. It even tells the depth of the discontinuity below the surface. By moving the transducer, the size of the flaw can be estimated. Moreover, it is possible to obtain a three-dimensional picture of the discontinuity by combining holography with ultrasonics.

2. In the through-transmission method, an ultrasonic pulse is generated at one transducer and detected at the opposite surface by a second transducer (Figure 163). The initial and transmitted pulses are displayed on an oscilloscope. The loss of energy from the initial to the transmitted pulse depends on whether or not a discontinuity is present in the metal.

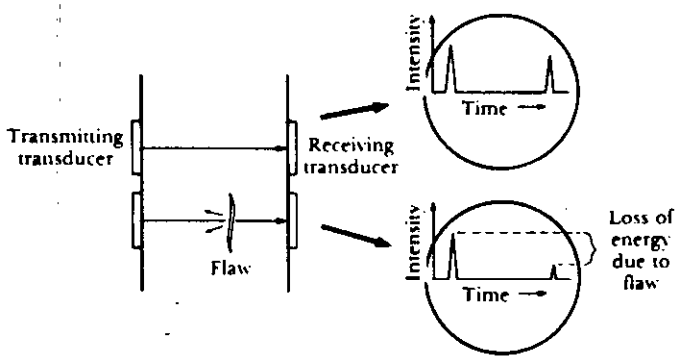


FIGURE 163. The through-transmission ultrasonic examination method.

In both the reflection and transmission methods, the phonon, or elastic wave, is regarded as an energy particle.

3. In the resonant method, the wavelike nature of the phonon is utilized. A continuum of pulses is induced and travels as an elastic wave through the material (Figure 164).

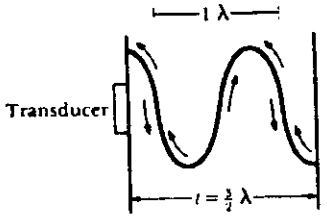


FIGURE 164. Ultrasonic resonance method.

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By choosing a wavelength or frequency so that the material thickness is a whole number of half-wavelengths, a stationary elastic wave is generated. A flaw in the material prevents the occurrence of resonance. This method is usually employed for determining the material thickness rather than detecting discontinuities.

In general, the ultrasonic examination technique requires that the discontinuity be perpendicular to the beam for detection.

26.6. Acoustic Emission Examination

Acoustic emissions are high-frequency elastic stress waves generated in a material by such phenomena as plastic deformation, crack growth or phase transformation. The emissions indicate initiation and progagation of deformation, and can be used to locate the origin of deformation by means of triangulation. Acoustic emissions can be used to monitor the various phenomena that can occur during welding or in service of a component. Continuous monitoring of events in a material can be done by automatic feedback controls and readout instruments.

26.7. Eddy-Current Examination

Eddy-current examination involves an alternating current flowing in a coil, which produces an impedance. Impedance is a vector quantity defined by the reactance and the resistance of the coil. Placing an energized coil in close proximity to a conductive material, the electric field produces eddy-currents. The factors that affect the impedance of the coil are the electrical conductivity, discontinuities, permeability, size, shape of the material, the closeness of the probe to the material, coil design, location of the material within the coil, and the frequency as well as the magnitude of the current. Therefore, interpretation of the eddy-current test results is very difficult. The eddy-current technique is used on thin-walled pipes and tubing.

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27. HEALTH AND SAFETY

27.1 Health

27.1.1 Introduction

A heightened awareness of the importance of health and safety associated with materials joining and cutting processes have in recent years prompted many independent studies. The primary purpose of these investigations has been to establish a cause-and-effect relationship. In general, health and safety programs entail the identification of elements of hazard, effects of exposure and safety measures.

27.1.2 Elements of Hazard

The elements of health hazard generated by materials joining and cutting processes include fumes, gases, radiation, noise and heat.

27.1.2.1 Fumes

Fumes are essentially particulate matter produced during welding, cutting, brazing and soldering. The amount of fume generated is dependent on the particular process, base and filler metals involved. In the case of arc welding, the factors affecting fume generation consist of type, polarity and diameter of the electrode, welding current, voltage and type as well as the surface condition of the base metal. Oxyfuel gas welding produces a significant quantity of fumes with base metals of galvanized steel and aluminum. The particulate matter most often associated with joining and cutting processes are chromium, copper, manganese, nickel, fluorides and silicon compounds.

Chromium in hexavalent, trivalent and divalent forms (Cr^{+6}, Cr^{+3}, Cr^{+2}) are normally present when stainless steel is welded. The hexavalent chromium is known to have a high potential hazard to human beings. Its presence in the welding environment should therefore be minimized. Fumes produced during shielded metal-arc welding (SMAW) of stainless steel has been reported to contain 3-7% chromium, of which 60-90% was the water soluble hexavalent type. Fumes generated by gas metal-arc welding (GMAW) of stainless steel contained approximately 15% chromium, of which 1% or less was identified as water soluble hexavalent chromium.

Copper is present in fumes produced from welding copper-base alloys and from the use of copper-coated electrodes. Studies have indicated that electrodes containing 0.18% copper did not result in excessive copper in the fumes. Manganese can be found in welding fumes as manganese oxide. Its source can be the coated, and flux-cored electrodes as well as the base metal. Nickel in the fume comes from welding stainless steel and nickel-base alloys. Fluorides constitute about 3-30% of the total fumes produced from basic covered electrodes. They are present in the form of sodium, potassium and calcium fluorides.

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The health hazard potential from fumes is a function of composition, concentration, particle size and exposure time. The types of joining process, filler and base metals and their conditions influence fume composition, concentration and particle size. The size of the particulate matter present in fumes is important because it determines the site of deposition in the respiratory tract. Most of the particles range in size from 0.5 - 0.7 m, which could possibly reach the air cells of the lung. Particles of larger size are mainly trapped in the upper respiratory passage and the mouth, particularly the teeth and gum areas.

27.1.2.2 Gases

Gases present in materials joining and cutting processes originate from atmospheric gases, shielding gases and products of decomposition. Examples of these various gases include ozone, nitrogen oxides, argon, helium, nitrogen, carbon dioxide and decomposition products from solvents and coatings on surfaces. In plasma-arc cutting, the concentration of ozone (O_3) is low due to its reaction with nitrous oxide and disintegration caused by molecular attachment to dust particles. The formation of ozone in welding is due to the intense ionization of the air. The amount of ozone produced is to a large extent process dependent. Other gases that may be produced in welding are phosgene ($ClCO_2$), phosphine (PH_3).

27.1.2.3 Radiation

There are three distinct types of electromagnetic radiation produced during welding: visible, ultraviolet and infrared. Visible light is predominantly in the short wavelength spectrum of 420-430 µm. Ozone is normally generated by ultraviolet emission at some distance from the arc.
The magnitude to which welders may be exposed can range from 20 millirem to 1 (one) rem.

27.1.2.4 Noise

Welders are exposed to excessive noise levels in welding, chipping and grinding. Other shop operations nearby may also expose welders to objectionable noise levels. Noise levels produced by electron beam welding was found to be proportional to the beam voltage and current. Excessive exposure to noise beyond a certain decibel level can be detrimental to hearing capacity.

27.1.2.5 Heat

The level of temperature produced during welding and thermal cutting depends on the specific process utilized. Extremely high temperatures are generated in some of these processes. Most of the skin conditions in welders are the result of flying hot metal particles and ultraviolet radiation. A lack of safety measures or observations thereof are often the cause of spark burns.

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27.1.3 Effects of Exposure

In the absence of appropriate safety measures, materials joining and cutting processes can have certain harmful effects or consequences on the health and safety of human beings. The effect(s) may be readily apparent or may require considerable time to notice. This then makes the general field of health and safety and range-and-effect difficult to define. In terms of health and safety requirements, the selection and classification of electrodes in Sweden for example are done according to rate of fume generation, threshold limit index and hygienic arc requirements. Each of these three requirements are quantified by determining specific values, respectively. Electrodes with "low yield" fume emission have recently been manufactured in England.

Without adequate ventilation and protective equipment, fumes and gases may find their way into the welder's lung, where they may induce acute or chronic effects. The effects may be pulmonary irritation (pneumoconosis) and systemic toxicity. Acute respiratory insufficiency can be induced by fumes and gases.

Lung injury may result from very small particles (inhalant) deposited on bronchial (alveolar) surfaces and retained long enough to cause lung damage. Chromium metal particles have been identified to cause chromic dermatitis, toxic effects and asthmatic attacks. A combination of chromium and nickel particulates also yielded asthmatic attack. Fumes from mild steel welding did not produce asthmatic response.

Chronic obstructive airway diseases can also be caused by smoking tobacco or non-tobacco products.

Gases are generally classified as irritants and nonirritants. Ozone, nitrogen oxide, phosgene, phosphine are considered as irritants. Carbon di- and mon-oxides are the nonirritant types. A group study of welder trainees between the ages of 18-21 where one-half of the trainees had previous welding experience involved the effects of ozone exposure. A maximum ozone level after 12 weeks of training was found to be 0.015 ppm both in the ambient air of the welding booths and in the respiratory zone. In the first six (6) weeks, there was a slight rise in the percentage of cells with single chromatid and isochromatid gaps, which then declined in the following six (6) weeks.

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Particulates are divided into those that cause pneumoconioses, which is caused by habitual inhalation of irritant metallic and mineral particles, and those that cause pulmonary irritation. Particles of copper, beryllium, tin, iron and aluminum can cause pneumoconioses. Pulmonary irritation can be induced by small particles of cadmium, chromium, lead, fluorides, manganese, mercury, magnesium, nickel, molybdenum, titanium, vanadium and zinc. The system of a human body can eject foreign particles that may be deposited on the respiratory membranes through the pathways illustrated in Figure 165.

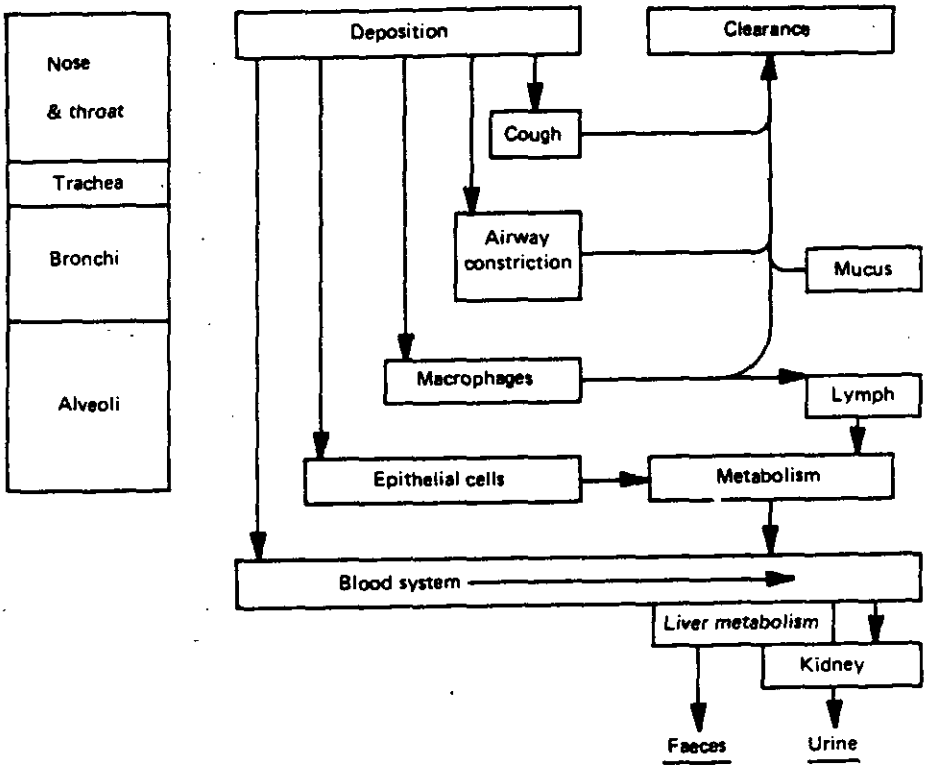


FIGURE 165. Pathways of elimination of fumes.(21)

Continuous exposure to fumes and gases could result in a chronic respiratory disease. As an example, a welder with 10 years of arc welding experience developed a diffused interstitial fibrosis of the lung, which is a formation of fibrous material in the small spaces of the tissue structure. Pulmonary siderosis, which is iron content within the lung, is a common occurrence in welders. Silicon dioxide has been known to cause interstitial fibrosis. Inflammatory changes in the mucous membrane of the upper respiratory tract were detected in 70% of the welders involved in a study dealing with the occupational effects of welding.

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Several methods can be used to assess the respiratory function of welder's lung. These are preliminary examination, respiratory questionnaire, clinical examination, lung function test and x-ray examination. The lung function test is used to determine the effects of acute and chronic exposure to fumes. Routine and periodic examinations of welders have been recommended along a scheme indicated in Figure 166.

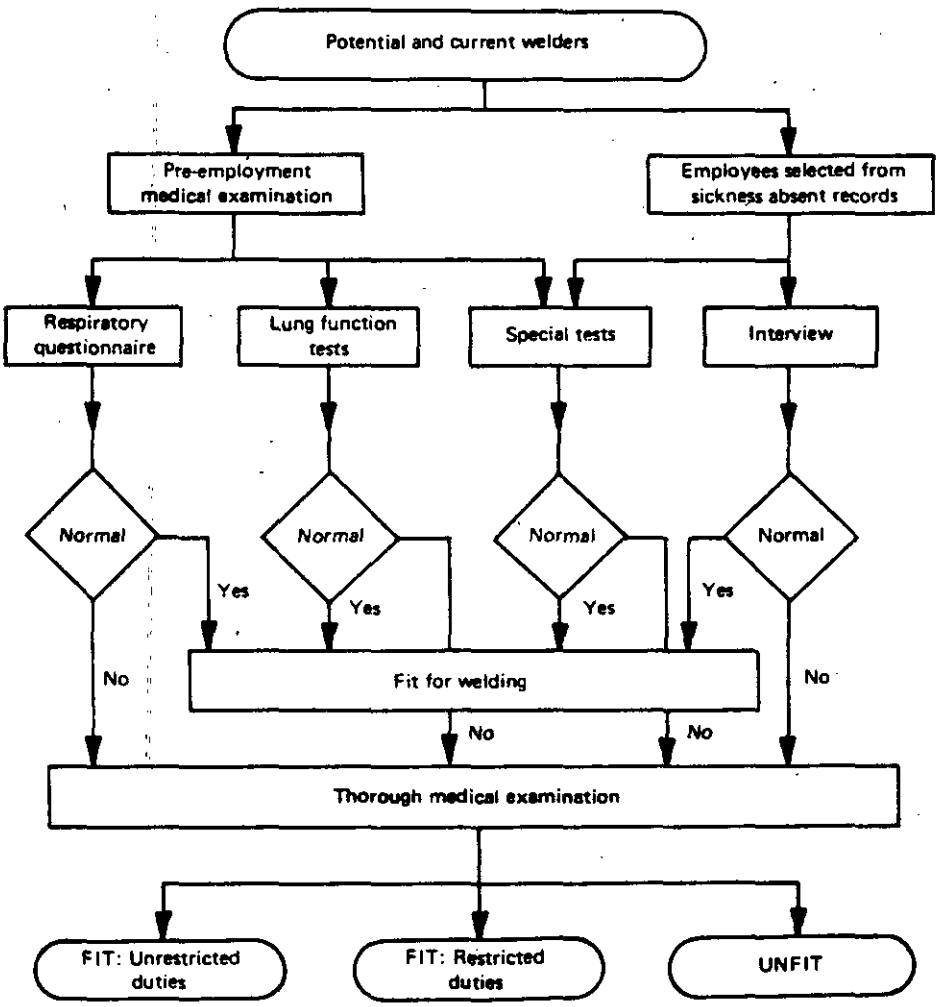


FIGURE 166. Health examination scheme for welders.(21)

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Radiographic examination of a group of welders ranging in age from 16 to 73 with an average age of 36.8 years showed that very few respiratory abnormalities were noted in workers with up to 15 years of welding experience. Thereafter, the prevalence of pneumoconiosis steadily increased until it passed 30% with 50 years of experience. This study did not take into account such variables as exposure time, welding variables and variations in working conditions.

In the absence of protective measure, eye injuries may occur during joining and cutting processes. Manifestations of eye injuries may be irritation, sluggishness of eye adaptation, arc eye, retinal damage, corneal and conjunctive injuries, cataracts and loss of vision in the extreme case.

Fumes can cause eye irritation, especially those containing basic material. Thermal damage of the eye is caused by visible and infrared radiation. The retina is the most sensitive part of the eye to the thermal effects of radiation. The tolerance level of the retina is extremely low to absorbed thermal energy. Exposure to relatively high intensity beam even for short periods of time can bring about serious injury to the retina.

Ultraviolet radiation for a brief period has no significant consequence. However, a severe exposure can give rise to irreversible photochemical reaction in the pigment layer. Permanent damage is mostly the combined effect of thermal energy and photochemical reaction.

The most dangerous effect of infrared radiation is the development of cataracts, which was found to be caused by wavelengths below 1400 m. Gas metal-arc welding was determined to produce the highest amount of radiation, while plasma-arc welding the least amount. Radiation intensity was proportional to the welding current.

The level of radiation in welding aluminum was related to the magnesium content of the aluminum. Magnesium-free alloys produced lower levels of radiation than magnesium containing aluminum alloys.

Thoriated tungsten electrodes constitute a radiation hazard due to the presence of radioactive thorium. The adverse biological effects of radiation are contained in table XXXVI.

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TABLE XXXVI
Adverse Biological Effects Due to Radiation.

Radiation	Wavelength (μm)	Adverse biological effects
UV-C	100-280	Arc flash and slight erythema
UV-B	280-315	Arc flash and slight erythema
UV-A	315-400	Arc flash, erythema, cataracts, and skin cancer
Visible	400-700	Photochemical or thermal retinal injury, and thermal skin burns
IR-A	700-1,400	Retinal burns, thermal skin burns, and cataracts
IR-B	1,400-3000	Thermal skin burns

The effects of electromagnetic radiation exposure are presented in table XXXVII.

TABLE XXXVII.
Effects of Electromagnetic Radiation Exposure.

Microwave and radio frequency radiation	Optical radiation	Ionizing radiation
Genetic effects	Skin cancer	Cancer
Teratogenic	Squamous cell epithelioma	Leukemia
Behavioral changes	Basal cell epithelioma	Sterility
Cataract	Herpes Simplex lesions	Cataract
Headache	Cataract	Radiodermatitis
Nervous system disorders	Retinal damage	Ulcerations
Loss of memory	Genetic alteration	Erythema
Dullness	Erythema	Genetic alteration
Sleepiness	Kerato conjunctivitis	Blood cell changes
Sinus arrhythmias	Heart stress	Malaise
	Cutaneous burns	Fever
	Asthenopia	Nausea
		Weakness
		Shock
		Diarrhea
		Life span shortening
		Anorexia

There have been some studies to determine the degree of protection offered by transparent curtains. The results showed that a slightly gray-colored curtain would be the best choice. Yellow curtains gave the best visibility. Wavelengths under 340 μm were surpassed in almost every curtain tested.

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A phenomenon called magnetophosphene or light streaking produced by low frequency (20-25Hz) magnetic field during spot welding can lead to a temporary vision impairment.

Local discolorations of the skin, similar to freckles, have been observed on the forearms of operators of spot welding machines. No discolorations or pigmentations were noticed on the operators' hands or wrists, which were normally protected by leather gloves. This type of skin condition does not result in pain or hinderance in work. The smaller brown spots of hyperpigmentation can disappear within a few years of cessation of exposure. The larger spots of skin discoloration may persist for as many as 8 years after exposure termination. Cases of contact eczema on unprotected areas of the body have been reported with welding of nickel. Also the nickel concentration in the urine of nickel-exposed workers was found to be considerably higher than that in unexposed workers.

The lead concentration in the blood of welders examined in recent studies conducted in Europe and in this country indicated a range of 20-25 g/100 ml. This is significantly below the maximum tolerance level of 70 g/ml set by occupational health organizations.

Permissible noise level for an eight-hour period has been set at 90 dB by OSHA. Evidence shows that ozone can induce point and chromosomal mutations.

27.2 Safety

Safety is for the protection of persons from injury and illness and the protection of equipment as well as property from damage by fire and explosions stemming from materials joining, cutting and related processes.

27.2.1 Equipments for Materials Joining

The safety aspects of materials joining should be given consideration when selecting an equipment for a job to be done. Assurance of consideration of safety in design is obtainable by choosing equipment complying with the Requirements for Electric Arc-Welding Apparatus, ANSI Standard C87.1-1971, National Electric Manufacturers Association, or the Safety Standard for Transformer-Type Arc-Welding Machines, ANSI Standard C33.2-1956, Underwriters' Laboratories. Special purpose machines not covered by the above-listed standards should conform in all respects to the standard set forth in appropriate safety publications.

Standard machines for arc welding purposes should be designed and constructed to carry their rated load with rated temperature rises, where the temperature of the cooling air does not exceed 40°C (104°F) and, where the altitude does not exceed 1000 m (3300 ft) and should be suitable for operation in atmospheres containing gases, dust and light rays generated by the welding arc.

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Unusual conditions may exist, in which case the equipment should be especially designed to meet safety requirements of the service. Examples of unusual service conditions are:

- Operation in ambient temperatures in excess of 40°C (104°F) or lower than 0°C (32°F)
- Operation of altitudes in excess of 1000 m (3300 ft)
- Exposure to highly corrosive vapors
- Exposure to steam or excessive humidity
- Exposure to excessive oil vapor or flammable gases
- Exposure to excessive dust
- Exposure to severe weather, seacoast or shipboard conditions
- Exposure to nuclear radiation.

If an arc welding machine is operated without being connected to a load, the open circuit voltage should not exceed the values given in table XXXVIII when the rated voltage is applied to the primary winding or when a generator-type arc welding machine is operating at maximum rated no-load speed.

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TABLE XXXVIII
Maximum Open Circuit Voltages of Welding Machines.

Welding Current	Maximum Open Cirucit (no-load) Voltage	
	Manual & Semiautomatic Machines	Automatic Machines
ac	80 rms	100 rms
dc 10% Ripple voltage	80 rms	100 average
dc 10% Ripple voltage	100 average	100 average

A controller integrally mounted in an electric motor driven welder should have a capacity for carrying rated motor current and be capable of making and interrupting stalled rotor current of the motor. It may also serve as the running overcurrent device if provided with the number of overcurrent units as specified by the National Electric Code, ANSI Standard C1-1971, Table 430-37. Control apparatus on all types of arc welding machines should be enclosed except for the operating wheels, levers or handles. Input power terminals, tap change devices and live metal parts connected to input circuits should be enclosed completely.

Welding cables should be flexible and constructed to meet the rigors of the welding environment. It should be adequately sized for the current and duty cycles expected.

Installation of welding equipment including incoming power lines and grounding of the frame of the welding machine should meet ANSI Standard C1-1971. Chains, wire ropes, cranes and elevators should not be used to carry current. Ground connection should be mechanically strong and electrically adequate for the required current. A disconnecting switch should be provided at every welding machine which is not equipped with such a device.

Leakage of cooling water, shielding gas and engine fuel should be prevented because they may affect the welder's safety. The electrode holder should be de-energized by disconnecting the power supply before the welder leaves his work station or stops working for any appreciable time. The welder should never permit the live metal parts of an electrode holder to touch his bare skin or any wet covering of his/her body. Wet machines should be dried prior to their use. Hand gloves and electrode holders should be dry. Water-cooled electrode holders should neither leak nor develop condensation which would adversely affect personal safety. The welder should not coil the welding cable around parts of his body.

Equipment used for materials joining, cutting and allied processes should be maintained in a safe working condition at all times. Any equipment defect, failure or safety hazard should be reported immediately. Defective equipment should not be used.

Resistance welding equipment should be installed by personnel qualified by the National Electric Code, ANSI Standard C-1, or its equivalent. All external weld-initiating control circuits should operate on low voltage, not over 120 volts for stationary equipment and not over 36 volts for portable

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equipment. All doors and access panels of all resistance welding machines including their control panels should be kept locked to prevent access by unauthorized persons to live components of the equipment. Protection from flying sparks should be provided in the form of fire-resistant guards or through the use of safety glasses. An emergency stop button should be provided on all multi-spot welding machines at each operator's position.

27.2.2 Goggles, Helmets and Shields

Eye protection is one of the most important safety considerations in the various joining and cutting processes. Helmets or hand shields should be used by personnel viewing these processes in action. Safety spectacles or goggles should also be worn to provide protection from flying objects and injurious rays from adjacent materials joining and cutting operations. The spectacles or goggles may have either clear or colored glass depending on type and amount of exposure to elements of hazard. Spectacles with suitable filter lenses and without side shields may be permitted for use during gas welding, brazing, soldering and nondestructive examination.

Helmets, hand and face shields as well as goggles should be made of materials that are thermally and electrically insulating, noncombustible and opaque to visible, ultraviolet and infrared radiation.

Personnel adjacent to welding areas should be protected from radiant energy by noncombustible screens or shields and should wear suitable eye protection. Booths and screens should permit air circulation at floor level.

Appropriate protective clothing required for joining and cutting operations should be determined in accordance with the size, nature and location of the work. Persons doing welding, cutting and related operations should wear protective gloves.

27.2.3 Ventilation

Ventilation in confined spaces should be a prerequisite to work. Gas cylinders and power sources should be placed on the outside of confined spaces. Where a welder must enter a confined space through a manhole or other small opening, means should be provided for rapid removal of him/her in case of emergency. When safety belts and lifelines are used, they should be so attached to the welder's body that the body cannot be jammed in a small exit opening. A helper with a preplanned rescue procedure should be stationed outside to observe the welder at all times and be ready to intervene.

Brazing and gas welding fluxes containing fluorine compounds should have a cautionary wording on tags or boxes to indicate that they contain fluorine compounds. A number of potentially hazardous materials are employed in fluxes, coatings, coverings and filler metals that are released to the atmosphere during joining and cutting. Suppliers of these materials should determine the hazardous properties, if any, of their materials.

Local exhaust or general ventilation systems should be provided in order to keep the amount of toxic fumes, gases and dust below acceptable concentrations as established by ANSI Z37 or Public Law 91-596 (OSHA 1970). Compliance should be determined by sampling of the atmosphere.

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In circumstances where it is impossible to provide ventilation, air-supplied respirators (self-contained breathing apparatus) or hose masks approved by appropriate agencies should be used. The need for ventilation in open spaces should be evaluated on the specific merits of the individual circumstances.

Brazing atmospheres in furnaces or retorts that are flammable should be purged to remove oxygen and air by one of the following methods:

1) Below 760°C (1400°F) the oxygen should be reduced below the explosive level by either evacuating below 660 Pa (5 mm mercury) and backfill with the flammable atmosphere to the working pressure or by purging with a neutral or inert atmosphere for 4 to 6 volumes followed by 4 to 6 volumes of the flammable atmosphere.

2) Above 760°C (1400°F), heated furnaces or retorts may be purged with a flammable gas atmosphere with the atmosphere entering the 760°C chamber area. Under these conditions, the air will be consumed progressively thereby avoiding any explosive condition.

In all cases, adequate area ventilation should be provided to exhaust to the out-of-doors all explosive or toxic gases which may emanate from furnace purging and brazing operations. Cleaning materials may be toxic or flammable. Therefore, appropriate safety measures recommended by the manufacturer's instructions should be adhered to. Degreasing or other cleaning operations involving chlorinated hydrocarbons should be so located that no vapors will be drawn into the atmosphere surrounding the molten metal. Trichloroethylene and perchloroethylene should be kept out of atmospheres penetrated by ultraviolet radiation of welding processes.

27.2.4 Fire Prevention

Fire prevention requires certain basic precautions before materials joining and cutting processes are commenced. The atmosphere should be nonflammable. Combustibles should be separated or protected from fire hazards. Suitable fire extinguishing equipment should be maintained ready for use. During welding cutting and allied processes, fire watchers should be assigned to locations where other than a minor fire might develop such as in the proximity of combustibles.

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APPENDIX A

GLOSSARY

A

ABRASION SOLDERING - A soldering process in which the faying surface of the base metal is mechanically abraded during soldering.

ABRASIVE - Material such as sand, crushed chilled cast iron, crushed steel grit, aluminum oxide silicon carbide, flint, garnet, or crushed slag used for cleaning or surface roughening.

ABRASIVE BLASTING - See preferred term blasting.

ABSORPTIVE LENS (eye protection) - A filter lens whose physical properties are designed to attenuate the effects of glare and reflected and stray light. See also filter plate.

ACCELERATING POTENTIAL - The potential in electron beam welding that imparts the velocity to the electrons, thus giving them energy.

ACCEPTABLE WELD - A weld that meets all the requirements and the acceptance criteria prescribed by the welding specifications.

ACID CORE SOLDER - See cored solder.

ACTIVATED ROSIN FLUX - A rosin or resin base flux containing the additive which increases wetting by the solder.

ACTUAL THROAT - See throat of a fillet weld.

ADHESIVE BOND - Attractive forces, generally physical in character between an adhesive and the base materials. Two principal interactions that contribute to the adhesion are van der Waals bonds and dipole bonds. See van der Waals bond and permanent dipole bond.

ADHESIVE BONDING (ABD) - A materials joining process in which an adhesive, placed between the faying surfaces, solidifies to produce an adhesive bond. See adhesive bond.

AIR ACETYLENE WELDING (AAW) - A fuel gas welding process in which coalescence is produced by heating with a gas flame or flames obtained from the combustion of acetylene with air, without the application of pressure and with or without the use of filler metal. (This process is believed to be of minor industrial significance.)

AIR CAP (thermal spraying) - A device for forming, shaping and directing an air pattern for the atomization of wire or ceramic rod.

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AIR CARBON ARC CUTTING (AAC) - An arc cutting process in which metals to be cut are melted by the heat of a carbon arc and the molten metal is removed by a blast of air.

AIR FEED - A thermal spraying process variation in which an air stream carries the powdered material to be sprayed through the gun and into the heat source.

ALL-WELD-METAL TEST SPECIMEN - A test specimen with the reduced section composed wholly of weld metal.

ALTERNATE POLARITY OPERATION - A resistance welding process variation in which succeeding welds are made with pulses of alternating polarity.

ANGLE OF BEVEL - See preferred term bevel angle.

ARC BLOW - The deflection of an electric arc from its normal path because of magnetic forces.

ARC BRAZING (AB) - A brazing process in which the heat required is obtained from an electric arc. See twin carbon arc brazing.

ARC CUTTING (AC) - A group of cutting processes which melts the metals to be cut with the heat of an arc between an electrode and the base metal. See carbon arc cutting, metal arc cutting, gas metal arc cutting, gas tungsten arc cutting, plasma arc cutting, and air carbon arc cutting. Compare with oxygen arc cutting.

ARC FORCE - The axial force developed by a plasma.

ARC GOUGING - An arc cutting procedure used to form a bevel or groove.

ARC OXYGEN CUTTING - See preferred term oxygen arc cutting.

ARC PLASMA - See plasma.

ARC SEAM WELD - A seam weld made by an arc welding process.

ARC SPOT WELD - A spot weld made by an arc welding process.

ARC TIME - The time during which an arc is maintained in making an arc weld.

ARC STRIKE - A discontinuity consisting of any localized remelted metal, heat-affected metal, or change in the surface profile of any part of a weld or base metal resulting from an arc.

ARC VOLTAGE - The voltage across the welding arc.

ARC WELDING (AW) - A group of welding processes which produces coalescence of metals by heating them with an arc, with or without the application of pressure and with or without the use of filler metal.

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ARC WELDING ELECTRODE - See electrode.

ARM (resistance welding) - A projecting beam extending from the frame of a resistance welding machine, which transmits the electrode force and may conduct the welding current.

AS-BRAZED - The condition of brazements, prior to any subsequent thermal, mechanical or chemical treatments.

AS-WELDED - The condition of weld metal, welded joints, and weldments after welding but prior to any subsequent thermal, mechanical, or chemical treatments.

ATOMIC HYDROGEN WELDING (AHW) - An arc welding process which produces coalescence of metals by heating them with an electric arc maintained between two metal electrodes in an atmosphere of hydrogen. Sheilding is obtained from the hydrogen. Pressure may or may not be used and filler metal may or may not be used.

ATOMIZATION (thermal spraying) - The division of molten material at the end of the wire or rod into fine particles.

AUTOGENOUS WELD - A fusion weld made without the addition of filler metal.

AUTOMATIC BRAZING - Brazing with equipment which performs the brazing operation without constant observation and adjustment by a brazing operator. The equipment may or may not perform the loading and unloading of the work. See machine brazing.

AUTOMATIC GAS CUTTING - See preferred term automatic oxygen cutting.

AUTOMATIC OXYGEN CUTTING - Oxygen cutting with equipment which performs the cutting operation without constant observation and adjustment of the controls by an operator. The equipment may or may not perform loading and unloading of the work. See machine oxygen cutting.

AUTOMATIC WELDING - Welding with equipment which performs the welding operation without adjustment of the controls by a welding operator. The equipment may or may not perform the loading and unloading of the work. See machine welding.

B

BACK BEAD - See preferred term back weld.

BACKFIRE - The momentary recession of the flame into the welding tip or cutting tip followed by immediate reappearance or complete extinction of the flame.

BACK GAUGING - The removal of the weld metal and base metal from the other side of a partially welded joint to assure complete penetration upon subsequent welding from that side.

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BACKHAND WELDING - A welding technique in which the welding torch or gun is directed opposite to the progress of welding. Sometimes referred to as the "pull gun technique" in GMAW and FCAW. See travel angle, work angle, and drag angle.

BACKING - A material (base metal, weld metal, carbon, or granular material) placed at the root of a weld joint for the purpose of supporting molten weld metal.

BACKING BEAD - See preferred term backing weld.

BACKING FILLER METAL - See consumable insert.

BACKING PASS - A pass made to deposit a backing weld.

BACKING RING - Backing in the form of a ring, generally used in the welding of piping.

BACKING STRAP - See preferred term backing strip.

BACKING STRIP - Backing in the form of a strip.

BACKING WELD - Backing in the form of a weld.

BACKSTEP SEQUENCE - A longitudinal sequence in which the weld bead increments are deposited in the direction opposite to the progress of welding the joint. See block sequence, cascade sequence, continuous sequence, joint building sequence, and longitudinal sequence.

BACKUP (flash and upset welding) - A locator used to transmit all or a portion of the upsetting force to the workpieces or to aid in preventing the workpieces from slipping during upsetting.

BACK WELD - A weld deposited at the back of a single groove weld.

BALLING UP - The formation of globules of molten brazing filler metal or flux due to lack of wetting of the base material.

BARE ELECTRODE - See electrode.

BARE METAL ARC WELDING (BMAW) - An arc welding process which produces coalescence of metals by heating them with an electric arc between a bare or lightly coated metal electrode and the work. Neither shielding nor pressure is used and filler metal is obtained from the electrode.

BASE METAL (material) - The metal (material) to be welded, brazed, soldered, or cut. See also substrate and parent metal (material).

BASE METAL TEST SPECIMEN - A test specimen composed wholly of base metal.

BEAD - See preferred term surfacing weld.

BEVEL - An angular type of edge preparation.

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BEVEL ANGLE - The angle formed between the prepared edge of a member and a plane perpendicular to the surface of the member.

BIT (soldering) - That part of the soldering iron, usually made of copper, which actually transfers heat (and sometimes solder) to the joint.

BIT SOLDERING - See preferred term iron soldering.

BLACKSMITH WELDING - See preferred term forge welding.

BLASTING - A method of cleaning or surface roughening by a forcibly projected stream of sharp angular abrasive.

BLIND JOINT - A joint, no portion of which is visible.

BLOCK BRAZING (BB) - A brazing process is which the heat required is obtained from heated blocks applied to the parts to be joined.

BLOCK SEQUENCE - A combined longitudinal and buildup sequence for a continuous multiple pass weld in which separated lengths are completely or partially built up in cross section before intervening lengths are deposited. See backstep sequence, longitudinal sequence, etc.

BLOWHOLE - See preferred term porosity.

BLOWPIPE

WELDING AND CUTTING BLOWPIPE - See preferred term welding torch or cutting torch.

BRAZING AND SOLDERING BLOWPIPE - A device used to obtain a small, accurately directed flame, for fine work, such as in the dental and jewelry trades. Any flame may be used, a portion of it being blown to the desired location for the required time by the blowpipe which is usually mouth operated.

BOND - See adhesive bond, mechanical bond and metallic bond.

BOND COAT (thermal spraying) - A preliminary (or prime) coat of material which improves adherence of the subsequent thermal spray deposit.

BONDING FORCE - The force that holds two atoms together; it results from a decrease in energy as two atoms are brought closer to one another.

BOND LINE - In a weldment, the junction between weld metal (material), brazing filler metal or solder and base material, between thermal spray deposits and substrate, or between base metal (material) parts when filler metal is not used.

BOTTLE - See preferred term cylinder.

BOXING - The continuation of a fillet weld around a corner of a member as an extension of the principal weld.

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BRAZE - A weld produced by heating an assembly to suitable temperatures and by using a filler metal, having a liquidus above 450°C (840°F) and below the solidus of the base materials. The filler metal is distributed between the closely fitted surfaces of the joint by capillary action.

BRAZEABILITY - The capacity of a metal to be brazed under the fabrication conditions imposed into a specific suitably designed structure and to perform satisfactorily in the intended service.

BRAZEMENT - An assembly whose component parts are joined by brazing.

BRAZER - One who performs a manual or semi-automatic brazing operation.

BRAZE WELDING - A method of welding by using a filler metal, having a liquidus above 450°C (840°F) and below the solidus of the base metals. Unlike brazing, in braze welding, the filler metal is not distributed in the joint by capillary action.

BRAZING (B) - A group of welding processes which produces coalescence of materials by heating them to a suitable temperature and by using a filler metal, having a liquidus above 450°C (840°F) and below the solidus of the base materials. The filler metal is distributed between the closely fitted surfaces of the joint by capillary action.

BRAZING ALLOY - See preferred term brazing filler metal.

BRAZING FILLER METAL - The metal which fills the capillary gap and has a liquidus above 450°C (840°F) but below the solidus of the base material.

BRAZING OPERATOR - One who operates machine or automatic brazing equipment.

BRAZING PROCEDURE - The detailed methods and practices including all joint brazing procedures involved in the production of a brazement. See joint brazing procedure.

BRAZING SHEET - Brazing filler metal in sheet form.

BRAZING TECHNIQUE - The details of a brazing operation which, within the limitations of the prescribed brazing procedure, are controlled by the brazer or the brazing operator.

BRAZING TEMPERATURE - The temperature to which the base maaterials are heated to enable the filler metal to wet the bae materials and form a brazed joint.

BRAZING TEMPERATURE RANGE - The temperature range within which brazing can be conducted.

BRIDGE SIZE (eye protection) - The distance between lenses on the nose side of each eye, expressed in millimetres.

BRONZE WELDING - A term erroneously used to denote braze welding. See braze welding.

BUILDUP SEQUENCE - See joint buildup sequence.

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BURNBACK TIME - See preferred term meltback time.

BURNING - See preferred term oxygen cutting.

BURNING IN - See preferred term flow welding.

BURNOFF RATE - See preferred term melting rate.

BURN-THRU - A term erroneously used to denote excessive melt-thru or a hole. See melt-thru.

BURN-THRU WELD - A term erroneously used to denote a seam weld or spot weld.

BUTTERING - A form of surfacing in which one or more layers of weld metal are deposited on the groove face of one member (for example, a high alloy weld deposit on steel base metal which is to be welded to a dissimilar base metal). The buttering provides a suitable transition weld deposit for subsequent completion of the butt weld.

BUTT JOINT - A joint between two members aligned approximately in the same plane.

BUTTON - That part of a weld, including all or part of the nugget, which tears out in the destructive testing of spot, seam, or projection welded specimens.

BUTT WELD - An erroneous term for a weld in a butt joint. See butt joint.

C

CAPILLARY ACTION - The force by which liquid, in contact with a solid, is distributed between closely fitted faying surfaces of the joint to be brazed or soldered.

CARBON ARC CUTTING (CAC) - An arc cutting process in which metals are severed by melting them with the heat of an arc between a carbon electrode and the base metal.

CARBON ARC WELDING (CAW) - An arc welding process which produces coalescence of metals by heating them with an arc between a carbon electrode and the work. No shielding is used. Pressure and filler metal may or may not be used.

CARBON ELECTRODE - See electrode.

CARBONIZING FLAME - See preferred term reducing flame.

CARBURIZING FLAME - See preferred term reducing flame.

CARRIER GAS (thermal spraying) - The gas used to carry powdered materials from the powder feeder or hopper to the gun.

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CASCADE SEQUENCE - A combined longitudinal and buildup sequence during which weld beads are deposited in overlapping layers. See backstep sequence, block sequence, buildup sequence, and longitudinal sequence.

CAULK WELD - See preferred term seal weld.

CERAMIC ROD FLAME SPRAY GUN - A flame spraying device using heat provided by an oxyfuel gas flame. The material to be sprayed is in ceramic rod form.

CERAMIC ROD FLAME SPRAYING - A method of flame spraying which provides heat through an oxyfuel gas flame and the material to be sprayed is in ceramic rod form.

CHAIN INTERMITTENT WELDS - Intermittent welds on both sides of a joint in which the weld increments on one side are approximately opposite those on the other side.

CHAMFER - See preferred term bevel.

CHEMICAL FLUX CUTTING (FOC) - An oxygen cutting process in which metals are severed using a chemical flux to facilitate cutting.

CHILL RING - See preferred term backing ring.

CHILL TIME - See preferred term quench time.

CIRCULAR ELECTRODE - See electrode - resistance welding electrode.

CIRCULAR RESISTANCE SEAM WELDING - See preferred term transverse resistance seam welding.

CIRCUMFERENTIAL RESISTANCE SEAM WELDING - See preferred term transverse resistance seam welding.

CLAD BRAZING SHEET - A metal sheet on which one or both sides are cladded with brazing filler metal.

CLAD METAL - A composite metal containing two or three layers that have been welded together. The welding may have been accomplished by roll welding, arc welding, casting, heavy chemical deposition, or heavy electroplating. See cladding, surfaceing.

CLADDING - A relatively thick layer (less than 1 mm [0.04]) of material applied by surfacing for the purpose of improved corrosion resisance or other properties (see coating, surfacing, hardfacing).

COALESCENCE - The growing together or growth into one body of the materials being welded.

COATED ELECTRODE - See preferred terms covered electrode and lightly coated electrode. See electrode.

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COATING - A relatively thin layer (1 mm [0.04 in.]) of material applied by surfacing for the purpose of corrosion prevention, resistance to high temperature scaling, wear resistance, lubrication, or other purposes. See cladding, surfacing, and hardfacing.

COATING DENSITY (thermal spraying) - The ratio of the determined density of a thermal spraying coating to the theoretical density of the material used in the coating process. Usually expressed as per density.

COEXTRUSION WELDING (CEW) - A solid-state welding process which produces coalescence of the faying surfaces by heating and forcing materials through an extrusion die.

COIL WITHOUT SUPPORT - A filler metal package type consisting of a continuous length of electrode in coil form without an internal support. It is appropriately bound to maintain its shape.

COIL WITH SUPPORT - A filler meal package type consisting of a continuous length of electrode in coil form wound on an internal support which is a simple cylindrical section without flanges.

COLD SOLDERED JOINT - A joint with incomplete coalescence caused by insufficient application of heat to the base metal during soldering.

COLD WELDING (CW) - A solid state welding process in which pressure is used at room temperature to produce coalescence of metals with substantial deformation at the weld. Compare hot pressure welding, diffusion welding, and forge welding.

COLLAR - The reinforcing metal of a non-pressure thermit weld.

COLLARING (thermal spraying) - Adding a shoulder to a shaft or similar component as a protective confining wall for the thermal spray deposit.

COMMUTATOR-CONTROLLED WELDING (resistance welding) - The making of a number of spot or projection welds. Several electrodes, in simultaneous contact with the work, function progressively under the control of an electrical commutating device.

COMPLETE FUSION - Fusion which has occurred over the entire base material surfaces intended for welding, and between all layers and weld beads.

COMPLETE JOINT PENETRATION - Joint penetration in which the weld metal completely fills the groove and is fused to the base metal throughout its total thickness.

COMPLETE PENETRATION - See preferred term complete joint penetration.

COMPOSITE ELECTRODE - See electrode.

COMPOSITE JOINT - A joint produced by welding used in conjunction with a non-welding process. (See weldbonding.)

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CONCAVE FILLET WELD - A fillet weld having a concave face

CONCAVE ROOT SURFACE - A root surface which is concave.

CONCAVITY - The maximum distance from the face of a concave fillet weld perpendicular to a line joining the toes.

CONCURRENT HEATING - The application of supplemental heat to a structure during a welding or cutting operation.

CONE - The conical part of an oxyfuel gas flame next to the orifice of the tip.

CONSTRICTED ARC (plasma arc welding and cutting) - A plasma arc column that is shaped by a constricting nozzle orifice.

CONSTRICTING NOZZLE (plasma arc welding and cutting) - A water cooled copper nozzle surrounding the electrode and containing the constricting orifice.

CONSTRICTING ORIFICE (plasma arc welding and cutting) - The hole in the constricting nozzle through which the arc passes.

CONSUMABLE GUIDE ELECTROSLAG WELDING - An electroslag welding process variation in which filler metal is supplied by an electrode and its guiding member. See electroslag welding (ESW).

CONSUMABLE INSERT - Preplaced filler metal which is completely fused into the root of the joint and becomes part of the weld.

CONTACT TUBE - A device which transfers current to a continuous electrode.

CONTINUOUS SEQUENCE - A longitudinal sequence in which each pass is made continuously from one end of the joint to the other . See backstep sequence, longitudinal sequence, etc.

CONVEX FILLET WELD - A fillet weld having a convex face.

CONVEX ROOT SURFACE - A root surface which is convex.

CONVEXITY - The maximum distance from the face of a convex fillet weld perpendicular to a line joining the toes.

COOL TIME (resistance welding) - The time interval between successive heat times in multiple-impulse welding or in the making of seam welds.

COPPER BRAZING - A term improperly used to denote brazing with a copper filler metal. See preferred terms furnace brazing and braze welding.

CORED SOLDER - A solder wire or bar containing flux as a core.

CORNER-FLANGE WELD - A flange weld with only one member flanged at the location of welding.

CORNER JOINT - A joint between two members located approximately at right angles to each other.

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CORONA (resistance welding) - The area sometimes surrounding the nugget of a spot weld at the faying surfaces which provides a degree of solid state welding.

CORRECTIVE LENS (eye protection) - A lens ground to the wearer's individual corrective prescription.

CORROSIVE FLUX - A flux with a residue that chemically attacks the base metal. It may be composed of inorganic salts and acids, organic salts and acids, or activated rosins of resins.

COSMETIC PASS - A weld pass made primarily for the purpose of enhancing appearance.

CO_2 WELDING - See preferred term gas metal arc welding.

COVALENT BOND - A primary bond arising from the reduction in energy associated with overlapping half-filled orbitals of two atoms.

COVERED ELECTRODE - See electrode.

COVER LENS (eye protection) - A round cover plate.

COVER PLATE (eye protection) - A removable pane of colorless glass, plastic-coated glass, or plastic that covers the filter plate and protects it from weld splatter, pitting, or scratching when used in a helmet, hood, or goggle.

CRACK - A fracture-type discontinuity characterized by a sharp tip and high ratio of length and width to opening displacement.

CRATER - In arc welding, a depression at the termination of a weld bead or in the molten weld pool.

CRATER CRACK - A crack in the crater of a weld bead.

CRATER FILL CURRENT - The current value during crater fill time.

CRATER FILL TIME - The time interval following weld time but prior to meltback time during which arc voltage or current reaches a preset value greater or less than welding values. Weld travel may or may not stop at this point.

CRATER FILL VOLTAGE - The arc voltage value during crater fill time.

CROSS WIRE WELD - A weld made between crossed wires or bars.

CUP - See preferred term nozzle.

CUTTING ATTACHMENT - A device for converting an oxyfuel gas welding torch into an oxygen cutting torch.

CURRING HEAD - The part of a cutting machine or automatic cutting equipment in which a cutting torch or tip is incorporated.

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CUTTING NOZZLE - See preferred term cutting tip.

CUTTING PROCESS - A process which brings about the severing or removal of metals. See arc cutting and oxygen cutting.

CUTTING TIP - That part of an oxygen cutting torch from which the gases issue.

CUTTING TORCH (oxyfuel) - A device used for directing the preheating flame produced by the controlled combustion of fuel gases and to direct and control the cutting oxygen.

CUTTING TORCH (plasma arc) - A device used for plasma arc cutting to control the position of the electrode, to transfer current to the arc, and to direct the flow of plasma and shielding gas.

CYLINDER MANIFOLD - See preferred term manifold.

D

DEFECT - A discontinuity or discontinuities which by nature or accumulated effect (for example, total crack length) render a part or product unable to meet minimum applicable acceptance standards or specifications. This term designates rejectability. See discontinuity and flaw.

DEFECTIVE WELD - A weld containing one or more defects.

DEPOSIT (thermal spraying) - See preferred term spray deposit.

DEPOSITED METAL - Filler metal that has been added during a welding operation.

DEPOSITION EFFICIENCY (arc welding) - The ratio of the weight of deposited metal to the net weight of filler metal consumed, exclusive of stubs.

DEPOSITION EFFICIENCY (thermal spraying) - The ratio usually expressed in percent of the weight of spray deposit to the weight of the material sprayed.

DEPOSITION RATE - The weight of material deposited in a unit of time. It is usually expressed as kilograms per hour (kg/h) (pounds per hour [lb/h]).

DEPOSITION SEQUENCE - The order in which the increments of weld metal are deposited. See longitudinal sequence, buildup sequence, and pass sequence.

DEPOSIT SEQUENCE - See preferred term deposition sequence.

DEPTH OF FUSION - The distance that fusion extends into the base metal or previous pass from the surface melted during welding.

DETONATION GUN (thermal spraying) - The device used in detonation flame spraying.

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DETONATION FLAME SPRAYING - A thermal spraying process variation in which the controlled explosion of a mixture of fuel gas, oxygen, and powdered coating material is utilized to melt and propel the material to the workpiece.

DIE

RESISTANCE WELDING DIE - A member usually shaped to the work contour to clamp the parts being welded and to conduct the welding current.

FORGE WELDING DIE - A device used in forge welding primarily to form the work while hot and apply the necessary pressure.

DIE WELDING - See preferrred terms forge welding and - command welding.

DIFFUSION AID - A solid filler metal applied to the faying surfaces to assist in diffusion welding.

DIFFUSION BONDING - See preferred terms diffusion brazing and diffusion welding.

DIFFUSION BRAZING (DFB) - A brazing process which produces coalescence of metals by heating them to suitable temperatures and by using a filler metal or an in situ liquid phase. The filler metal may be distributed by capillary action or may be placed or formed at the faying surfaces. The filler metal is diffused with the base metal to the extend that the joint properties have been changed to approach those of the base metal. Pressure may or may not be applied.

DIFFUSION WELDING (DFW) - A solid state welding process which produces coalescence of the faying surfaces by the application of pressure and elevated temperature. The process does not involve macroscopic deformation, melting, or relative motion of parts. A solid filler metal (diffusion aid) may or may not be inserted between the faying surfaces. See also forge welding, hot pressure welding, and cold welding.

DILUTION - The change in chemical composition of a welding filler material caused by the admixture of the base material or previously deposited weld material in the deposited weld bead. It is normally measured by the percentage of base material or previously deposited weld material in the weld bead.

DIP BRAZING (DB) - A brazing process in which the the heat required is furnished by a molten chemical or metal bath. When a molten chemical bath is used, the bath may act as a flux. When a molten metal bath is used, the bath provides the filler metal.

DIRECT CURRENT ELECTRODE NEGATIVE (DCEN) - The arrangement of direct current arc welding leads in which the work is the positive pole and the electrode is the negative pole of the welding arc. See also straight polarity.

DIRECT CURRENT ELECTRODE POSITIVE (DCEP) - The arrangement of direct durrent arc welding leads in which the work is the negative pole and the electrode is the positive pole of the welding arc. See also reverse polarity.

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DIRECT CURRENT REVERSE POLARITY (DCRP) - See reverse polarity and direct current electrode positive.

DIRECT CURRENT STRAIGHT POLARITY (DCSP) - See straight polarity and direct current electrode negative.

DISCONTINUITY - An interruption of the typical structure of a weldment, such as a lack of homogeneity in the mechanical, metallurgical, or physical characteristics of the material or weldment. A discontinuity is not necessarily a defect. See defect, flaw.

DOPED SOLDER - A solder containing a small amount of an element intentionally added to insure retention of one or more characteristics of the materials on which it is used.

DOUBLE ARCING (plasma arc welding and cutting) - A condition in which the main arc does not pass through the constricting orifice but transfers to the inside surface of the nozzle. A secondary arc is simultaneously established between the outside surface of the nozzle and the workpiece. Double arcing usually damages the nozzle.

DOUBLE-WELDED JOINT - In arc and oxyfuel gas welding, any joint welded from both sides.

DOVETAILING (thermal spraying) - A method of surface roughening involving angular undercutting to interlock the spray deposit.

DOWNHAND - See preferred term flat position.

DOWNSLOPE TIME (resistance welding) - The time during which the welding current is continuously decreased.

DOWNSLOPE TIME (automatic arc welding) - The time during which the current is changed continuously from final taper current or welding current to final current.

DRAG TIME (thermal cutting) - The offset distance between the actual and the theoretical exit points of the cutting oxygen stream measured on the exit surface of the material.

DRAG ANGLE - The travel angle when the electrode is pointing backward. See also backhand welding.

Note: This angle can be used to define the position of welding guns, welding torches, high energy beams, welding rods, thermal cutting and thermal spraying guns.

DROP-THRU - An undesirable sagging or surface irregularity, usually encountered when brazing or welding near the solidus of the base metal caused by overheating with rapid diffusion or alloying between the filler metal and the base metal.

DRUM - A filler metal package type consisting of a continuous length of electrode wound or coiled within an enclosed cylindrical container.

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DUTY CYCLE - The percentage of time during an arbitrary test period, usually 10 min, during which a power supply can be operated at its rated output without overloading.

DYNAMIC ELECTRODE FORCE - See electrode force.

E

EDGE-FLANGE WELD - A flange weld with two members flanged at the location of welding.

EDGE JOINT - A joint between the edges of two or more parallel or nearly parallel members.

EDGE PREPARATION - The surface prepared on the edge of a member for welding.

EDGE WELD - A weld in an edge joint.

EFFECTIVE LENGTH OF WELD - The length of weld throughout which the correctly proportioned cross section exists. In a curved weld, it shall be measured along the axis of the weld.

EFFECTIVE THROAT - The minimum distance from the root of a weld to its face less any reinforcement.

ELECTRIC ARC GUN (thermal spraying) - A device where the material to be sprayed is in the form of two consumable electrodes. An arc is struck and maintained between the ends of the two electrodes, and provides the heat for melting.

ELECTRIC ARC SPRAYING (EASP) (thermal spraying) - A process using as a heat source, an electric arc between two consumable electrodes of a coating material, and a compressed gas which is used to atomize and propel the material to the substrate.

ELECTRIC BONDING (thermal spraying) - See preferred term surfacing.

ELECTRIC BRAZING - See preferred terms resistance brazing and arc brazing.

ELECTRODE

ARC WELDING ELECTRODE - A component of the welding circuit through which current is conducted between the electrode holder and the arc. See arc welding.

BARE ELECTRODE - A filler metal electrode consisting of a single metal or alloy that has been produced into a wire, strip, or bar form and that has had no coating or covering applied to it other than that which was incidental to its manufacture or preservation.

CARBON ELECTRODE - A non-filler material electrode used in arc welding or cutting, consisting of a carbon or graphite rod, which may be coated with copper or other coatings.

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COMPOSITE ELECTRODE - Any of a number of multicomponent filler metal electrodes in various physical forms such as stranded wires, tubes, and covered wire.

COVERED ELECTRODE - A composite filler metal electrode consisting of a core of a bare electrode or metal cored electrode to which a covering sufficient to provide a slag layer on the weld metal has been applied. The covering may contain materials providing such functions as shielding from the atmosphere, deoxidation, and arc stabilization and can serve as a source of metallic additions to the weld.

FLUX CORED ELECTRODE - A composite filler metal electrode consisting of a metal tube or other hollow configurations containing ingredients to provide such functions as shielding atmosphere, deoxidation, arc stabilization and slag formation. Alloying materials may be included in the core. External shielding may or may not be used.

METAL CORD ELECTRODE - A composite filler metal electrode consisting of a metal tube or other hollow configuration containing alloying ingredients. Minor amounts of ingredients providing such functions as arc stabilization and fluxing of oxides may be included. External shielding gas may or may not be used.

STRANDED ELECTRODE - A composite filler metal electrode consisting of stranded wires which may mechanically enclose materials to improve properties, stabilize the arc, or provide shielding.

EMISSIVE ELECTRODE - A filler metal electrode consisting of a core of a bare electrode or a composite electrode to which a very light coating has been appllied to produce a stable arc.

LIGHTLY COATED ELECTRODE - A filler metal electrode conssting of a metal wire with a light coating applied subsequent to the drawing operation, primarily for stabilizating the arc.

METAL ELECTRODE - A filler or non-filler metal electrode, used in arc welding or cutting, consisting of a metal wire or rod that has been manufactured by any method and that is either bare or covered with a suitable covering or coating.

TUNGSTEN ELECTRODE - A non-filler metal electrode used in arc welding or cutting, made principally of tungsten.

ELECTROSLAG WELDING ELECTRODE - A filler metal component of the welding circuit through which current is conducted between the electrode guiding member and the molten slag.

RESISTANCE WELDING ELECTRODE - The part or parts of a resistance welding machine through which the welding current and, in most cases, pressure, are applied directly to the work. The electrode may be in the form of a rotating wheel, rotating roll, bar, cylinder, plate, clamp, chuck, or modification thereof. See resistance welding.

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ELECTRODE EXTENSION (gas metal arc welding, flux cored arc welding, submerged arc welding) - The length of unmelted electrode extending beyond the end of the contact tube during welding.

ELECTRODE FORCE

DYNAMIC ELECTRODE FORCE (resistance welding) - The force in newtons (pounds force) between the electrodes during the actual welding cycle in making spot, seam, or projection welds by resistance welding.

STATIC ELECTRODE FORCE (resistance welding) - The force between the electrodes in making spot, seam, or projection welds by resistance welding under welding conditions, but with no current flowing and no movement in the welding machine.

THEORETICAL ELECTRODE FORCE (resistance welding) - The force, neglecting friction and inertial, in making spot, seam, or projection welds by resistance welding, available at the electrodes of a resistance welding machine by virtue of the initial force application and the theoretical mechanical advantage of the system.

ELECTRODE HOLDER - A device used for mechanically holding the electrode while conducting current to it.

ELECTRODE LEAD - The electrical conductor between the source of arc welding current and the electrode holder.

ELECTRODE SETBACK (plasma arc welding and cutting) - The distance the electrode is recessed behind the constricting orifice measured from the outer face of the nozzle.

ELECTRODE SKID (resistance welding) - The sliding of an electrode along the surface of the work during the making of spot, weam, or projection welds by resistance welding.

ELECTROGAS WELDING (EGW) - An arc welding process which produces coalescence of metals by heating them with an arc between a continuous filler metal (consumable) electrode and the work. Molding shoe(s) are used to confine the molten weld metal for vertical position welding. The electrodes may be either flux cored or solid. Shielding may or may not be obtained from an externally supplied gas or mixture.

ELECTRON BEAM CUTTING (EBC) - A cutting process which uses the heat obtained from a concentrated beam composed primarily of high velocity electrons which impinge upon the workpieces to be cut; it may or may not use an externally supplied gas.

ELECTRON BEAM GUN - A device for producing and accelerating electrons. Typical components include the emitter (also called the filament or cathode) which is heated to produce electrons via thermionic emission, a cup (also called the grid or grid cup), and the anode.

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ELECTRON BEAM GUN COLUMN - The electron beam gun plus auxiliary mechanical and electrical components which may include beam alignment, focus, and deflection coils.

ELECTRON BEAM WELDING (EBW) - A welding process which produces coalescence of metals with the heat obtained from a concentrated beam composed primarily of high velocity electrons impinging upon the joint. See electron beam welding -- high vacuum, electron beam welding -- medium vacuum, and electron beam welding -- nonvacuum.

ELECTRON BEAM WELDING -- HIGH VACUUM (EBW-HV) - An electron beam welding process variation in which welding is accomplished at a pressure of approximately 10^{-6} to 10^{-3} torr.

ELECTRON BEAM WELDING -- MEDIUM VACUUM (EBW-MV) - An electron beam welding process variation in which welding is accomplished at a pressure of approximately 10^{-3} to 25 torr. The term "medium vacuum" encompasses the range of pressures often referred to as "soft" and "partial vacuum."

ELECTRON BEAM WELDING -- NONVACUUM (EBW-NV) - An electron beam welding process variation in which welding is accomplished at atmospheric pressure.

ELECTRONIC HEAT CONTROL (resistance welding) - A device for adjusting the heating value (rms value) of the current in making a resistance weld by controlling the ignition or firing of the electronic devices in an electronic contactor. The current is initiated each half-cycle at an adjustable time with respect to the zero point on the voltage wave.

ELECTROSLAG WELDING (ESW) - A welding process producing coalescence of metals with molten slag which melts the filler metal and the surfaces of the work to be welded. The molten weld pool is shielded by this slag which moves along the full cross section of the joint as welding progresses. The process is initiated by an arc which heats the slag. The arc is then extinguished and the conductive slag is maintained in a molten condition by its resistance to electric current passing between the electrode and the work. See electroslag welding electrode and consumable guide electroslag welding.

ELECTROSLAG WELDING ELECTRODE - See electrode.

END RETURN - See preferred term boxing.

EROSION (brazing) - A condition caused by dissolution of the base metal by molten filler metal resulting in a postbraze reduction in the thickness of the base metal.

EXHAUST BOOTH - A mechanically ventilated, semi-enclosed area in which an air flow across the work area is used to remove fumes, gases, and material particles.

EXPLOSION WELDING (EXW) - A solid state welding process in which coalescence is effected by high-velocity movement together of parts to be joined produced by a controlled detonation.

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EYE SIZE (eye protection) - The nominal size of the lens-holding section of an eye frame expressed in millimetres.

F

FACE FEED - The application of filler metal to the joint, usually by hand, during brazing and soldering.

FACE OF WELD - The exposed surface of a weld on the side from which welding was done.

FACE REINFORCEMENT - Reinforcement of weld at the side of the joint from which welding was done. See also roof reinforcement.

FACE SHIELD (eye protection) - A device positioned in front of the eyes and a portion of, or all of, the face, whose predominant function is protection of the eyes and face. See also hand shield and helmet.

FAYING SURFACE - That mating surface of a member which is in contact or in close proximity with another member to which it is to be joined

FEED RATE (thermal spraying) - The rate at which a material passes through the gun in a unit of time. A synonym for spray rate.

FERRITE NUMBER - An arbitrary, standardized value designating the ferrite content of an austenitic stainless steel weld metal. It should be used in place of percent ferrite or volume percent ferrite on a direct one to one replacment basis.

FILLER METAL (material) - The metal (material) to be added in making a welded, brazed, or soldered joint. See electrode, welding rod, backing filler metal, brazing filler metal, diffusion aid, solder, and spray deposit.

FILLET WELD - A weld of approximately triangular cross section joining two surfaces approximately at right angles to each other in a lap joint, T-joint or corner joint.

FILLET WELD SIZE - See preferred term size of weld.

FILTER GLASS - See preferred term filter plate.

FILTER LENS (eye protection) - A round filter plate.

FILTER PLATE (eye protection) - An optical material which protects the eyes against excessive ultraviolet, infrared, and visible radiation.

FINES - Any or all material finer than a particular mesh under consideration.

FIRECRACKER WELDING - A variation of the shielded metal arc welding process in which a length of covered electrode is placed along the joint in contact with the parts to be welded; during the welding operation, the stationary electrode is consumed as the arc travels the length of the electrode.

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FISHEYE - A discontinuity found on the fracture surface of a weld in steel that consists of a small pore or inclusion surrrounded by an approximately round, bright area.

FISSURE - A small crack-like discontinuity with only slight separation (opening displacement) of the fracture surfaces. The prefixes macro or micro indicate relative size.

FIXTURE - A device designed to hold parts to be joined in proper relation to each other.

FLAME CUTTING - See preferred term oxygen cutting.

FLAME SPRAYING (FLSP) - A thermal spraying process in which an oxyfuel gas flame is the source of heat for melting the coating material. Compressed gas may or may not be used for atomizing and propelling the material to the substrate.

FLAME WELD - A weld made on the edges of two or more members to be joined, usually light gage metal, at least one of the members being flanged.

FLARE-BEVEL GROOVE WELD - A weld in a groove formed by two members with curved surfaces.

FLARE-V-GROOVE WELD - A weld in a groove formed by two members with curved surfaces.

FLASH - The material which is expelled or squeezed out of a weld joint and forms around the weld.

FLASHBACK ARRESTOR - A device to limit damage from a flashback by preventing propagation of the flame front beyond the point at which the arrester is installed.

FLASH COAT - A thin coating usually less than 0.05 mm (0.002 in.) in thickness.

FLASH BUTT WELDING - See preferred term flash welding.

FLASHING TIME - The time during which the flashing action is taking place in flash welding.

FLASH-OFF TIME - See preferred term flashing time.

FLASH WELD - A weld made by flash welding.

FLASH WELDING (FW) - A resistance welding process which produces coalescence simultaneously over the entire area of abutting surfaces, by the heat obtained from resitance to electric current between the two surfaces, and by the application of pressure after heating is substantially completed. Flashing and upsetting are accompanied by expulsion of metal from the joint.

FLAT POSITION - The welding position used to weld from the upper side of the joint; the face of the weld is approximately horizontal.

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FLAW - A near synonym for discontinuity but with an undesirable connotation. See defect, discontinuity.

FLOWABILITY - The ability of molten filler metal to flow or spread over a metal surface.

FLOW BRAZING (FLB) - A brazing process which produces coalescence of metals by heating them with molten nonferrous filler metal poured over the joint until brazing temperature is attained. The filler metal is distributed in the joint by capillary action.

FLOW BRIGHTENING (soldering) - Fusion (melting) of a chemically or mechanically desposited metallic coating on a substrate.

FLOW WELDING (FLOW) - A welding process which produces coalescence of metals by heating them with molten filler metal, poured over the surfaces to be welded until the welding temperature is attained and untile the required filler metal has been added. The filler metal is not distributed in the joint by capillary action.

FLUX - Material used to prevent, dissolve, or facilitate removal of oxides and other undesirable surface substances.

FLUX CORED ARC WELDING-ELECTROGAS (FCAW-EG) - A variation of the flux cored arc welding process in which molding shoes are used to confine the molten weld metal for vertical position welding. Additional shielding may or may not be obtained from an externally supplied gas or gas mixture.

FLUX CORED ELECTRODE - See electrode.

FLUX COVER - In metal bath dip brazing and dip soldering a cover of flux over the molten filler metal bath.

FLUX OXYGEN CUTTING - See preferred term chemical flux cutting.

FOCAL POINT - See preferred term focal spot.

FOCAL SPOT (electron beam welding and laser beam welding) - A spot at which an energy beam has the most concentrated energy level and the smallest cross sectional area.

FOREHAND WELDING - A welding technique in which the welding torch or gun is directed toweard the progress of welding. See also travel angle, work angle, push angle.

FORGE-DELAY TIME (resistance welding) - The time elapsing between the beginning of weld time or weld interval and the instant of application of forging force to the electrodes.

FORGE WELDING (FOW) - A solid state welding process which produces coalescence of metals by heating them in air in a forge and by applying pressure or blows sufficient to cause permanent deformation at the interface. Compare cold welding, roll welding, diffusion welding, and hot pressure welding.

FRICTION SOLDERING - See preferred term abrasion soldering.

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FRICTION WELDING (FRW) - A solid state welding process which produces coalescence of materials by the heat obtained from a mechanically induced sliding motion between rubbing surfaces. The work parts are held together under pressure.

FUEL GASES - Gases usually used with oxygen for heating such as acetylene, natural gas, hydrogen, propane, methylacetylene propadiene, and other synthetic fuels and hydrocarbons.

FULL FILLET WELD - A fillet whose size is equal to the thickness of the thinner member joined.

FURNACE BRAZING (FB) - A brazing process in which the parts to be joined are placed in a furnace heated to a suitable temperature.

FURNACE SOLDERING (FS) - A soldering process in which the parts to be joined are placed in a furnace heated to a suitable temperature.

FUSED SPRAY DEPOSIT (thermal spraying) - A self-fluxing spray deposit which is subsequently heated to coalescence within itself and with the substrate.

FUSED ZONE - See preferred terms fusion zone, nugget and bond line.

FUSING (thermal spraying) - See preferred term fusion.

FUSION - The melting together of filler metal and base metal (substrate), or of base metal only, which results in coalescence. See depth of fusion.

FUSION FACE - A surface of the base metal which will be melted during welding.

FUSION WELDING - Any welding process or method which uses fusion to complete the weld.

FUSION ZONE - The area of base metal melted as determined on the cross section of a weld.

G

GAS BRAZING - See preferred term torch brazing.

GAS CARBON ARC WELDING (CAW-G) - A carbon arc welding process variation which produces coalescence of metals by heating them with an electric arc between a single carbon electrode and the work. Shielding is obtained from a gas or gas mixture.

GAS CUTTER - See preferred oxygen cutter.

GAS CUTTING - See preferred oxygen cutting.

GAS GOUGING - See preferred term oxygen gouging.

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GAS METAL ARC CUTTING (GMAC) - An arc cutting process used to sever metals by melting them with the heat of an arc between a continuous metal (consumable) electrode and the work. Shielding is obtained entirely from an externally supplied gas or gas mixture.

GAS METAL ARC WELDING (GMAW) - An arc welding process which produces coalescence of metals by heating them with an arc between a continuous filler metal (consumable) electrode and the work. Shielding is obtained entirely from an externally supplied gas or gas mixture.

GAS METAL ARC WELDING-ELECTROGAS (GMAW-EG) - A variation of the gas metal arc welding process using molding shoes to confine the molten weld metal for vertical position welding.

GAS METAL ARC WELDING-PULSED ARC (GMAW-P) - A variation of the gas metal arc welding process in which the current is pulsed. See also pulsed power welding.

GAS METAL ARC WELDING-SHORT CIRCUITING ARC (GMAW-S) - A variation of the gas metal arc welding process in which the consumable electrode is deposited during repeated short circuits. See also gas circuiting transfer.

GAS POCKET - See preferred term porosity.

GAS REGULATOR - See preferred term regulator.

GAS SHIELDED ARC WELDING - A general term used to describe gas metal arc welding, gas tungsten arc welding, and flux cored arc welding when gas shielding is employed.

GAS SHIELDED STUD WELDING - See stud arc welding.

GAS TORCH - See preferred term welding torch and cutting torch.

GAS TUNGSTEN ARC CUTTING (GTAC) - An arc cutting process in which metals are severed by melting them with an arc between a single tungsten (nonconsumable) electrode and the work. Shielding is obtained from a gas or gas mixture.

GAS TUNGSTEN ARC WELDING (GTAW) - An arc welding process which produces coalescence of metals by heating them with an arc between a tungsten (nonconsumable) electrode and the work. Shielding is obtained from a gas or gas mixture. Pressure may or may not be used and filler metal may or may not be used. (This process has sometimes been called TIG welding.)

GAS TUNGSTEN ARC WELDING-PULSED ARC (GTAW-P) - A variation of the gas tungsten arc welding process in which the current is pulsed. See also pulsed power welding.

GAS WELDING - See preferred term oxyfuel gas welding (OFW).

GLOBULAR TRANSFER (arc welding) - A type of metal transfer in which molten filler metal is transferred across the arc in large droplets.

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GOUGING - The forming of a bevel or groove by material removal. See also back gouging, arc gouging, and oxygen gouging.

GRADATED COATING - A thermal sprayed deposit composed of mixed materials in successive layers which progressively change in composition from the constituent material of the substrate to the surface of the sprayed deposit.

GRIT - See preferred term abrasive.

GROOVE - An opening or channel in the surface of a part or between two components which provides space to contain a weld.

GROOVE AND ROTARY ROUGHENING (thermal spraying) - A method of surface roughening in which grooves are made and the original surface roughened and spread.

GROOVE ANGLE - The total included angle of the groove between parts to be joined by a groove weld.

GROOVE FACE - That surface of a member included in the groove.

GROOVE RADIUS - The radius used to form the shape of a J- or U-groove weld joint.

GROOVE TYPE - The geometric configuration of a groove.

GROOVE WELD - A weld made in the groove between two members to be joined. The standard types of groove welds are as follows:

double-bevel-groove weld
double-flare-bevel-groove weld
double-flare-V-groove weld
double-J-groove weld
double-U-groove weld
double-V-groove weld
single-bevel-groove weld
single-flare-bevel-groove weld
single-flare-V-groove weld
single-J-groove weld
single-U-groove weld
single-V-groove weld
single-groove weld

GROUND CONNECTION - An electrical connection of the welding machine frame to the earth for safety. See also work connection and work lead.

GROUND LEAD - See preferred term work lead.

GUN

ARC WELDING GUN - In semiautomatic machine, and automatic welding, a manipulating device to transfer current and guide the electrode into the arc. It may include provisions for shielding and arc initiation.

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RESISTANCE WELDING GUN - A manipulating device to transfer current and provide electrode force to the weld area (usually in reference to a portable gun).

THERMAL SPRAYING GUN - A thermal spraying device.

GUN EXTENSION (thermal spraying) - The extension tube attached in front of the thermal spraying device to permit spraying within confined areas or deep recesses.

H

HAMMER WELDING - See preferred terms forge welding and cold welding.

HAND SHIELD - A protective device, used in arc welding, for shielding the eyes, face, and neck. A hand shield is equipped with a suitable filter plate and is designed to be held by hand.

HARD FACING - A particular form of surfacing in which a coating or cladding is appplied to a substrate for the main purpose of reducing wear or loss of material by abrasion, impact, erosion, galling, and cavitation. See coating, cladding, and surfacing.

HARD SOLDER - A term erroneously used to denote silver-base brazing filler metals.

HARD SURFACING - See preferred terms surfacing or hard surfacing.

HEAD - See welding head and cutting head.

HEAT-AFFECTED ZONE - The portion of the base metal which has not been melted, but whose mechanical properties or microstructure have been altered by the heat of welding, brazing, soldering, or cutting.

HEATING GATE - The opening in a thermit mold through which the parts to be welded are preheated.

HEAT TIME (resistance welding) - The duration of current flow during any one impulse in multiple-impulse welding, or the duration of current flow when making a simple welding cycle.

HELMUT (eye protection) - A protection device, used in arc welding, for shielding the eyes, face, and neck. A helmet is equppped with a suitable filter plate and is designed to be worn on the head.

HIGH FREQUENCY RESISTANCE WELDING (HFRW) - A resistance welding process which produces coalescence of metals with the heat generated from the resistance of the workpieces to a high frequency alternating current in the 10 to 500 kHz range and the rapid application of an upsetting force after heating is substantially completed. The path of the current in the work-piece is controlled by the use of the proximity effect (the feed current follows closely the return current conductor.)

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HIGH PULSE CURRENT - The current levels during the high pulse time which produces the high heat level.

HIGH PULSE TIME - The duration of the high current pulse time.

HOLD TIME (resistance welding) - The duration of force application at the point of welding after the last impulse of current ceases.

HOLDING TIME - In brazing and soldering, the amount of time a joint is held within a specified temperature range.

HORIZONTAL FIXED POSITION (pipe welding) - In pipe welding the position of a pipe joint in which the axis of the pipe is approximately horizontal and the pipe is not rotated during welding.

HORIZONTAL POSITION

FILLET WELD - The position in which welding is performed on the upper side of an approximately horizontal surface and against an approximately vertical surface.

GROOVE WELD - The position of welding in which the axis of the weld lies in an approximately horizontal plane and the face of the weld lies in an approximately vertical plane.

HORIZONTAL ROLLED POSITION (pipe welding) - The position of a pipe joint in which the axis of the pipe is approximately horizontal, and welding is performed in the flat position by rotating the pipe.

HORN (resistance welding) - An essentially cylindrical arm or extension of an arm of a resistance welding machine which transmits the electrode force and usually conducts the welding current. See Arm.

HORN SPACING (resistance welding) - The distance between adjacent surfaces of the horns of a resistance welding machine.

HOT ISOSTATIC PRESSURE WELDING - A diffusion welding process variation which produces coalescence of materials by heating and applying hot inert gas under pressure.

HOT PRESSURE WELDING (HPW) - A solid state welding process which produces coalescence of materials with heat and application of pressure sufficient to produce macrodeformation of the base material. Vacuum or other shielding media may be used. See also forge welding and diffusion welding.

HOT START CURRENT - A very brief current pulse at arc initiation to stabilize the arc quickly.

HOT WIRE WELDING - A variation of arc welding processes (GTAW, PAW, SAW) in which a filler metal wire is resistance-heated as it is fed into the molten weld pool.

HYDROGEN BRAZING - A term erroneously used to denote any brazing process which takes place in a hydrogen or hydrogen-containing atmosphere.

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HYDROMATIC WELDING - See preferred term pressure-controlled welding.

I

IMPULSE (resistance welding) - An impulse of welding current consisting of a single pulse or a series of pulses, separated only by an interpulse time.

INADEQUATE JOINT PENETRATION - Joint penetration whih is less than that specified.

INCLINED POSITION - The position of a pipe joint in which the axis of the pipe is approximately at an angle of 45° to the horizontal and the pipe is not rotated during welding.

INCLINED POSITION (with restriction ring) - The position of a pipe joint in which the axis of the pipe is approximately at an angle of 45° to the horizontal and a restriction ring is located near the joint. The pipe is not rotated during welding.

INCLUDED ANGLE - See preferred term groove angle.

INCOMPLETE FUSION - Fusion which is less than complete.

INDENTATION - In a spot, seam, or projection weld, the depression on the exterior surface or surfaces of the base metal.

INDUCTION BRAZING (IB) - A brazing process in which the heat required is obtained from the resistance of the work to induced electric current.

INDUCTION SOLDERING (IS) - A soldering process in which the heat required is obtained from the resistance of the work to induced electric current.

INDUCTION WELDING (IW) - A welding process which produces coalescence of metals through the heat obtained from resistance of the work to induced electric current, with or without the application of pressure.

INDUCTION WORK COIL - See preferred term work coil.

INERT GAS - A gas which does not normally combine chemically with the base metal or filler metal. See also protective atmosphere.

INERT GAS METAL ARC WELDING - See preferred term gas metal arc welding.

INERT GAS TUNGSTEN ARC WELDING - See preferred term gas tungsten arc welding.

INFRARED BRAZING (IRB) - A brazing process in which the heat required is furnished by infrared radiation.

INFRARED RADIATION - Electromagnetic energy with wavelengths from 770 to 12,000 nanometers.

INFRARED SOLDERING (IRS) - A soldering process in which the heat required is furnished by infrared radiation.

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INITIAL CURRENT - The current after starting, but before establishment of welding current.

INTERGRANULAR PENETRATION - The penetration of a filler metal along the grain boundaries of a base metal.

INTERMEDIATE FLUX - A soldering flux with a residue that generally does not attack the base metal. The original composition may be corrosive.

INTERMITTENT WELD - A weld in which the continuity is broken by recurring unwelded spaces.

INTERPASS TEMPERATURE - In a multiple pass weld, the temperature (minimum or maximum as specified) of the deposited weld metal before the next pass is started.

INTERPULSE TIME (resistance welding) - The time between successive pulses of current within the same impulse.

INTERRUPTED SPOT WELDING - See preferred term multiple-impulse welding.

IONIC BOND - A primary bond arising from the electrostatic attraction between two oppositely charged ions.

IRON SOLDERING (INS) - A soldering process in which the heat required is obtained from a soldering iron.

J

JOINT - The junction of members or the edges of members which are to be joined or have been joined.

JOINT BRAZING PROCEDURE - The materials, detailed methods, and practices employed in the brazing of a particular joint.

JOINT BUILDUP SEQUENCE - The order in which the weld beads of a multiple-pass weld are deposited with respect to the cross section of the joint. See also block sequence and longitudinal sequence.

JOINT CLEARANCE - The distance between the faying surfaces of a joint. In brazing, this distance is referred to as that which is present before brazing, at the brazing temperature, or after brazing is completed.

JOINT DESIGN - The joint geometry together with the required dimensions of the welded joint.

JOINT EFFICIENCY - The ratio of the strength of a joint to the strength of the base metal (expressed in percent).

JOINT GEOMETRY - The shape and dimensions of a joint in cross section prior to welding.

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JOINT PENETRATION - The minimum depth a groove or flange weld extends from its face into a joint, exclusive of reinforcement. Joint penetration, root penetration, and effective throat.

JOINT WELDING SEQUENCE - See preferred term joint buildup sequence.

K

KERF - The width of the cut produced during a cutting process.

KEYHOLE - A technique of welding in which a concentrated heat source penetrates completely through a workpiece forming a hole at the leading edge of the molten weld metal. As the heat source progresses, the molten metal fills in behind the hole to form the weld bead.

KEYING (thermal spraying) - See preferred term mechanical bond.

KNEE (resistance welding) - The lower arm supporting structure in a resistance welding machine.

KNURLING (thermal spraying) - See preferred terms rotary roughening, groove and rotary roughening, and threading and knurling.

L

LACK OF FUSION - See preferred term incomplete fusion.

LACK OF JOINT PENETRATION - See preferred term inadequate joint penetration.

LAND - See preferred term root face.

LAP JOINT - A joint between two overlapping members.

LASER BEAM CUTTING (LBC) - A cutting process which severs materials with the heat obtained from the application of a concentrated coherent light beam impinging upon the workpiece to be cut. The process can be used with or without an externally supplied gas.

LASER BEAM WELDING (LBW) - A welding process which produces coalescence of materials with the heat obtained from the application of a concentrated coherent light beam impinging upon the members to be joined.

LAYER - A stratum of weld metal or surfacing material. The layer may consist of one or more weld beads laid side by side.

LAYER LEVEL WOUND - See preferred term level wound.

LAYER WOUND - See preferred term level wound.

LEAD ANGLE - See preferred term travel angle.

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LEAD BURNING - An erroneous term used to denote the welding of lead.

LEG OF A FILLET WELD - The distance from the root of the joint to the toe of the fillet weld.

LENS - See preferred term filter lens.

LEVEL WOUND - Spooled or coiled filler metal that has been wound in distinct layers such that adjacent turns touch.

LIGHTLY COATED ELECTRODE - See electrode.

LIQUATION - The separation of a low melting constituent of an alloy from the remaining constituents, usually apparent in alloys having a wide melting range.

LIQUIDUS - The lowest temperature at which a metal or an alloy is completely liquid.

LOCAL PREHEATING - Preheating a specific portion of a structure.

LOCAL STRESS RELIEF HEAT TREATMENT - Stress relief heat treatment of a specific portion of a structure.

LOCKED-UP STRESS - See preferred term residual stress.

LONGITUDINAL RESISTANCE SEAM WELDING - The making of a resistance seam weld in a direction essentially parallel to the throat depth of a resistance welding machine.

LONGITUDINAL SEQUENCE - The order in which the increments of a continuous weld are deposited with respect to its length. See backstep sequence, block sequence, etc.

LOW FREQUENCY CYCLE (resistance welding) - One positive and one negative pulse of current within the same weld or heat time at a frequency from which it is obtained.

LOW PULSE CURRENT - The current levels during the low pulse time which produce the low heat levels.

LOW PULSE TIME - The duration of the low current pulse time.

M

MACHINE BRAZING - Brazing with equipment which performs the brazing operation under the constant observation and control of a brazing operator. The equipment may or may not perform the loading and unloading of the work. See automatic brazing.

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MACHINE OXYGEN CUTTING - Oxygen cutting with equipment which performs the cutting operation under the constant observation and control of an oxygen cutting operator. The equipment may or may not perform the loading and unloading of the work. See automatic oxygen cutting.

MACHINE WELDING - Welding with equipment which performs the welding operation under the constant observation and control of a welding operator. The equipment may or may not perform the loading and unloading of the work. See automatic welding.

MANIFOLD - A multiple header for interconnection of gas or fluid souces with distribution points.

MANUAL BRAZING - A brazing operation performed and controlled completely by hand. See automatic brazing and machine brazing.

MANUAL OXYGEN CUTTING - A cutting operation peformed and controlled completely by hand. See automatic oxygen cutting and machine oxygen cutting.

MANUAL WELDING - A welding operation performed and controlled completely by hand. See automatic welding, machine welding, and semiautomatic arc welding.

MASH RESISTANCE SEAM WELDING - A resistance seam welding process variation in which a seam weld is made in a lap joint primarily by high temperature plastic working and diffusion as opposed to melting and solidification. The joint thickness after welding is less than the original assembled thickness.

MASK (thermal spraying) - A device for protecting a surface from the effects of blasting or coating adherence or coalescence with the substrate. Masks are generally of two types: reusable or disposable.

MATRIX (thermal spraying) - The major continuous substance of a thermal sprayed coating as opposed to inclusions or particles of materials having dissimilar characteristics.

MECHANICAL BOND (thermal spraying) - The adherence of a thermal sprayed deposit to a roughened surface by the mechanism of particle interlocking.

MELTBACK TIME - The time interval at the end of crater fill time to arc outage during which electrode feed is stopped. Arc voltage and arc length increase and current decreases to zero to prevent the electrode from freezing in the weld deposit.

MELTING RANGE - The temperature range between solidus and liquidus.

MELTING RATE - The weight or length of electrode melted in a unit of time.

MELT-THRU - Complete joint penetration for a joint welded from one side. Visible root reinforcement is produced.

MELT ARC CUTTING (MAC) - Any of a group of arc cutting processes which severs metals by melting them with the heat of an arc between a metal electrode and the base metal. See shielded metal arc cutting and gas metal arc cutting.

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METAL ARC WELDING - See shielded metal arc welding, flux cored arc welding, gas metal arc welding, gas tungsten arc welding, submerged arc welding, plasma arc welding, and stud arc welding.

METAL CORED ELECTRODE - See electrode.

METAL ELECTRODE - See electrode.

METALLIC BOND - The principal bond which holds metal together and which is formed between base metals and filler metals in all welding processes. This is a primary bond arising from the increased spatial extension of the valence electron wave functions when an aggregate of metal atoms is brought close together. See bonding force, covalent bond, and ionic bond.

METALLIZING - See preferred term thermal spraying.

METALLURGICAL BOND - See preferred term metallic bond.

METAL POWDER CUTTING (POC) - An oxygen cutting process which severs metals through the use of powder, such as iron, to facilitate cutting.

MIG WELDING - See preferred terms gas metal arc welding and flux cored arc welding.

MIXING CHAMBER - That part of a welding or cutting torch in which a fuel gas and oxygen are mixed.

MOLTEN CHEMICAL-BATH DIP BRAZING - A dip brazing process variation. See dip brazing (DB).

MOLTEN METAL FLAME SPRAYING - A thermal spraying process variation in which the metallic material to be sprayed is in the molten condition. See flame spraying (FLSP).

MOLTEN WELD POOL - The liquid state of a weld prior to solidification as weld metal.

MULTIPLE-IMPULSE WELDING - A resistance welding process variation in which welds are made by more than one impulse of current.

MULTIPLE-IMPULSE WELD TIMER (resistance welding) - A device for multiple impulse welding which controls only the heat time, the cool time, and the weld interval or the number of heat times.

MULTIPLE NOZZLE (plasma arc welding and cutting) - A constricting nozzle containing two or more orifices located in a configuration to achieve a degree of control over the arc shape.

N

NEUTRAL FLAME - An oxyfuel gas flame in which the portion used is neither oxidizing nor reducing.

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NONCORROSIVE FLUX - A soldering flux which in neither its original nor residual form chemically attacks the base metal. It usually is composed of rosin or resin-based materials.

NONSYNCHRONOUS INITIATION (resistance welding). The initiation or termination of the welding transformer primary current at any random time with respect to the voltage wave.

NONSYNCHRONOUS TIMING - See preferred term nonsynchronous initiation.

NONTRANSFERRED ARC (plasma arc welding and cutting, and thermal spraying) - An arc established between the electrode and the constricting nozzle. The workpiece is not in the electrical cirucit. See transferred arc.

NOZZLE - A device which directs shielding media.

NUGGET (resistance welding) - The weld metal joining the parts in spot, roll spot, seam, or projection welds.

NUGGET SIZE - The diameter or width of the nugget measured in the plane of the interface between the pieces joined.

O

OFF TIME (resistance welding) - The time during which the electrodes are off the work. This term is generally used when the welding cycle is repetitive.

OPEN-CIRCUIT VOLTAGE - The voltage between the output terminals of the welding machine when no current is flowing in the welding circuit.

ORIFICE GAS plasma arc welding and cutting) - The gas that is directed into the torch to surround the electrode. It becomes ionized in the arc to form the plasma, and issues from the orifice in the torch nozzle as the plasma jet.

OFFICE THROAT LENGTH (plasma arc welding and cutting) - The length of the constricting orifice.

OUT-OF-POSITION - The term includes horizontal, vertical and overhead positions.

OVEN SOLDERING - See preferred term furnace soldering.

OVERHEAD POSITION - The position in which welding is performed from the underside of the joint.

OVERLAP - The protrusion of weld metal beyond the toe, face, or root of the weld; in resistance seam welding, the area in the preceding weld remelted by the succeeding weld.

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OVERLAYING - See preferred term surfacing.

OXIDIZING FLAME - An oxyfuel gas flame having an oxidizing effect (excess oxygen).

OXYACETYLENE CUTTING (OFC-A) - An oxyfuel gas cutting process used to sever metals by means of the chemical reaction of oxygen with the base metal at elevated temperatures. The necessary temperature is maintained by gas flames resulting from the combustion of acetylene with oxygen.

OXYACETYLENE WELDING (OAW) - An oxyfuel gas welding process which produces coalescence of metals by heating them with a gas flame or flames obtained from the combustioon of acetylene with oxygen. The process may be used with or without the application of pressure and with or without the use of filler metal.

OXYFUEL GAS CUTTING (OFC) - A group of cutting processes used to sever metals by means of the chemical reaction of oxygen with the base metal at elevated temperatures. The necessary temperature is maintained by means of gas flames obtained from the combustion of a specified fuel gas and oxygen. See oxygen cutting, oxyacetylene cutting, oxyhydrogen cutting, oxynatural gas cutting, and oxypropane cutting.

OXYFUEL GAS WELDING (OFW) - A group of welding processes which produces coalescence by heating materials with an oxyfuel gas fleme or flames, with or without the application of pressure and with or without the use of filler metal.

OXYFUEL GAS SPRAYING - See preferred term flame spraying.

OXYGAS CUTTING - See preferred term oxygen cutting.

OXYGEN ARC CUTTING (AOC) - An oxygen cutting process used to sever metals by means of the chemical reaction of oxygen with the base metal at elevated temperatures. The necessary temperature is maintained by an arc between a consumable tubular electrode and the base metal.

OXYGEN CUTTER - One who performs a manual oxygen cutting operation.

OXYGEN CUTTING (OC) - A group of cutting processes used to sever or remove metals by means of the chemical reaction of oxygen with the base metal at elevated temperatures. In the case of oxidation-resistant metals, the reaction is facilitated by the use of a chemical flux or metal powder. See oxygen arc cutting, oxyfuel gas cutting, oxygen lance cutting, chemical flux cutting, and metal powder cutting.

OXYGEN CUTTING OPERATOR - One who operates machine or automatic oxygen cutting equipment.

OXYGEN GOUGING - An application of oxygen cutting in which a bevel or groove is formed.

OXYGEN GROOVING - See preferred term oxygen gouging.

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OXYGEN LANCE - A length of pipe used to convey oxygen to the point of cutting in oxygen lance cutting.

OXYGEN LANCE CUTTING (LOC) - An oxygen cutting process used to sever metals with oxygen supplied through a consumable lance; the preheat to start the cutting is obtained by other means.

OXYGEN LANCING - See preferred term oxygen lance cutting.

OXYHYDROGEN CUTTING (OFC-H) - An oxyfuel gas cutting process used to sever metals by means of the chemical reaction of oxygen with the base metal at elevated temperatures. The necessary temperature is maintained by gas flames resulting from the combustion of hydrogen with oxygen.

OXYHYDROGEN WELDING (OHW) - An oxyfuel gas welding process which produces coalescence of materials by heating them with a gas flame or flames obtained from the combustion of hydrogen with oxygen, without the application of pressure and with or without the use of filler metal.

OXYNATURAL GAS CUTTING (OFC-N) - An oxyfuel gas cutting process used to sever metals by means of the chemical reaction of oxygen with the base metal at elevated temperatures. The necessary temperature is maintained by gas flames resulting from the combustion of natural gas with oxygen.

OXYPROPANE CUTTING (OFC-P) - An oxyfuel gas cutting process used to sever metals by means of the chemical reaction of oxygen with the base metal at elevated temperatures. The necessary temperature is maintained by gas flames resulting from the combustion of propane with oxygen.

D

PARENT METAL - See preferred term base metal.

PARTIAL JOINT PENETRATION - Joint penetration which is less than complete. See also complete joint penetration.

PASS - See preferred term weld pass.

PASS SEQUENCE - See deposition sequence.

PASTE BRAZING FILLER METAL - A mixture of finely divided brazing filler metal with an organic or inorganic flux or neutral vehicle or carrier.

PEEL TEST - A destructive method of inspection which mechanically separates a lap joint by peeling.

PEENING - The mechanical working of metals using impact blows.

PENETRATION - See preferred terms joint penetration and root penetration.

PERCENT FERRITE - See preferred term ferrite number.

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PERCUSSION WELD - A weld made by percussion welding.

PERCUSSION WELDING (PEW) - A resistance welding process which produces coalescence of the abutting members using heat from an arc produced by a rapid discharge of electrical energy. Pressure is applied percussively during or immediately following the electrical discharge.

PERMANENT DIPOLE BOND - A secondary bond arising from the attaction between dipoles, the oppositely charged ends of which are electronegative and electropositive atoms.

PILOT ARC (plasma arc welding) - A low current continuous arc between the electrode and the constricting nozzle to ionize the gas and facilitate the start of the main welding arc.

PIANO LENS (eye protection) - A lens which does not incorporate correction.

PLASMA - A gas that has been heated to an at least partially ionized condition, enabling it to conduct an electric current.

PLASMA ARC CUTTING (PAC) - An arc cutting process which severs metal by melting a localized area with a constricted arc and removing the molten material with a high velocity jet of hot, ionized gas issuing from the orifice.

PLASMA ARC WELDING (PAW) - An arc welding process which produces coalescence of metals by heating them with a constricted arc between an electrode and the workpiece (transferred rc) or the electrode and the constricting nozzle (nontransferred arc). Shielding is obtained from the hot, ionized gas issuing from the orifice which may be supplemented by an auxiliary source of shielding gas. Shielding gas may be an inert gas or a mixture of gases. Pressure may or may not be used, and filler metal may or may not be supplied.

PLASMA METALLIZING (PSP) - A thermal spraying process in which a nontransferred arc is utilized as the source of heat for melting and propelling the coating material to the workpiece.

PLATEN (resistance welding) - A member with a substantially flat surface to which dies, fixtures, backups, or electrode holders are attached, and which transmits the electrode force or upsetting force. One platen usually is fixed and the other moveable.

PLATEN FORCE (resistance welding) - The force available at the moveable platen to cause upsetting in flash or upset welding. This force may be dynamic, theoretical, or static.

PLATEN SPACING - The distance between adjacent surfaces of the platens in a resistance welding machine.

PLENUM CHAMBER - See plenum.

PLENUM (plasma arc welding and cutting, and thermal spraying) - The space between the inside wall of the constricting nozzle and the electrode.

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PLUG WELD - A circular weld made through a hole in one member of a lap or T-joint fusing that member to the other. The walls of the hole may or may not be parallel and the hole may be partially or completely filled with the weld metal. (A fillet welded hole or a spot weld should not be construed as conforming to this definition.)

POKE WELD - See preferred term push weld.

POKE WELDING - See preferred term push welding.

POLARITY - See direct current electrode negative, direct current electrode positive, straight polarity, and reverse polarity.

POROSITY - Cavity type discontinuities formed by gas entrapment during solidification.

POSITIONED WELD - A weld made in a joint which has been so placed as to facilitate making the weld.

POSITION OF WELDING - See flat position, horizontal position, horizontal fixed position, horizontal rolled position, inclined position, overhead position, and vertical position.

POSTFLOW TIME - The time interval from current shut-off to shielding gas and/or cooling water shut-off.

POSTHEAT CURRENT (resistance welding) - The current through the welding circuit during postheat time in resistance welding.

POSTHEATING - The application of heat to an assembly after a welding, brzing, soldering, thermal spraying, or cutting operation. See postweld heat treatment.

POSTHEAT TIME (resistance welding) - The time from the end of weld heat time to the end of weld time.

POSTWELD HEAT TREATMENT - Any heat treatment subsequent to welding.

POSTWELD INTERVAL (resistance welding) - The total time elapsing between the end of welding current and the start of hold time.

POWDER CUTTING - See preferred terms chemical flux cutting and metal powder cutting.

POWDER FEEDER (thermal spraying) - A device for supplying powdered materials to thermal spraying equipment.

POWDER FLAME SPRAYING - A thermal spraying process variation in which the material to be sprayed is in powder form. See flame spraying (FLSP).

POWDER METALLIZING - See preferred term powder flame spraying.

PRECOATING - Coating the base metal in the joint by dipping, electroplating, or other applicable means prior to soldering or brazing.

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PREFLOW TIME - The time interval between start of shielding gas flow and arc starting.

PREFORM - Brazing or soldering filler metal fabricated in a shape or form for a specific application.

PREHEAT - See preferred term preheat temperature.

PREHEAT CURRENT (resistance welding) - An impulse or series of current impulses which occurs prior to and is separated from welding current.

PREHEATING - The application of heat to the base metal immediately before welding, brazing, soldering, thermal spraying, or cutting operation. See postweld heat treatment.

POSTHEAT TIME (resistance welding) - The time from the end of weld heat time to the end of weld time.

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PREFORM - Brazing or soldering filler metal fabricated in a shape or form for a specific application.

PREHEAT - See preferred term preheat temperature.

PREHEAT CURRENT (resistance welding) - An impulse or series of current impulses which occurs prior to and is separated from welding current.

PREHEATING - The application of heat to the base metal immediately before welding, brazing, soldering, thermal spraying, or cutting.

PREHEAT TEMPERATURE - A specified temperature that the base metal must attain in the welding, brazing, soldering, thermal spraying, or cutting area immediately before these operations are performed.

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PREHEAT TIME (resistance welding) - A portion of the preweld interval during which preheat current occurs.

PRESSURE-CONTROLLED WELDING - A resistance welding process variation in which a number of spot or projection welds are made with several electrodes functioning progressively under the control of a pressure-sequencing device.

PRESSURE GAS WELDING (PGW) - An oxyfuel gas welding process which produces coalescence simultaneously over the entire area of abutting surfaces by heating them with gas flames obtained from the combustion of a fuel gas with oxygen and by the application of pressure, without the use of filler metal.

PRESSURE WELDING - See preferred terms solid-state welding, hot pressure welding, forge welding, diffusion welding, pressure gas welding, and cold welding.

PRETINNING - See preferred term precoating.

PREWELD INTERVAL (resistance welding) - The time elapsing between the end of squeeze time and the beginning of welding current in making spot welds and in projection or upset welding. In flash welding, it is the time during which the material is preheated.

PROCEDURE - The detailed elements (with prescribed values or ranges of values) of a process or method used to produce a specific result.

PROCEDURE QUALIFICATION - The demonstration that welds made by a specific procedure can meet prescribed standards.

PROCEDURE QUALIFICATION RECORD (PQR) - A document providing the actual welding variables used to produce an acceptable test weld and the results of tests conducted on the weld for the purpose of qualifying a welding procedure specification.

PROGRESSIVE BLOCK SEQUENCE - A block sequence during which successive blocks are completed progressively along the joint, either from one end to the other or from the center of the joint toward either end.

PROJECTION WELD - A weld made by projection welding.

PROJECTION WELDING (RPW) - A resistance welding process which produces coalescence of metals with the heat obtained from resistance to electric current through the work parts held together under pressure by electrodes. The resulting welds are localized at predetermined points by projections, embossments, or intersections.

PROTECTIVE ATMOSPHERE - A gas envelope surrounding the part to be brazed, welded, or thermal sprayed, with the gas composition controlled with respect to chemical composition, dew point, pressure, flow rate, etc. Examples are inert gases, combusted fuel gases, hydrogen, and vacuum.

PUDDLE - See preferred term molten weld pool.

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PULL GUN TECHNIQUE - See backhand welding.

PULSATION WELDING - See preferred term multiple-impulse welding.

PULSATION WELD TIMER - See preferred term multiple-impulse weld timer.

PULSE (resistance welding) - A current of controlled duration through a welding circuit.

PULSED POWER WELDING - An arc welding process variation in which the power is cyclically programmed to pulse so that effective but short duration values of a parameter can be utilized. Such short duration values are significantly different from the average value of the parameter. Equivalent terms are pulsed voltage or pulsed current welding; see also pulsed spray welding.

PULSED SPRAY WELDING - An arc welding process variation in which the current is pulsed to utilize the advantages of the spray mode of metal transfer at average currents equal to or less than the globular to spray transition current.

PULSE START DELAY TIME - The time interval from current initiation to the beginning of current pulsation, if pulsation is used.

PULSE TIME (resistance welding) - The duration of a pulse.

PUSH ANGLE - The travel angle when the electrode is pointing forward. See also forehand welding. Note: This angle can be used to define the position of welding guns, welding torches, high energy beams, welding rods, thermal cutting and thermal splraying torches, and thermal spraying guns.

PUSH WELD (resistance welding) - A spot or projection weld made by push welding.

PUSH WELDING - A resistance welding process variation in which spot or projection welds are made by manually applying force to one electrode and using the work or a backing as the other electrode.

Q

QUALIFICATION - See preferred terms welder performance qualification and procedure qualification.

QUENCH TIME (resistance welding) - The time from the end of weld time to the beginning of temper time.

R

RANDOM INTERMITTENT WELDS - Intermittent welds on one or both sides of a joint in which the weld increments are deposited without regard to spacing.

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RANDOM SEQUENCE - A longitudinal sequence in which the weld bead increments are deposited at random.

RANDOM WOUND - Spooled or coiled filler metal that has not been wound in distinct layers.

RATE OF DEPOSITION - See deposition rate.

RATE OF FLAME PROPAGATION - The speed at which a flame travels through a mixture of gases.

REACTION FLUX (soldering) - A flux composition in which one or more of the ingredients reacts with a base metal upon heating to deposit one or more metals.

REACTION SOLDERING - A soldering process variation in which a reaction flux is used.

REACTION STRESS - The residual stress which could not otherwise exist if the members or parts being welded were isolated as free bodies without connection to other parts of the structure.

REACTOR (arc welding) - A device used in arc welding circuits for the purpose of minimizing irregularities in the flow of welding current.

REDUCING ATMOSPHERE - A chemically active protective atmosphere which at elevated temperature will reduce metal oxides to their metallic state. (Reducing atmosphere is a relative term and such an atmosphere may be reducing to one oxide but not to another oxide.)

REDUCING FLAME - A gas flame having a reducing effect. (Excess fuel gas.)

REFLOW SOLDERING - A nonpreferred term used to describe a soldering process variation in which preplaced solder is melted to produce a soldered joint or coated surface.

REFLOWING (soldering) - See preferred term flow brightening.

REGULATOR - A device for controlling the delivery of gas at some substantially constant pressure.

REINFORCEMENT OF WELD - Weld metal in excess of the quantity required to fill a joint. See face reinforcement and root reinforcement.

RESIDUAL STRESS - Stress remaining in a structure or member as a result of thermal or mechanical treatment or both. Stress arises in fusion welding primarily because the weld metal contracts on cooling from the solidus to room temperature.

RESISTANCE BRAZING (RB) - A brazing process in which the heat required is obtained from the resistance to electric current in a circuit of which the work is a part.

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RESISTANCE BUTT WELD - See preferred terms upset weld and flash weld.

RESISTANCE BUTT WELDING - See preferred terms upset welding and flash welding.

RESISTANCE SEAM WELD TIMER - In resistance seam welding, a device which controls the heat times and the cool times.

RESISTANCE SEAM WELDING (RSEW) - A resistance welding process which produces coalescence at the faying surfaces by the heat obtained from resistance to electric current through the work parts held together under pressure by electrodes. The resulting weld is a series of overlapping resistance spot welds made progressively along a joint by rotating the electrodes.

RESISTANCE SOLDERING (RS) - A soldering process in which the heat required is obtained from the resistance to electric current in a circuit of which the work is a part.

RESISTANCE SPOT WELDING (RSW) - A resistance welding process which produces coalescence at the faying surfaces in one spot by the heat obtained from the resistance to electric current through the work parts held together under pressure by electrodes. The size and shape of the individually formed welds are limited primarily by the size and contour of the electrodes.

RESISTANCE WELDING (RW) - A group of welding processes which produces coalescence of metals with the heat obtained from resistance of the work to electric current in a circuit of which the work is a part, and by the application of pressure. See resistance welding electrode.

RESISTANCE WELDING ELECTRODE - See electrode.

RESISTANCE WELDING GUN - A manipulating device to transfer current and provide electrode force to the weld area (usually in reference to a portable gun).

REVERSE POLARITY - The arrangement of direct current arc welding leads with the work as the negative pole and the electrode as the positive pole of the welding arc. A synonym for direct current electrode positive.

ROLL RESISTANCE SPOT WELDING - A resistance welding process variation in which separated spot welds are made with one or more rotating circular electrodes. The rotation of the electrodes may or may not be stopped during the making of a weld.

ROLL WELDING (ROW) - A solid-state welding process which produces coalescence of metals by heating and by applying pressure with rolls sufficient to cause deformation at the faying surfaces. See also forge welding.

ROOT - See preferred terms root of joint and root of weld.

ROOT HEAD - A weld deposit that extends into or includes part or all of the root of the joint.

ROOT GAP - See preferred term root opening.

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ROOT OF JOINT - That portion of a joint to be welded where the members approach closest to each other. In cross section, the root of the joint may be either a point, a line, or an area.

ROOT OF WELD - The points, as shown in cross section, at which the back of the weld intersects the base metal surfaces.

ROOT OPENING - The separation between the members to be joined at the root of the joint.

ROOT PENETRATION - The depth that a weld extends into the root of a joint measured on the centerline of the root cross section.

ROOT RADIUS - See preferred term groove radius.

ROOT REINFORCEMENT - Reinforcement of weld at the side other than that from which welding was done.

ROOT SURFACE - The exposed surface of a weld on the side other than from which welding was done.

ROTARY ROUGHENING (thermal spraying) - A method of surface roughening in which a revolving roughening tool is pressed against the surface being prepared, while either the work or the tool, or both, move.

ROUGH THREADING (thermal spraying) - A method of surface roughening which consists of cutting threads with the sides and tops of the threads jagged and torn.

S

SALT-BATH DIP BRAZING - See preferred term molten chemical-bath dip brazing.

SANDWICH BRAZE - A brazed assembly of dissimilar materials using a preplaced shim, other than the filler metals, as a transition layer to minimize thermal stresses.

SCARF - See preferred term edge preparation.

SCARF JOINT - A form of butt joint.

SEAL COAT (thermal spraying) - Material applied to infiltrate the pores of a thermal spray deposit.

SEAL WELD - Any weld designed primarily to provide a specific degree of tightness against leakage.

SEAM WELD - A continuous weld made between or upon overlapping members, in which coalescence may start and occur on the faying surfaces, or may have proceeded from the surface of one member. The continuous weld may consist of a single weld bead or a series of overlapping spot welds.

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SEAM WELDING - The making of seam welds.

SECONDARY CIRCUIT - That portion of a welding machine which conducts the secondary current between the secondary terminals of the welding transformer and the electrodes, or electrode and work.

SELECTIVE BLOCK SEQUENCE - A block sequence in which successive blocks are completed in a certain order selected to create a predetermined stress pattern.

SELF-FLUXING ALLOYS (thermal spraying) - Certain materials that "wet" the substrate and coalesce when heated to their melting point, without the addition of a fluxing agent.

SEMIAUTOMATIC ARC WELDING - Arc welding with equipment which controls only the filler metal feed. The advance of the welding is manually controlled.

SEMIAUTOMATIC BRAZING - Brazing with equipment which controls only the brazing filler metal feed. The advance of the brazing is manually controlled.

SEMIBLIND JOINT - A joint in which one extremity of the joint is not visible.

SEQUENCE TIME (automatic arc welding) - See preferred term weld cycle.

SEQUENCE TIMER (resistance welding) - A device for controlling the sequence and duration of any or all of the elements of a complete welding cycle except weld time or heat time.

SEQUENCE WELD TIMER (resistance welding) - A device which performs the functions of a sequence timer and the functions of either a weld timer or a multiple-impulse weld timer.

SERIES SUBMERGED ARC WELDING (SAW-S) - A submerged arc welding process variation in which electric current is established between two (consumable) electrodes which meet just above the surface of the work. The work is not in the electrical circuit.

SERIES WELDING (resistance welding) - A resistance welding process variation in which two spot or seam welds or two or more projection welds are made simultaneously with three electrodes, forming a series circuit.

SHADOW MASK (thermal spraying) - A thermal spraying process variation in which an area is partially shielded during the thermal spraying operation, thus permitting some overspray to produce a feathering at the coating edge.

SHEET SEPARATION (resistance welding) - The gap surrounding the weld, between faying surfaces, after the joint has been welded in spot, seam, or projection welding.

SHIELDED CARBON ARC WELDING (CAW-S) - A carbon arc welding process variation which produces coalescence of metals by heating them with an electric arc between a carbon electrode and the work. Shielding is obtained from the combustion of a solid material fed into the arc or from a blanket of flux on the work, or both. Pressure may or may not be used and filler metal may or may not be used.

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SHIELDED METAL ARC CUTTING (SMAC) - A metal arc cutting process in which metals are severed by melting them with the heat of an arc between a covered metal electrode and the base metal.

SHIELDED METAL ARC WELDING (SMAW) - An arc welding process which produces coalescence of metals by heating them with an arc between a covered metal electrode and the work. Shielding is obtained from decomposition of the electrode covering. Pressure is not used and filler metal is obtained from the electrode.

SHIELDING GAS - Protective gas used to prevent atmospheric contamination.

SHORT CIRCUITING ARC WELDING - See gas metal arc welding -- short circuit arc (GMAW-S).

SHORT CIRCUITING TRANSFER (arc welding) - Metal transfer in which molten metal from a consumable electrode is deposited during repeated short circuits.

SHOULDER - See preferred term root face.

SHRINKAGE STRESS - See preferred term residual stress.

SHRINKAGE VOID - A cavity type discontinuity normally formed by shrinkage during solidification.

SIEVE ANALYSIS - A method of determining particle size distribution, usually expressed as the weight percentage retained upon each of a series of standard screens of decreasing mesh size.

SILVER SOLDERING, SILVER ALLOY BRAZING - Nonpreferred terms used to denote brazing with a silver-base filler metal. See preferred terms furnace brazing, induction brazing, and torch brazing.

SINGLE-IMPULSE WELDING (resistance welding) - A resistance welding process variation in which spot, projection, or upset welds are made by a single impulse of current. When alternating current is used, an impulse may consist of a fraction of a cycle or a number of cycles.

SINGLE-PORT NOZZLE - A constricting nozzle containing one orifice, located below and concentric with the electrode.

SINGLE-WELDED JOINT - In arc and gas welding, any joint welded from one side only.

SIZE OF WELD

GROOVE WELD - The joint penetration (depth of bevel plus the root penetration when specified). The size of a groove weld and its effective throat are one and the same.

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FILLET WELD - For equal leg fillet welds, the leg lengths of the largest isosceles right triangle which can be inscribed within the fillet weld cross section. For unequal leg fillet welds, the leg lengths of the largest right triangle which can be inscribed within the fillet weld cross section. Note: When one member makes an angle with the other member greater than 105 degrees, the leg length (size) is of less significance than the effective throat which is the controlling factor for the strength of a weld.

FLANGE WELD - The weld metal thickness measured at the root of the weld.

SKIP WELD - See preferred term intermittent weld.

SKULL - The unmelted residue from a liquated filler metal.

SLAG INCLUSION - Nonmetallic solid material entrapped in weld metal or between weld metal and base metal.

SLOT WELD - A weld made in an elongated hole in one member of a lap or T-joint joining that member to that portion of the surface of the other member which is exposed through the hole. The hole may be open at one end and may be partially or completely filled with weld metal. (A fillet welded slot should not be construed as conforming to this definition.)

SLUGGING - The act of adding a separate piece or pieces of material in a joint before or during welding that results in a welded joint not complying with design, drawing, or splecification requirements.

SOFT SOLDER - See preferred term solder.

SOLDER - A filler metal used in soldering which has a liquidus not exceeding 450°C (840°F)

SOLDERABILITY - The capacity of a material to be soldered under the fabrication conditions imposed upon a specific, suitably designed structure.

SOLDERING (S) - A group of welding processes which produces coalescence of materials by heating them to a suitable temperature and by using a filler metal having a liquidus not exceeding 840°F and below the solidus of the base metals. The filler metal is distributed between the closely fitted laying surfaces of the joint by capillary action.

SOLDERING GUN - An electrical soldering iron with a pistol grip and a quick heating, relatively small bit.

SOLDERING IRON - A soldering tool having an internally or externally heated metal bit usually made of copper.

SOLID-STATE WELDING (SSW) - A group of welding processes which produces coalescence at temperatures essentially below the melting point of the base metal being joined, without the addition of a brazing filler metal. Pressure may or may not be used.

SOLIDUS - The highest temperature at which a metal or alloy is completely solid.

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SPACER STRIP - A metal strip or bar prepared for a groove weld and inserted in the root of a joint to serve as a backing and to maintain root opening during welding. It can also bridge an exceptionally wide gap due to poor fitup.

SPATTER - The metal particles expelled during welding and which do not form a part of a weld.

SPATTER LOSS - Metal lost due to spatter.

SPIKING (electron beam welding and laser welding) - A condition where the depth of penetration is nonuniform and changes abruptly over the length of the weld.

SPIT - See preferred term flash.

SPLIT PIPE BACKING - Backing in the form of a pipe segment used for welding round bars.

SPOOL - A type of filler metal package consisting of a continuous length of electrode wound on a cylinder (called the barrel) which is flanged at both ends. The flange extends below the inside diameter of the barrel and contains a spindle hole.

SPOT WELD - A weld made between or upon overlapping members in which coalescence may start and occur on the faying surfaces or may proceed from the surface of one member. The weld cross section (plan view) is approximately circular. See also arc spot weld and resistance spot welding.

SPOT WELDING - The making of spot welds.

SPRAY DEPOSIT (thermal spraying) - The coating applied by a thermal spraying process.

SPRAYING SEQUENCE (thermal spraying) - The order in which different layers of similar or different materials are applied in a planned relationship, such as overlapped, superimposed, or at certain angles.

SPRAY RATE - A synonym for feed rate. See feed rate.

SPRAY TRANSFER (arc welding) - Metal transfer in which molten metal from a consumable electrode is propelled axially across the arc in small droplets.

SEQUENCE TIME (resistance welding) - The time interval between the initial application of the electrode force on the work and the first application of current in making spot and seam welds and in projection or upset welding.

STACK CUTTING - Thermal cutting of stacked metal plates arranged so that all the plates are severed by a single cut.

STAGGERED INTERMITTENT WELDS - Intermittent welds on both sides of a joint in which the weld increments on one side are alternated with respect to those on the other side.

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STANDOFF DISTANCE - The distance between a nozzle and the base metal.

START CURRENT - The current value during start time interval.

START TIME - The time interval prior to weld time during which arc voltage and current reach a preset value greater or less than welding values.

START VOLTAGE - The arc voltage during the start time.

STATIC ELECTRODE FORCE - See electrode force.

STEPBACK SEQUENCE - See preferred term backstep sequence.

STEP BRAZING - The brazing of successive joints on a given part with filler metals of successively lower brazing temperatures so as to accomplish the joining without disturbing the joints previously brazed. A similar result can be achieved at a single brazing temperature if the remelt temperature of prior joints is increased by metallurgical interaction.

STEP SOLDERING - The soldering of successive joints on a given part with solders of successively lower soldering temperatures so as to accomplish the joining without disturbing the joints previously soldered.

STICK ELECTRODE - See covered electrode.

STICK ELECTRODE WELDING - See preferred term shielded metal arc welding.

STICKOUT - See preferred term electrode extension.

STITCH WELD - See preferred term intermittent weld.

STOPOFF - A material used on the surfaces adjacent to the joint to limit the spread of soldering or brazing filler metal.

STORED ENERGY WELDING (resistance welding) - A resistance welding process variation in which welds are made ewith electrical energy accumulated electrostatically, electromagnetically, or electrochemically at a relatively low rate and made available at the required welding rate.

STRAIGHT POLARITY - The arrangement of direct current arc welding leads in which the work is the positive pole and the electrode is the negative pole of the welding arc. A synonym for direct current electrode negative.

STRANDED ELECTRODE - See electrode.

STRESS CORROSION CRACKING - Failure of metals by cracking under combined action of corrosion and stress, residual or applied. In brazing, the term applies to the cracking of stressed base metal due to the presence of a liquid filler metal.

STRESS RELIEF CRACKING - Intergranular cracking in the heat-affected zone or weld metal that occurs during the exposure of weldments to elevated temperatures during postweld heat treatment or high temperature service.

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STRESS RELIEF HEAT TREATMENT - Uniform heating of a structure or a portion thereof to a sufficient temperature to relieve the major portion of the residual stresses, followed by uniform cooling.

STRINGER BEAD - A type of weld bead made without appreciable weaving motion. See also weave bead.

STUD ARC WELDING (SW) - An arc welding process which produces coalescence of metals by heating them with an arc between a metal stud, or similar part, and the other work part. When the surfaces to be joined are properly heated, they are brought together under pressure. Partial shielding may be obtained by the use of a ceramic ferrule surrounding the stud. Shielding gas or flux may or may not be used.

STUD WELDING - A general term for the joining of a metal stud or similar part to a workpiece. Welding may be accomplished by arc, resistance, friction, or other suitable process with our without external gas shielding.

SUBMERGED ARC WELDING (SAW) - An arc welding process which produces coalescence of metals by heating them with an arc or arcs between a bare metal electrode or electrodes and the work. The arc and molten metal are shielded by a blanket or granular, fusible material on the work. Pressure is not used, and filler metal is obtained from the electrode and sometimes from a supplement source(s) (welding rod, flux, or metal granules).

SUBSTRATE - Any base material to which a thermal sprayed coating or surfacing weld is applied.

SUCK-BACK - See preferred term concave root surface.

SURFACE PREPARATION - The operations necessary to produce a desired or specified surface condition.

SURFACE ROUGHENING - A group of procedures for producing irregularities on a surface. See blasting, dovetailing, groove and rotary roughening, rotary roughening, rough threading, and threading and knurling.

SURFACING - The deposition of filler metal (material) on a base metal (substrate) to obtain desired properties or dimensions. See also buttering, cladding, coating and hardfacing.

SURFACING WELD - A type of weld composed of one or more stringer or weave beads deposited on an unbroken surface to obtain desired properties or dimensions.

SWEAT SOLDERING - A soldering process variation in which two or more parts are reheated and assembled into a joint without the use of additional solder.

SYNCHRONOUS INITIATION (resistance welding) - The initiation and termination of each half-cyle of welding transformer primary current so that all half-cycles of such current are identical in making spot and seam welds or in making projection welds.

SYNCHRONOUS TIMING - See preferred term synchronous initiation.

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T

TACK WELD - A weld made to hold parts of a weldment in proper alignment until the final welds are made.

TAPER DELAY TIME - The time interval after upslope during which the current is constant.

TAPER TIME - The time interval when current increases or decreases continuously from the welding current to final taper current.

TAPS - Connections to a transformer winding which are used to vary the transformer turns ratio, thereby controlling welding voltage and current.

TEMPER TIME (resistance welding) - That part of the postweld interval following quench time to the beginning of hold time in resistance welding.

TEMPORARY WELD - A weld made to attach a piece or pieces to a weldment for temporary use in handling, shipping, or working on the weldment.

THEORETICAL ELECTRODE FORCE - See electrode force.

THEORETICAL THROAT - See throat of a fillet weld.

THERMAL CUTTING - A group of cutting processes which melts the metal (material) to be cut. See arc cutting, oxygen cutting, electron beam cutting, and laser beam cutting.

THERMAL SPRAYING (THSP) - A group of welding or allied processes in which finely divided metallic or nonmetallic materials are deposited in a molten or semi-molten condition to form a coating. The coating matrial may be in the form of powder, ceramic rod, wire, or molten materials. See flame spraying, plasma spraying, and electric arc spraying.

THERMAL SPRAYING GUN - A device for heating, feeding, and directing the flow of a thermal spraying material.

THERMAL STRESSES - Stresses in metal resulting from nonuniform temperature distributions.

THERMAL CRUCIBLE - The vessel in which the thermit reaction takes place.

THERMAL MIXTURE - A mixture of metal oxide and finely divided aluminum with the addition of alloying metals as required.

THERMIT MOLD - A mold formed around the parts to be welded to receive the molten metal.

THERMIT REACTION - The chemical reaction between metal oxide and aluminum which produces superheated molten metal and aluminum oxide slag.

THERMIT WELDING (TW) - A welding process which produces coalescence of metals by heating them with superheated liquid metal from a chemical reaction between a metal oxide and aluminum, with or without the application of pressure. Filer metal, when used, is obtained from the liquid metal.

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THERMOCOMPRESSION BONDING - See preferred term hot pressure welding.

THREADING AND KNURLING (thermal spraying) - A method of surface roughening in which spiral threads are prepared, followed by upsetting with a knurling tool.

THROAT DEPTH (resistance welding) - The distance from the center line of the electrodes or platens to the nearest point of interference for flat sheets in a resistance welding machine. In the case of a resistance seam welding machine with a universal head, the throat depth is measured with the machine arranged for transverse welding.

THROAT HEIGHT (resistance welding) - The unobstructed dimension between arms throughout the throat depth in a resistance welding machine.

THROAT OF A FILLET WELD

THEORETICAL THROAT - The distance from the beginning of the root of the joint perpendicular to the hypotenuse of the largest right triangle that can be inscribed within the fillet weld cross section. This dimension is based on the assumption that the root opening is equal to zero.

ACTUAL THROAT - The shortest distance from the root of weld to its face.

EFFECTIVE THROAT - The minimum distance minus any reinforcement from the root of weld to its face.

THROAT OF A GROOVE WELD - See preferred term size of weld.

THROAT OPENING - See preferred term horn spacing.

TIG WELDING - See preferred term gas tungsten arc welding.

TINNING - See preferred term precoating.

TIP SKID - See preferred term electrode skid.

T-JOINT - A joint between two members located approximately at right angles to each other in the form of a T.

TOE CRACK - A crack in the base metal occurring at the toe of a weld.

TOE OF WELD - The junction between the face of a weld and the base metal.

TORCH - See preferred terms cutting torch and welding torch.

TORCH BRAZING (TB) - A brazing process in which the heat required is furnished by a fuel gas flame.

TORCH SOLDERING (TS) - A soldering process in which the heat reauired is furnished by a fuel gas flame.

TORCH TIP - See preferred terms welding tip or cutting tip.

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TRANSFERRED ARC (plasma arc welding) - A plasma arc established between the electrode and the workpiece.

TRANSVERSE RESISTANCE SEAM WELDING - The making of a resistance seam weld in a direction essentially at right angles to the throat depth of a resistance seam welding machine.

TRAVEL ANGLE - The angle that the electrode makes with a reference line perpendicular to the axis of the weld in the plane of the weld axis. See also drag angle and push angle. Note: This angle can be used to define the position of welding guns, welding torches, high energy beams, welding rods, thermal cutting and thermal spraying torches, and thermal spraying guns.

TRAVEL ANGLE (pipe) - The angle that the electrode makes with a reference line extending from the center of the pipe through the molten weld pool in the plane of the weld axis. Note: This angle can be used to define the position of welding guns, welding torches, high energy beams, welding rods, thermal cutting and thermal spraying torches, and thermal spraying guns.

TRAVEL START DELAY TIME - The time interval from arc initiation to the start of work or torch travel.

TRAVEL STOP DELAY TIME - The time interval from beginning of downslope time or crater fill time to shut-off of torch or work travel.

TUNGSTEN ELECTRODE - See electrode: tungsten electrode.

TWIN CARBON ARC WELDING (CAW-T) - A carbon arc welding process variation which produces coalescence of metals by heating them with an electric arc between two carbon electrodes. No shielding is used. Pressure and filler metal may or may not be used.

U

ULTRASONIC COUPLER (ultrasonic soldering and ultrasonic welding) - Elements through which ultrasonic vibration is transmitted from the transducer to the tip.

ULTRASONIC SOLDERING - A soldering process variation in which high frequency vibratory energy is transmitted through molten solder to remove undesirable surface films and thereby promote wetting of the base metal. This operation is usually accomplished without a flux.

ULTRASONIC WELDING (USW) - A solid-state welding process which produces coalescence of materials by the local application of high frequency vibratory energy as the work parts are held together under pressure.

ULTRA-SPEED WELDING - See preferred term commutator-controlled welding.

UNDERBEAD CRACK - A crack in the heat-affected zone generally not extending to the surface of the base metal.

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UNDERCUT - A groove melted into the base metal adjacent to the toe or root of a weld and left unfilled by weld metal.

UNDERFILL - A depression on the face of the weld or root surface extending below the surface of the adjacent base metal.

UNIPOLARITY OPERATION - A resistance welding process variation in which succeeding welds are made with pulses of the same polarity.

UPSET - Bulk deformation resulting from the application of pressure in welding. The upset may be measured as a percent increase in interfacial area, a reduction in length, or a percent reduction in thickness (for lap joints).

UPSET BUTT WELDING - See preferred term upset welding.

UPSETTING FORCE - The force exterted at the faying surfaces during upsetting.

UPSETTING TIME - The time during upsetting.

UPSET WELD - A weld made by upset welding.

UPSET WELDING - (UW) - A resistance welding process which produces coalescence simultaneously over the entire area of abutting surfaces or progressively along a joint by the heat obtained from resistance to electric current through the area where those surfaces are in contact. Pressure is applied before heating is started and is maintained throughout the heating period.

UPSLOPE TIME (resistance welding) - The time during which the welding current continuously increases from the beginning of welding current.

UPSLOPE TIME (automatic arc welding) - The time during which the current changes continuously from initial current value to the welding value.

V

VAN DER WAALS BOND - A secondary bond arising from the fluctuating dipole nature of an atom with all occupied electron shells filled.

VACUUM BRAZING - A nonpreferred term used to denote various brazing processes which take place in a chamber or retort below atmospheric pressure.

VERTICAL POSITION - The position of welding in which the axis of the weld is approximately vertical.

VERTICAL POSITION (pipe welding) - The position of a pipe joint in which welding is performed in the horizontal position and the pipe may or may not be rotated.

VOLUME POSITION (pipe welding) - The position of a pipe joint in which welding is performed in the horizontal position and the pipe may or may not be rotated.

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VOLTAGE REGULATOR - An automatic electrical control device for maintaining a constant voltage supply to the primary of a welding transformer.

W

WATER WASH (thermal spraying) - The forcing of exhaust air and fumes from a spray booth through water so that the vented air is free of thermal sprayed particles or fumes.

WAVE SOLDERING (WS) - An automatic soldering process where work parts are automatically passed through a wave of molten solder. See also dip soldering.

WAX PATTERN (thermit welding) - Wax molded around the prts to be welded to the form desired for the completed weld.

WEAVE BEAD - A type of weld bead made with transverse oscillation.

WELD - A localized coalescence of metals or nonmetals produced either by heating the materials to suitable temperatures, with or without the application of pressure alone and with or without the use of filler material.

WELDABILITY - The capacity of a material to be welded under the fabrication conditions imposed into a specific, suitably designed structure and to perform satisfactorily in the intended service.

WELD BEAD - A weld deposit resulting from a pass. See stringer bead and weave bead.

WELDBONDING - A joining method which combines resistance seam welding with adhesive bonding. The adhesive may be applied to a faying surface before welding or may be applied to the area of sheet separation after welding.

WELD BRAZING - A joining method which combines resistance welding with brazing.

WELD CRACK - A crack in weld metal.

WELD-DELAY TIME (resistance welding) - The amount of time the beginning of welding current is delayed with respect to the initiation of the forge-delay timer in order to synchronize the forging force with welding current flow.

WELDER - One who performs a manual or semiautomatic welding operation. (Sometimes erroneously used to denote a welding machine.)

WELDER CERTIFICATION - Certification in writing that a welder has produced welds meeting prescribed standards.

WELDER PERFORMANCE QUALIFICATION - The demonstration of a welder's ability to produce welds meeting prescribed standards.

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WELDER REGISTRATION - The act of registering a welder certification or a photostatic copy thereof.

WELD GAGE - A device designed for checking the shape and size of welds.

WELD-HEAT TIME (resistance welding) - The time from the beginning of welding current to the beginning of post-heat time.

WELDING - A materials joining process used in making welds.

WELDING CURRENT - The current in the welding circuit during the making of a weld.

WELDING CURRENT (automatic arc welding) - The current in the welding circuit during the making of a weld, but excluding upslope, downslope, start, and crater fill current.

WELDING CURRENT (resistance welding) - The current in the welding circuit during the making of a weld, but excluding preweld or postweld current.

WELDING CYCLE - The complete series of events involved in the making of a weld.

WELDING ELECTRODE - See preferred term electrode.

WELDING FORCE - See preferred terms electrode force and platen force.

WELDING GENERATOR - A generator used for supplying ccurrent for welding.

WELDING GROUND - See preferred term work connection.

WELDING HEAD - The part of a welding machine or automatic welding equipment in which a welding gun or torch is incorporated.

WELDING LEADS - The work lead and electrode lead of an arc welding circuit.

WELDING MACHINE - Equipment used to perform the welding operation. For example, spot welding machine, arc welding machine, seam welding machine, etc.

WELDING OPERATOR - One who operates machine or automatic welding equipment.

WELDING POSITION - See flat position, horizontal position, horizontal fixed position, horizontal rolled postion, inclined position, overhead position, and vertical position.

WELDING PRESSURE - The pressure exerted during the welding operation on the parts being welded. See also electrode force and platen force.

WELDING PROCEDURE - The detailed methods and practices including all welding procedure specifications involved in the production of a weldment. See welding procedure specification.

WELDING PROCEDURE SPECIFICATION (WPS) - A document providing in detail the required variables for a specific application to assure repeatability by properly trained welders and welding operators.

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WELDING PROCESS - A materials joining process which produces coalescence of materials by heating them to suitable temperatures, with or without the application of pressure or by the application of pressure alone, and with or without the use of filler metal.

WELDING RECTIFIER - A device in a welding machine for converting alternating current to direct current.

WELDING ROD - A form of filler metal used for welding or brazing which does not conduct the electrical current.

WELDING SEQUENCE - The order of making the welds in a weldment.

WELDING TECHNIQUE - The details of a welding procedure which are controlled by the welder or welding operator.

WELDING TIP - A welding torch tip designed for welding.

WELDING TORCH (arc) - A device used in the gas tungsten and plasma arc welding processes to control the position of the electrode, to transfer current to the arc, and to direct the flow of shielding and plasma gas.

WELDING TORCH (oxyfuel gas) - A device used in oxyfueld gas welding, torch brazing, and torch soldering for directing the heating flame produced by the controlled combustion of fuel gases.

WELDING TRANSFORMER - A transformer used for supplying current for welding. See also reactor (arc welding).

WELDING VOLTAGE - See arc voltage.

WELDING WHEEL - See preferred term electrode.

WELDING WIRE - See preferred term electrode and welding rod.

WELD INTERFACE - The interface between weld metal and base metal in a fusion weld, between base metals in a solid-state weld without filler metal, or between filler metal and base metal in a solid-state weld with filler metal and in a braze.

WELD INTERVAL (resistance welding) - The total of all heat and cool time when making one multiple-impulse weld.

WELD INTERVAL TIMER (resistance welding) - A device which controls heat and cool times and weld interval when making multiple-impulse welds singly or simultaneously.

WELD LENGTH - See preferred term effective length of weld.

WELD LINE - See preferred term weld interface.

WELDMENT - An assembly whose component parts are joined by welding.

WELD METAL - That portion of a weld which has been melted during welding.

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WELD METAL AREA - The area of the weld metal as measured on the cross section of a weld.

WELDOR - See preferred term welder.

WELD PASS - A single progression of a welding or surfacing operation along a joint, weld deposit, or substrate. The result of a pass is a weld bead, layer, or spray deposit.

WELD PENETRATION - See preferred terms joint penetration and root penetration.

WELD SIZE - See preferred term size of weld.

WELD TAB - Additional material on which the weld may be initiated or terminated.

WELD TIME (automatic arc welding) - The time interval from the end of start time or end of upslope to beginning of crater fill time or beginning of downslope.

WELD TIME (resistance welding) - The time that welding current is applied to the work in making a weld by single-impulse welding or flash welding.

WELD TIMER - A device which controls only the weld time in resistance welding.

WELD VOLTAGE - See arc voltage.

WETTING - The phenomenon whereby a liquid filler metal or flux spreads and adheres in a thin continuous layer on a solid base metal.

WIPED JOINT - A joint made with solder having a wide melting range and with the heat supplied by the molten solder poured onto the joint. The solder is manipulated with a hand-held cloth or paddle so as to obtain the required size and contour.

WIRE FLAME SPRAYING - A thermal spraying process variation in which the material to be sprayed is in wire or rod form. See flame spraying (FLSP).

WIRE FEED SPEED - The rate of speed in mm/s or in./min at which filler metal is consumed in arc welding or thermal spraying.

WIRE STRAIGHTENER - A device used for controlling the case of coiled wire to enable it to be easily fed into the gun.

WORK ANGLE - The angle that the electrode makes with the referenced plane or surface of the base metal in a plane perpendicular to the axis of the weld. See also drag angle and push angle. Note: This angle can be used to define the position of welding guns, welding torches, high energy beams, welding rods, thermal cutting and thermal spraying torches, and thermal spraying guns.

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WORK ANGLE (pipe) - The angle that the electrode makes with the referenced plane extending from the center of the pipe through the molten weld pool. Note: This angle can be used to define the position of welding guns, welding torches, high energy beams, welding rods, thermal cutting and thermal spraying torches, and thermal spraying guns.

WORK COIL - The inductor used when welding, brazing, or soldering with induction heating equipment.

WORK CONNECTION - The connection of the work lead to the work.

WORK LEAD - The electric conductor between the source of arc welding current and the work.

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APPENDIX B
GOVERNMENT AND NONGOVERNMENT DOCUMENTS PERTAINING TO MATERIALS JOINING

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SOURCES

Government Specifications and Standards for use in connection with a bid or contract are furnished by:

Commanding Officer
Naval Publications and Forms Center
Attn: NPFC 105
5801 Tabor Avenue
Philadelphia, PA 19120

For other uses, copies of Government Documents may be purchased from:

Business Center
General Services Administration
Washington, DC 20405

Military Standards and Handbooks in book form may also be purchased from:

Superintendent of Documents
Government Printing Office
Washington, DC 20402

Nongovernment documents are available from the organizations that publish them. The addresses for organizations whose publications are listed in this Appendix are given below. The listing is not all inclusive. The reader is encouraged to obtain indexes of publications from the various organizations. The American Welding Society, for example, offers over 100 publications covering every conceivable phase of the welding process. These publications include codes, specifications, and recommended practices for all processes.

American Welding Society
550 LeJeune Road
Miami, FL 33126

ASTM
1916 Race Street
Philadelphia, PA 19103

SAE
400 Commonwealth Ave.
Warrendale, PA 15096

American Petroleum Institute
Publications & Distribution Section
1271 Avenue of the Americas
New York, NY 10020

American Society of Mechanical Engineers
United Engineering Center
345 East 47th Street
New York, NY 10017

National Fire Protection Association
60 Batterymarch Street
Boston, MA 02110

Compressed Gas Association, Inc.
500 Fifth Avenue
New York, NY 10036

American National Standards Institute
1430 Broadway
New York, NY 10018

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GOVERNMENT STANDARDIZATION DOCUMENTS PERTAINING
TO
MATERIALS JOINING

EQUIPMENT

Document Number	Title
GG-I-271	Igniter, Friction, Oxyacetylene - Torch
GGG-B-00690	Brazing and Soldering Set (Acetylene Gas)
GGG-H-35A	Hammer, Hand, Welder, Chipping
GGG-H-211C	Helmet, Welders, and Shield, Welding, Hand Held
GGG-S-595B	Soldering Iron, Nonelectric (Copper)
GGG-S-600	Soldering Iron, Self-heating (Without Electric or External Heat)
GGG-T-00570A(2)	Torch Kit, Soldering (Propane Gas)
KK-G-486D (1)	Gloves, Leather, Welders, Gauntlet
MIL-B-24302 (1)	Brazing System, Near Infrared Quartz Lamp
MIL-C-17223A (2)	Cleaner, Welding and Cutting Tip
MIL-C-18843D	Cutting Machine, Oxygen Fuel, Motor Driven, Portable
MIL-C-80111B	Cutting Machine, Shape, Oxyacetylene Motor Dirven, Portable
MIL-D-45582A	Desoldering and Resoldering Kits, Electric
MIL-F-80082A	Furnace, Heat Treating and Brazing, Electric Controlled Atmosphere, Pusher and Conveyor Belt Types
MIL-F-80113B	Furnace, Vacuum, Heat Treating and Brazing, Electrically Heated, Cold Wall Design
MIL-H-865	Holder, Electrode, Shielded Metal Arc Welding
MIL-M-3800C	Metallizing Outfits (Wire Gas), Guns & Accessories.

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EQUIPMENT (Continued)

Document Number	Title
MIL-M-80173A	Manipulator, Automatic Welding Head, Stationary Base and Traveling Boom, Electric Motor Driven
MIL-P-80068B	Positioner, Welding, Table Type, Gear Driven
MIL-P-80085A	Positioners, Welding, Headstock and Tailstock Type
MIL-P-80186	Positioner, Welding, Table Type, Cantilever Arm Design
MIL-R-5632B	Rods and Wire, Steel, Welding (For Aircraft)
MIL-R-6944B	Rods and Wire, Welding, Magnesium Alloys
MIL-S-45847C	Soldering and Brazing Outfit, Resistance Heating
MIL-STD-606	Helmet, Welders, Shield, Welding, Hand Held, and Lenses, Helmet, Welders
MIL-T-13880F	Torch, Oxyacetylene, Welding and Cutting
MIL-T-17636B	Tips, Welding Torch
MIL-T-17870B	Tip, Torch, Cutting, and Tip, Torch, Welding (Replacement) (Navy)
MIL-T-18068 (1)	Torches, Welding, and Torches, Cutting - Hand, Commercial, Oxyacetylene and Oxyhydrogen Gases (Navy)
MIL-T-18134B	Torch, Welding, Hand, Oxyacetylene, and Oxyhydrogen Gases, Light Duty (Navy)
MIL-T-18365A (1)	Torch, Brazing, Self-contained, Liquefied Petroleum Gas
MIL-T-18803C (1)	Torch, Outfit, Cutting and Welding, Oxyacetylene
MIL-T-45809D	Table, Welding
MIL-T-80131A	Turning Roll, Welding, Powered Type and Idler Type
MIL-W-4125C	Welding Kit, Gas, Portable (Supersedes 5-352 and 94-40660)
MIL-W-6712B	Wire Metallizing
MIL-W-7973E	Welding Machine, Electrical Resistance, Spot, Press Type, Three Phase

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EQUIPMENT (Continued)

Document Number	Title
MIL-W-8393	Welding, Resistance, Electronic Circuit Modules
MIL-W-19285A (1)	Welding Machines, Arc, Direct-current, Variable Voltage Transformer-rectifier Type
MIL-W-23680D	Welding Set, Arc, Stud
MIL-W-23739 (3)	Weld Preheat System (Strip Heater Type)
MIL-W-28502A	Welding Kit, Oxyacetylene, Power Weld, Metal Overlay
MIL-W-29383A	Welding Set, Arc, Ac/dc, Mobile Multiprocess, Diesel Engine Driven
MIL-W-45562A (3)	Welding and Soldering Equipment, Supplies and Accessories, Packaging of
MIL-W-45844B	Welding Machine, Resistance, Spot Type, Portable (Automotive Repair)
MIL-W-52161C	Welding Set, Inert Gas Shielded Arc, Consumable Metal electrode
MIL-W-52574A	Welding Process and Welding Procedure Requirements for Manufacture of Equipment Utilizing Steels
MIL-W-52580A	Welding Machine, Arc, Direct Current, Constant Current and Constant Potential, Gasoline Engine Driven, 300 Ampere Capacity, Skid Mounted
MIL-W-52629A	Welding Shop, Trailer Mounted, Lightweight
MIL-W-80019B	Welding Machine, Arc, Direct Current, Constant Current, Electrical Motor Driven
MIL-W-80024C	Welding Machine, Arc, Ac, Transformer, Constant Current
MIL-W-80027C	Welding Machine, Arc, Ac/dc, Transformer-rectifier, Constant Current
MIL-C-80036B	Cutting Machine, Oxygen Flame and Plasma Arc, Automatic, Stationary
MIL-C-80037C	Welding Machine, Arc, Dc, Constant Current, Internal Combustion Engine Driven
MIL-W-80040D	Welding Machines, Arc, Dc, Transformer-rectifier, Constant Voltage for Multi-operator Systems

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EQUIPMENT (Continued)

Document Number	Title
MIL-W-80042B	Welder Control Stations for Multi-operator Systems
MIL-W-80043C	Welding Machine, Arc, Dc, Transformer-rectifier Constant Current
MIL-W-80046B (1)	Welding Machine, Resistance, Stored Energy, Capacitor
MIL-W-80062B	Welding Machine, Arc, Dc, Transformer-rectifier, Adjustable Constant Voltage
MIL-P-80068B	Positioner, Welding, Table Type, Gear Driven
MIL-W-80070C	Welding Machine, Electrical, Resistance, Rocker, Spot, Single Phase
MIL-W-80075B	Welding Machine, Arc, Dc, Transformer-rectifier Constant Current, Eight Operator
MIL-W-80080B	Welding Machine, Arc, Ac/dc, Constant Current, Internal Combustion Engine Driven
MIL-P-80085A	Positioner, Welding, Headstock and Tailstock Types
MIL-W-80087B	Welding Machine, Resistance, Rocker Arm, Spot Type, Air Operated, Three Phase
MIL-C-80090B	Cutting System, Metal, Plasma Arc
MIL-F-80091	Flux Recovery Systems, and Flux Reclaiming Systems, Welding Submerged Arc
MIL-W-80102B (1)	Welding Set, Arc, Gas Shielded, Consumable Electrode
MIL-W-80103B	Welding Equipment, Arc, Dc, Pipe, Automatic
MIL-W-80105B	Welding Set, Arc, Gas Shielded Tungsten Electrode
MIL-P-80109C	Plasma Spray System, (Power) Guns, and Accessories
MIL-C-80111B	Cutting Machine, Shape, Oxygen and Fuel Gas, Motor Driven, Portable
MIL-W-80110	Welding Power Source, Stud
MIL-D-80118B	Deposition System, Vacuum Evaporation Resistance Type and Electron Beam Type
MIL-W-80117A	Welding Set, Submerged Arc

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EQUIPMENT (Continued)

Document Number	Title
MIL-W-80119A	Welding Machine, Electron Beam, Vacuum Type
MIL-W-80121A	Welding Machine, Resistance, Spot, Portable Electrode, Single Phase
MIL-W-80124B	Welding Machine (Microwelder), Resistance, Single Phase, Bench Mounted
MIL-T-80131A	Turning Roll, Welding, Powered Type and Idler Type
MIL-M-80141B	Metallizing Outfit, Power Gas, Guns and Accessories
MIL-G-80152A	Gouging and Cutting Machine, Direct Current, Electric Arc, Portable
MIL-W-80164A	Welding Head, Arc, Automatic, Consumable Wire Electrode
MIL-W-80165A	Welding Machine, DC, Constant Current, Constant Voltage, Internal Combustion Engine Driven
MIL-M-80173A	Manipulator, Automatic Welding Head, Stationary Base and Traveling Boom, Electric Motor Driven
MIL-W-80183A	Welding Machine, Electrical Resistance, Roll Spot and Seam, Press Type, Three Phase
MIL-W-80185	Welding Machines, Thermocompression, Diffusion Bonding, Miniature Wire
MIL-P-80186	Postitioner, Welding, Table Type, Cantilever ARm Design
MIL-W-80187	Welding Machines, Ultrasonic Bonding, Miniature Wire
MIL-W-80191A	Welding Machine, Arc, Dc, Transformer-rectifier, Constant Voltage, Constant Current
MIL-W-80192A	Welding Machine, Resistance, Spot and Seam, Metal Foil, Single Phase, Portable
MIL-W-80197	Manipulator Automatic, Welding Head Traveling Base and Traveling Boom, Electric Motor Driven
MIL-W-80198	Welding System, Metal, Plasma Arc
MIL-D-80200A	Deposition System, Vacuum, Sputtering Type

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EQUIPMENT (Continued)

Document Number	Title
MIL-W-80211A	Welding Machine (Microwelder), Resistance, Stored Energy, Battery Type, Bench Mounted
MIL-W-80214A	Welding Machine, Arc, Dc, Transformer-rectifier, Pulsating Current
MIL-M-80226	Metallizing System, Wire, Electric Arc, Guns and Accessories
MIL-W-80227	Welding Machine, Ultrasonic, Spot
MIL-D-80229	Deposition Machine, Carbide
MIL-W-80230	Welding System, Electronic Beam, Atmospheric
MIL-W-80232A (1)	Welding Machine, Arc, Ac/dc, Transformer-rectifier, Constant Current, Programmed
MIL-W-80235A	Welding Machine, Electric Resistance, Spot Press Type Air Operated, Single Phase, Alternating Current (Ac)
MIL-W-80243A	Welding Machine, Arc, Dc, Transformer-rectifier, Constant Current, Programmed
MIL-W-80244A	Welding Machines, Friction, Continuous Drive and Inertia Types
MIL-W-80253	Welding Machine, Electrical, Resistance, Spot, Press Type, Single Phase, Bench or Floor Mounted
MIL-W-80256	Welding Machines, Electrical, Resistance, Spot, Press Type, Air Operated, Single
MIL-W-83967 (1)	Welding Machine, Electron Beam, Vacuum Chamber
MIL-W-83972 (1)	Welding Machine, Electrical Resistance, Spot and Seam, Press Type, Three Phase
MS-15011B	Fitting, Hose, Welding and Cutting, Oxygen and Acetylene
MS-15013C	Tip for Military Standard Cutting Torch
W-S-564A	Soldering Gun
W-S-570A (2)	Soldering Iron (Electric)

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GOVERNMENT STANDARDIZATION DOCUMENTS PERTAINING TO MATERIALS JOINING

MATERIALS

Document Number	Title
A-A-50291	Brazing Alloys, Aluminum and Magnesium, Filler Metal
BB-A-106B	Acetylene, Dissolved, Technical Grade
BB-H-886C	Hydrogen
BB-O-925A	Oxygen, Technical, Gas and Liquid
MIL-A-81596 (1)	Aluminum Foil for Sandwich Construction
MIL-B-20148C	Brazing Sheet, Aluminum and Aluminum Alloy Steels and Plates, Aluminum Brazing Alloy Clad
MIL-C-21275C	Core Material, Metallic, Heat Resisting, for Structural Sandwich Construction
MIL-C-45948	Carbon Block, Carbon Paste, and Carbon Rods
MIL-E-278B (1)	Electrode, Welding, Covered, Aluminum Bronze
MIL-E-2156D (7)	Electrodes and Rods - Welding, Bare, Nickel Alloy
MIL-E-6843C	Electrodes, Welding, Covered, Low-Alloy Steel Primarily for Aircraft and Weapon Applications
MIL-E-8697E	Electrode, Welding, Coated, Low Hydrogen Heat Treatable Steel
MIL-E-8971A/1	Electrode, Graphite, Spectrometric Grade, Disc Type
MIL-E-13080D	Electrode, Welding, Covered, Austenitic Steel (19-9 Modified) for Armor Applications
MIL-E-13191B (1)	Electrode, Welding, Covered, Bronze, for General Use
MIL-E-15597E	Electrode, Welding, Covered, Coated, Aluminum and Aluminum Alloy
MIL-E-17764C	Electrode, Cutting, Tubular, Steel, Covered
MIL-E-17777D	Electrodes, Cutting and Welding, Carbon Graphite, Uncoated and Copper Coated
MIL-E-18193B (1)	Electrodes, Welding, Carbon Steel and Alloy Steel, Bare, Coiled
MIL-E-19141D	Electrode, Welding, Surfacing, Iron Base Alloy

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MATERIALS (Continued)

Document Number	Title
MIL-E-19933E (1)	Electrodes and Rods - Welding, Bare, Chromium and Chromium-Nickel Steels
MIL-E-21562D (7)	Electrodes and Rods - Welding, Bare, Nickel Alloy
MIL-E-22200F (3) SUPP1 INT AMD 2 (SH)	Electrodes, Welding, Covered, General Specification for
MIL-E-0022200F (3)	Electrodes, Welding, Covered, General Specification for (Use in lieu of MIL-E-22200D)
MIL-E-22200/1E INT AMD 2 (SH)	Electrodes, Welding, Mineral Covered, Iron Power, Low Hydrogen Medium and High Tensile Steel, As Welded or Stress Relieved Weld Application
MIL-E-0022200/1F	Electrode, Welding, Mineral Covered, Iron Power, Low Hydrogen Medium and High Tensile Steel, As Welded or Stress Relieved Weld Application
MIL-E-00222000/2C (2)	Electrode, Welding, Covered (Austenitic Chromium-Nickel Steel, for Corrosive and High Temperature Services) (Use in lieu of MIL-E-22200/2A)
MIL-E-22200/3E (1)	Electrodes, Welding, Covered, Nickel Base Alloy, and Cobalt Base Alloy
MIL-E-22200/4C INT AMD 3	Electrodes, Welding, Covered, Copper-Nickel Alloy
MIL-E-22200/5C	Electrodes, Welding, Mineral Covered, Iron Power, Low Hydrogen, High Tensile Low Alloy Steel - Heat Treatable Only
MIL-E-22200/6C	Electrodes, Welding, Mineral Covered, Low Hydrogen, Medium and High Tensile Steel
MIL-E-22200/7B	Electrodes, Welding, Covered, Molybdenum Alloy Steel Application
MIL-E-22200/8B (1)	Electrodes, Welding, Covered, Low-Hydrogen and Iron Powder, Low Hydrogen, Chromium-Molybdenum Alloy Steel and Corrosion Resisting Steel
MIL-E-22200/9B	Electrodes, Welding, Mineral Covered, Low Hydrogen or Iron Powder, Low Hydrogen, Nickel-Manganese-Chromium-Molybdenum Alloy Steel for Producing HY-130 Weldments for As Welded Applications

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MATERIALS (Continued)

Document Number	Title
MIL-E-0022200/10A (1)	Electrodes, Welding, Mineral Covered, Low-Hydrogen Iron Powder for Producing HY-100 Steel Weldments as Welded Applications
MIL-E-22200/11	Electrodes, Welding, Mineral Covered Iron Powder, Low Hydrogen and Nickel Chromium Molybdenum Vanadium Alloy Steel for Producing HY-130 Weldments to be Heat Treated
MIL-E-23765C (4) SUPP 1	Electrode and Rods - Welding, Bare, Solid, General Specification for
MIL-E-23765/1D	Electrode and Rod - Welding, Bare, Solid, Mild Steel
MIL-E-23765/2C (2)	Electrode and Rod - Welding, Bare, Solid, Low Alloy Steel
MIL-E-23765/3	Electrode, Welding, Bare, Copper and Copper Alloy
MIL-E-24355B	Electrodes, Welding, Bare, Solid, Nickel-Manganese-Chromium-Molybdenum Alloy Steel for Producing HY-130 Weldments for As Welded Applications
MIL-E-24403A (1)	Electrode - Welding, Flux, Cored General Specification for
MIL-E-24403/1C (1)	Eelctrodes - Welding, Flux Cored, Ordinary Strength & Low Alloy Steel
MIL-E-24403/2A	Electrode - Welding, Flux Cored, Low Alloy Steel
MIL-E-45829A	Electrode, Welding, Copper, Silicon-Deoxidized, Solid, Bare
MIL-F-6939B	Flux, Aluminum and Aluminum Alloy, Gas Welding
MIL-F-12784B	Flux, Soldering (Stearine Compound IC-3)
MIL-F-14256D (2)	Flux, Soldering, Liquid (Rosin Base)
MIL-F-18251C (2)	Fluxes, Welding, Submerged Arc Process Carbon and Low-alloy Steel Application
MIL-F-19922	Fluxes, Welding (Compositions), Submerged Arc Process with Type B, Electrodes, Carbon and Low-alloy Steel Application
MIL-F-20329A	Flux, Soldering, Rosin Base, General Purpose
MIL-I-23413C	Inserts, Welding, Coiled Filler Material, Solid Rings
MIL-I-25135C (3)	Inspection Materials Penetrant (Asg)

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MATERIALS (Continued)

Document Number	Title
MIL-L-741C	Lead, Electrical, Arc Welding
MIL-M-45558B (1)	Moisture Stabilizer, Welding Electrode
MIL-P-8853	Paste, Copper Brazing, Water Thinning (Asg)
MIL-R-5632B	Rod and Wire, Steel, Welding (for Aircraft)
MIL-R-6944B	Rod and Wire, Welding, Magnesium Alloys
MIL-R-17131C	Rod, Welding, Surfacing
MIL-R-19631B	Rods, Welding, Copper and Copper Alloy
MIL-R-81588B	Rod & Wire, Welding Titanium & Titanium Alloys
MIL-S-627 NOTICE-2	Solder, Low Melting Point
MIL-S-12204C	Solder, Aluminum Alloy
MIL-W-10430B (6)	Welding Rods and Electrodes, Preparation for Delivery of
MIL-W-16193B	Wire, Platinum Alloy
MIL-W-22248A	Weldments, Aluminum and Aluminum Alloy
O-F-499C (1)	Flux, Brazing, Silver Alloy, Low Melting Point
O-F-506B	Flux, Soldering, Paste and Liquid
QQ-B-650B	Brazing Alloys, Copper, Copper-Zinc, and Copper-Phosphorus
QQ-B-653B	Brazing Alloy, Gold
QQ-B-654A	Brazing Alloy, Silver
QQ-B-655C	Brazing Alloys, Aluminum and Magnesium Filler Metal
QQ-E-450A	Electrode, Welding Covered, Mild Steel
QQ-R-566B (1)	Rods, Welding, Aluminum and Aluminum Alloys
QQ-R-571C	Rod, Welding, Copper and Nickel Alloy
QQ-S-571E (2)	Solder, Tin Alloy, Lead-Tin Alloy and Lead Alloy
S-F-571C	Forge, Coal Burning (Portable and Stationary, Manual and Power Operated)

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GOVERNMENT STANDARDIZATION DOCUMENTS PERTAINING
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PROCESSES

Document Number	Title
MIL-B-7883B	Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys
MIL-B-9972A (1)	Brazing, Nickel Alloy, General Specification for
MIL-B-12672 (1)	Braze-welding, Oxyacetylene, of Built-up Metal Structures
MIL-B-12673 (1)	Brazing, Oxyacetylene, of Built-up Metal Structures
MIL-B-24302 (1)	Brazing System, Near Infrared Quartz Lamp
MIL-S-7811	Sandwich Construction, Aluminum Alloy Faces, Aluminum Foil Honeycomb Core
MIL-S-45743E (1)	Soldering, Manual Type, High Reliability, Electrical Connections, for Missile Systems, Procedures for
MIL-S-46844C	Solder, Bath Soldering of Printed Wiring Assemblies, Automatic Machine Type
MIL-STD-645B	Dip Brazing of Aluminum Alloys
MIL-STD-1261A	Welding Procedures for Constructional Steels
MIL-W-41B (1)	Welding of Armor, Metal Arc, Manual, with Austenitic Electrodes, for Aircraft
MIL-W-6858D	Welding, Resistance, Spot and Seam
MIL-W-6873B	Welding, Flash, Carbon and Alloy Steel
MIL-W-8604A	Welding, Fusion, Aluminum Alloys, Processing and Performance
MIL-W-8611A	Welding, Metal Arc and Gas, Steels, and Corrosion and Heat Resistant Alloys, Process for
MIL-W-8939	Welding, Resistance, Electronic Circuit Modules (Asg)
MIL-W-10430B (6)	Welding Rods and Electrodes, Preparation for Delivery of

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PROCESSES (Continued)

Document Number	Title
MIL-W-12332A (2)	Welding, Resistance, Spot, Seam and Projection, for Fabricating Assemblies of Low Carbon Steel
MIL-W-13773A (2)	Welding Repair of Readily Weldable Castings Other Than Armor, Metal-Arc, Manual
MIL-W-18326A	Welding of Magnesium Alloys, Gas and Arc, Manual and Machine Process for
MIL-W-21157A	Weldment, Steel, Carbon and Low Alloy (Yield Strength 30,000-60,000 Psi)
MIL-W-22248A	Weldment, Aluminum and Aluminum Alloy
MIL-W-23739 (3)	Weld Preheat System (Strip Heater Type)
MIL-W-45205B (1)	Welding, Inert Gas, Metal Arc, Aluminum Alloys, Readily Weldable for Structures Excluding Armor
MIL-W-45206A	Welding, Aluminum Alloy Armor
MIL-W-45210A	Welding, Resistance, Spot, Weldable Aluminum Alloys
MIL-W-45211A	Welding, Stud, Aluminum
MIL-W-45223A	Welding, Spot, Hardenable Steels
MIL-W-46086A (1)	Welding Homogeneous Armor, Metal Arc, Manual
MIL-W-46132	Welding Fusion, Electronic Beam Process for
MIL-W-46154	Welding, Resistance, Spot and Projection for Fabricating Assemblies of Carbon Steel Sheets
MIL-W-46870	Welding of Electronic, Circuitry, Process for
MIL-W-47091	Welding, Arc and Gas, for Fabricating Ground Equipment for Rockets and Guided Missiles
MIL-W-52574A	Welding Process and Welding Procedure Requirements for Manufacture of Equipment Utilizing Steels
MIL-W-62160	Welding, Flash, Standard Low Carbon Steel
MIL-STD-645B	Dip Brazing of Aluminum Alloys
MIL-STD-1941	Metal Arc Welding of Homogeneous Armor
MIL-STD-1183	Welding, Gas, Steels, Constructional, Readily Weldable for Low Stressed Joints

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PROCESSES (Continued)

Document Number	Title
MIL-STD-1185	Welding, High Hardness Armor
MIL-STD-1261A	Welding Procedures for Constructional Steels
MIL-STD-1877	Brazing, Nickel, High Temperature Vacuum
MIL-STD-1881	Brazing, Silver, General Process for

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GOVERNMENT STANDARDIZATION DOCUMENTS PERTAINING
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QUALITY ASSURANCE

Document Number	Title
DOD-STD-1866	Soldering Process, General (Non Electrical) (Metric)
FED-STD-151b	Metal, Test Methods
MIL-I-6866B (2)	Inspection, Penetrant Method of
MIL-I-6868E (3)	Inspection Process, Magnetic Particle
MIL-I-45208A (1)	Inspection System Requirements
MIL-P-52539	Pipeline Weld Set, Testing
MIL-Q-9858A (1)	Quality Program Requirements
MIL-R-11470A	Radiographic Inspection, Qualification of Equipment, Operators and Procedures
MIL-R-45774 (2)	Radiographic Inspection, Soudness Requirements for Fusion Welds in Aluminum and Magnesium Missile Components
MIL-STD-21A	Welded Joint Designs, Armored Tank Type
MIL-STD-22D	Welded Joint Design
MIL-STD-100C	Engineering Drawing Practices
MIL-STD-105D	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109B	Quality Assurance Terms and Definitions
MIL-STD-248C	Welding and Brazing Procedures and Performance Qualification
MIL-STD-271E	Nondestructive Testing Requirements for Metals
MIL-STD-278E	Fabrication Welding and Inspection, and Casting Inspection for Machinery, Piping and Pressure Vessels for Ships for the United States Navy
MIL-STD-401B	Sandwich Constructions and Core Materials, General Test Methods

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QUALITY ASSURANCE (Continued)

Document Number	Title
MIL-STD-410D	Qualification of Inspection Personnel (Magnetic Particle and Penetrant)
MIL-STD-414 Change 1	Sampling Procedures and Tables for Inspection by Variables for Percent Defective
MIL-STD-453	Inspection, Radiographic
MIL-STD-781C	Reliability Tests Exponential Distribution
MIL-STD-798B	Nondestructive Testing, Welding, Quality Control, Material Control and Identification and Hi-Shock Test Requirements for Piping System Components for Naval Shipboard Use
MIL-STD-810B	Environmental Test Methods
MIL-STD-1165	Glossary of Environmental Terms (Terrestrial)
MIL-STD-1235B	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1252	Inertia Friction Welding Process, Procedure and Performance Qualification
MIL-STD-1264	Radiographic Inspection for Soundness of Welds in Steel by Comparison to Graded ASTM E 390 Reference Radiographs
MIL-STD-1595A	Qualification of Aircraft; Missile and Aerospace Fusion Welders
MIL-STD-1688 Notice 1	Fabrication, Welding and Inspection of HY-80/100 Submarine Applications
MIL-STD-1689	Fabrication, Welding and Inspection of Ships Structure
MIL-STD-1692	Underwater Welding Requirements for Naval Facilities
MIL-STD-1693	Fabrication Welding and Inspection of Hyperbaric Chambers and Other Critical Land Based Structures

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QUALITY ASSURANCE (Continued)

Document Number	Title
MIL-STD-1699	Nondestructive Evaluations of Butt Welds in Crane and Railroad Rails
MIL-STD-1875	Ultrasonic Inspection, Requirements for
MIL-STD-2149	Standard Procedures for Explosion Testing Ferrous and Non-Ferrous Metallic Materials and Weldments
MIL-STD-2154	Inspection, Ultrasonic, Wrought Metals, Process for
MIL-T-24255A	Technical Repair Standards (for Submarines Only)
MS-33530	Tolerance, Welded Carbon and Alloy Steel Tubing (Asg)
MS-33531A	Tolerance, Welded Corrosion Resistant Steel Tubing (Asg)
Navships 250-643-7	Standard Terminology and Definitions for Weld Conditions and Defects
Navships 9099-003-9001	Radiographic Standards for Production and Repair Welds
NSF-Spec-130	Certification of Welding Machine Operators and Manual Welders, Specifications for

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MILITARY HANDBOOKS

MIL-HDBK Number	Title
5	Aerospace Vehicle Structures, Metallic Materials and Elements for
23A Notice 1	Structural Sandwich Composites
53/1A	Guide for Attribute Lot Sampling Inspection and MIL Std 105
105	Material and Material Processes Series Arc Welding
106	Multilevel Continuous Sampling Procedures and Table for Inspection by Attributes (Handbook H-106)
107	Inspection and Quality Control Single Level Continuous Sampling Procedures and Tables for Inspection by Attributes (See MIL Std 1235)
108	Quality Control and Reliability - Sampling Procedures and Tables for Life and Reliability Testing (Based on Exponential Distribution)
109	Quality Control and Reliability - Statistical Procedures for Determining Validity of Suppliers Attributes Inspection of Chains, Power Transmission and Conveying, Application, Installation and Maintenance
333	Handbook for Standardization of Nondestructive Testing Methods (Volumes 1 and 2)
693A	Magnesium and Magnesium Alloys
694A	Aluminum and Aluminum Alloys
697A	Titanium and Titanium Alloys
698A	Copper and Copper Alloys
723A	Steel and Wrought Iron Products
726	Ultrasonic Testing

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QUALIFIED PRODUCT LISTINGS

Document Number	Title
19933-52	Electrode and Rod, Welding, Bare, Chromium and Nickel Steels
22200-42	Electrode, Welding, Covered, General Specification for
45057 Notice 1	Electrodes, Welding, Mineral Covered, Low Hydrogen, for Repair Welding of Alloy Steel Castings

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USAF ANA STANDARDS

Document Number	Title
AN785 Rev 6	Coupling, Union, Brazing
AN871 Rev 7	Flange, Straight Thread Boss Welding

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OTHER DEPARTMENTAL DOCUMENTS

Document Number	Title
ANC23	Sandwich Construction for Aircraft - Materials Properties and Design Criteria

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OTHER MILITARY PUBLICATIONS

Document Number	Title
Navships 250-634-7	Standard Terminology and Definitions for Weld Conditions and Defects
Navships 0900-003-9001	Radiographic Standards for Production and Repair Welds
NSF-Spec-130	Certification of Welding Machine operators and Manual Welders, Specification for

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NONGOVERNMENT STANDARDIZATION DOCUMENTS PERTAINING TO MATERIALS JOINING

AMERICAN WELDING SOCIETY STANDARDS

AWS Document Number	Title
A1.1-80	Metric Practice Guide for the Welding Industry
A2.4-79	Welding and Nondestructive Testing, Symbols for - 31 Aug 78
A3.0-85	Welding Terms and Definitions - 15 Aug 79
A4.2	Calibrating Magnetic INstruments to Measure the Delta Ferrite Content of Austenitic Stainless Steel Weld Metal, Standard Procedures for
A5.0-83	Filler Metal Comparison Charts
A5.01-78	Filler Metal Procurement Guidelines
A5.1-81	Specification for Carbon Steel Covered Arc Welding Electrodes
A5.2-80	Rods, Oxyfuel Gas Welding, Iron and Steel - 15 May 80
A5.3-80	Specification for Aluminum and Aluminum Alloy Covered Arc Welding Elctrode
A5.4-81	Electrode, Steel Welding, Chromium and Chromium Nickel, Corrosion Resisting - 8 Oct 81
A5.5-81	Specification for Low-alloy Steel Covered Arc-welding Electrodes
A5.6-84	Electrodes, Specification for Copper and Copper Alloy Covered - 27 Jul 76
A5.7-84	Copper and Copper Alloy Bare Welding Rods and Electrodes, Specification for - 11 Feb 77
A5.8-81	Metal, Brazing Filler Metal - 20 Jul 81
A5.9-81	Welding Electrodes and Welding Rods, Corrosion Resisting Chromium and Chromium Nickel, Steel Bare and Composite Metal Cored and Stranded - 7 Oct 81
A5.10-80	Welding Rod and Electrode Aluminum and Aluminum Bare - 7 Jul 80
A5.11-84	Nickel and Nickel Alloy Covered Welding Electrodes, Specification for
A5.12-80	Specification for Tungsten Arc Welding Electrodes

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AMERICAN WELDING SOCIETY STANDARDS (continued)

AWS Document Number	Title
A5.13-80	Rod and Electrodes Welding, Solid Surfacing - 25 Apr 80
A5.14-84	Nickel and Nickel Alloy Bare Welding Rods and Electrodes, Specification for
A5.15-82	Iron, Cast, Welding Rods and Covered Electrodes for - 29 Jul 82
A5.16-70R73	Rod, Welding and Electrodes, Titanium and Titanium Alloy Bare - 19 Apr 73
A5.17-80	Specification for Carbon Steel Electrodes and Fluxes for Submerged Arc Welding
A5.18-79	Specification for Carbon Steel Filler Metals for Gas Shielded Arc Welding
A5.19-80	Welding Rods and Bare Electrodes, Magnesium Alloy - 25 Jul 80
A5.20-79	Specification for Carbon Steel Electrodes for Flux Cored Arc Welding
A5.21-80	Composite Surfacing Welding Rods and Electrodes - 25 Apr 80
A5.22-80	Electrode, Flux Cored Corrosion Resisting Chromium and Chromium Nickel Steel - 22 Aug 80
A5.23-80	Specification for Low Alloy Steel Electrodes and Fluxes for Submerged Arc Welding
A5.24-79	Rods and Electrodes, Zirconium and Zirconium Alloy Bare Welding - 27 Mar 79
A5.25-78	Steel, Low Alloy, Consumables Used for Electroslag Welding of Carbon and High Strength - 18 Feb 77
A5.26-78	Steel, Low Alloy, Consumables Used for Electrogas Welding of Carbon and High Strength - 18 Feb 77
A5.27-85	Rod, Welding, Copper and Copper Alloy Gas - 3 Mar 78
A5.28-79	Specification for Low Alloy Steel Filler Metals for Gas Shieled Arc Welding
A5.29-80	Specification for Low Alloy Steel Eelectrodes for Flux Cored Arc Welding
A5.30-79	Specification for Consumable Inserts

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AMERICAN WELDING SOCIETY STANDARDS (continued)

AWS Document Number	Title
B1.0-77	Guide for the Nondestructive Inspection of Welds
B2.1-84	Standard for Welding Procedure and Performance Qualification
B2.2-85	Standard for Brazing Procedure and Performance Qualification
B3.0-77	Standard Qualification Procedure
B4.0-85	Standard Methods for Mechanical Testing of Welds
C1.1-66	Recommended Practices for Resistance Welding
C1.3-70	Recommended Practices for Resistance Welding Coated Low Carbon Steels
C2.1-73	Recommended Safe Practices for Thermal Spraying
C2.2-67	Recommended Practices for Metallizing with Aluminum and Zinc for Protection of Iron and Steel
C2.13-70	Flame Spraying of Ceramics
C2.14-74	Corrosion Tests of Flame-Sprayed Coated Steel
C2.15-75	Recommended Practices for Fused Thermal Sprayed Deposits
C2.16-78	Guide for Thermal Spray Operator and Equipment Qualification
C2.17-78	Recommended Practices for Electric Arc Spraying
C3.2-82	Standard Method for Evaluating the Strength of Brazed Joints
C3.3-80	Recommended Practices for Design, Manufacture, and Inspection of Critical Brazed Components
C5.1-73	Recommended Practices for Plasma-Arc Welding
C5.2-83	Recommended Practices for Plasma-Arc Cutting
C5.3-82	Recommended Practices for Air Carbon-Arc Gouging and Cutting
C5.4-84	Recommended Practices for Stud Welding

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AMERICAN WELDING SOCIETY STANDARDS (continued)

AWS Document Number	Title
C5.5-80	Recommended Practices for Gas Tungsten Arc Welding
C5.6-79	Recommended Practices for Gas Metal Arc Welding
C5.7-81	Recommended Practices for Electrogas Welding
D1.1-85	Structural Welding Code - Steel
D1.2-83	Structural Welding Code - Aluminum
D1.3-81	Specification for Welding Sheet Steel in Structures
D1.4-79	Structural Welding Code - Reinforcing Steel
D3.5-85	Guide for Steel Hull Welding
D3.6-83	Specification for Underwater Welding
D3.7-83	Aluminum Hull Welding, Guide for
D5.2-84	Welded Steel Elevated Tanks, Standpipes, and Reservoirs for Water Storage, Standard for
D8.4-61	Recommended Practices for Automotive Welding Design
D8.5-69	Recommend Practices for Automotive Portable Gun Resistance Spot Welding
D8.6-77	Standard for Automotive Resistance Spot Welding Electrodes
D8.7-78	Specification for Automotive Weld Quality - Resistance Spot Welding
D8.8-79	Specification for Automotive Frame Weld Quality - Arc Welding
D9.1-84	Specification for Welding of Sheet Metal
D10.4-79	Recommended Practices for Welding Austenitic Chromium Nickel Stainless Steel Piping and Tubing
D10.6-85	Gas Tungsten Arc-Welding of Titanium Piping and Tubing
D10.7-60	Recommended Practices for Gas Shielded Arc Welding of Aluminum and Aluminum Alloy Pipe
D10.8-78	Recommended Practices for Welding of Chromium-Molybdenum Steel Piping and Tubing

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AMERICAN WELDING SOCIETY STANDARDS (continued)

AWS Document Number	Title
D10.9-80	Specification for Qualification of Welding Procedures and Welders for Piping and Tubing
D10.10-75	Local Heat Treatment of Welds in Piping and Tubing
D10.11-80	Recommended Practices for Root Pass Welding and Gas Purging
D10.12-79	Recommended Practices and Procedures for Welding Plain Carbon Steel Pipe
D11.1-65	A Test Program on Welding Iron Castings
D14.1-85	Specification for Welding Industrial and Mill Cranes
D14.2-71	Specification for Metal Cutting Machine Tool Weldments
D14.3-82	Specification for Welding Earthmoving and Construction Equipment
D14.4-77	Classification and Application of Welded Joints for Machinery and Equipment
D14.5-80	Specification for Welding of Presses and Press Components
D14.6-81	Specification for Rotating Elements of Equipment
F1.1-85	Method for Sampling Airborne Particulates Generated by Welding and Allied Processes
F1.2-85	Laboratory Method for Measuring Fume Generation Rates and Total Fume Emission of Welding and Allied Processes
F1.3-83	Evaluating Contaminants in the Welding Environment: A Sampling Strategy Guide
F2.1-78	Recommended Safe Practices for Electron Beam Welding and Cutting
F2.2-84	Lens Shade Selector
F4.1-80	Recommended Safe Practices for the Preparation for Welding and Cutting Containers and Piping that have held Hazardous Substances
F6.1-78	Method for Sound Level Measurement of Manual Arc Welding and Cutting Processes
Z49.1-83	Safety in Welding and Cutting

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AMERICAN SOCIETY FOR TESTING AND MATERIALS DOCUMENTS

ASTM Number	Title
A275-83	Steel Forging, Magnetic Particle Examination of - 30 Jul 82
A370-77	Mechanical Testing of Steel Products
A539-85	Electric Resistance - Welded Coiled Steel Tubing for Gas and Fuel Oil Lines - 27 Jul 79
A733-76R81	Welded & Seamless Carbon Steel & Austenitic Stainless Steel Pipe Nipples - 15 Oct 76
B32-83	Metal, Solder - 30 Apr 76
B284-79	Solder, Rosin Flux Core - 26 Jan 79
B361-81	Fitting, Factory Made Wrought Aluminum & Aluminum Alloy Welding - 27 Feb 81
B486-74R85	Solder, Paste - 27 Dec 74
B594-82	Aluminum Alloy Wrought Products for Aerospace Applications, Ultrasonic Inspection of - 27 Aug 82
E94-84a	Radiographic Testing, Recommended Practice for - 25 Aug 77 (Reapproved 1983)
E142-85	Radiographic Testing, Controlling Quality of - 27 May 77
E164-81	Weldments, Ultrasonic Contact Examination of - 27 Mar 81
E165-80	Liquid Penetrant Inspection Method - 15 Dec 80
E268-84a	Electromagnetic Testing - 2 Jul 81
E269-84a	Magnetic Particle Inspection - 29 Dec 80
E270-84	Penetrant, Liquid, Inspection, Definition of Terms Relating to - 26 Nov 82

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AMERICAN SOCIETY FOR TESTING AND MATERIALS DOCUMENTS (Continued)

ASTM Number	Title
E273-83	Ultrasonic Welds, Longitudinal & Spiral, Inspection for, of Welded Pipe & Tubing
E390-75R84	Steel Fusion Welds, Standard Reference Radiographs for - 26 Dec 75
E500-85	Ultrasonic Testing, Standard Definitions of Terms Relating to - 29 Apr 74
E709-80R85	Examination, Magnetic Particle - 9 Jan 80

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AEROSPACE MATERIAL SPECIFICATIONS

AMS Document Number	Title
2631	Titanium Alloys, Ultrasonic Inspection of
2664D	Brazing - Silver, For Use Up To 800°F (425°C)
2665D	Brazing - Silver, For Use Up To 400°F (205°C)
2667C	Brazing - Silver, For Flexible Metal Hose - 600°F (315°C) Max Operating Temp.
2668C	Brazing - Silver, For Flexible Metal Hose - 400°F (200°C) Max Operating Temp.
2669C	Brazing - Silver, For Flexible Metal Hose - 800°F (425°C) Max Operating Temp.
2670F	Brazing - Copper Furnace, Carbon and Low-Alloy Steels
2671E	Brazing - Copper Furnace, Corr. & Heat Resistant Steels and Alloys
2672D	Brazing - Aluminum
2673B	Brazing - Aluminum Molten Flux (Dip)
2675E	Brazing - Nickel Alloy
2680	Welding, Electron Beam, for Fatigue Critical Applications
2681	Electron Beam Welding
2685	Welding, Metal Arc, Inert Gas, Nonconsumable Electrode (GTAW Method)
2689	Fusion Welding, Titanium and Titanium Alloys
2690A	Welding (Parallel Gap) of Microelectronic Interconnections to Thin Film Substrates
2694A	Repair Welding of Aerospace Castings
2813D	Wire, Welding, Standard Method, Packaging of
2814D	Wire, Welding, Premium Quality, Packaging of

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AEROSPACE MATERIAL SPECIFICATIONS (Continued)

AMS Document Number	Title
2815T	Wire, Welding, Line Code System
2816R	Welding Wire, Color Code System - Identification
3410F	Flux - Brazing, Silver
3411B	Flux - Brazing, Silver, High Temperature
3412B	Flux - Brazing, Aluminum
3414A	Flux - Welding, Aluminum
3415A	Flux - Aluminum Dip Brazing, 1030°F Fusion Point
3416A	Flux - Aluminum Dip Brazing, 1090°F Fusion Point
3430A	Paste, Copper, Brazing Filler Metal, Water Thinning
4184C	Wire Brazing - 10Si 4 Cu
4185B	Wire Brazing 12Si
4188	Welding Wire
4188/1	Wire, Welding, Aluminum Alloy, 4.5 Cu-0.70 Ag - 0.30 Mn - 0.25 Mg - 0.25 Ti (A201.0)
4188/2	Wire, Welding, Aluminum Alloy, 4.6 Cu - 0.35 Mn - 0.25 Mg - 0.22 Ti (A206.0)
4188/3	Wire, Welding, Aluminum Alloy, 5.0 Si - 1.2 Cu - 0.50 Mg (C355.0)
4188/4	Wire, Welding, Aluminum Alloy, 7.0 Si - 0.30 Mg (A356.0)
4188/5	Wire, Welding, Aluminum Alloy, 7.0 Si - 0.52 Mg (357.0)
4189B	Wire, Welding - 4.1Si 0.2 Mg
4190C	Wire, Welding, Aluminum Alloy, 5.2 Si (4043)
4191B	Wire, Welding - 6.3Cu 0.3Mn 018Zr 0.15Ti 0.10V
4395C	Wire, Welding, Magnesium Alloy, 9.0 Al - 2.0 Zn (AZ92A)

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AEROSPACE MATERIAL SPECIFICATIONS (Continued)

AMS Document Number	Title
4396B	Wire, Welding - 3.3Ce 2.5Zn 0.72Zr
4740	Powder, Copper 99.0 Min Cu
4750B	Solder, Tin Lead 45 Sn - 55 Pb
4751	Solder, Tin-Lead, Eutectic, 63Sn 37Pb
4755A	Solder, Lead-Silver, 94Pb 5.5 Ag
4756	Solder - 97.5Pb 1.5Ag 1Sn
4764	Brazing Alloy Copper Base 39 Mn
4765B	Silver, Brazing Filler Metal 56 Ag
4766D	Silver, Brazing Alloy 85 Ag - 15 Mn
4767A4	Silver, Brazing Alloy, 92.5 Ag - 7.2 Cu - 0.22 Li
4768E	Brazing Filler Metal - Silver, 35Ag 26Cu 217n 18Cd, 1125° - 1295°F (605° - 700°C) Solidus-Liquidus Range
4769C	Brazing Filler Metal - Silver, 45Ag 24Cd 16Zn 15Cu, 1125°-1145°F (605° - 620°C) Solidus-Liquidus Range
4770G	Metal, Brazing Filler, Silver 50 Ag - 18 Cd - 16.5 Zn - 15.5 Cu - 1160 Deg. F - 1175 Deg F (625 Deg. C - 635 Deg. C) Solidus Liquidus Range
4771E	Brazing Filler Metal - Silver, 50Ag 15Cd 15.5Zn 15.5Cu 3.0Ni, 1170° - 1270°F (630°-690°C) Solidus-Liquidus Range
4772F	Silver, Brazing Filler Metal, 54 Ag - 40 Cu, - 5.0 Zn - 1.0 Ni 1325 - 1575 F Melt, Range
4773C	Silver, Brazing Alloy 60 Ag - 30 Cu - 10 Sn -
4774C	Silver, Brazing Alloy 63 Ag - 28.5 Cu - 6.0 Sn - 2.5 Ni
4775C	Metal, Brazing Filler, Nickel 73 Ni - 4.5 Si - 14 Cr - 3.18 - 4.5 Fe 1790 Deg. F - 1970 Deg. F (975 Deg. C - 1075 Deg C) Solidus Liquidus Range

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AEROSPACE MATERIALS SPECIFICATIONS (Continued)

AMS Document Number	Title
4776B	Base, Brazing Alloy Nickel - 4 Si - 16.5 Cr - 4 Fe - 3.88 - Low Carbon
4777B	Base, Brazing Alloy Nickel - 4 Si - 7 Cr - 3 Fe - 38
4778C	Base, Brazing Alloy Nickel - 4.5 Si - 2.98
4779B	Brazing Filler Metal, Nickel, 94Ni 3.5Si 1.8B, 1800° - 1950°F (980° - 1065°C) Solidus-Liquidus Range
4780B	Base, Brazing Alloy Manganese - 16 Ni - 16 Co - 0.88
4782A	Metal, Brazing Filler, Nickel, 71 Ni - 10 Si - 19 Cr 1975 Deg - 2075 Deg. F (1080 DEg - 1135 Deg. C) Solidus Liquidus Range
4783A	Metal, Brazing Filler, High Temperature 50, Co-8.0 Si 19 Cr - 17 Ni - 0.0 W - 0.80 B 2050 Deg - 2100 Deg F - (1120 Deg - 1150 Deg C) Solidus Liquidus Range - 15 Jul 78
4784B	Metal, Brazing Filler, High Temperature 50 Au - 25 Pd - 25 Ni - 2015 Deg 2050 Deg F (1100 Deg 1120 Deg C) Solidus Liquidus Range
4785B	Metal, Brazing Filler, High Temperature 30 Au - 34 Pd - 36 Ni - 2075 Deg 2130 Deg F (1135 Deg 1165 Deg C) Solidus Liquidus Range
4786B	Metal, Brazing Filler, High Temperature 70 Au - 8.0 Pd - 22 Ni - 1845 Deg 1915 Deg F (1005 Deg C - 1045 Deg C) Solidus Liquidus Range
4787B	Metal, Brazing Filler, High Temperature 82 Au - 18 Ni 1740 Deg F (950 Deg C) Solidus Liquidus Temperature
4951E	Wire, Titanium, Welding
4953B	Wire, Welding - 5Al 2.5Sn
4954D	Wire, Welding - 6Al 4V
4995B	Wire, Welding - 8Al 1Mo 1V

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AEROSPACE MATERIALS SPECIFICATIONS (Continued)

AMS Document Number	Title
4956B	Wire, Welding - 6Al 4V, Extra Low Interstitial, Environment Controlled Packaging
5027B	Wire, Welding 1.05 Cr - 0.55 Ni - 1.0 Mo - 0.70V (0.26 - 0.32 C) Vacuum Melted, Environment Controlled Packaged
5028A	Wire, Steel Welding 1.05 Cr - 0.55 Mo - 0.07 V (0.34 - 0.40C) Vacuum Melted, Environment Controlled Packaged
5029B	Wire, Welding - 0.78Cr 1.8Ni 0.35Mo 0.20V (0.33-0.38C), Vacuum Melted, Environment Controlled Packaging
5030D	Wire, Welding - Low Carbon
5031D	Welding, Electrodes, Covered, Steel - 0.007-0.15C
5675B	Alloy Welding Wire, Corrosion & Heat Resistant 70 Ni - 2.4 Mn - 15.5 Cr - 30.1 I - 7.0 Fe
5676B	Wire, Welding, Alloy, Corrosion & Heat Resistant 80 Ni - 20 Cr
5677A	Electrodes, Covered Welding - 75Ni 19.5Cr 1.6(Cb + TA)
5679D	Alloy Welding Wire, Corrosion & Heat Resistant 73 Ni - 15.5 Cr - 2.2 (Cb + Ta) (SAE 30347)
5680E	Steel Welding Wire, Corrosion & Heat Resistant 18.5 Cr - 11 Ni - 0.40 (Cb + Ta) (SAE 30347)
5681C	Eelectrodes, Covered Welding - 19.5Cr 10.5Ni 0.60 (CB + Ta)
5683D	Wire, Welding - 75Ni 15.5Cr 8Fe
5684D	Electrodes, Covered, Alloy, Welding, 72Ni 15Cr 9Fe 2.8 (Cb + Ta)
5691E	Electrodes,1 Covered, Steel, Welding 18.5Cr 12.5Ni 2.2Mo
5692	Wire, Steel Welding, Corrosion & Heat Resistant 19 Cr - 12.5 Ni - 2.50 Mo - 1 Oct 82
5693	Wire, 18Cr9.0Ni Cryogenically Produced High Tensile Strength

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AEROSPACE MATERIALS SPECIFICATIONS (Continued)

AMS Document Number	Title
5694E	Steel Welding Wire, Corrosion & Heat Resistant 27 Cr - 21.5 Ni
5695D	Electrodes, Covered Welding - 25Cr 20Ni
5696	Wire, Welding, 19Cr 12.5Ni 2.5Mo, High Ferrite Grade
5774B	Wire, Welding - 16.5Cr 4.5Ni 2.9Mo 0.1N
5775B	Electrodes, welding, Covered, Steel, Corrosion & Moderate Heat Resistant - 16.5 Cr - 4.5 Ni - 2.9 Mo - 0.10 N - 15
5776B	Steel Welding Wire, Corrosion & Moderate Heat Resistant - 12.5 Cr (SAE 51410)
5777B	Electrodes, Steel Welding, Covered, Corrosion Resistant - 12.5 Cr
5778B	Alloy Welding Wire, Corrosion & Heat Resistant - 72 Ni - 15.5 Cr - 1.0 (Cb & Ta) - 2.4 Ti - 0.70 Al - 7.0 Fe
5779A	Alloy Welding Electrodes, Covered, Corrosion & Heat Resistant - 75 Ni - 5 Cr - 1.5 (Cb plus Ta) - 1.9 Ti - 0.55 A - 5.5 Fe
5780A	Wire, Steel, Welding, Corrosion & Moderate Heat Resistant - 15.5 Cr - 4.5 Ni - 2.9 Mo - 0.10 N - 16
5781B	Welding Electrodes, Covered, Steel Corrosion & Moderate Heat Resistant - 15.5 Cr - 4.5 Ni - 2.9 Mo - 0.10 N
5782C	Wire, Welding - 20.5Cr 9.0Ni 0.50Mo 1.5W 1.2 (Cb + Ta) 0.20Ti
5783C	Electrodes, Covered Welding - 19.5Cr 8.8Ni 0.50Mo 1.5W 1.0 (Cb +Ta)
5784A	Wire, Steel, Welding, Corrosion & Heat Resistant - 29 Cr - 9.5 Ni
5785C	Steel Welding Electrodes, Covered, Corrosion & Heat Resistant - 28.5 Cr - 9.5 Ni
5786B	Alloy Welding Wire, Corrosion & Heat Resistant - 62.5 Ni - 5.0 Cr - 24.5 Mo - 5.5 Fe

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AEROSPACE MATERIALS SPECIFICATIONS (Continued)

AMS Document Number	Title
5787A	Electrode, Covered Wedling, 63Ni 5Cr 24.5Mo 5.5Fe
5788B	Coating Alloy - Cobalt Base 6200 - 28 Cr, - 4.5 W
5789A	Wire, Alloy, Welding, Corrosion & Heat Resistant - 54 Co - 25.5 Cr - 10.5 Ni - 7.5 W
5790	Wire, Welding, 20Cr 10Ni 0.75 (Cb-Ta), High Ferrite Grade
5794B	Alloy Welding, Wire, Corrison & Heat Resistant - 31 Fe - 21 Cr - 20 Ni - 20 Ni - 20 Co - 3.0 Mo - 2.5 W - 1.0 (Cb & Ta) - 0.15 n
5795C	Electrodes, Covered Welding - 31Fe 20Cr 20Ni 20Co 3.0Mo 2.5W 1.0 (Cb + Ta)
5796A	Alloy Welding Wire, Corrosion & Heat Resistant - 52 Co - 20 Cr - 10 Ni - 15 W
5797A	Electrodes, Covered Welding - 51.5Co 20Cr 10Ni 15W
5798C	Wire, Alloy Welding, Corrosion & Heat Resistant - 47.5 Ni - 22 Cr - 1.5 Co j- 9.0 Mo - 0.60 W - 18.5 Fe
5799B	Electrode, Alloy welding, Covered, Corrosion & Heat Resistant - 48 Ni - 22 Cr - 1.5 Co - 9.0 Mo - 0.60 W - 18.5 Fe
5800B	Wire, Alloy Welding, Corrosion & Heat Resistant - 54 Ni - 19 Cr - 11 Co - 10 Mo - 3.2 Ti - 1.5 Al - 0.0068 Vacuum Melted
5801A	Alloy Welding Wire, Corrosion & Heat Resistant - 39 Co - 22 Cr - 22 Ni - 14.5 W - 0.07 La
5804C	Steel Welding Wire, Corrosion & Heat Resistant - 15 Cr - 25.5 Ni - 1.3 Mo - 2.2 Ti - 0.0068 - 0.30 V
5805D	Steel Welding Wire, Corrosion & Heat Resistant - 15 Cr - 25.5 Ni - 1.3 Mo - 2.2 Ti - 0.0068 - 0.30 V Vacuum Induction Melted
5812C	Steel Wire, Welding, Corrosion & Moderate Heat Resistant - 15 Cr - 7.1 Ni - 2.4 Mo - 1.0 Al - Vacuum Melted
5817B	Wire, Steel Welding, Corrosion & Moderate Heat Resistant - 13 Cr - 2 Ni - 3 W

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AEROSPACE MATERIALS SPECIFICATIONS (Continued)

AMS Document Number	Title
5821B	Wire, Steel Welding, Corrosion & Moderate Heat Resistant - 12 Cr (SAE 51410 Modified) Ferrite Control Grade
5823A	Wire, Welding - 1kl.8Cr 2.8Ni 1.6Co 1.8Mo 0.32V
5824A	Wire, Welding - 17 Cr 7.1Ni 1.0Al
5825C	Steel Welding Wire, Corrosion Resistant - 16.5 Cr - 4.8 Ni - 0.22 (Cb plus Ta) - 3.6 Cu
5826	Wire, Welding - 15Cr 5.1Ni 0.30 (Cb + Ta) 3.2Cu
5827B	Electrodes, Welding, Covered Steel - 16.4Cr 4.8Ni 0.22 (Cb + Ta) 3.6 Cu
5828C	Alloy Welding Wire, Corrosion & Heat Resistant - 57 Ni - 19.5 Cr - 13.5 Co - 4.2 Mo - 3.1 Ti - 1.4 Al - 0.006B, Vacuum Induction Melted
5829A	Wire, Welding 56Ni 19.5Cr 18Co 2.5Ti 1.5Al, Vac. Induc. Melted
5832C	Alloy Welding Wire, Corrosion & Heat Resistant - 52.5 Ni - 19 Cr - 3.0 Mo - 5.1 (Cb plus Ta) - 0.90 Ti - 0.50 Al - 18 Fe - Consumable Electrode or Vacuum Induction Melted
5835	Wire, Welding, Alloy, 72Ni 3.2Mn 20Cr 2.5 (Cb + Ta) 0.48Ti, Vacuum Induction Melted, Environment Controlled Packaged
5837C	Alloy Welding Wire, Corrosion & Heat Resistant - 62 Ni - 21.5 Cr - 9.0 MO - 3.7 (Cb plus Ta)
5838A	Wire, Welding, Alloy - 65Ni 16Cr 15Mo 0.30Al 0.06La
5840A	Wire, Steel Welding, Corrosion & Moderate Heat Resistant - 13 Cr - 8.0 Ni - 2.3 Mo - 1.1 Al Vacuum Melted
6452	Wire, Welding, 0.95Cr 0.20Mo (0.38-0.43C), Vacuum Melted, Environment Controlled Packaging
6457	Wire, Welding - 0.95Cr 0.20Mo (0.28-0.33C), Vac. Melted
6458E	Steel Wire, Welding - 0.65 Si - 1.25 Cd - 0.50 Mo - 0.30 V (0.28-0.33C) - Vacuum Melted

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AEROSPACE MATERIALS SPECIFICATIONS (Continued)

AMS Document Number	Title
6459A	Wire, Steel Welding - 1.0 Cr - 1.0 Mo - 0.12 V (0.18-0.23C) - Vaccum Induction Melted
6460C	Wire, Welding, Steel - 0.75 Si - 0.62 Cr - 0.20 Mo - 0.10 Zr (0.10-0.17C)
6461F	Wire, Welding - 0.95Cr 0.20V, Vacuum Melted (0.28-0.33C)
6462E	Wire, Welding - 0.95Cr 0.20V (0.28-0.33)
6463A	Wire, Welding Steel - 18.5 Ni - 8.5 Co - 5.25 Mo - 0.72 Ti - 0.10 Al - Vacuum Melted, Environment Controlled
6464C	Electrodes, Covered Welding - 1.5Mo 0.20V (0.06-0.12C)
6465A	Wire, Welding - 2.0 Cr - 10 Ni - 8.0 Co - 1.0 Mo - 0.02 Al - Vacuum Melted
6466C	Steel Welding Wire, Corrosion Resistant - 5.2 Cr - 0.55 Mo
6467C	Electrodes, Covered Welding Steel - 5Cr 0.55Mo
6468A	Wire, Welding - 1.0Cr 10Ni 3.8Co 0.45Mo 0.08V (0.14-0.17C), Vacuum Melted, Environment Controlled Packaging
6501	Wire, Welding, Maraging, 18Ni 8.0Co 4.9Mo 0.40Ti, Vacuum Melted, Environment Controlled Packaging

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AEROSPACE RECOMMENDED PRACTICE

ARP Number	Title
ARP 1317	Electron Beam Welding (Sept 75)
ARP 1330	Welding of Structures for Ground Support Equipment (Mar 74)
ARP 1331	The Chemical Milling Process (Mar 74)
ARP 1332	Wave Soldering Practice (Mar 74)
ARP 1333	Nondestructive Testing of Electron Beam Welded Joints in Titanium-Base Alloys (Mar 74)

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AMERICAN PETROLEUM INSTITUTE

API Number	Title
2550-80	Cylindrical Tanks, Upright, Standard Method for Measurement and Calibration of - 1980

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ASME BOILER AND PRESSURE VESSEL CODE

ASME Number	Title
IX-74	ASME Boiler and Pressure Vessel Code Welding & Brazing Qualifications (cancelled item)

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COMPRESSED GAS ASSOCIATION, INC.

CGA Number	Title
V1-77	Compressed Gas Cylinder Value Outlet and Inlet Connections - 1977

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AMERICAN NATIONAL STANDARDS

ANSI Number	Title
B16.9-78	Fittings, Factory Made Wrought Steel Butt Welding - 17 Jan 78
B16.18-78	Fitting, Pressure, Cast Copper Alloy Solder Point - 21 Nov 78
B16.23-76	Fittings, DWV Drainage, Cast Copper Alloy Solder Joint - 2 Feb 76
B16.29-73	Fittings, Wrought Copper & Wrought Copper Solder Joint - Drainage - 4 Sep 73
B16.29-80	Fittings, DWY, Wrought Copper & Wrought Copper Alloy Solder Joint, Drainage - 31 Jul 80
B16.34-81	Valves, Flanged & Butt Welding - 81 (w/errata 5/83)
Z49.1-67	Safety in Welding and Cutting, USA Standard - 1967

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APPENDIX C

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